Chapter 6

PHASE DIAGRAMS OF BINARY AND TERNARY FLUORIDE SYSTEMS

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1. INTRODUCTION

Investigations of the equilibrium phase diagrams of metal fluoride systems have existed as a classical scientific effort since Gibbs, Roozeboom, *et al.* made their major contributions to science around the turn of the century. It remained for the workers in the Oak Ridge Aircraft Reactor Program and later in the Molten Salt Reactor Program to develop the means to acquire definitive data in massive amounts from a variety of interrelated techniques before the complex phase diagrams pertaining to the materials aspects of the Program could be defined with rapidity and precision. Because of their success in developing the experimental methods for investigation of the fluoride systems, the Oak Ridge group is responsible for a considerable fraction of the binary and ternary systems of the metal fluorides for which detailed phase diagrams appear in the literature.

In the earlier stages of the Molten Salt Reactor Program, research in fluoride systems prior to its fruition proceeded not only in the United States, but also in Western Europe and Russia. As a consequence, reports of the metal fluoride systems began to appear in the literature at an unprecedented rate over the last few decades. These investigations were conducted to define the phase relationships in numerous binary, ternary, and quaternary systems of the metal fluorides as a means of evaluating their potential as nuclear-fuel-enriching solutions, fuel solvents, coolants,

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and fuel-reprocessing media. Within this scope lay systems that could have application to single- or two-region breeder and converter reactors, to advanced-design reactors that employ liquid lead in direct contact with fuel salt, to extractive schemes for separation of rare earths from mixed salts, and to the assessment of accident conditions that might exist if fuel solutions and coolant media were mixed. Obviously, the scope of these goals is so broad that their pursuit was necessarily intermittent; their dynamic character, inherent in such a technology, caused some of the investigations to proceed through only limited stages of development.

With the culmination of the research and development efforts required to select the most suitable combinations of fluorides for application in the technology, it is fitting that the cumulative information which has been developed be recapitulated in summary form. To this end, all of the phase diagrams known to the author that can be regarded as condensed systems, and that delineate the liquidus, have been collated and included in the present review. Excluded from the collection are the recently reported systems in which ammonium fluoride, hydrazinium fluoride, water, or the highly volatile fluorides are components, for, in general, reports on these systems are fragmentary. Information is included, if known, pertaining to structural properties of the intermediate compounds that form in binary and ternary fluoride systems, and in some cases where such information comprises the only known information for a particular system. It will be noted that occasionally in this collection structural data, obtained subsequent to the termination of phase investigations with a particular system, indicate stoichiometry of some compounds to be different from that reported previously. Since such structural determinations originate from single-crystal studies, the compound stoichiometry designated by the results of these determinations is probably correct.

The following sections of this review are developed essentially as a catalog, similar to the format of the early report, *Phase Diagrams of Nuclear Reactor Materials*, ORNL-2548 (1959). They are arranged in group combinations, I–I, I–II, I–III, etc., and by increasing atomic numbers and oxidation states within groups. Many of the phase diagrams in the current review have been included in *Phase Diagrams for Ceramists*, published by the American Ceramic Society. A number of the diagrams, especially those originating in the Russian literature, were available for reproduction here through the courtesy of the American Ceramic Society. Much of the structural data that have been collected is summarized in the report, *Crystallographic Data for Some Metal Fluorides, Chlorides, and Oxides*, D. D. Brunton, H. Insley, and R. E. Thoma, ORNL-3761 (1965).

It has not been possible to assess the accuracy of the information pertaining to the phase transitions listed in the following sections, for this can only be done experimentally. Similarly, no effort has been given to appraisal of the various experimental methods which have been employed in the determination of the phase diagrams. Except in a few instances, the data are less precise than is desired in the sophisticated researches that are now commonplace with these systems. The accuracy of most of the transitions range from ± 3 to $\pm 10°$. The reader will generally find that the accuracies given are within 5°.

Annotation of the invariant and singular equilibrium points in the phase diagrams of the fluoride systems follows an abbreviated style in which the temperature and composition of these invariant equilibrium points are designated. For example, $E = 500°$, 50% MF indicates that within the system a eutectic occurs at 500°C containing 50 mole % MF. All other designations follow the same pattern, which designates temperature in degrees centigrade and composition in mole percent, exclusively. Where crystal lattice dimensions are given, the units are in angstroms.

In the most current report of a system one may occasionally find a review of the previous investigations pertaining to the system, thus affording the reader the opportunity to trace the evolutionary stages of a system. Therefore, the practice has been adopted to include in the present collection only the most recent phase diagram of a system unless significantly disparate versions have been reported.

The reader is cautioned to note that it is commonplace in the Russian literature for authors to construct phase diagrams in terms of equivalent compositions, e.g., the composition of LiF–NaF–CaF_2 will be given diagrammatically in terms of Li_2F_2, Na_2F_2, and CaF_2. Phase diagrams for such systems are shown in their original form, for otherwise it would entail complete reconstruction (with possible attendant interpretational errors) of the phase diagram to conform to Western style.

2. BINARY SYSTEMS

The System LiF–NaF. See Fig. 1. Source: J. L. Holm, *Acta Chem. Scand.* **19**:638 (1965). $E = 649°$, 39% NaF. Results of earlier work by A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:753 (1941), are in good agreement, showing the eutectic at 40 mole % NaF with m.p. 652°C. Holm reported the maximum solid solubility of LiF in NaF to be 4 ± 0.5 mole %, as compared with the value $8 \pm 1\%$ reported by J. M. Short and R. Roy, *J. Am. Ceram. Soc.* **47**:149 (1964).

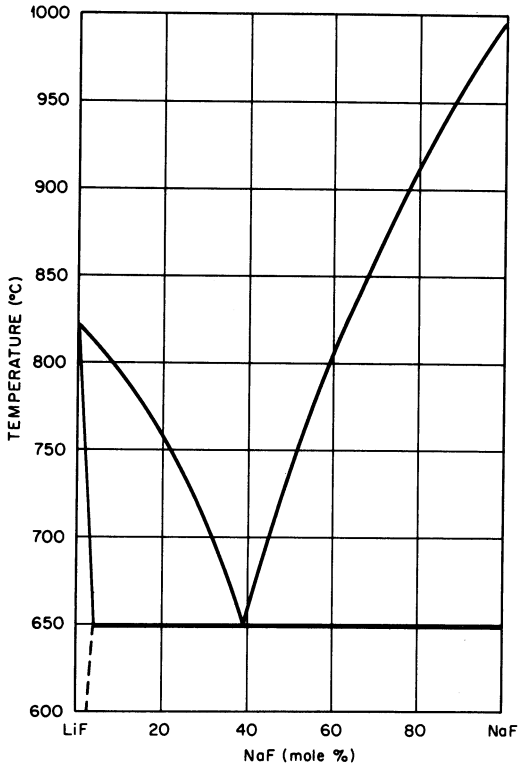


Fig. 1. The system LiF–NaF.

The System LiF–KF. See Fig. 2. Source: A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:753 (1941). $E = 492°$, 50% KF.

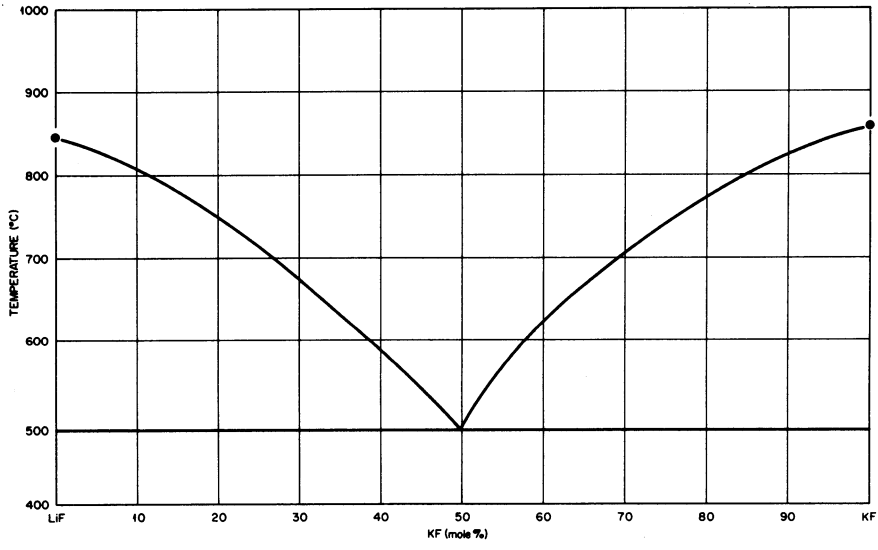


Fig. 2. The system LiF–KF.

The System LiF–RbF. See Fig. 3. Source: C. J. Barton, T. N. McVay, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1954). The eutectic is at 44 mole % LiF, m.p. 470°; the peritectic is at 47 mole % LiF, and at 475°. The data used for the construction of this phase diagram were obtained from cooling-curve experiments. The general

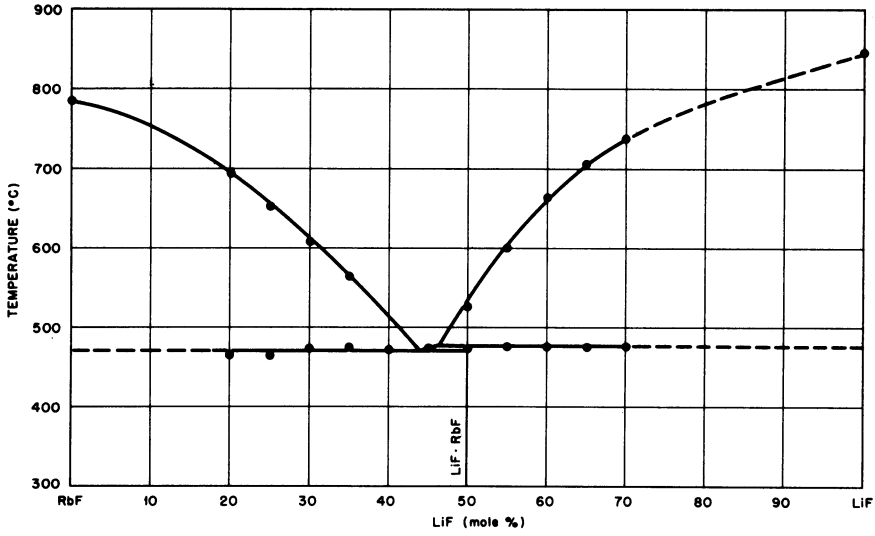


Fig. 3. The system LiF–RbF.

characteristics of the system were reported earlier by E. P. Dergunov, *Dckl. Akad. Nauk SSSR* **58**:1369 (1947). The structure of the intermediate compound, LiF · RbF, was established by Burns and Busing, *Inorg. Chem.* **4**:1510 (1965). Crystals of the compound are monoclinic, space group $C2/c$, $a = 5.83$, $b = 11.16$, $c = 7.86$, $\beta = 94°55'$.

The System LiF–CsF. See Fig. 4. Source: C. J. Barton, L. M. Bratcher, T. N. McVay, and W. R. Grimes, ORNL, unpublished work (1954), $E = 495 \pm 5°$, 55% CsF; $E = 475 \pm 5°$, 63% CsF. The data for construction of this phase diagram were obtained from cooling-curve analysis and thermal gradient quenching experiments (designated hereafter as TGQ). The structure of the intermediate compound, LiF · CsF, has been determined by J. H. Burns and W. R. Busing, *Inorg. Chem.* **4**:1510 (1965), S. G. $C2/c$, $a = 6.01$, $b = 11.64$, $c = 8.18$, $\beta = 90°45'$.

A phase diagram of the system was published by G. A. Bukhalova and D. V. Sementsova, *Zh. Neorg. Khim.* **10**:1880 (1965), showing LiF · CsF as a congruently melting compound, and with $E = 490°$, ~48% CsF, and $E = 479°$, 60% CsF. More recent phase diagrams of ternary systems limited

by the LiF · CsF binary system (Figs. 153, 164, 166, 179–181) indicate that LiF · CsF melts congruently at 494°. Although the phase diagrams of the ternary systems reported by Bukhalova and co-workers indicate conclusively that congruent melting occurs, they are based on thermal analysis, whereas inference of incongruency was based on petrographic

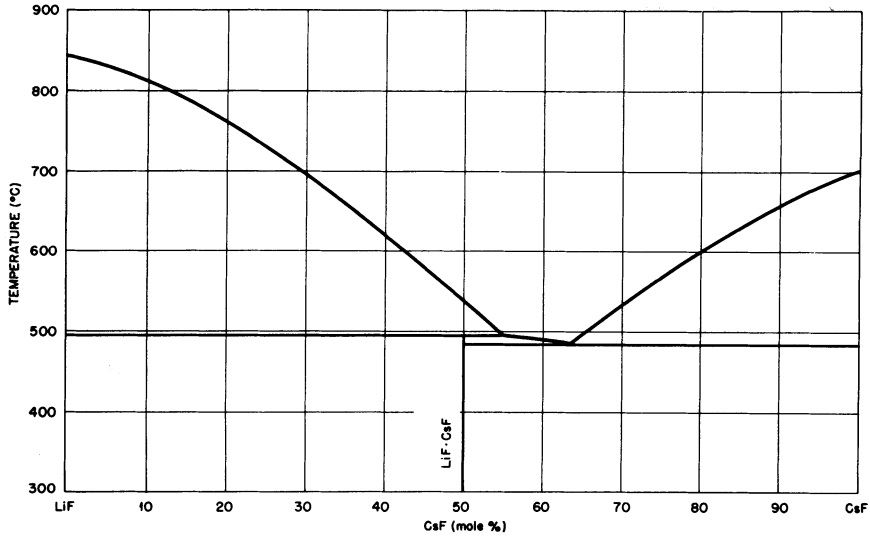


Fig. 4. The system LiF–CsF.

examination of quenched specimens as well as on thermal analysis of cooling-curve data. Evidence of the existence of two eutectics adjacent to Alkamede lines, as shown in Figs. 153, 164, and 166 leading from LiF · CsF, is strongly indicative of congruent melting of LiF · CsF. However, because of the frequency of experimental artifacts in fluoride systems, it is the reviewer's conclusion that the melting relations of LiF · CsF have not yet been established unequivocally.

The System NaF–KF. See Fig. 5. Source: J. L. Holm, *Acta Chem. Scand.* **19**:638 (1965). $E = 721°$, 60% KF. The phase diagram indicates the eutectic temperature to be 11° higher than that found by A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:753 (1941), although of the same composition. In contrast to previous investigators, Holm found that KF can contain $5 \pm 0.5\%$ NaF at the eutectic temperature. No solid solubility of KF in NaF was found.

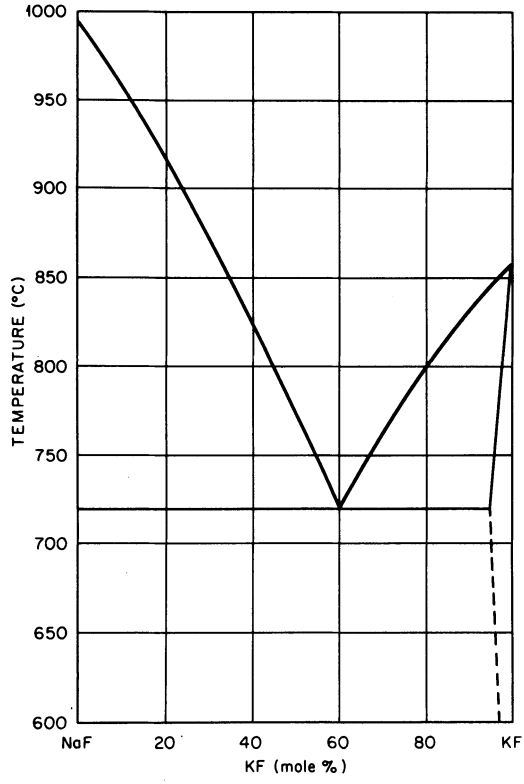


Fig. 5. The system NaF–KF.

The System NaF–RbF. See Fig. 6. Source: C. J. Barton, L. M. Bratcher and W. R. Grimes, ORNL, unpublished work (1951). Barton

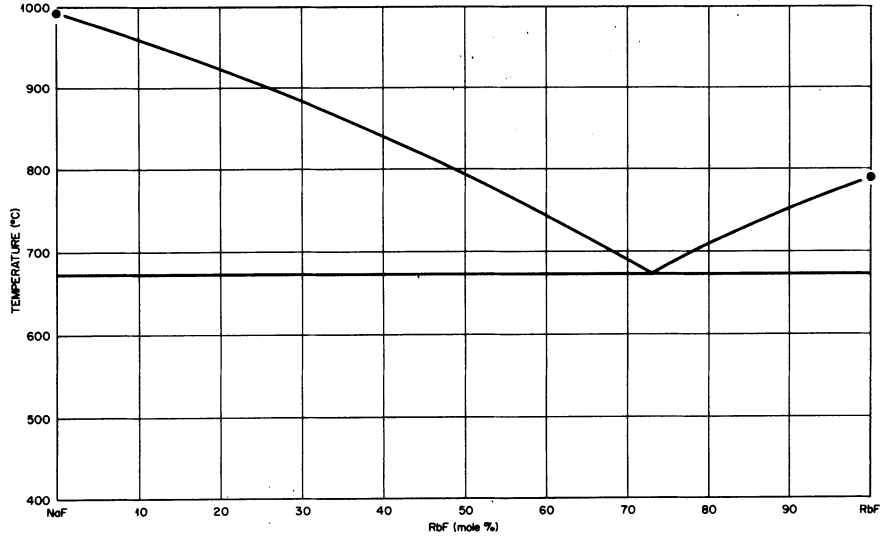


Fig. 6. The system NaF–RbF.

and co-workers found the eutectic at 73 mole % RbF, 675 ± 10°. In a more recent examination, Holm [*Acta Chem. Scand.* **19**:639 (1965)] concluded that the eutectic exists at 667° and at 67.2 mole % RbF.

The System NaF–CsF. See Fig. 7. Source: D. L. Deadmore and J. S. Machin, *J. Phys. Chem.* **64**:824 (1960). The eutectic composition was not reported, but is at approximately 24 mole % NaF. A report by R. G. Samuseva and V. E. Plyushchev, *Russ. J. Inorg. Chem.* (*English Transl.*) **6**:1093 (1961), indicates that the eutectic occurs at 20 mole % NaF and at 630°.

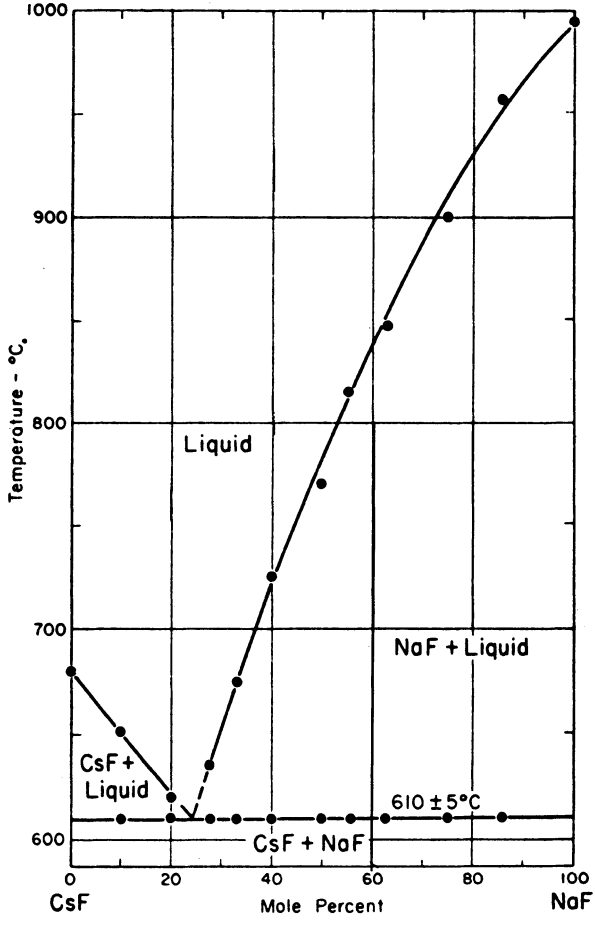


Fig. 7. The system NaF–CsF.

The phase diagram shown in this latter work exhibits anomalous negative deviation from ideality at high concentrations of NaF. The anomaly was not discussed by the authors, nor did they indicate cognizance of the 1960 diagram shown here.

The System KF–RbF. See Fig. 8. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1951). The system was found to exhibit continuous solid solution of the components with no evidence of immiscibility in the solid state. The minimum temperature of the liquidus is 770 ± 10° at 72% RbF.

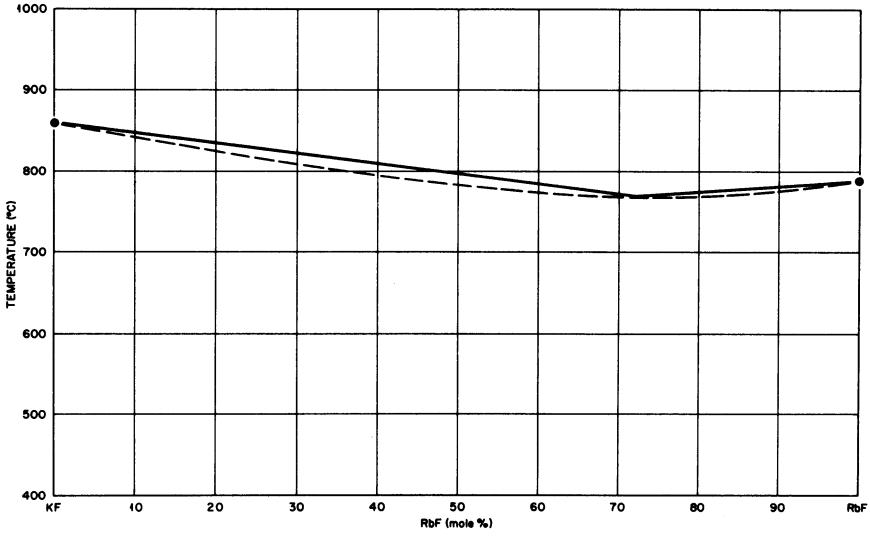
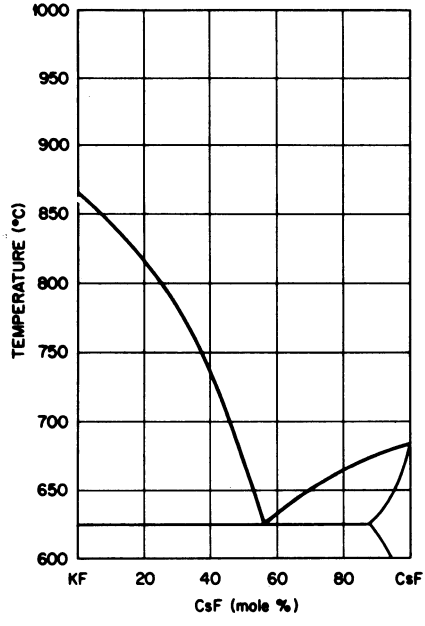


Fig. 8. The system KF–RbF.

The System KF–CsF. See Fig. 9. Source: R. G. Samuseva and V. E. Plyushchev, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:689 (1965). The

Fig. 9. The system KF–CsF.



authors inferred that CsF can accommodate up to 15 mole % KF in solid solution at the eutectic temperature.

The System RbF–CsF. See Fig. 10. Source: R. G. Samuseva and V. E. Plyushchev, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:689 (1965). The authors deduced that RbF and CsF form a continuous series of solid

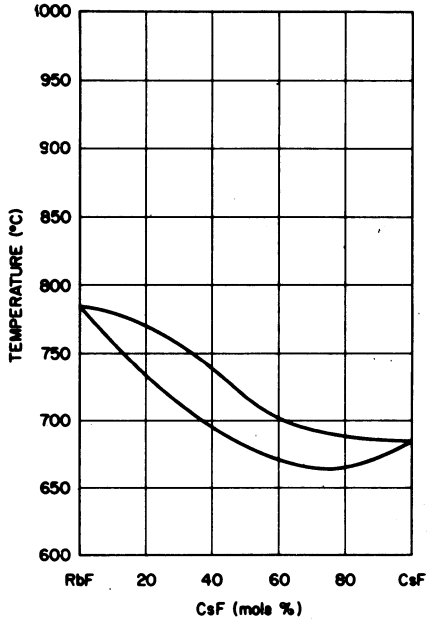


Fig. 10. The system RbF–CsF.

solutions with a temperature minimum in the liquidus. As constructed, the proposed phase diagram violates the phase rule.

The System LiF–BeF_2. See Fig. 11. Refinement of the LiF–BeF_2 phase diagram has most recently been accomplished by K. A. Romberger, J. Braunstein, and R. E. Thoma, *J. Phys. Chem.* **76**:1454 (1972) and by R. E. Thoma, H. Insley, H. A. Friedman, and G. M. Hebert, *J. Nucl. Mater.* **27**:166 (1968). In the most recent description of the structure of Li_2BeF_4, J. H. Burns and G. K. Gordon, *Acta Cryst.* **20**:135 (1966) reported Li_2BeF_4 as hexagonal, $R\bar{3}$, $a = 13.29$, $c = 8.91$. The structure of $LiBeF_3$ has not been established. Its powder pattern is listed in ORNL-3761.

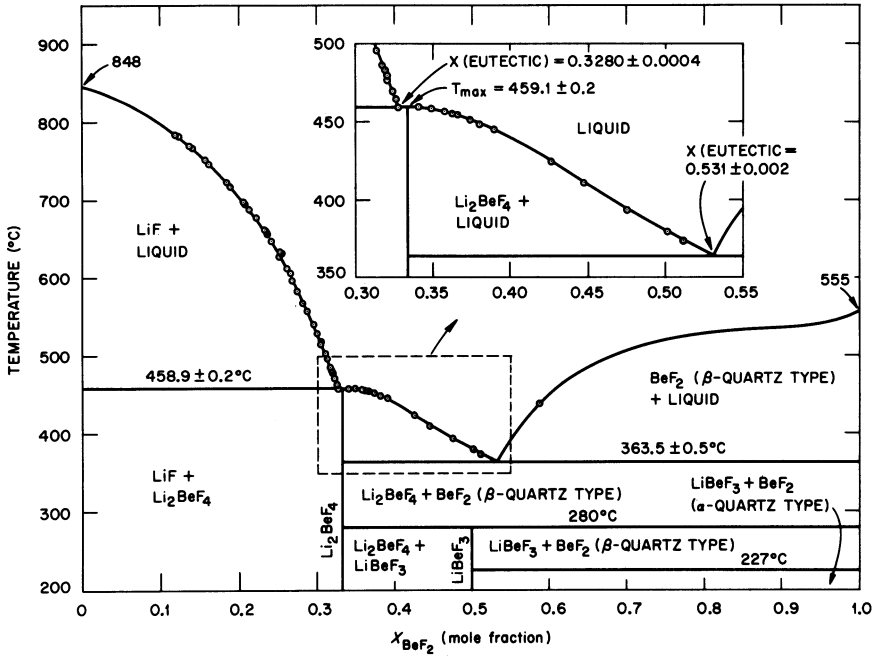


Fig. 11. The system LiF–BeF_2.

The System NaF–BeF_2. See Fig. 12. Source: D. M. Roy, R. Roy, and E. F. Osborn, *J. Am. Ceram. Soc.* **36**:185 (1953). These authors did

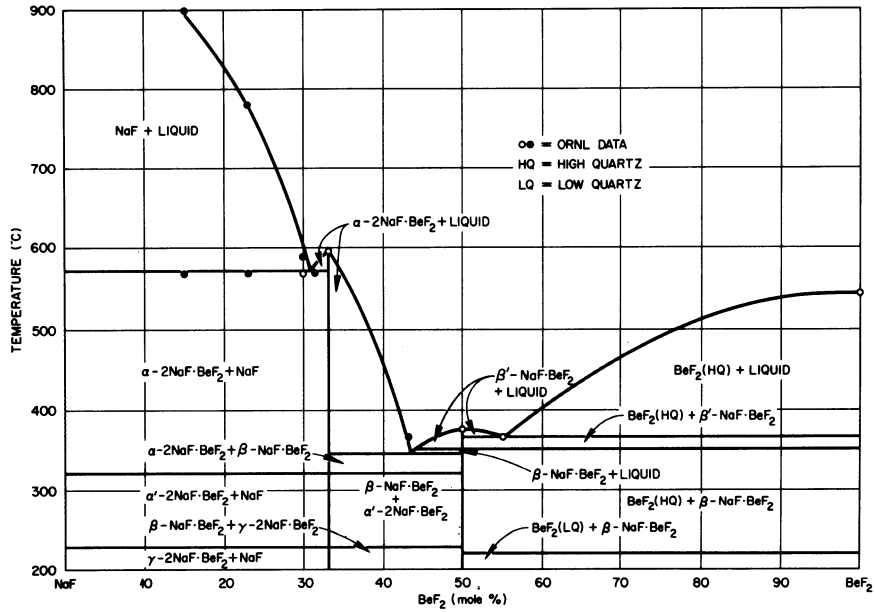


Fig. 12. The system NaF–BeF_2.

not list invariant equilibria; shown below are estimated values from the work reported and from experiments performed at ORNL. $E = 570°$, 31% BeF_2; m.p. Na_2BeF_4 600°; α-$Na_2BeF_4 \rightleftarrows \alpha'$-$Na_2BeF_4$ 320°; α'-$Na_2BeF_4 \rightleftarrows \gamma$-$Na_2BeF_4$ 225°; $E = 340°$, 43% BeF_2; m.p. $NaBeF_3$ 376 ± 5°; β-$NaBeF_3 \rightleftarrows \beta'$-$NaBeF_3 \approx 370°$; $E = 365°$, 55% BeF_2.

The System KF–BeF_2. See Fig. 13. Source: R. E. Moore, C. J. Barton, T. N. McVay, and W. R. Grimes, ORNL (unpublished), 1955. Further characterization of the phase diagram has been reported by M. P. Borsenkova, A. V. Novoselova, Yu. P. Simanov, V. I. Chernikh, and E. I. Yarembash, *Zh. Neorg. Khim.* **1**:2071 (1956), and by A. V. Novoselova, Yu. M. Korenev, and M. P. Borzenkova, *Zh. Neorg. Khim.* **9**:2042 (1964), who

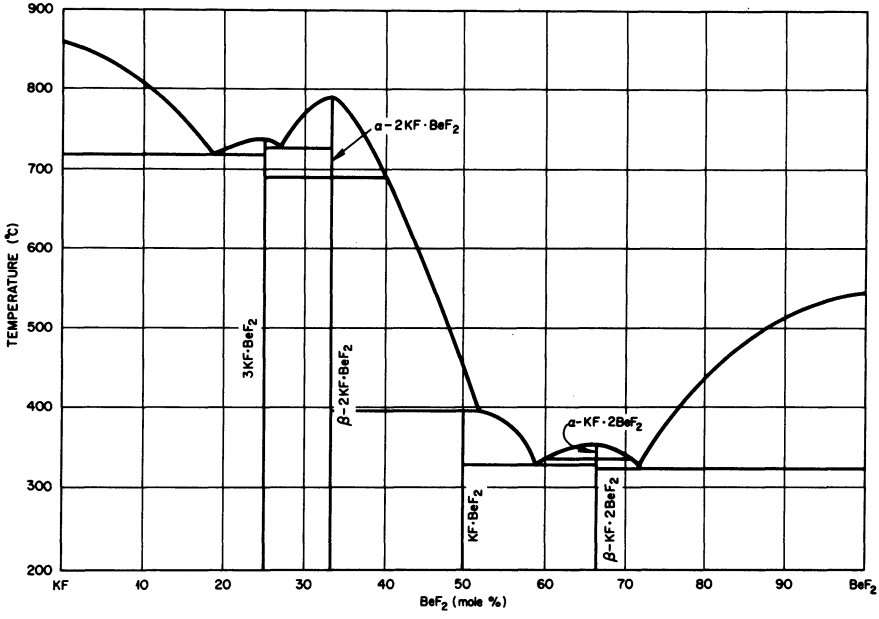


Fig. 13. The system KF–BeF_2.

described the system from 50 to 100% BeF_2. $E = 720°$, 19% BeF_2; m.p. $3KF \cdot BeF_2$ 740°; $E = 730$, 27% BeF_2; m.p. $2KF \cdot BeF_2$ 787°; α-$2KF \cdot BeF_2 \rightleftarrows \beta$-$2KF \cdot BeF_2$ 685°; $P = 390°$, 52% BeF_2; $E = 330$, 59% BeF_2; m.p. $KF \cdot 2BeF_2$ 358°; α-$KF \cdot 2BeF_2 \rightleftarrows \beta$-$KF \cdot 2BeF_2$; $E = 323°$, 72.5% BeF_2 (Borzenkova *et al.*).

Structural data: β-2KF · BeF_2: orthorhombic, $a = 5.70$, $b = 9.90$, $c = 7.28$, H. O'Daniel and L. Tscheischwili, *Z. Krist.* **104**:350 (1942).

The System RbF–BeF_2. See Fig. 14. Sources: R. E. Moore, C. J. Barton, L. M. Bratcher, T. N. McVay, G. D. White, R. J. Sheil, W. R. Grimes, and R. E. Meadows, ORNL, unpublished work (1955–1956), and R. G. Grebenshchikov, *Dokl. Akad. Nauk SSSR* **114**:317 (1957).

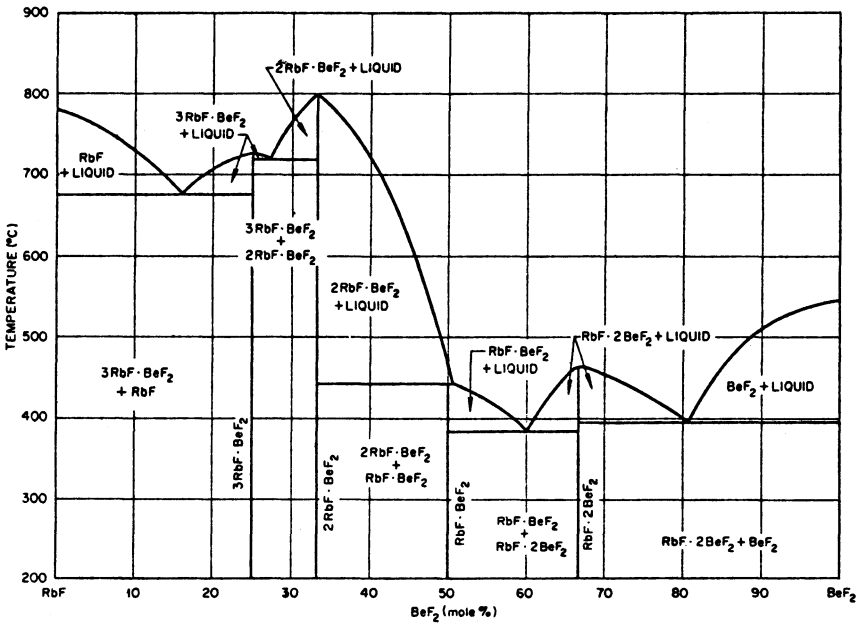


Fig. 14. The system RbF–BeF_2.

The two phase diagrams are in fair agreement, although the latter investigators have concluded that the solid-state relations are more complex than were deduced by the earlier group of workers. Invariant and singular equilibrium points found by Moore and co-workers are listed below.

The System RbF–BeF_2. See Fig. 15. $E = 675°$, 16% BeF_2; m.p. 3RbF · BeF_2 725°; $E = 720°$, 27% BeF_2; m.p. 2RbF · BeF_2 800°; $P = 442°$, 50.5% BeF_2; $E = 383°$, 61% BeF_2; m.p. RbF · $2BeF_2$ 464°; $E = 397°$, 81% BeF_2. The phase diagram as reported by Grebenshchikova is shown in Fig. 15.

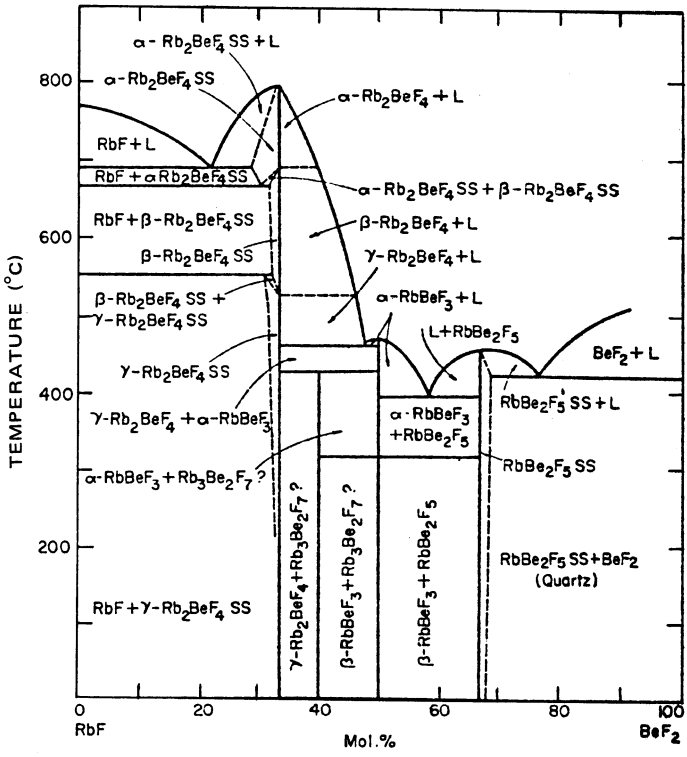


Fig. 15. The system RbF–BeF_2.

Structural data: RbF · BeF_2: triclinic, P_1 or P-1, $a = 4.69$, $b = 4.61$, $c = 6.12$, V. Ilyukhin and B. Belov, *Dokl. Akad. Nauk SSSR* **140**:1066 (1961).

The System CsF–BeF_2. See Fig. 16. Source: O. N. Breusov, A. V. Novoselova, and Yu. P. Simanov, *Dokl. Akad. Nauk SSSR* **118**:935 (1958). $E = 598°$, 14% BeF_2; $P = 659°$, 23.5% BeF_2; α-3CsF · $BeF_2 \rightleftarrows \beta$-3CsF · BeF_2 617°; m.p. 2CsF · BeF_2 793°; α-2CsF · $BeF_2 \rightleftarrows \beta$-2CsF · BeF_2 404°; $E = 449°$, 48% BeF_2; m.p. CsF · BeF_2 475°; α-CsF · $BeF_2 \rightleftarrows \beta$-CsF · BeF_2 360°; β-CsF · $BeF_2 \rightleftarrows \alpha$-CsF · BeF_2 140°; $E = 393°$, 58.4% BeF_2; m.p. CsF · $2BeF_2$ 480°; α-CsF · $2BeF_2 \rightleftarrows \beta$-CsF · $2BeF_2$ 450°; $E = 367°$, 77.5% BeF_2.

Structural data: The compound $CsBeF_3$ was reported by H. Steinfink and G. D. Brunton, *Acta Cryst.* **B24**:807 (1968) to crystallize in the space group *Pnma* with $a = 4.828$, $b = 6.004$, and $c = 12.794$.

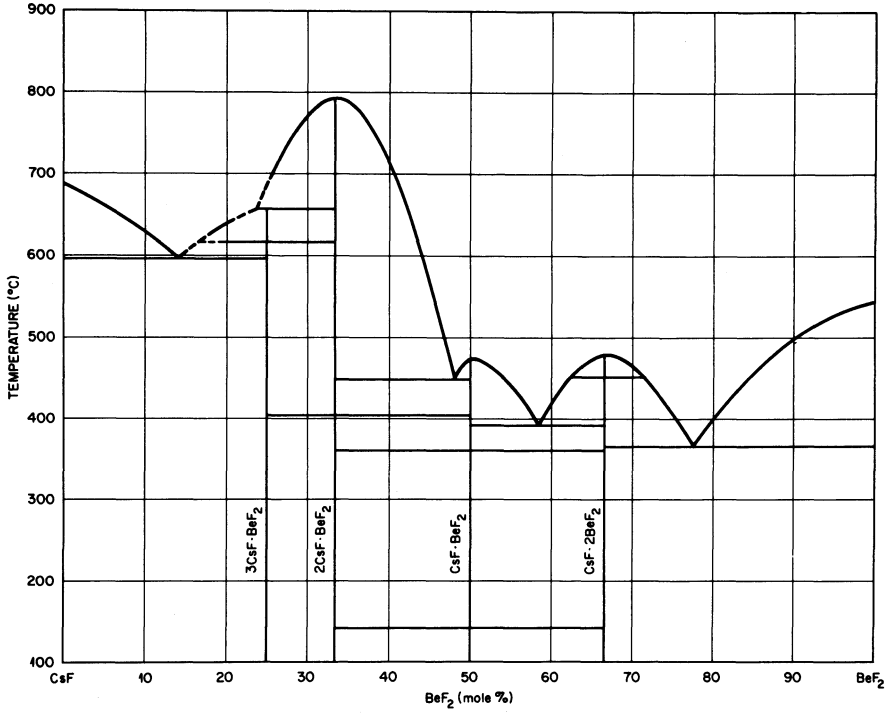


Fig. 16. The system $CsF–BeF_2$.

The System LiF–MgF_2. See Fig. 17. Source: W. E. Counts, Rustum Roy, and E. F. Osborn, *J. Am. Ceram. Soc.* **36**, 15 (1953). $E = 735°$, 36% MgF_2. The mutual solubility limits of the components were not given. The solidus between ~32 and ~38% MgF_2 should be regarded as horizontal so as to conform to the phase rule.

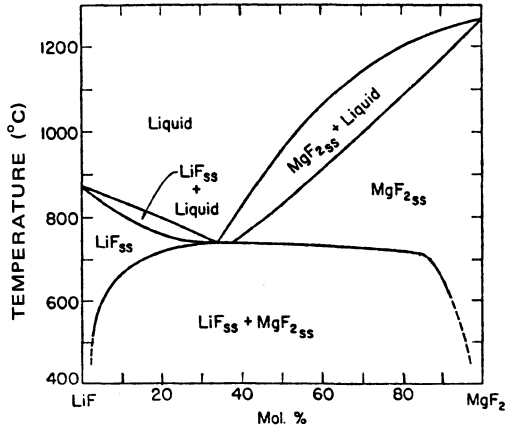


Fig. 17. The system $LiF–MgF_2$.

The System NaF–MgF_2. See Fig. 18. Source: A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:755 (1941). $E = 830°$, 25% MgF_2; m.p. NaF · MgF_2 1030°; $E = 1000°$, 64% MgF_2.

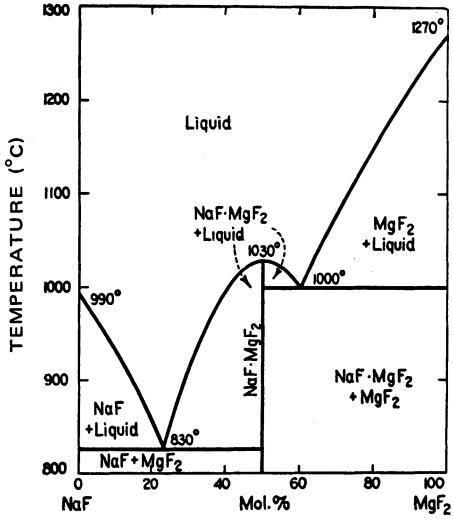


Fig. 18. The system NaF–MgF_2.

The System KF–MgF_2. See Fig. 19. Source: R. C. DeVries and Rustum Roy, *J. Am. Chem. Soc.* **75**:2841 (1953). $E = 778$, 15% MgF_2;

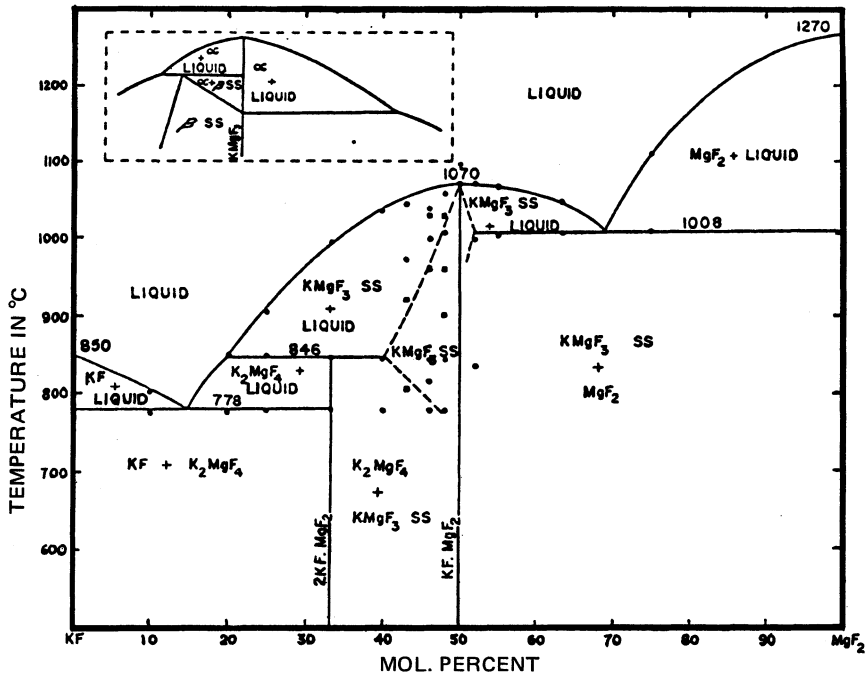


Fig. 19. The system KF–MgF_2.

$P = 846°$, 20.5% MgF_2; congruent m.p. $KF \cdot MgF_2$ 1070°; $E = 1008°$, 69% MgF_2. A possible alternative to the phase relationships of $KF \cdot MgF_2$ is shown in the inset.

Structural data: The compound $KF \cdot MgF_2$ was reported by W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), as a monoclinic structure, $a = b = c = 7.98$, $\beta = 91°18'$.

The System CsF–MgF_2. The CsF–MgF_2 phase diagram has not been described. The structure of the intermediate compound, $4CsF \cdot 3MgF_2$, has been reported by H. Steinfink and G. D. Brunton, *Inorg. Chem.* **8**:1665 (1968), as orthorhombic *Cmca*, $a = 6.1333$, $b = 14.561$, $c = 13.653$. W. L. W. Ludekens and A. J. E. Welch have reported the structure of $CsF \cdot MgF_2$, *Acta Cryst.* **5**:841 (1952), as tetragonal, $a = 9.37$, $c = 8.70$.

The System LiF–CaF_2. See Fig. 20. Source: D. L. Deadmore and J. S. Machin, *J. Phys. Chem.* **64**:824 (1960). The eutectic composition was not reported in the source reference; it is approximately 23 mole % CaF_2.

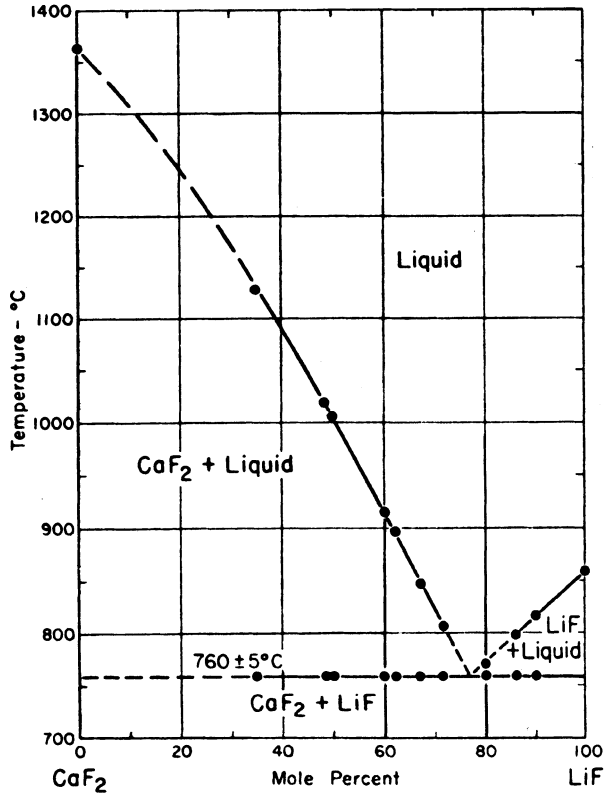


Fig. 20. The system LiF–CaF_2.

A phase diagram of the system was reported earlier by W. E. Roake, *J. Electrochem. Soc.* **104**:661 (1957). The reference diagram differs from the earlier version principally by adoption of ~1360°C as the m.p. of CaF_2 rather than ~1430°C as used by Roake.

The System NaF–CaF_2. See Fig. 21. Source: C. J. Barton, L. M. Bratcher and W. R. Grimes, ORNL, unpublished work (1955–1956).

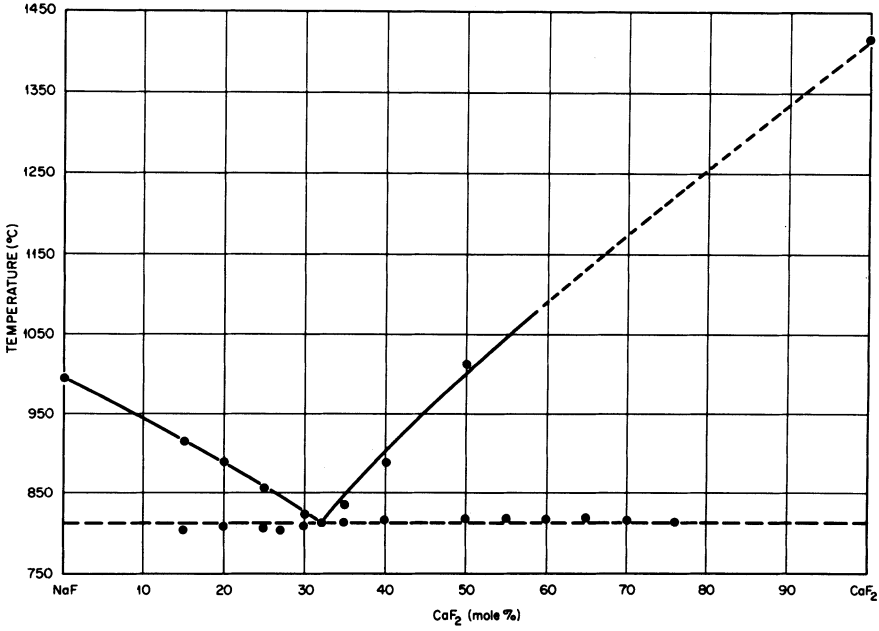


Fig. 21. The system NaF–CaF_2.

$E = 818°$, 32.5% CaF_2. In earlier work, P. P. Fedotieff and W. P. Iljinskii, *Z. Anorg. Chem.* **129**:101 (1923), reported a phase diagram of the system that is in close agreement with Fig. 21.

The System KF–CaF_2. See Fig. 22. Source: Mohammed Ishaque, *Bull. Soc. Chim. France* **1952**:130. $E = 782°$, 15.4% CaF_2; Cong. m.p. of $KF \cdot CaF_2$, 1068°; $E = 1060$, 62% CaF_2.

Structural data: The compound $KF \cdot CaF_2$ was reported by W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), as monoclinic, $a = b = c = 8.78$, $\beta = 92°36'$.

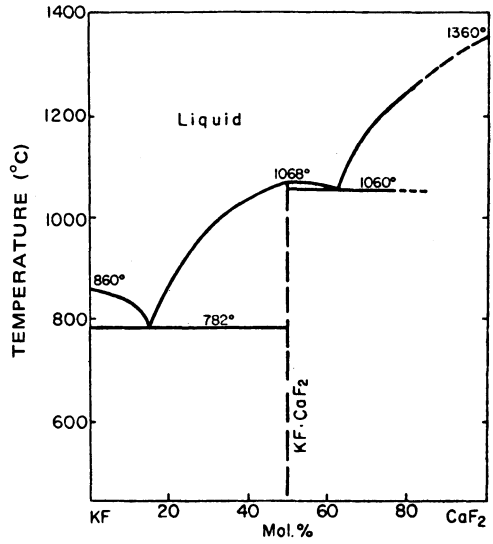


Fig. 22. The system $KF–CaF_2$.

The System $RbF–CaF_2$. See Fig. 23. Source: C. J. Barton, L. M. Bratcher, R. J. Sheil, and W. R. Grimes, ORNL, unpublished work (1956). $E = 7600$, 9% CaF_2; m.p. $RbF \cdot CaF_2$ 1110°; $E = 1090°$, 57% CaF_2.

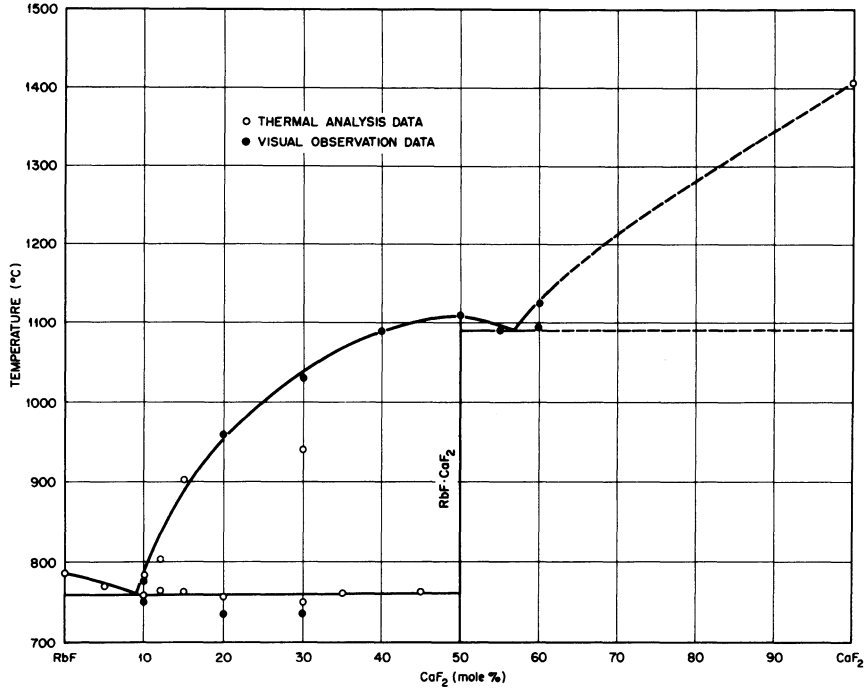


Fig. 23. The system $RbF–CaF_2$.

Structural data: The compound $RbF \cdot CaF_2$ was reported by W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), as cubic, with $a = 4.434$.

The System LiF–ZnF_2. See Fig. 24. Source: O. Schmitz-Dumont and Horst Bornefeld, *Z. Anorg. Allgem. Chem.* **287**:121 (1956). $E = 620°$, 64% ZnF_2. The minimum in the solid solution curve ("eutectic") is listed

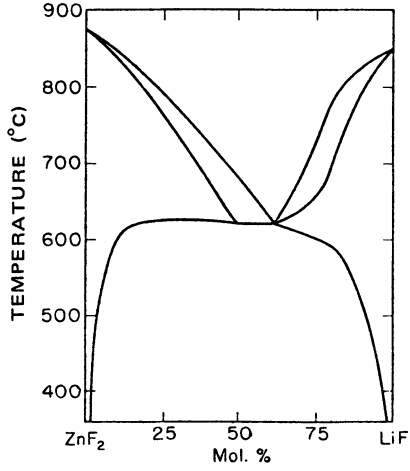


Fig. 24. The system LiF–ZnF_2.

as 620°C. The reviewer has estimated the compositions to be 64 mole % ZnF_2. The solid exsolution curve between ~10 and 20% LiF should be regarded as having zero or negative slope so as to conform to the phase rule.

The System NaF–ZnF_2. See Fig. 25. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1952). Invariant and singular equilibria: $E = 6350$, 33% ZnF_2; m.p. $NaF \cdot ZnF_2$ 748 ± 10%; $E = 685°$, 69% ZnF_2. In the initial report of phase equilibria in the NaF–ZnF_2 system, N. Puschin and A. Baskow, *Z. Anorg. Chem.* **81**:347 (1913), described $NaF \cdot ZnF_2$ as melting incongruently. A paper by O. Schmitz-Dumont and Horst Bornefeld, *Z. Anorg. Allgem. Chem.* **287**:120 (1956), includes x-ray diffraction data for the intermediate compound, $NaF \cdot ZnF_2$. This compound was reported by W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), to crystallize with a tetragonal structure, $a = 7.72$, $c = 8.11$.

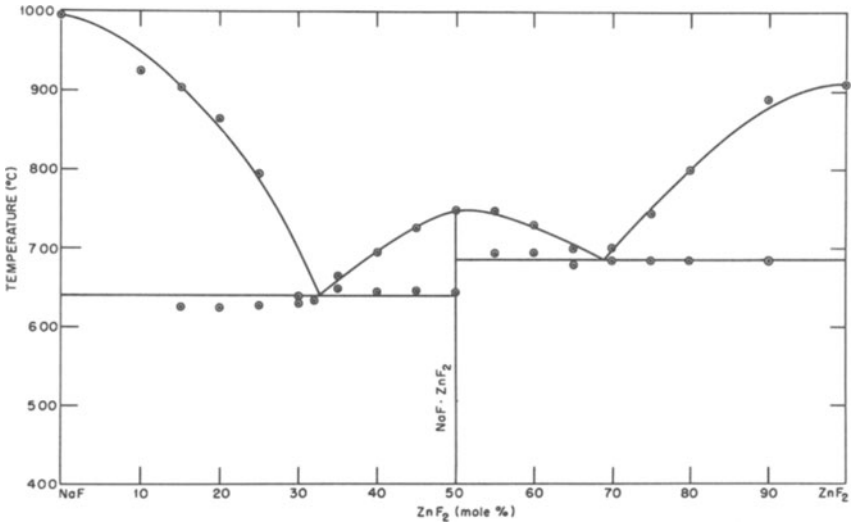


Fig. 25. The system $NaF–ZnF_2$.

The System $KF–ZnF_2$. See Fig. 26. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1952). Invariant

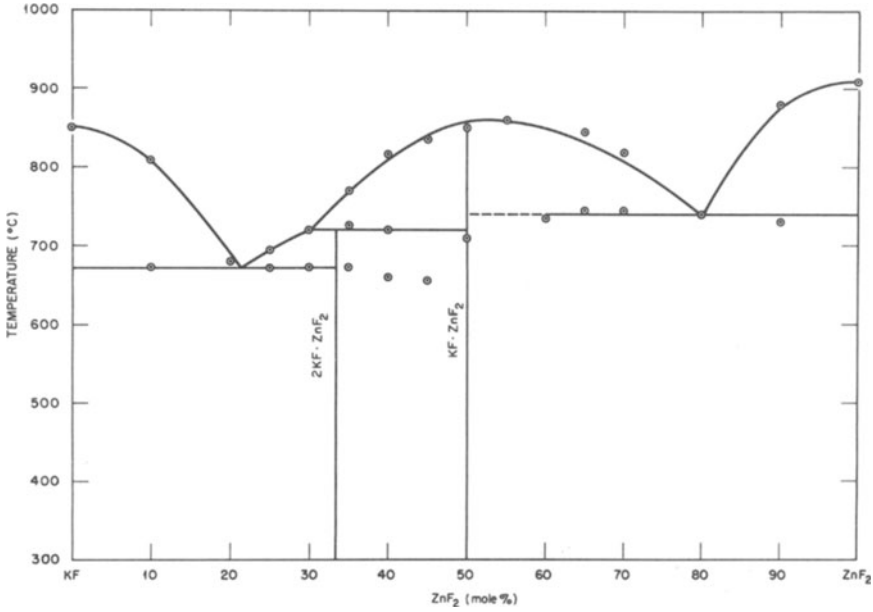


Fig. 26. The system $KF–ZnF_2$.

and singular equilibria: $E = 670$, 21% ZnF_2; $P = 720 \pm 10°$, 30% ZnF_2; m.p. $KF \cdot ZnF_2$ 850 $\pm$ 10°; $E = 740 \pm 5°$, 80% ZnF_2. In a subsequent investigation, O. Schmitz-Dumont and Horst Bornefeld, *Z. Anorg. Allgem. Chem.* **287**:122 (1956), concluded that $KZnF_3$ melts congruently at 870°C, and that K_2ZnF_4 melts incongruently at 737°C. They indexed the x-ray powder diffraction pattern of K_2ZnF_4 as tetragonal, $a = 4.009$, $c = 13.02$. G. D. Brunton *et al.* (ORNL-3761) reported $KZnF_3$ as cubic, $Pm3m$, $a = 4.056$. W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), have reported the structure of $KZnF_3$ as tetragonal, $a = 8.47$, $c = 8.07$.

The System RbF–ZnF_2. See Fig. 27. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1952). Invariant and singular equilibria: $E = 595 \pm 10°$, 21% ZnF_2; $P = 620 \pm 10°$, 32% ZnF_2; m.p. $RbF \cdot ZnF_2$ 730 $\pm$ 10%; $E = 650$, 70% ZnF_2. In a subsequent investigation, O. Schmitz-Dumont and Horst Bornefeld, *Z. Anorg. Allgem. Chem.* **287**:122 (1956), reported the melting temperatures of $2RbF \cdot ZnF_2$ and $RbF \cdot ZnF_2$ to be 640° and 740°, respectively. The liquidus profile reported by Dumont and Bornefeld is in close agreement with Fig. 27, with respect to compositions of the invariant equilibrium points.

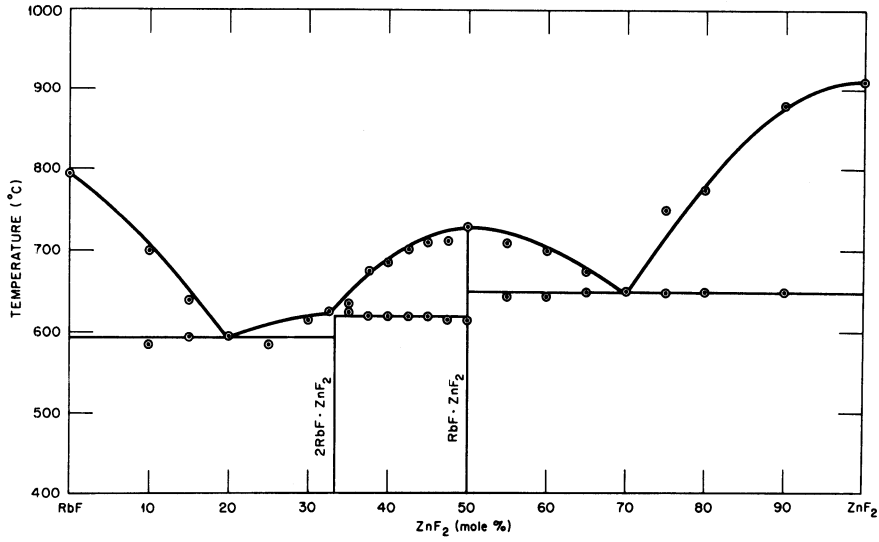


Fig. 27. The system RbF–ZnF_2.

Structural data: W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), have reported the structure of $RbZnF_3$ as tetragonal, $a = 8.67$, $c = 9.01$.

The System $CsF–ZnF_2$. See Fig. 28. Source: O. Schmitz-Dumont and Horst Bornefeld, *Z. Anorg. Allgem. Chem.* **287**:122 (1956). Invariant equilibria described by the source reference include only the following: $CsZnF_3$–β-Cs_2ZnF_4 eutectic occurs at 43 mole % CsF and at 554°; incongruent melting point of $CsZn_2F_5$ 618°; Cs_2ZnF_4 melting point 633°; Cs_2ZnF_4 crystal transition temperature 633°.

Structure data: W. L. W. Ludekens and A. J. E. Welch, *Acta Cryst.* **5**:841 (1952), have reported the structure of $CsZnF_3$ as tetragonal, $a = 9.86$, $c = 9.01$.

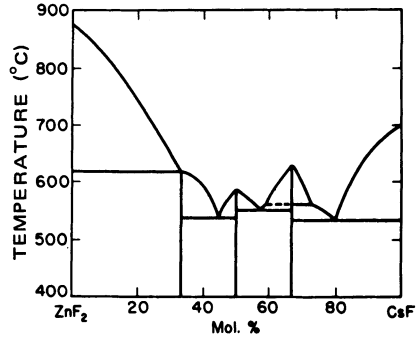


Fig. 28. The system $CsF–ZnF_2$.

The System $AgF–ZnF_2$. See Fig. 29. Source: R. C. DeVries and Rustum Roy, *J. Am. Chem. Soc.* **75**:24 (1953). The minimum in the solid solution formed from AgF and $AgF \cdot ZnF_2$ occurs at 380° and at 14 mole % ZnF_2. $E = 630$, 58% ZnF_2.

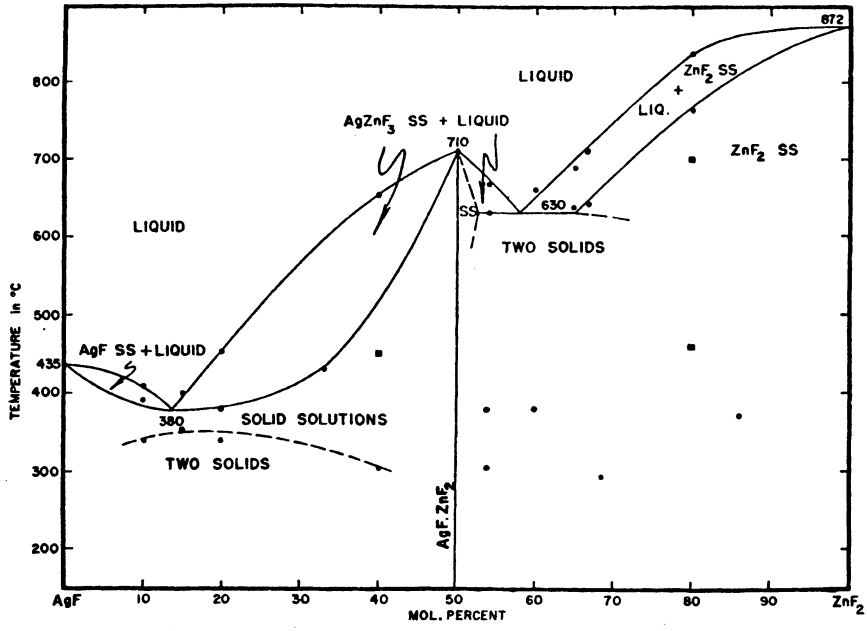


Fig. 29. The system $AgF–ZnF_2$.

The Systems $RbF–SrF_2$ and $CsF–SrF_2$. See Figs. 30 and 31. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English*

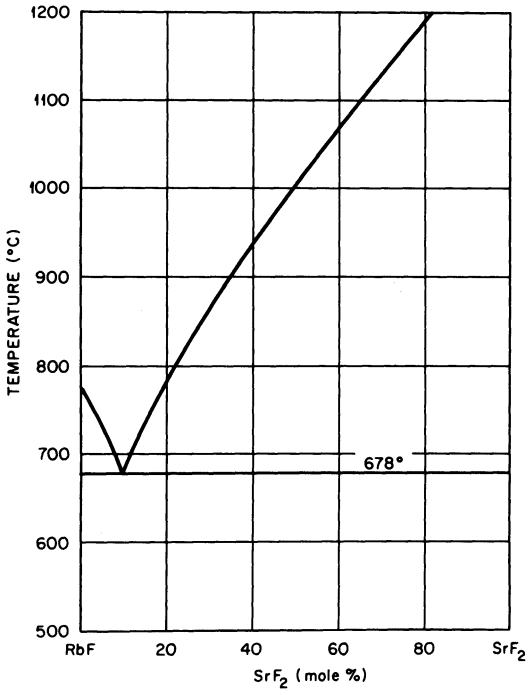


Fig. 30. The system $RbF–SrF_2$.

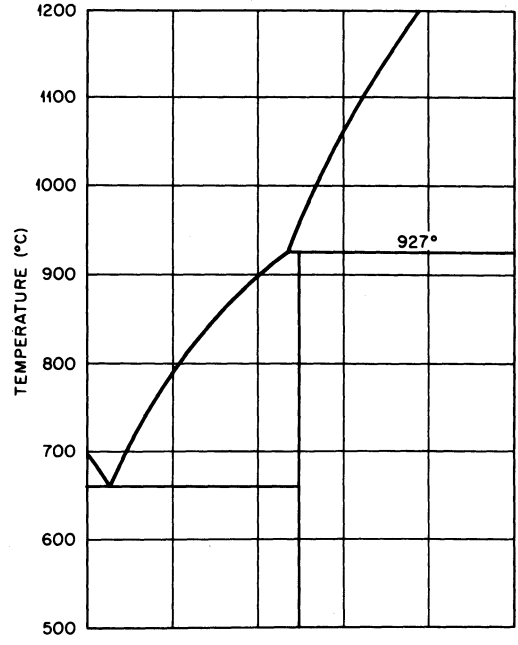


Fig. 31. The system $CsF–SrF_2$.

Transl.) **12**:1148 (1967). $RbF–SrF_2$: $E = 678°$, 10% SrF_2 (est.). $CsF–SrF_2$: $E = 660°$, 5% SrF_2, $P = 927°$, 48% SrF_2 (est.).

The System $NaF–CdF_2$. See Fig. 32. Source: N. A. Puschin and A. V. Baskow, *Z. Anorg. Chem.* **81**:359 (1913). $E = 660°$, 47.5% CdF_2.

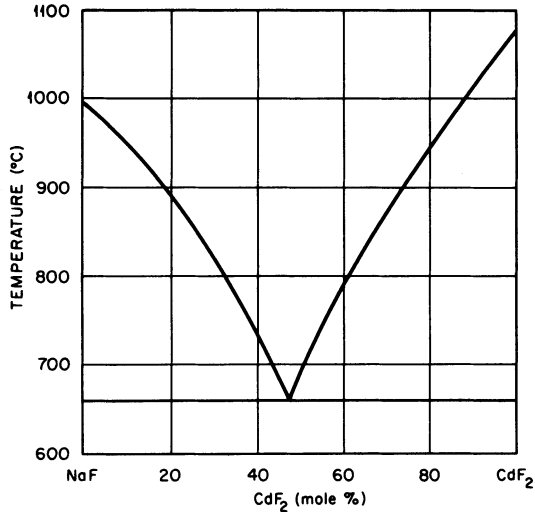


Fig. 32. The system $NaF–CdF_2$.

The System $KF–CdF_2$. See Fig. 33. Source: J. C. Cousseins, *Compt. Rend. Acad. Sci. Paris, Ser. C* **263**:585 (1966). $E = 694°$, 21% CdF_2;

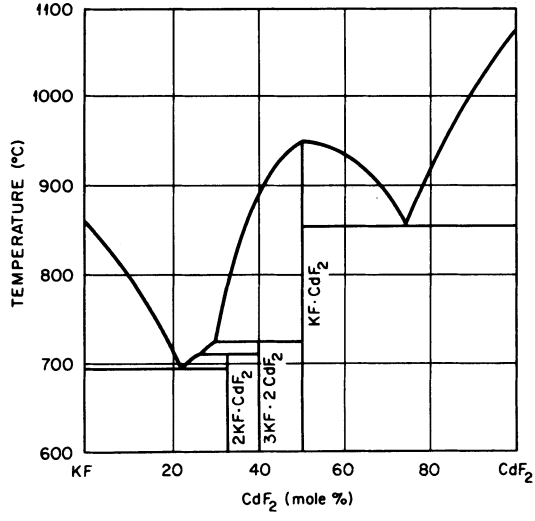


Fig. 33. The system $KF–CdF_2$.

$P = 710°$, 23% CdF_2; $P = 725°$, 25% CdF_2; m.p. $KF \cdot CdF_2$ 949°; $E = 855°$, 77.5% CdF_2.

The System RbF–CdF_2. See Fig. 34. Source: J. C. Cousseins and C. P. Perez, *Compt. Rend. Acad. Sci. Paris, Ser. C* **264**:2060 (1967). $E = 713°$, 12% CdF_2; $P = 792°$, 32% CdF_2; $P = 800°$, 32.5% CdF_2; m.p. $RbF \cdot CdF_2$ 1018°; $E = 903°$, 77% CdF_2. The phase diagram is based on

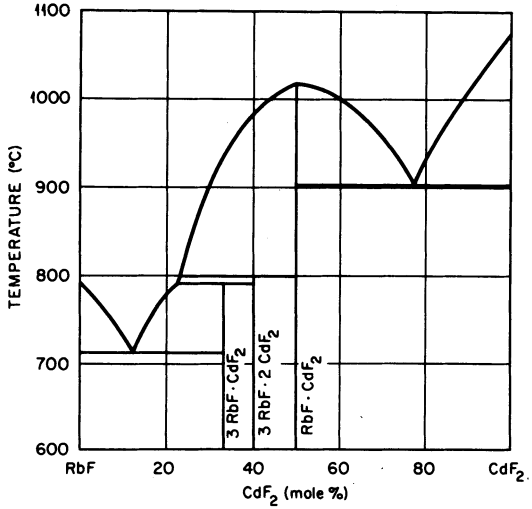


Fig. 34. The system RbF–CdF_2.

DTA and x-ray diffraction analysis. The compounds $2RbF \cdot CdF_2$ and $3RbF \cdot 2CdF_2$ are described as tetragonal, while $RbF \cdot CdF_2$ is noted to be cubic (perovskite structure), $a = 4.398$.

The System MF–BaF_2. See Figs. 35–38. Source: G. A. Bukhalova and E. S. Yagub'yan, *Izv. Akad. Nauk SSSR, Neorg. Mater.* **3**:1096 (1967), and V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **12**:1148 (1967). LiF–BaF_2: $E = 765°$, 16.5% BaF_2; NaF–BaF_2: $E = 812°$, 26% BaF_2 (est.); KF–BaF_2: $E = 729°$, 30% BaF_2 (est.); CsF–BaF_2: $E = 628°$, 16% BaF_2.

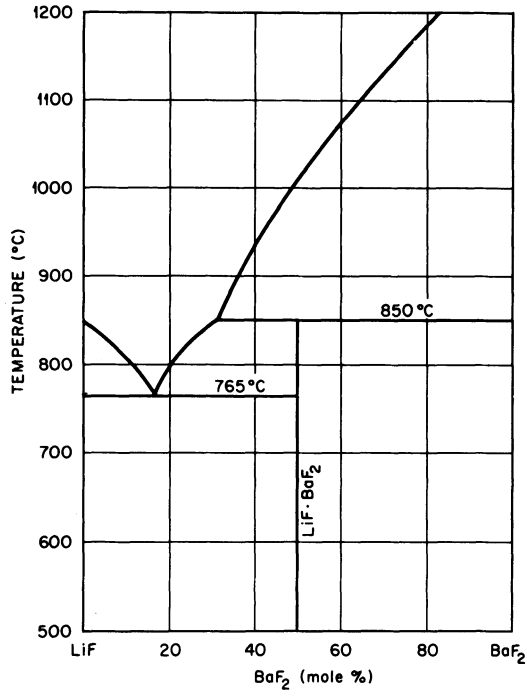


Fig. 35. The system $LiF–BaF_2$.

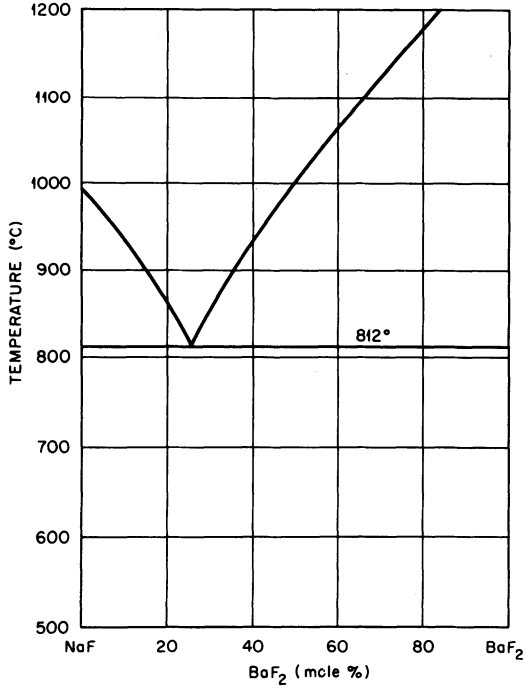


Fig. 36. The system $NaF–BaF_2$.



Fig. 37. The system $KF–BaF_2$.

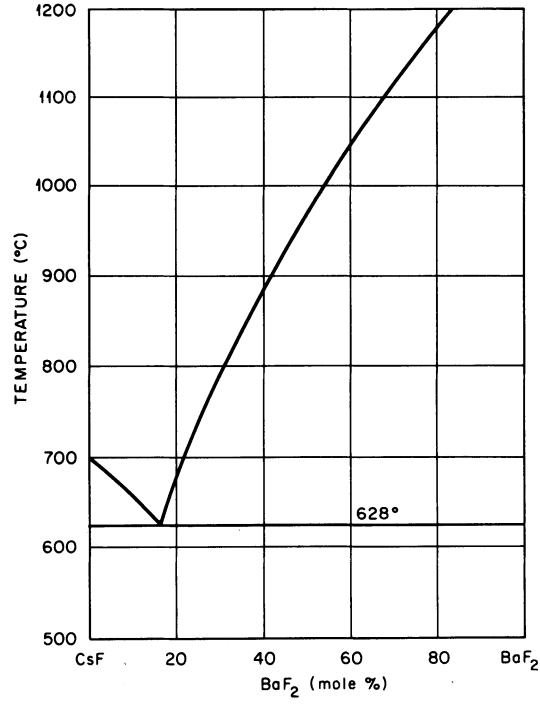


Fig. 38. The system $CsF–BaF_2$.

The System NaF–SnF_2. See Fig. 39. Source: J. D. Donaldson *et al.*, *J. Chem. Soc.* **1965**(714):3876. Invariant and singular equilibria were not

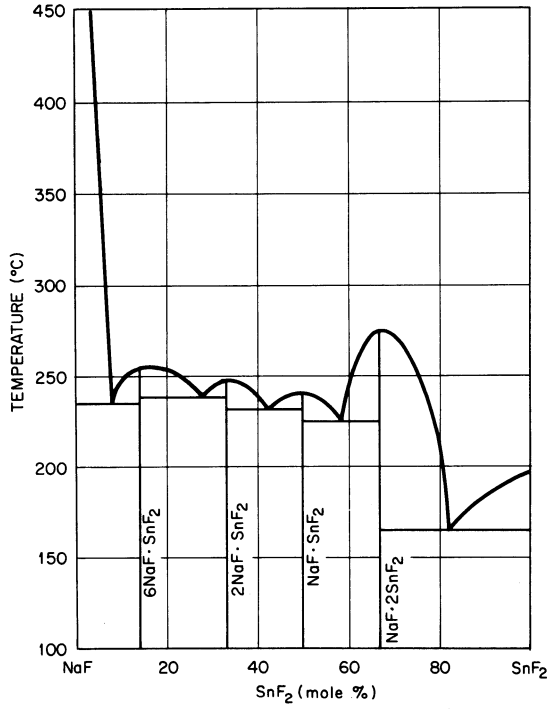


Fig. 39. The system NaF–SnF_2.

defined. Melting points of the intermediate compounds were given as follows: $6NaF \cdot SnF_2$ 279°; $2NaF \cdot SnF_2$ 273°; $NaF \cdot SnF_2$ 265°; $NaF \cdot 2SnF_2$ 299°.

The System KF–SnF_2. See Fig. 40. Source: J. D. Donaldson *et al.*, *J. Chem. Soc.*, **1965**(714):3876. Invariant and singular equilibria were not defined. Melting points of the intermediate compounds were given as follows: $6KF \cdot SnF_2$ 325°; $2KF \cdot SnF_2$ 323°; $KF \cdot SnF_2$ 310°; $KF \cdot 2SnF_2$ 354°.

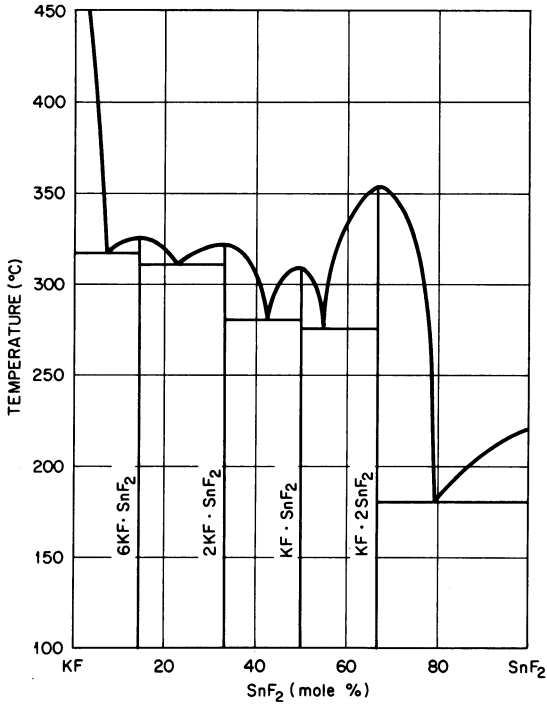


Fig. 40. The system $KF–SnF_2$.

The System $NaF–PbF_2$. See Fig. 41. Source: O. Schmitz-Dumont and Gunter Bergerhoff, *Z. Anorg. Allgem. Chem.* **283**:317 (1956). $E = 520°$, 66% PbF_2.

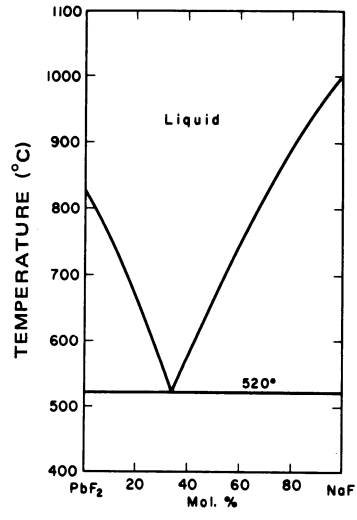


Fig. 41. The system $NaF–PbF_2$.

The System KF–PbF_2. See Fig. 42. Source: O. Schmitz-Dumont and Gunter Bergerhoff, *Z. Anorg. Allgem. Chem.* **283**:317 (1956). $P = 475°$, composition not given; $E = 465°$, 42 % PbF_2; $P = 645°$, composition not given. The maximum solubility of KF in crystalline PbF_2 is 15 mole %.

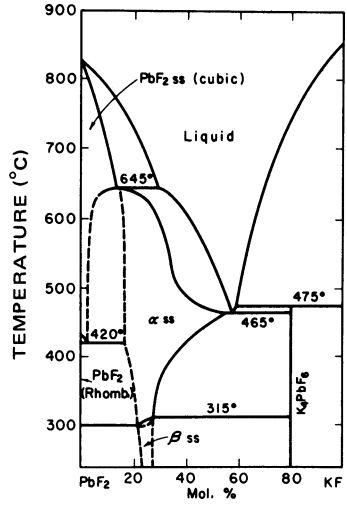


Fig. 42. The system KF–PbF_2.

The System RbF–PbF_2. See Fig. 43. Source: O. Schmitz-Dumont and Gunter Bergerhoff, *Z. Anorg. Allgem. Chem.* **283**:318 (1956). $P = 600°$, ~70 mole % PbF_2; $E = 550°$, 55 % PbF_2; congruent m.p. of RbF · PbF_2 560°; $E = 485°$, ~30 % PbF_2.

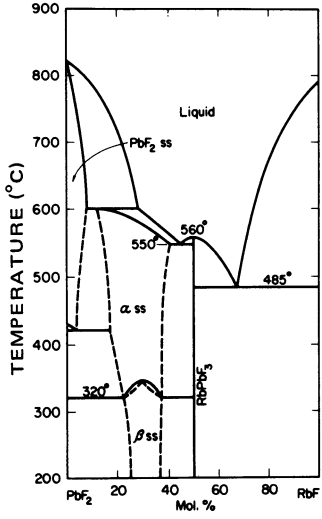


Fig. 43. The system RbF–PbF_2.

The System CsF–PbF_2. See Fig. 44. Source: O. Schmitz-Dumont and Gunter Bergerhoff, *Z. Anorg. Allgem. Chem.* **283**:319 (1956). $E = 500°$, 28% PbF_2; $P = 615°$, 40% PbF_2; congruent m.p. of $CsF \cdot PbF_2$ 725°; $P = 615°$, 60% PbF_2; $E = 565°$, 71% PbF_2.

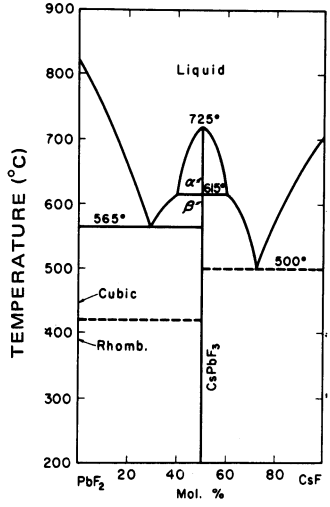
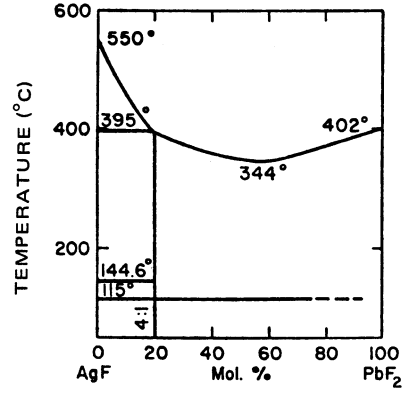


Fig. 44. The system CsF–PbF_2.

The System AgF–PbF_2. See Fig. 45. Source: C. Tubandt and S. Eggert, *Z. Anorg. Allgem. Chem.* **110**:223 (1920). The phase diagram as constructed in the source reference does not obey the phase rule.

Fig. 45. The system AgF–PbF_2.



The Alkali Fluoride–MnF_2 Binary Systems. See Figs. 46–50. Source: I. N. Belyaev and O. Ya. Revina, *Russ. J. Inorg. Chem.* (*English Transl.*)

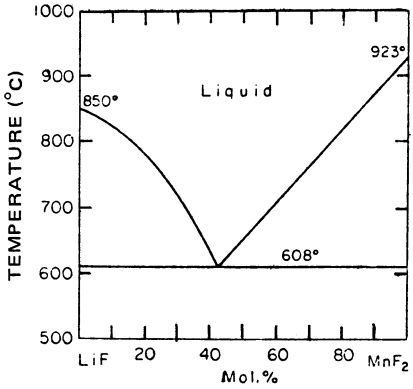


Fig. 46. The system $LiF–MnF_2$.

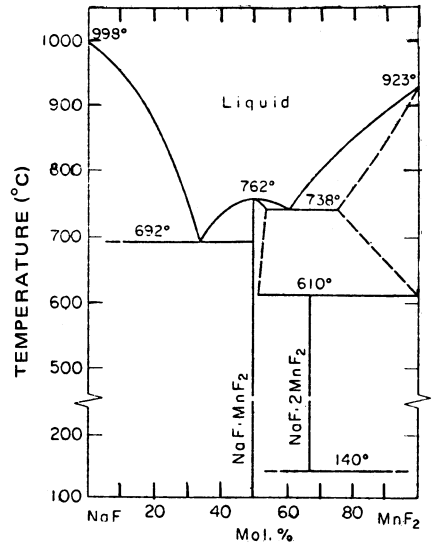
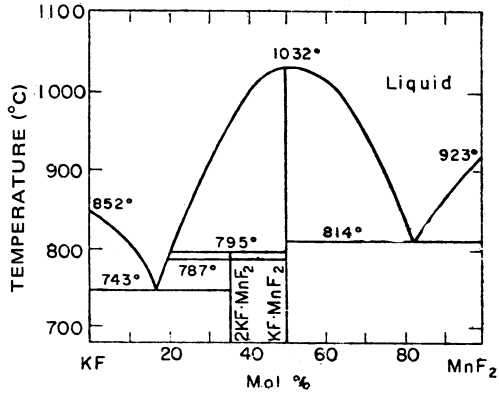


Fig. 47. The system $NaF–MnF_2$.

Fig. 48. The system $KF–MnF_2$.



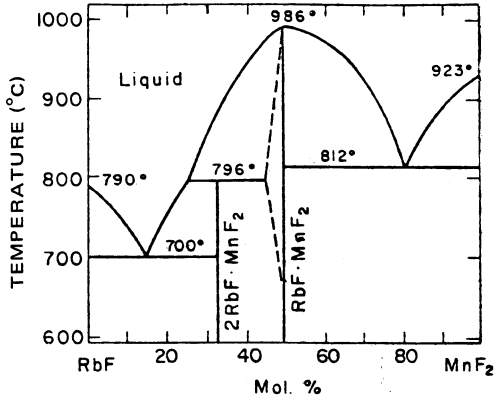


Fig. 49. The system $RbF–MnF_2$.

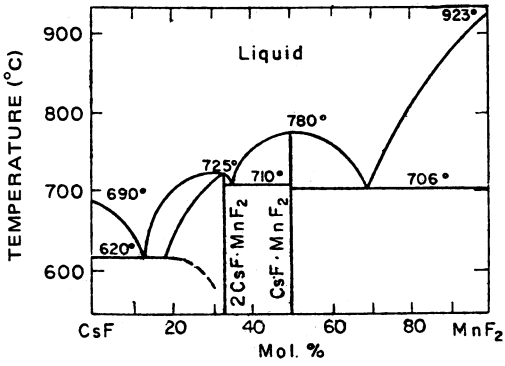


Fig. 50. The system $CsF–MnF_2$.

11:774 (1966). The invariant and singular equilibrium points are given in the table.

Mole %		Temperature	Type of equilibrium
MF	MnF_2		
LiF	43	608	Eutectic
NaF	34	692	Eutectic
	50	762	Congruent m.p.
	66	738	Eutectic
		610	Upper temp. of stability of $NaF \cdot 2MnF_2$
		140	Solid-state inversion temp. of $NaF \cdot 2MnF_2$
	55	738	Solid-state solubility limit of MnF_2 in $NaF \cdot MnF_2$
	77	738	Solid-state solubility limit of $NaF \cdot MnF_2$ in MnF_2
KF	17	743	Eutectic
		795	Incongruent melting temp. of $2KF \cdot MnF_2$
		787	Polymorphic transition temp. of $2KF \cdot MnF_2$
	84	814	Eutectic
RbF	16	700	Peritectic
	24	796	Peritectic
	(46)[a]	796	See footnote
	81	814	Eutectic
CsF	14	620	Eutectic
	20	620	Solid-state solubility limit of CsF in $2CsF \cdot MnF_2$
	33.3	725	Congruent m.p.
	35	710	Eutectic
	50	780	Congruent m.p.
	71	706	Eutectic

[a] The source report states that "a small region of restricted solid solutions of RbF in $RbMnF_3$ extends up to 54 mole % MnF_2," although the phase diagram, as drawn, indicates that the limit should be about 46 mole % MnF_2.

The System NaF–FeF_2. See Fig. 51. Source: R. E. Thoma, H. A. Friedman, B. S. Landau, and W. R. Grimes, ORNL, unpublished work (1957). $E = 680°$, 30% FeF_2; m.p. NaF · FeF_2, 783°; $E = 475°$, 63% FeF_2. The phase diagram was based on cooling-curve analysis, TGQ

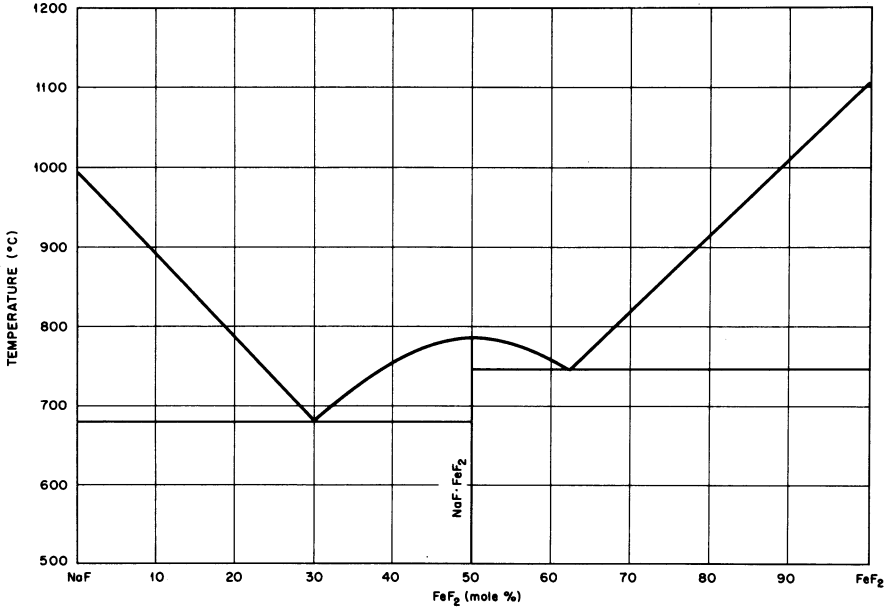


Fig. 51. The system NaF–FeF_2.

data, and x-ray diffraction analysis. A partial phase diagram of the system NaF–FeF_3 has been reported by N. Puschin and A. Baskow, *Z. Anorg. Chem.* **81**:361 (1913), showing a eutectic at ~890° and at ~35% FeF_3.

The System KF–CoF_2. See Fig. 52. Source: I. N. Belyaev and S. A. Shilov, *Russ. J. Inorg. Chem.* (*English Transl.*) **14**:1046 (1969). Invariant and singular equilibrium points: $E = 750°$, 19 mole % CoF_2; congruent m.p. of 2KF · CoF_2 858°; $E = 848°$, 37.5 mole % CoF_2; congruent m.p. of KF · CoF_2 1032°; $E = 945°$, 76 % CoF_2. Relative values of the heat evolved on cooling KF–CoF_2 mixtures are shown graphically on the phase diagram and should not be construed to signify phase transformations.

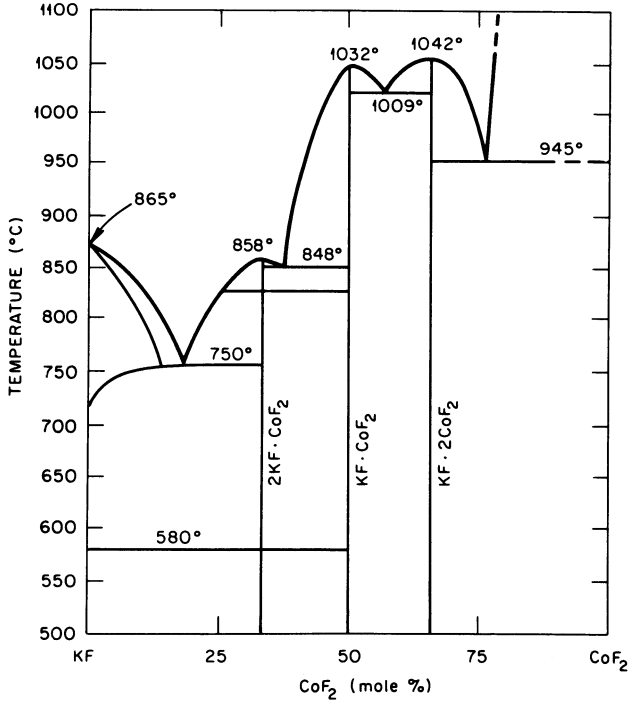
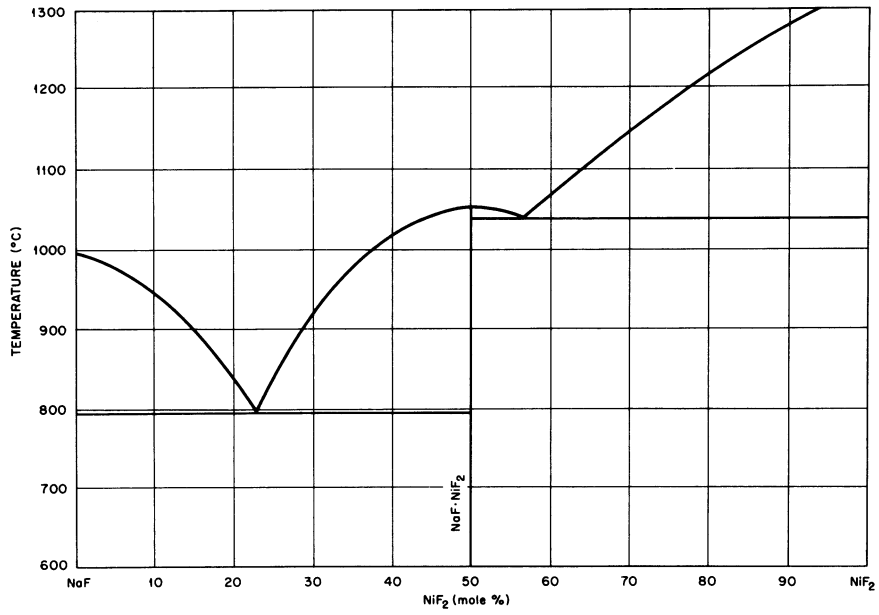


Fig. 52.
The system KF–CoF_2.

The System NaF–NiF_2. See Fig. 53. Source: R. E. Thoma, H. A. Friedman, B. S. Landau, and W. R. Grimes, ORNL, unpublished work



(1957). $E = 795°$; 23% NiF_2; m.p. $NaF \cdot NiF_2$ 1045°; $E = 1040°$, 57% NiF_2. The phase diagram was based on cooling-curve analysis, TGQ and x-ray diffraction analysis.

The System KF–NiF_2. See Fig. 54. Source: G. Wagner and D. Balz, *Z. Electrochem.* **56**:576 (1952). $E = 797°$, 9.1% NiF_2; $P = 930°$, 23.3% NiF_2; m.p. $KF \cdot NiF_2$ 1130 $\pm$ 2°; $E = 1084°$, 65.5% NiF_2.

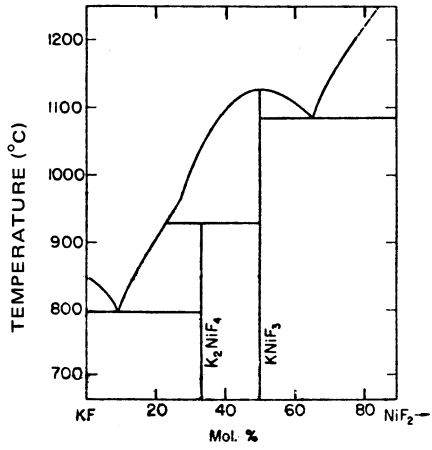


Fig. 54. The system KF–NiF_2.

The System LiF–BF_3. See Fig. 55. Source: A. S. Dworkin and M. A. Bredig, in: Molten Salt Reactor Program Semiannual Progress Report for Period Ending August 31, 1972, ORNL-4832, p. 58. Dworkin and Bredig did not state the composition of the eutectic in their report, although they did list activity coefficients for LiF at mole fractions X_{BF_4} of 0.2, 0.4, and 0.6 as 1.16, 1.32, and 1.140, respectively.

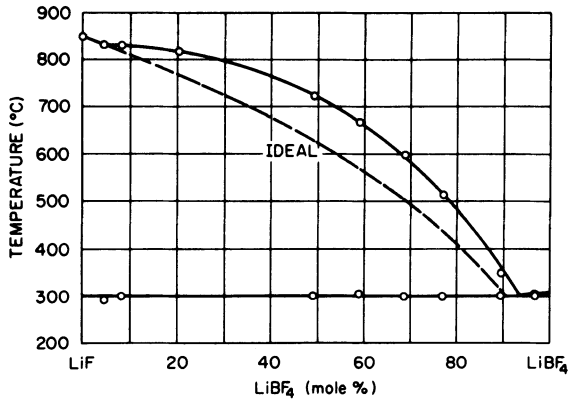


Fig. 55. The system LiF–BF_3.

The System $NaF–BF_3$. See Fig. 56. Source: C. J. Barton, L. O. Gilpatrick, J. A. Bornmann, H. H. Stone, T. N. McVay, and H. Insley, *J. Inorg. Nucl. Chem.* **33**:337 (1971). The eutectic is located at 92 ± 1 mole % $NaBF_4$ and melts at $384 \pm 2°$. An earlier report by V. G. Selivanov and V. V. Stender, *Zh. Neorg. Khim.* **3**:448 (1958), indicates the eutectic composition to be at 60 mole % $NaBF_4$ and the melting temperature to be 304°. The results are probably erroneous because of oxide impurities.

The structure of $NaBF_4$ has been reported by G. D. Brunton, *Acta Cryst.* **B24**:1703 (1968), as orthorhombic, *Cmcm*, $a = 6.8368$, $b = 6.2619$, $c = 6.7916$.

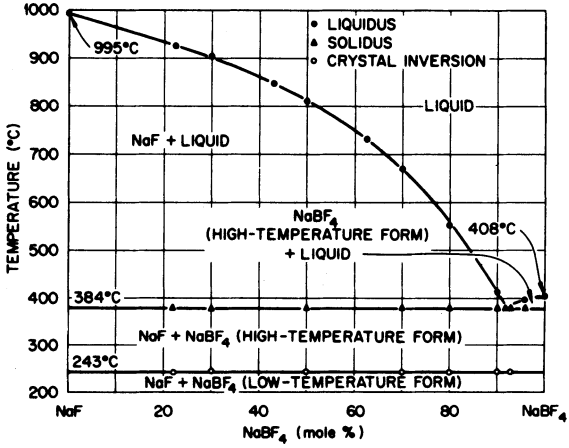


Fig. 56. The system $NaF–BF_3$.

The System $KF–BF_3$. See Fig. 57. Source: C. J. Barton *et al.*, *J. Inorg. Nucl. Chem.* **33**, 337 (1971). The eutectic formed by KF and KBF_4 occurs at 74.5 ± 1 mole % KBF_4, and at $460 \pm 2°$.

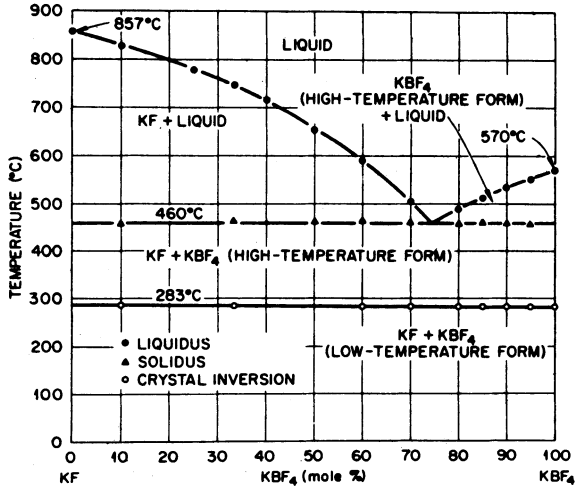


Fig. 57. The system $KF–BF_3$.

Structural data: KBF_4, orthorhombic, *Pnma*, $a = 8.6588$, $b = 5.4800$, $c = 7.0299$ [G. D. Brunton, *Acta Cryst.* **B25**:2161 (1969)].

The System RbF–BF_3. See Fig. 58. Source: L. O. Gilpatrick and C. J. Barton, *J. Inorg. Nucl. Chem.* **36**:725 (1974). The authors of the source reference deduced that the components, RbF and $RbBF_4$, form a simple eutectic system. $E = 442 \pm 2°$, 68.5% $RbBF_4$.

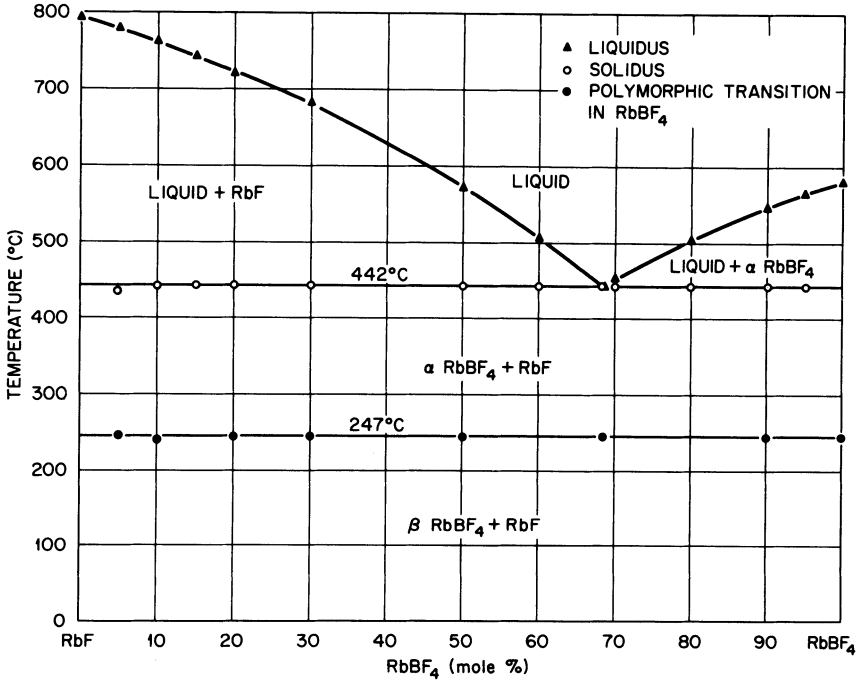


Fig. 58. The system RbF–BF_3.

Structural data: $RbBF_4$ crystallizes with an orthorhombic structure, *Pbnm*, $a = 7.29$, $b = 9.10$, $c = 5.13$ [M. J. R. Clark and H. Lynton, *Can. J. Chem.* **47**:2579 (1969)].

The System LiF–AlF_3. See Fig. 59. Source: E. P. Dergunov, *Dokl. Akad. Nauk, SSSR* **60**:1185 (1948). The system was described earlier, and more completely by P. P. Fedotieff and K. Timofieff, *Z. Anorg. Allgem. Chem.* **206**:266 (1937). Figure 59 is the phase diagram as proposed by Fedotieff and Timofieff.

Structural data: α-$3LiF \cdot AlF_3$: orthorhombic, $a = 9.510$, $b = 8.2295$, $c = 4.8762$, *Pna*21 [J. H. Burns, A. C. Tennissen, and G. D. Brunton, *Acta Cryst.* **B24**, 225 (1968). The compound $3LiF \cdot AlF_3$ undergoes at

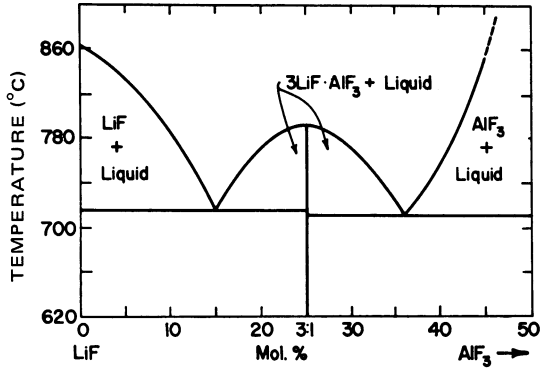


Fig. 59. The system $LiF–AlF_3$.

least one structural transition on heating and cooling. The structure given above is that of the room-temperature modification.

The viscosity of $LiF–AlF_3$ and $LiF–NaF–AlF_3$ melts was reported by M. M. Vetnikov and G. I. Sipniga, *Zh. Prikl. Khim.* **36**(9):905 (1963).

The System $NaF–AlF_3$. See Fig. 60. Source: K. Grojtheim, *Kgl. Norske Videnskap. Selskabs. Skrifter*, 5 (1956). Numerous examinations of the system $NaF–AlF_3$ have been reported, including phase diagrams by the following investigators: P. P. Fedotieff and W. P. Iljinskii, *Z. Anorg. Chem.* **80**:121 (1913); see also N. A. Puschin and A. V. Baskow, *ibid.*, **81**:350 (1913); E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **60**(7):1185 (1948); Giiejiro Fuseya, Chuzo Sugihara, Nobuo Nagao, and Ichiro Teraoka, *J. Electrochem. Soc. Japan* **18**(2):66 (1950). A paper by P. A. Foster, Jr., *J. Am. Ceram. Soc.* **51**:107 (1968) discusses the melting point

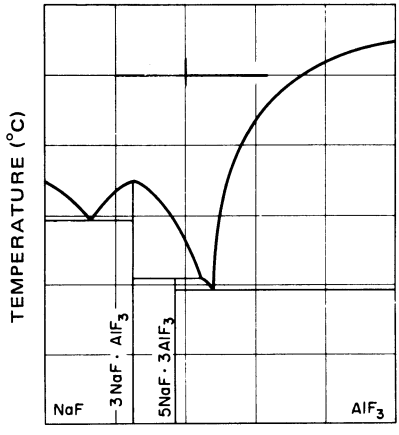


Fig. 60. The system $NaF–AlF_3$.

of Na_3AlF_6. The constitution of cryolite and $NaF–AlF_3$ melts has been discussed by W. B. Frank and L. M. Foster, *J. Phys. Chem.* **64**:310 (1960). A review of the structural modifications of Na_3AlF_6 may be found in A. F. Wells, *Structural Inorganic Chemistry*, 3rd ed., Oxford University Press, London (1962).

The System $KF–AlF_3$. See Fig. 61. Source: Bert Phillips, C. M. Warshaw, and J. Mockrin, *J. Am. Ceram. Soc.* **49**:633 (1966). The phase diagram as reported by these authors indicates that $KAlF_4$ is dimorphic and inverts on cooling from a cubic to an orthorhombic form at ~15°. The authors do not list the compositions of the invariant equilibrium points in the systems, but do state that they are in good agreement with those listed by the previous investigators, P. P. Fedotieff and K. Timofieff, *Z. Anorg. Allgem. Chem.* **206**:263–66 (1932).

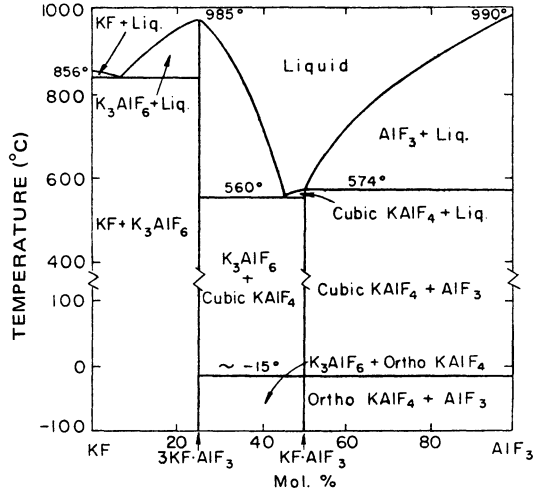


Fig. 61. The system $KF–AlF_3$.

Structural data: $KF \cdot AlF_3$: tetragonal, $P4/mmm$, $a = 3.550$, $c = 6.139$ [G. D. Brunton *et al.*, ORNL-3761].

The System $RbF–AlF_3$. See Fig. 62. Sources: N. Puschin and A. Baskow, *Z. Anorg. Chem.* **81**:356 (1913), and E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **50**:1185 (1948). Figure 62 is from Puschin and Baskow. $E = 790°$, 6.5 mole % AlF_3 (Puschin and Baskow); $E = 740°$, 7 % AlF_3 (Dergunov); congruent m.p. $3RbF \cdot AlF_3$, 985° (Puschin and Baskow);

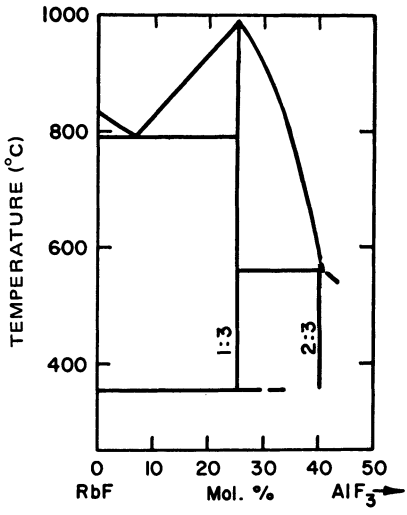


Fig. 62. The system RbF–AlF_3.

$P = 560°$, 41.5% AlF_3; congruent m.p. 3RbF · AlF_3, 914° (Dergunov). The source reference indicates a lower temperature of stability for 3RbF · AlF_3 of 350°, and the occurrence of the intermediate compound, 3RbF · $2AlF_3$, melting incongruently at 560°.

The System CsF–AlF_3. See Fig. 63. Sources: N. Puschin and A. Baskow, *Z. Anorg. Chem.* **81**:356 (1913), and E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **50**:1185 (1948). Figure 63 is from Puschin and Baskow. Invariant and singular equilibrium points: $E = 685°$, 5.5 % AlF_3 (Puschin and Baskow); $E = 663°$, $\simeq$6 % AlF_3 (Dergunov); congruent m.p. 3CsF · AlF_3, 825° (Puschin and Baskow); congruent m.p. 3CsF · AlF_3, 808° (Dergunov). The eutectic at some composition richer in AlF_3 than 30 mole % melts at 490°.

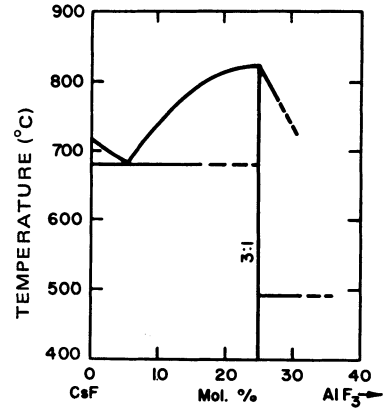


Fig. 63. The system CsF–AlF_3.

The System LiF–ScF_3. See Fig. 64. Source: E. P. Babaeva and G. A. Bukhalova, *Zh. Neorg. Khim.* **11**:648 (1966). $E = 606°$, 29% ScF_3.

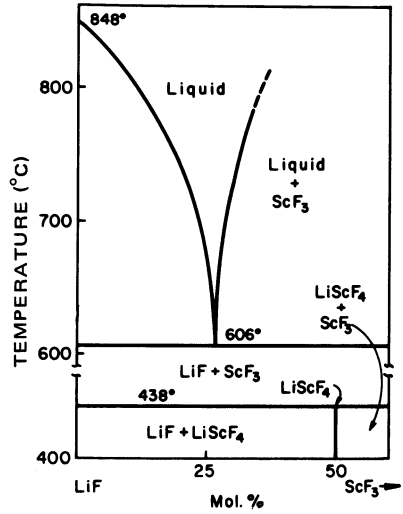


Fig. 64. The system LiF–ScF_3.

The System NaF–ScF_3. See Fig. 65. Source: R. E. Thoma and R. H. Karraker, *Inorg. Chem.* **5**:1933 (1966). Invariant and singular equi-

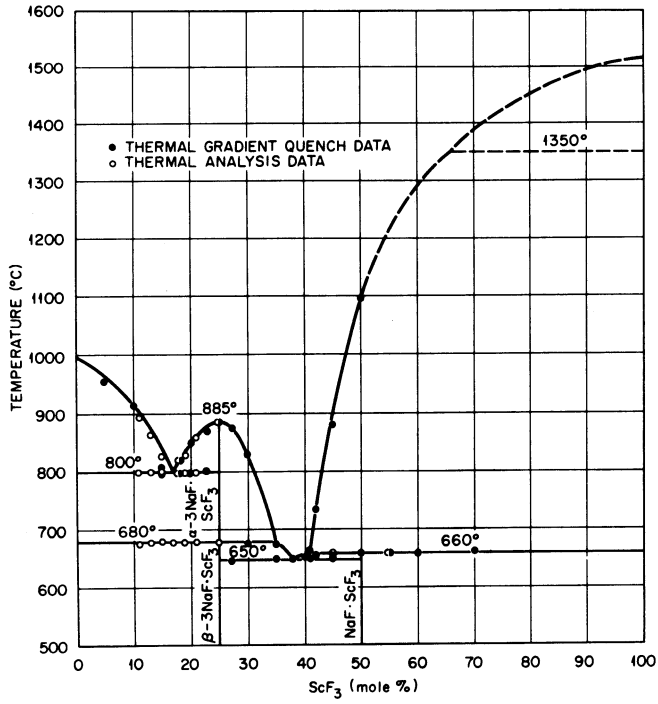


Fig. 65. The system NaF–ScF_3.

librium points and structural data are given in the tables.

Mole % ScF_3	Invariant temperature	Type of equilibrium
17	800	Eutectic
25	885	Congruent melting point
35	680	Peritectic inversion of $3NaF \cdot ScF_3$
38	650	Eutectic
42	660	Peritectic

Property	β-$3NaF \cdot ScF_3$	$NaF \cdot ScF_3$
Equilibrium stability range	$<680°$	$<660°$
Optical character	Uniaxial (+), $2V \simeq 0°$	Uniaxial (−)
Refractive indices	$N_\omega = 1.368$, $N_\varepsilon = 1.374$	$N_\omega = 1.431$
	From X-ray diffraction data	
Symmetry	Monoclinic, $P2_1/n$	Hexagonal, $P3_1 k2m$, $P3_2 k2m$, $P3_1 21$, or $P3_2 21$
Parameters	$a = 5.60 \pm 0.02$, $b = 5.81 \pm 0.02$, $c = 8.12 \pm 0.02$, $\beta = 90°45' \pm 5'$	$a = 12.97 \pm 0.03$, $c = 9.27 \pm 0.02$
Density	2.87 (calc.), $z = 2$	3.01 (calc.), $z = 18$

The System CsF–ScF_3. See Fig. 66. Source: E. P. Babaeva and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:352 (1966). $E = 680°$, 4% ScF_3; m.p. Cs_3ScF_6 1084°; $E = 798°$, 33% ScF_3.

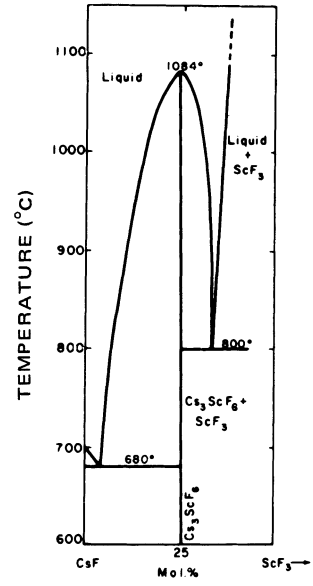


Fig. 66. The system $CsF–ScF_3$.

The System $TlF–ScF_3$. See Fig. 67. Source: J. Chassaing, *Rev. Chim. Min.* **9**:265 (1972). Invariant and singular equilibrium points:

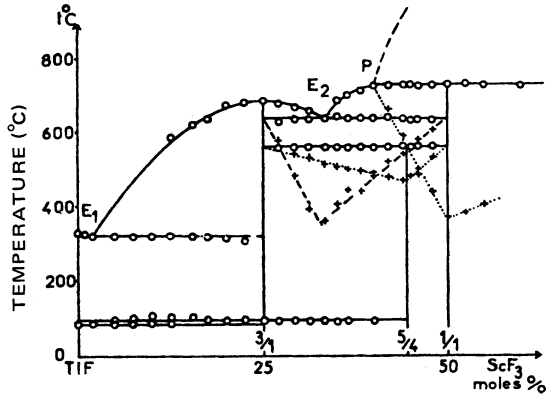


Fig. 67. The system $TlF–ScF_3$.

$E = 320°$, 2% ScF_3; m.p. Tl_3ScF_6 686°; $E = 638°$, 33% ScF_3; $P = 732°$, 40% ScF_3; $Tl_5Sc_4F_{17}$ upper temperature of stability 563°; $TlScF_4$ upper temperature of stability 732°.

The System LiF–YF_3. See Fig. 68. Source: R. E. Thoma *et al.*, *J. Phys. Chem.* **65**:1096 (1961). Invariant equilibria: $E = 695°$, 19 % YF_3;

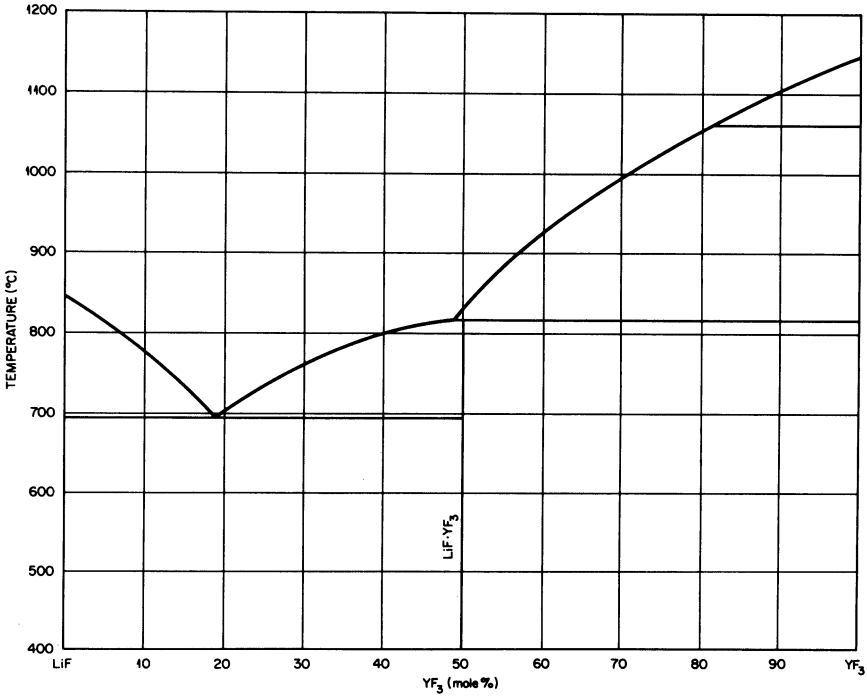


Fig. 68. The system LiF–YF_3.

$P = 819°$, 49%. The compound $LiYF_4$ is described in the source reference as tetragonal, $a = 5.26 \pm 0.03$, $c = 10.94 \pm 0.03$, S.G. $I4_1/a$. A solid state transition in YF_3 was shown to occur at 1052°.

The System NaF–YF_3. See Fig. 69. Source: R. E. Thoma, G. M. Hebert, H. Insley, and C. F. Weaver, *Inorg. Chem.* **2**:1005 (1963). A = liquid, B = NaF + liquid, C = NaF + hex. NaF · YF_3, D = hex. NaF · YF_3 + liquid, E = (fluorite) cubic NaF · YF_3–5NaF · $9YF_3$ solid solution + liquid, F = (fluorite) cubic NaF · YF_3–5NaF · $9YF_3$ ss, G = hex. NaF · YF_3 ss + cubic NaF · YF_3–5NaF · $9YF_3$ ss, H = cubic NaF · YF_3–5NaF · $9YF_3$ ss + ordered 5NaF · $9YF_3$ ss, I = hex. NaF · YF_3 ss, J = hex. NaF · YF_3 ss + ordered 5NaF · $9YF_3$ ss, K = ordered 5NaF · $9YF_3$ ss, L = hex. NaF · YF_3 ss + orthorhombic YF_3, M = ordered 5NaF · $9YF_3$ + ortho-YF_3, N = cubic 5 NaF · $9YF_3$ + ortho-YF_3, O = ortho-YF_3 +

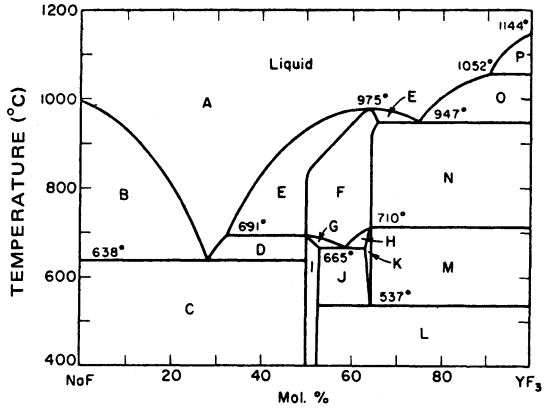


Fig. 69. The system NaF–YF_3.

liquid, and P = high-temperature YF_3 + liquid. Invariant and singular equilibrium points are given in the table.

Mole % YF_3	Temperature	Type of equilibrium	Equilibrium reaction
28	638	Eutectic	L $\rightleftarrows$ NaF + hexagonal NaF · YF_3
32	691	Peritectic	Cubic NaF · YF_3 + L $\rightleftarrows$ hexagonal NaF · YF_3 + L
58.5	665	Eutectoid	Fluoride ss $\rightleftarrows$ hexagonal NaF · YF_3 ss + orthorhombic YF_3
64.28	975	Congruent m.p. for 5NaF · $9YF_3$	L $\rightleftarrows$ cubic 5NaF · $9YF_3$
64.28	710	Inversion of 5NaF · $9YF_3$	Disordered cubic 5NaF · $9YF_3$ $\rightleftarrows$ ordered 5NaF · $9YF_3$
64.28	537	Dec. of 5NaF · $9YF_3$	5NaF · $9YF_3$ $\rightleftarrows$ hexagonal NaF · YF_3 ss + YF_3 (orthorhombic)
75	947	Eutectic	L $\rightleftarrows$ cubic 5NaF · $9YF_3$ ss + orthorhombic YF_3
91	1052	Peritectic	L + high-temperature YF_3 $\rightleftarrows$ L + orthorhombic YF_3

Structural data: NaF · YF_3 hexagonal, $a = 5.95$, $c = 3.52$, $P63/m$ [J. H. Burns, ORNL-3262 (1962)]. Lattice constants for the cubic solid solutions range from 5.447 to 5.530 at the 5NaF · $9YF_3$ composition limit.

The System KF–YF_3. See Fig. 70. Source: G. A. Bukhalova and E. P. Babaeva, *Zh. Neorg. Khim.* **11**:644 (1966). The binary system, as reported in the source reference, was adopted as the limiting binary system diagram for the NaF–KF–YF_3 system shown in Fig. 183. Invariant and singular equilibrium points: $E = 760°$, 13 % YF_3; congruent m.p. K_3YF_6 980°; $E = 750°$, 43 % YF_3. Upper temperature limit of stability of KYF_4 702°.

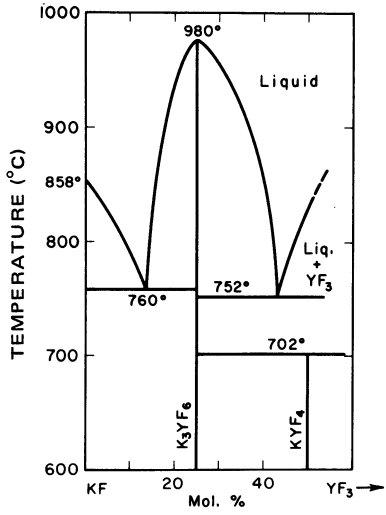


Fig. 70. The system KF–YF_3.

The System RbF–YF_3. See Fig. 71. Source: E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **60**:1186 (1948). Compositions of the invariant equilibrium points were not given in the source reference. The eutectics are estimated to occur at 9 and 40 mole % YF_3.

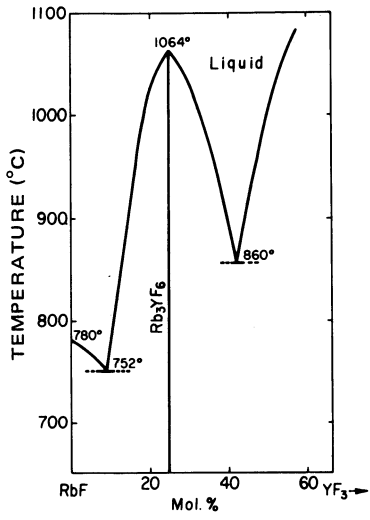


Fig. 71. The system RbF–YF_3.

The System CsF–YF_3. See Fig. 72. Source: G. A. Bukhalova and E. P. Babaeva, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:219 (1966). The phase diagram reported in the source reference agrees with that of E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **60**:1185 (1948). The eutectics occur at 4 mole % YF_3, 670°; and at 37.5 mole % YF_3, 890°.

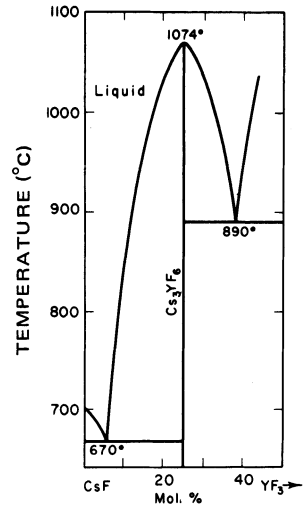


Fig. 72. The system CsF–YF_3.

The LiF–LnF_3 Systems. See Fig. 73. Source: R. E. Thoma, G. D. Brunton, R. A. Penneman and T. K. Keenan, *Inorg. Chem.* **9**:1096 (1970).

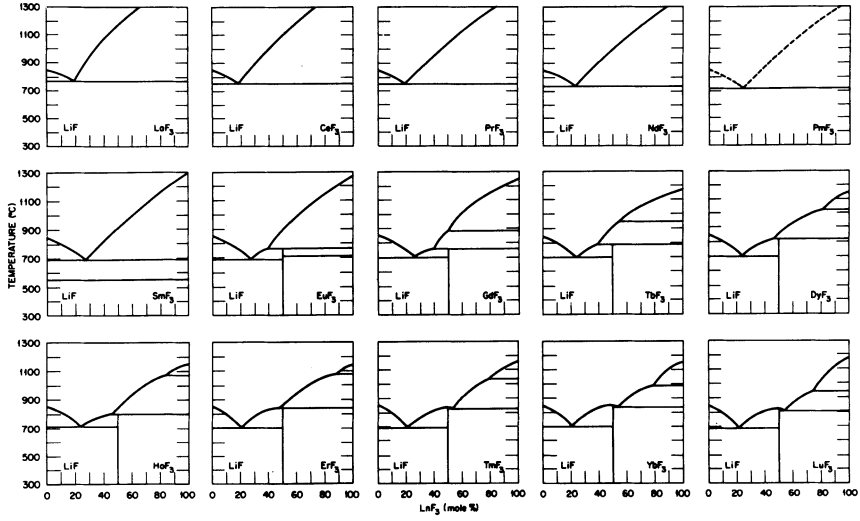


Fig. 73. The LiF–LnF_3 systems.

Invariant and singular equilibrium points are given in the table.

Ln	Mole % LnF_3	Temperature	Type of equilibrium
La	20[a]	770	Eutectic
Ce	19[b]	755	Eutectic
Pr	19	750	Eutectic
Nd	23	738	Eutectic
Sm	27	698	Eutectic
Eu	27	688	Eutectic
	30	710	Peritectic
	40	760	Peritectic
Gd	26	700	Eutectic
	39	755	Peritectic
	50	875	Peritectic
Tb	24	700	Eutectic
	39	790	Peritectic
	(54)[c]	950	Peritectic
Dy	(24)	700	Eutectic
	46	820	Peritectic
	(81)	1030	Peritectic
Ho	(24)	(710)	Eutectic
	46	798	Peritectic
	(84)	1070	Peritectic
Er	21	700	Eutectic
	48	840	Peritectic
	(88)	1075	Peritectic
Tm	21	700	Eutectic
	50	(835)	Congruent m.p.
	53	824	Eutectic
	(78)	1030	Peritectic
Yb	21	700	Eutectic
	50	850	Congruent m.p.
	53.5	840	Eutectic
	(79)	985	Peritectic
Lu	22	695	Eutectic
	50	(825)	Congruent m.p.
	54	810	Eutectic
	74	945	Peritectic

[a] Reported as LiF–LaF_3 (86–14 mole %), 758°, by G. A. Bukhalova and E. P. Babaev, *Zh. Neorgan. Khim.* **10**:1883 (1965).
[b] From C. J. Barton and R. A. Strehlow, *J. Inorg. Nucl. Chem.* **20**:45 (1961).
[c] Estimated value.

The compounds of the $LiLnF_4$ class are all isostructural and isomorphous with $LiYF_4$ [R. E. Thoma *et al.*, *Inorg. Chem.* **65**:1096 (1961)]. Details of the structure were first established through the work of Brunton, in a determination of the structure of $LiYbF_4$ from single crystals (see source reference).

The NaF–LnF_3 Systems. See Fig. 74. Source: R. E. Thoma, H. Insley, and G. M. Hebert, *Inorg. Chem.* **5**:1222 (1966). Invariant and sin-

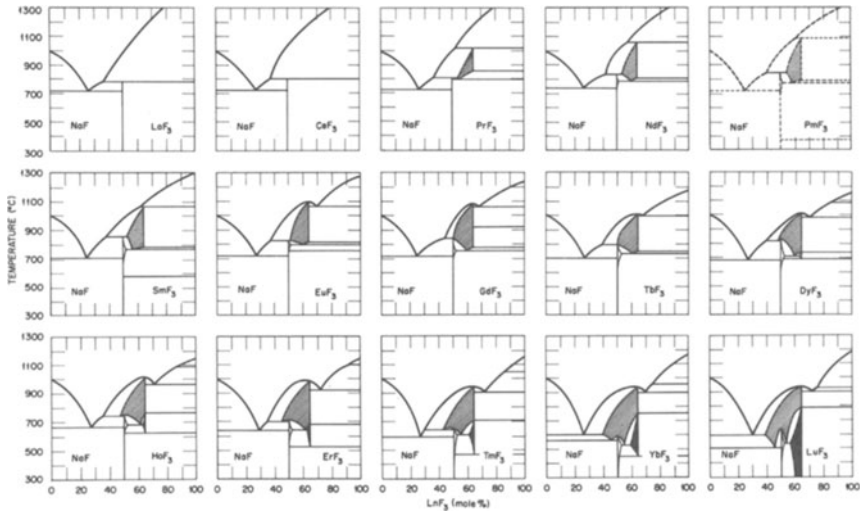


Fig. 74. The NaF–LnF_3 systems.

gular equilibrium points and structural data are given in the tables.

Ln	Mole % LnF_3	Temperature	Type of equilibrium
La	29	730[a]	Eutectic
	43	810[a]	Peritectic
Ce[b]	28	726	Eutectic
	38	810	Peritectic
Pr	27	733	Eutectic
	37	810	Peritectic
	52	1025	Peritectic
Nd	26	732	Eutectic

(*Continued*)

Ln	Mole % LnF_3	Temperature	Type of equilibrium
Nd	42	850	Peritectic
	58	1056	Peritectic
Sm	25	725	Eutectic
	38	834	Peritectic
	62	1060	Peritectic
Eu	25	718	Eutectic
	39	825	Peritectic
	64.3	1091	Congruent m.p.
	70	1060	Eutectic
Gd	25	712	Eutectic
	43	835	Peritectic
	64.3	1080	Congruent m.p.
	68	1058	Eutectic
Tb	27	696	Eutectic
	39	789	Peritectic
	64.3	1031	Congruent m.p.
	68	994	Eutectic
Dy	27	680	Eutectic
	38	780	Peritectic
	64.3	1020	Congruent m.p.
	70	982	Eutectic
Ho	27	663	Eutectic
	35	745	Peritectic
	64.3	1010	Congruent m.p.
	71	962	Eutectic
Er	27	643	Eutectic
	35	702	Peritectic
	64.3	975	Congruent m.p.
	72	920	Eutectic
Tm	27	592	Eutectic
	29	632	Peritectic
	64.3	940	Congruent m.p.
	~70	~905	Eutectic
Yb	26	594	Eutectic
	64.3	928	Congruent m.p.
	~70	893	Eutectic
Lu	29	595	Eutectic
	64.3	930	Congruent m.p.
	72	873	Eutectic

[a] Eutectic and peritectic temperatures were reported by E. Mathes and S. Holz, *Z. Chem.* **2**:22 (1962), to be about 630 and 770°C, respectively.

[b] C. J. Barton, J. D. Redman and R. A. Strehlow, *J. Inorg. Nucl. Chem.* **20**:45 (1961).

Compound formula	Symmetry	Space group or type	Lattice constants (Å)		Refractive indices	
			a_0	c_0	N_ε	N_ω
$NaF \cdot LaF_3$	Hexagonal	$P\bar{6}$	6.157	3.822	1.500	1.486
$NaF \cdot CeF_3$	Hexagonal	$P\bar{6}$	6.131	3.776	1.514	1.493
$NaF \cdot PrF_3$[a]	Cubic	$Fm3m$	5.695–5.702			
$NaF \cdot PrF_3$	Hexagonal	$P\bar{6}$	6.123	3.822	1.516	1.494
$NaF \cdot NdF_3$	Cubic	$Fm3m$	5.670–5.678			
$NaF \cdot NdF_3$	Hexagonal	$P\bar{6}$	6.100	3.711	1.515	1.493
$NaF \cdot SmF_3$	Cubic	$Fm3m$	5.605–5.628			
$NaF \cdot SmF_3$	Hexagonal	$P\bar{6}$	6.051	3.640	1.516	1.492
$NaF \cdot EuF_3$	Cubic	$Fm3m$	5.575–5.605			
$NaF \cdot EuF_3$	Hexagonal	$P\bar{6}$	6.044	3.613	1.516	1.492
$NaF \cdot GdF_3$	Cubic	$Fm3m$	5.552–5.583			
$NaF \cdot GdF_3$	Hexagonal	$P\bar{6}$	6.020	3.601	1.507	1.483
$NaF \cdot TbF_3$	Cubic	$Fm3m$	5.525–5.563			
$NaF \cdot TbF_3$	Hexagonal	$P\bar{6}$	6.008	3.580	1.506	1.486
$NaF \cdot DyF_3$	Cubic	$Fm3m$	5.508–5.545			
$NaF \cdot DyF_3$	Hexagonal	$P\bar{6}$	5.985	3.554	1.510	1.486
$NaF \cdot HoF_3$	Cubic	$Fm3m$	5.480–5.526			
$NaF \cdot HoF_3$	Hexagonal	$P\bar{6}$	5.991	3.528	1.510	1.486
$NaF \cdot ErF_3$	Cubic	$Fm3m$	5.455–5.510			
$NaF \cdot ErF_3$	Hexagonal	$P\bar{6}$	5.959	3.514	1.504	1.482
$NaF \cdot TmF_3$	Cubic	$Fm3m$	5.435–5.495			
$NaF \cdot TmF_3$	Hexagonal	$P\bar{6}$	5.953	3.494	1.496	1.476
$NaF \cdot YbF_3$	Cubic	$Fm3m$	5.418–5.480			
$NaF \cdot YbF_3$	Hexagonal	$P\bar{6}$	5.929	3.471	~1.490	
$NaF \cdot LuF_3$	Cubic	$Fm3m$	5.400–5.465			
$NaF \cdot LuF_3$	Hexagonal	$P\bar{6}$	5.912	3.458	1.428	1.484

[a] J. H. Burns, *Inorg. Chem.* **4**:881 (1965). X-ray data for the hydrated double fluorides of NaF and the lanthanide trifluorides were reported by L. R. Batsanova, L. M. Yankovskaya, and L. V. Lukina, *Russ. J. Inorg. Chem.* (*English Transl.*) **17**:654 (1972).

The KF–LnF_3 Systems. See Fig. 75. Sources: E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **60**:1185 (1948); E. P. Dergunov, *ibid.*, **85**:1025 (1952); C. J. Barton, L. O. Gilpatrick, G. D. Brunton, D. Hsu, and H. Insley, *J. Inorg. Nucl. Chem.* **33**:325 (1971); B. S. Zakharova, L. P. Reshetnikova, and A. V. Novoselova, *Vestn. Mosk Univ., Ser. II* **22**:102 (1967); and

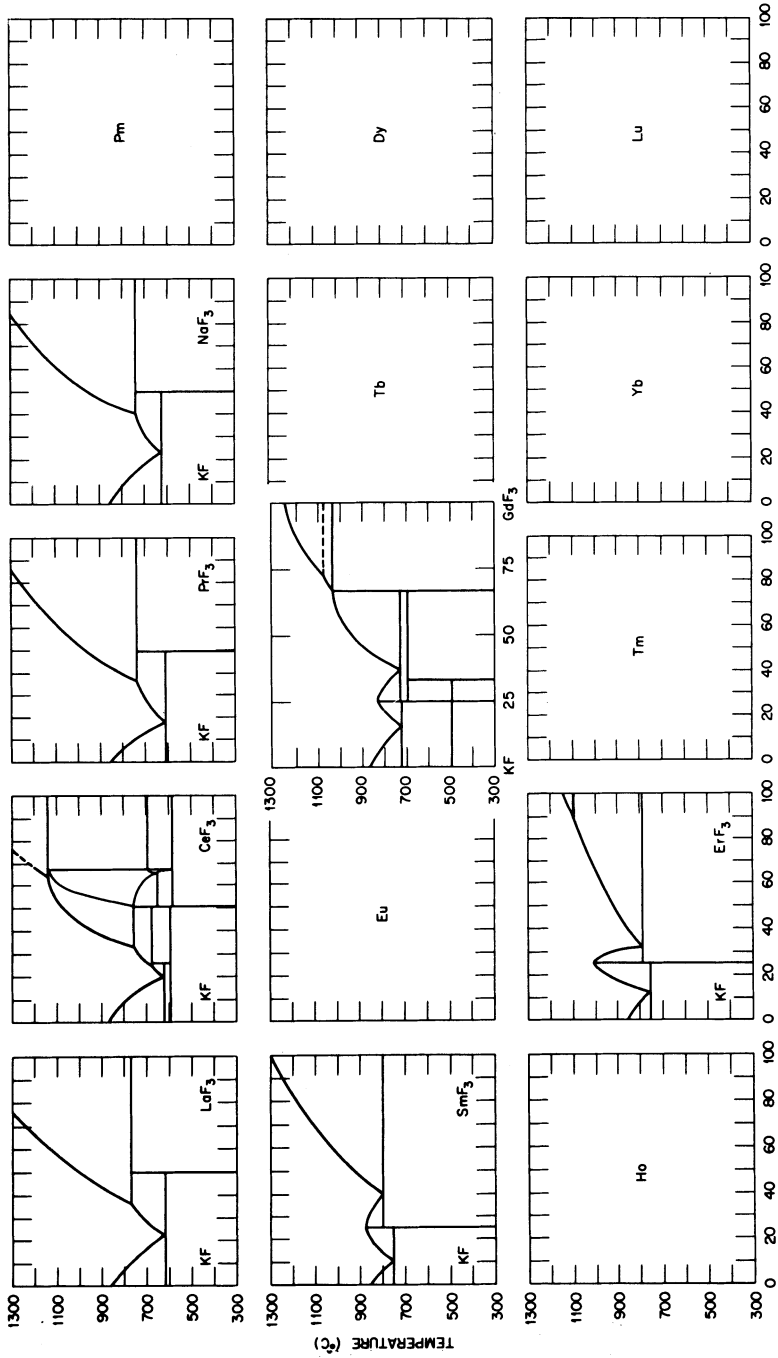


Fig. 75. The KF–LnF_3 systems.

A. Kozak, M. Samouel and A. Chretien, *Rev. de Chim. Min.* **10**:259 (1973). Invariant and singular equilibrium points are given in the table.

Ln	Mole % LnF_3	Temperature	Type of equilibrium
La[a]	22	620	Eutectic [Dergunov (1948)]
	36	770	Peritectic
Ce	19	620 ± 3	Eutectic [Barton *et al.* (1971)]
	24	765 ± 10	Peritectic
	32	755 ± 5	Peritectic
	63	1135 ± 15	Peritectic
	65	650 ± 12	Eutectoid
Pr[a]	23	610	Eutectic [Dergunov (1952)]
	63	740	Peritectic
Nd	23	625	Eutectic [Zakharov *et al.* (1967)]
		550	Upper limit of stability of $3KF \cdot NdF_3$
	40	740	Peritectic
Sm	10	758	Eutectic [Dergunov (1952)]
	25	882	Congruent m.p.
	40	800	Eutectic
Gd	5	672	Eutectic [Kozak *et al.* (1973)]
	25	997	Congruent m.p.
	37	880	Eutectic
	66.7	1160	Congruent m.p.
	74	1128	Eutectic
Er[a]	12	756	Eutectic [Dergunov (1952)]
	25	1012	Congruent m.p.
	32	790	Eutectic

[a] Compositions were not listed in the source reference. Values given below are estimated.

Structural data: $3KF \cdot CeF_3$: orthorhombic, *Pnma*, $a = 6.2895$, $b = 3.8040$, $c = 15.596$ [G. D. Brunton, *Acta Cryst.* **B25**:600 (1969). $KF \cdot LaF_3$: hexagonal, *P*-62*m*, $a = 6.526$, $c = 3.791$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948).

The RbF–LnF_3 Systems. See Fig. 76. Sources: See KF–LnF_3 systems. Invariant and singular equilibrium points are given in the table.

Structural data: $RbF \cdot LaF_3$: tetragonal, $a = 8.11$, $c = 6.5$ [E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **60**:1185 (1948).

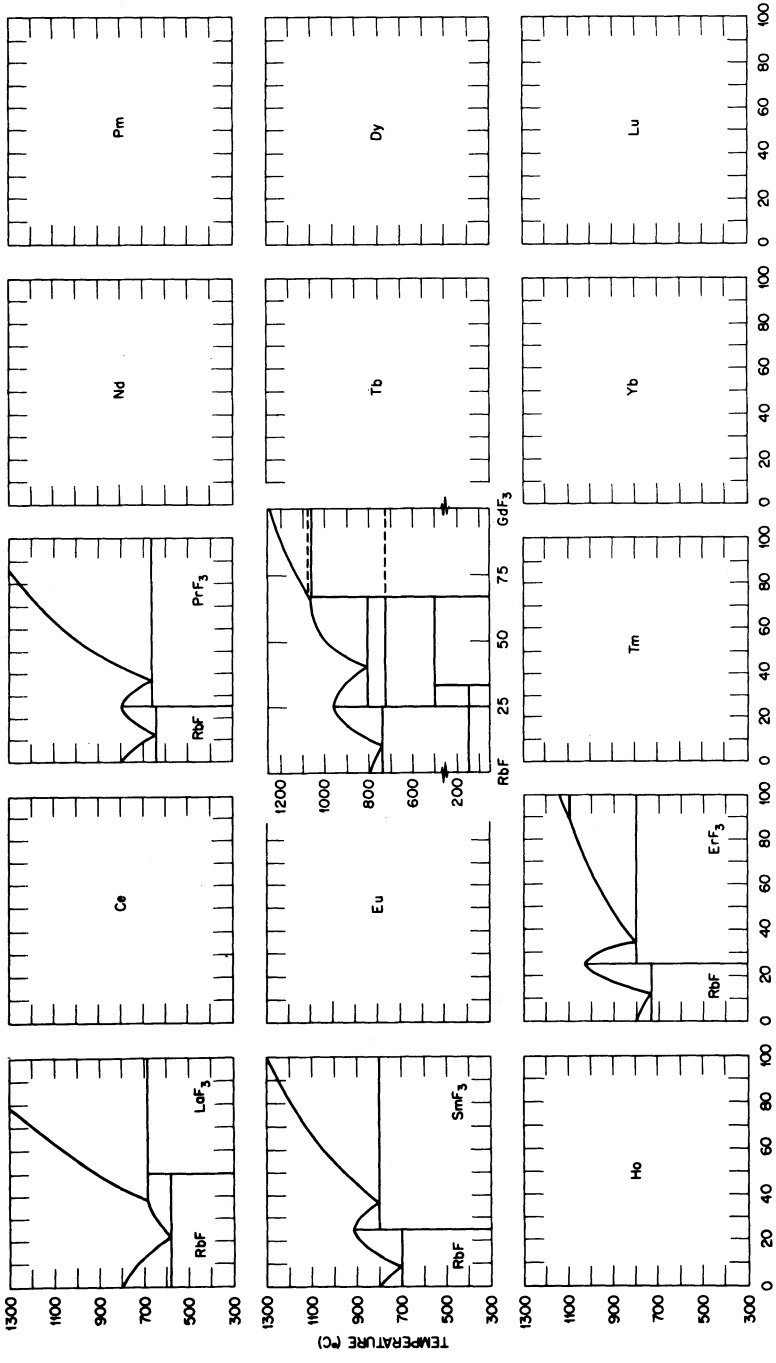


Fig. 76. The RbF–LnF_3 systems.

Ln	Mole %	Temperature	Type of equilibrium
La[a]	22	582	Eutectic [Dergunov (1948)]
	38	684	Peritectic
Pr[a]	22	640	Eutectic [Dergunov (1952)]
	25	797	Congruent m.p.
	37	660	Eutectic
Sm[a]	9	700	Eutectic [Dergunov (1952)]
	25	916	Congruent m.p.
	37	800	Eutectic
Gd	10	796	Eutectic [Kozak *et al.* (1973)]
	25	962	Congruent m.p.
		500	Upper temperature limit of stability of $2RbF \cdot GdF_3$
	40	803	Eutectic
	65	1060	Peritectic
Er[a]	12	732	Eutectic [Dergunov (1952)]
	25	1034	Congruent m.p.
	33.3	800	Eutectic

[a] Compositions were not listed in the source reference. Values listed here are estimated.

The CsF–LnF_3 Systems. See Fig. 77. Sources: See KF–LnF_3 systems. Invariant and singular equilibrium points are given in the table.

Ln	Comp. mole %	Temperature	Type of equilibrium
La[a]	12	600	Eutectic [Dergunov (1952)]
	25	795	Congruent m.p.
	33	726	Eutectic
Pr[a]	8	654	Eutectic [Dergunov (1952)]
	25	920	Congruent m.p.
	41	783	Eutectic
Gd	5	672	Eutectic [Kozak *et al.* (1973)]
	25	997	Congruent m.p.
	37	880	Eutectic
	66.7	1160	Congruent m.p.
	76	1128	Eutectic
Er[a]	5	634	Eutectic [Dergunov (1952)]
	25	1048	Congruent m.p.
	42	800	Eutectic

[a] Compositions were not listed in the source reference. Values listed here are estimated.

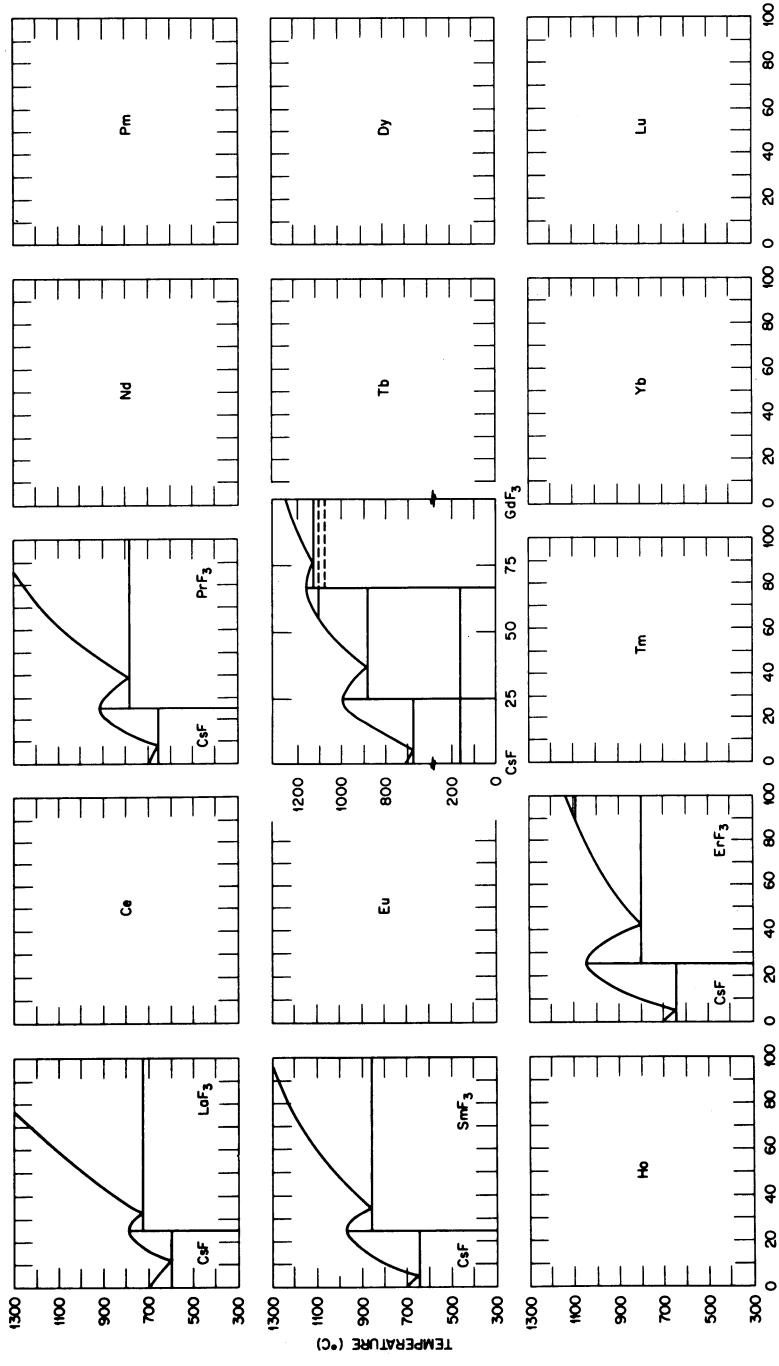


Fig. 77. The CsF–LnF_3 systems.

The System RbF–VF_3. See Fig. 78. Source: J. C. Cousseins and J. C. Cretenet, *Compt. Rend.* **265**:1464 (1967). Invariant and singular equilibrium

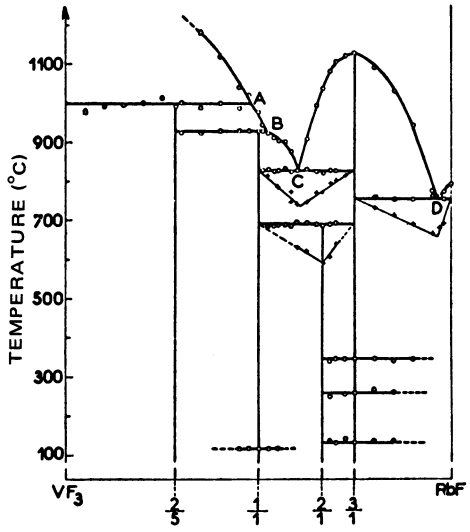


Fig. 78. The system RbF–VF_3.

points are given in the table.

Mole % ThF_4	Temperature	Type of equilibrium
4.5	755	Eutectic
25	1128	m.p. $3RbF \cdot VF_3$
	345	$3RbF \cdot VF_3$ inversion temp.
	258	" " "
	130	" " "
	690	Upper temp. of stability of $2RbF \cdot VF_3$
39.5	828	Eutectic
48	928	Peritectic
	115	$RbF \cdot VF_3$ inversion temp.
52	996	Peritectic

The source paper lists the x-ray powder diffraction data for four of the intermediate compounds formed in the system.

The System LiF–UF_3. See Fig. 79. Source: C. J. Barton, V. S. Coleman, L. M. Bratcher, and R. R. Grimes, ORNL, unpublished work (1953–1954). $E = 770°$, 27% UF_3.

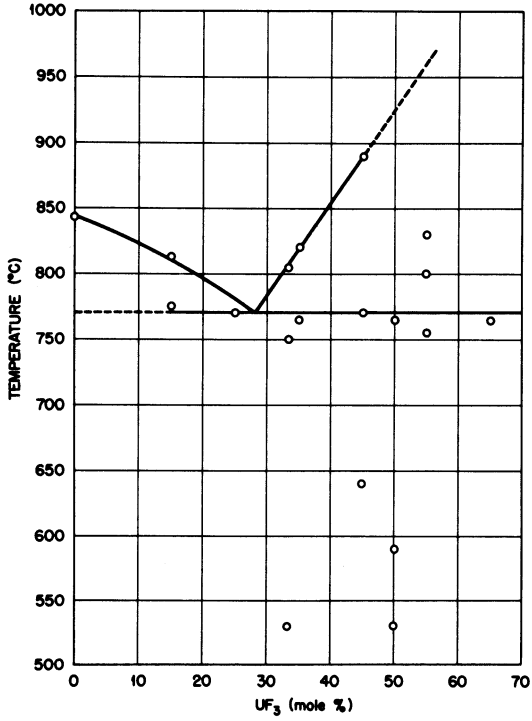


Fig. 79. The system LiF–UF_3.

The System LiF–PuF_3. See Fig. 80. Source: C. J. Barton and R. A. Strehlow, *J. Inorg. Nucl. Chem.* **18**:143 (1961). $E = 743°$, 19.5% PuF_3.

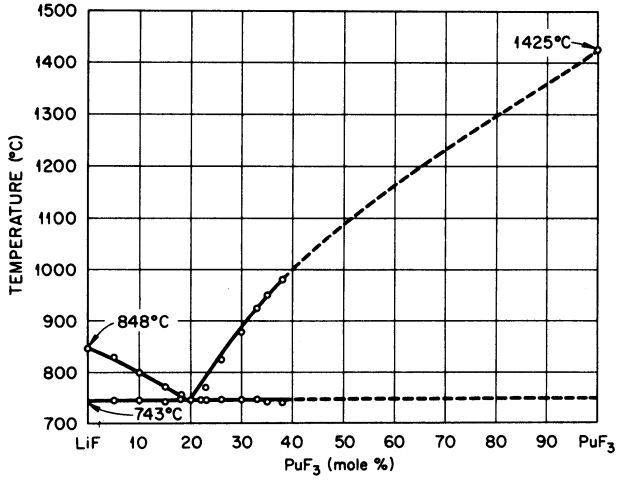


Fig. 80. The system LiF–PuF_3.

The System NaF–UF_3. See Fig. 81. Source: C. J. Barton, V. S. Coleman, T. N. McVay, and W. R. Grimes, ORNL, unpublished work (1953–1954). $E = 715°$, 27 % UF_4; $P = 775°$, 35 % UF_4; α-NaF · $UF_3 \rightleftarrows$ β-NaF · UF_3, 595°.

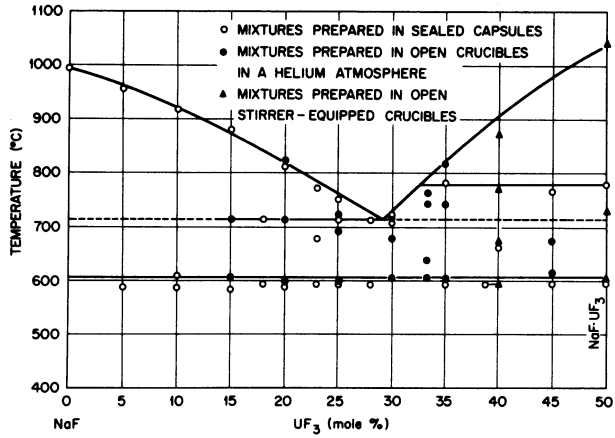


Fig. 81. The system NaF–UF_3.

Structural data: NaF · UF_3: hexagonal, $P\bar{6}2m$, $a = 6.167$, $c = 3.77$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948)].

The System NaF–PuF_3. See Fig. 82. Source: C. J. Barton, J. D. Redman, and R. A. Strehlow, *J. Inorg. Nucl. Chem.* **20**:45 (1961). Invariant equilibrium points: $E = 727°$, 23% PuF_3; $P = 842°$, 39% PuF_3.

Structural data: NaF · PuF_3: hexagonal, $P\bar{6}2m$, $a = 6.117$, $c = 3.746$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948)].

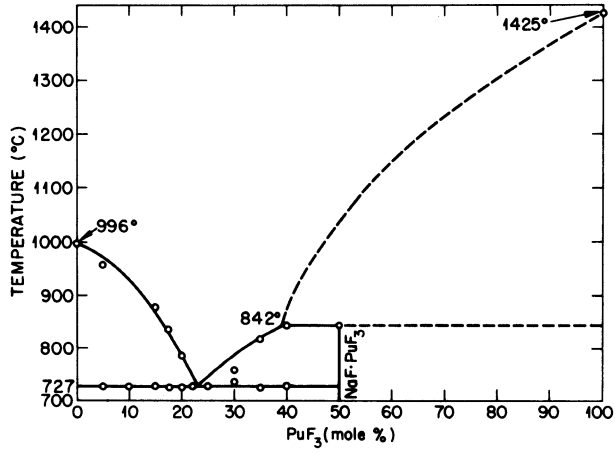


Fig. 82. The system NaF–PuF_3.

The System LiF–ZrF_4. See Fig. 83. Source: R. E. Thoma, H. Insley, H. A. Friedman, and G. M. Hebert, *J. Chem. Eng. Data* **10**:219 (1965). $E = 598°$, 21 % ZrF_4; m.p. $3LiF \cdot ZrF_4$ 662°; α-$3LiF \cdot ZrF_4 \rightleftarrows \beta$-$3LiF \cdot ZrF_4$, 475°; β-$3LiF \cdot ZrF_4 \rightleftarrows LiF + 2LiF \cdot ZrF_4$, 470°; $E = 570°$, 29.5 % ZrF_4; m.p. $2LiF \cdot ZrF_4$ 596°; $E = 507°$, 49 % ZrF_4; $P = 520°$, 51.5 % ZrF_4; $3LiF \cdot 4ZrF_4 \rightleftarrows 2LiF \cdot ZrF_4 + ZrF_4$, 466°. Liquidus curves of the binary systems LiF–, NaF–, and RbF–ZrF_4 were reported by K. A. Sense, R. W. Stone, and R. B. Filbert, Jr., USAEC Report BMI-1199, 31 (1957).

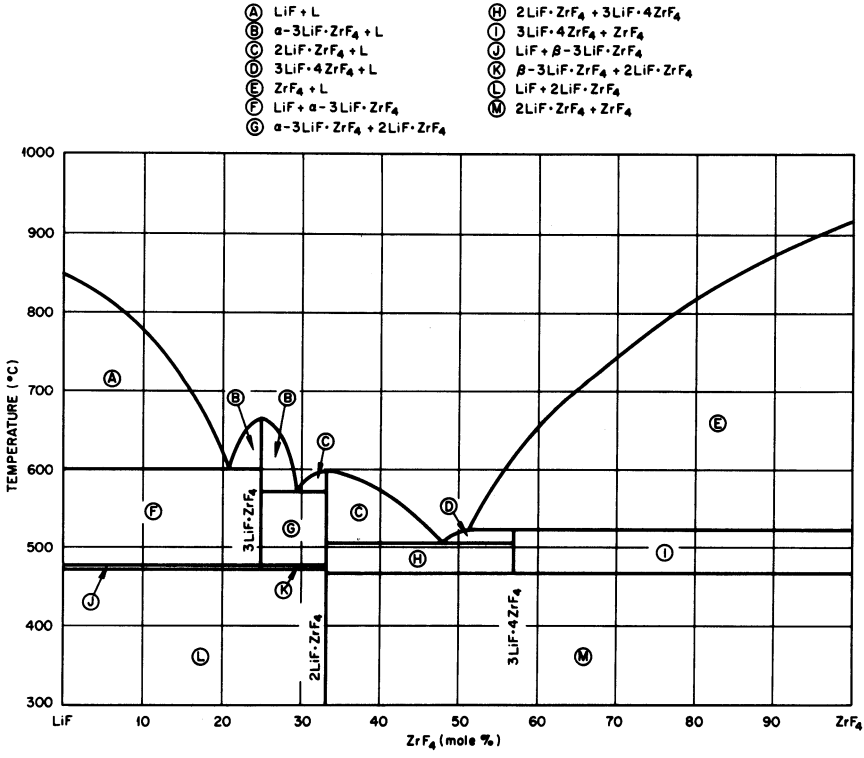


Fig. 83. The system LiF–ZrF_4.

Structural data: Li_2ZrF_6 crystallizes as a hexagonal phase, $P\bar{3}1m$, $a = 4.9733$, $c = 4.658$ [G. Brunton, *Acta Cryst.* **B29**:2294 (1973); R. Hoppe and W. Dahne, *Naturwissenschaften* **47**:397 (1960)].

The System NaF–ZrF_4. See Fig. 84. Source: C. J. Barton *et al.*, *J. Phys. Chem.* **62**:665 (1958). Invariant and singular equilibrium points are given in the table.

Mole % ZrF_4 in liquid	Temperature	Type of equilibrium
20	747	Eutectic
25	850	Congruent m.p.
	523	Inversion
	500	Eutectoid
34	640	Peritectic
39.5	544	Peritectic
40	533	Inversion
	487	Decomposition
40.5	505	Inversion
40.5	500	Eutectic
46.2	525	Congruent m.p.
49.5	512	Eutectic
56.5	537	Peritectic

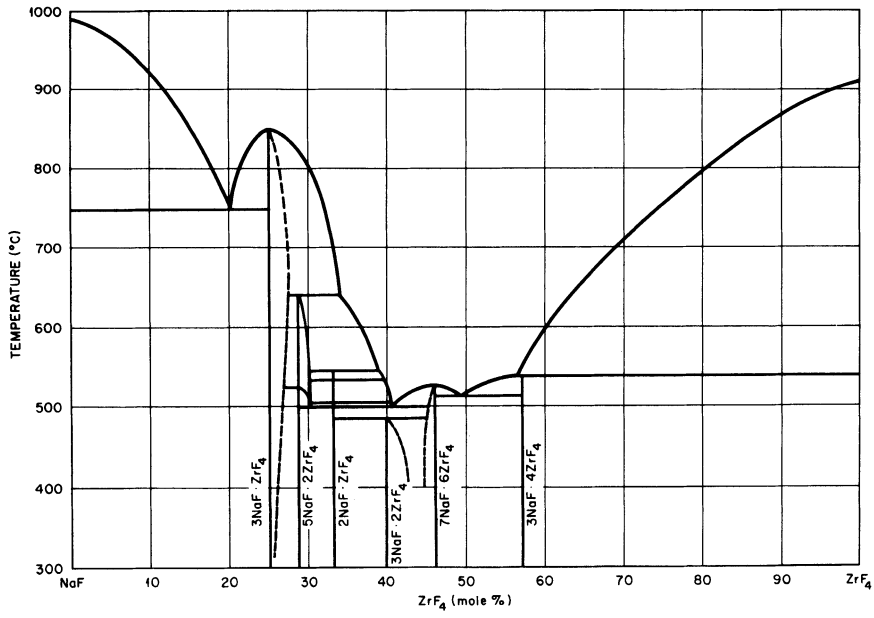


Fig. 84. The system $NaF–ZrF_4$.

A phase diagram of the system was reported by R. Winand, *Compt. Rend. Congr. Intern. Chimie Ind., 31^e, Liege*, 744 (1958), approximately concurrently with the publication of the source reference. It bears only gross resemblance to Fig. 84 and has not been evaluated. A liquidus curve for the ZrF_4 primary phase field was computed by K. A. Sense, R. W. Stone, and O. B. Filbert, Jr., USAEC Report BMI-1199, 31 (1957); see Fig. 1509, *Phase Diagrams for Ceramists* (1964).

Structural data: $3NaF \cdot ZrF_4$: tetragonal, $I4/mmm$, $a = 5.31$, $c = 10.50$ [L. A. Harris, *Acta Cryst.* **12**:172 (1959)]; β-$2NaF \cdot ZrF_4$: tetragonal, $a = 13.85$, $c = 1023$ [G. D. Brunton *et al.*, ORNL-3761]; γ-$NaF \cdot ZrF_4$: monoclinic, $P21/c$, $a = 5.5562$, $b = 5.4069$, $c = 16.073$, $\beta = 95.886°$ [G. D. Brunton *et al.*, ORNL-3761]; $7NaF \cdot 6ZrF_4$: hexagonal, R-3, $a = 13.807$, $c = 9.429$ [J. H. Burns, R. D. Ellison, and H. A. Levy, *Acta Cryst.* **B24**:230 (1968); $5NaF \cdot 2ZrF_4$: monoclinic, $C2/m$, $a = 11.62$, $b = 5.49$, $c = 8.44$, $\beta = 97.7°$ (corresponds to low-temperature form shown in phase diagram) [R. M. Herak *et al.*, *Bilten Inst. Boris Kidric* **14**:21 (1963)].

The System KF–ZrF_4. See Figs. 85 and 86. Sources: C. J. Barton, H. Insley, R. P. Metcalf, R. E. Thoma, and W. R. Grimes, ORNL, unpublished work (1951–1955), and A. V. Novoselova, Yu. M. Korenev, and

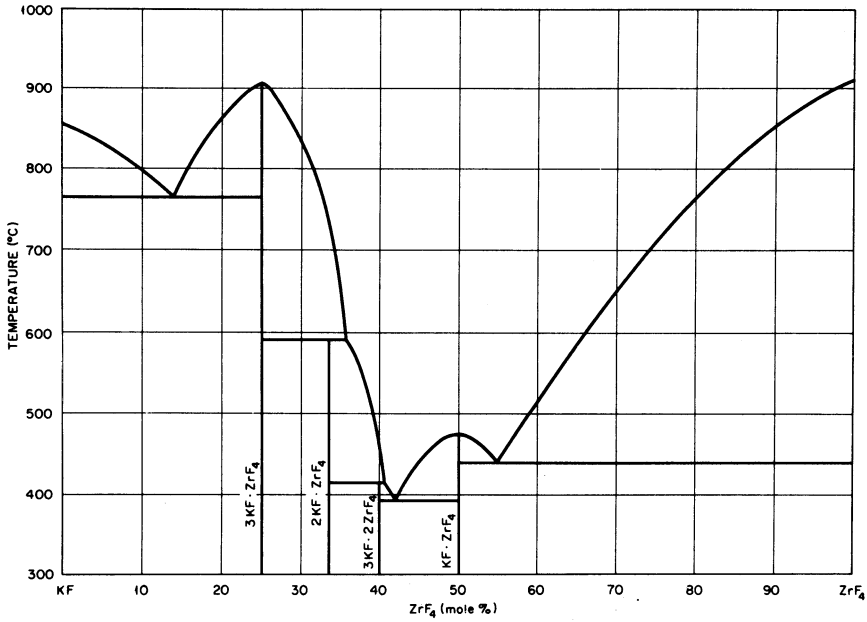


Fig. 85. The system KF–ZrF_4.

Yu. P. Simanov, *Dokl. Akad. Nauk SSSR* **139**:893 (1961). Invariant equilibria are listed below from the ORNL work. $E = 765$, 14% ZrF_4; congruent m.p. $3KF \cdot ZrF_4$ 910°; $P = 590°$, 36% ZrF_4; $P = 412°$, 40.5% ZrF_4: $E = 390°$, 42% ZrF_4; congruent m.p. $KF \cdot ZrF_4$ 475°; $E = 440°$, 55% ZrF_4. The complex diagram reported by Novoselova *et al.* has not been evaluated by subsequent investigators.

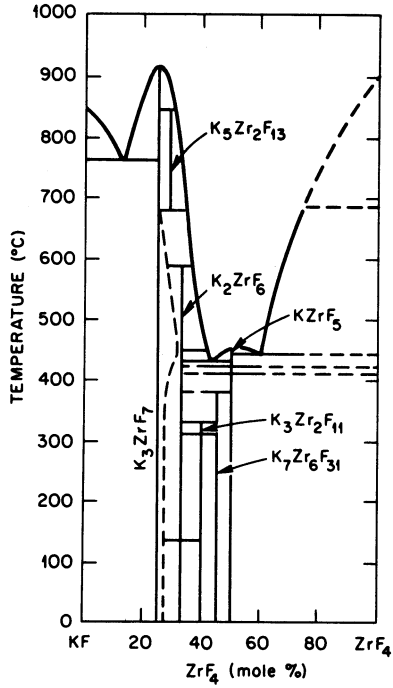


Fig. 86. The system $KF–ZrF_4$.

Structural data: $3KF \cdot ZrF_4$: cubic, *Fm3m*, $a = 8.988$ [G. D. Brunton *et al.*, ORNL-3761]. $2KF \cdot ZrF_4$: orthorhombic, *Immm*, $a = 6.58$, $b = 11.42$, $c = 6.94$ [V. Bode and G. Teufer, *Z. Anorg. Allgem. Chem.* **283**:18 (1956).

The System $RbF–ZrF_4$. See Fig. 87. Source: R. E. Moore, R. E. Thoma, C. J. Barton, W. R. Grimes, B. S. Landau, and H. A. Friedman, ORNL, unpublished work (1955–1956). Invariant and singular equilibrium points are given in the table.

Vapor–liquid equilibria were described by K. A. Sense, R. W. Stone, and R. B. Filbert, USAEC Report BMI-1199, 17 (1957).

Structural data: $3RbF \cdot ZrF_4$: cubic, *Fm3m*, $a = 9.31$ [G. D. Brunton *et al.*, ORNL-3761]. $2RbF \cdot ZrF_4$: hexagonal, $P\bar{3}m$, $a = 6.16$, $c = 4.82$ [V. Bode and G. Teufer, *Z. Anorg. Allgem. Chem.* **283**:18 (1956)]. $5RbF \cdot$

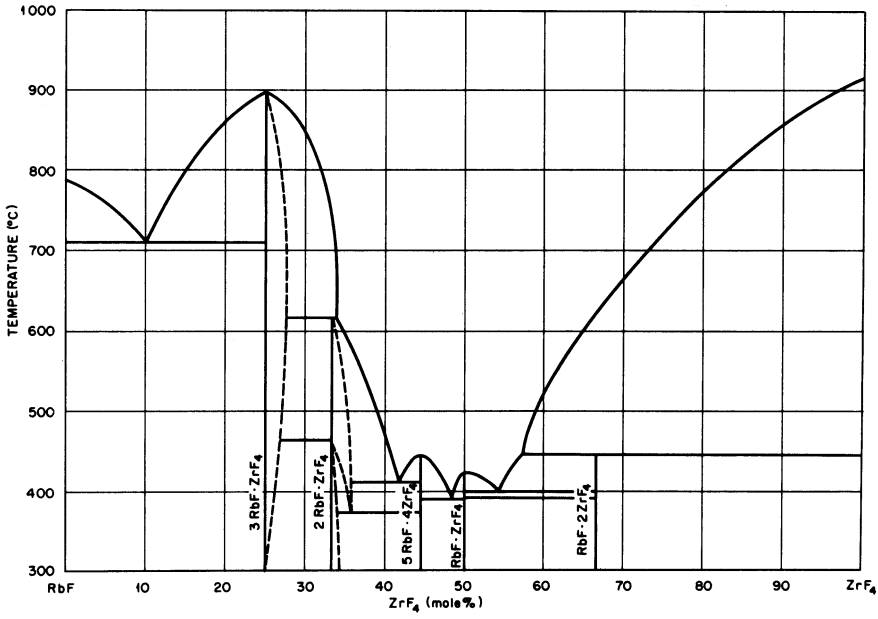


Fig. 87. The system RbF–ZrF_4.

$4ZrF_4$: George Brunton, *Acta Cryst.* **B27**:1944 (1971), has reported that the compound crystallizes with space group $P2_1$; $a = 11.520$, $b = 11.222$, $c = 7.868$, and $\cos\beta = 0.1445$.

Mole % ZrF_4 in liquid	Temperature	Type of equilibrium
10	710	Eutectic
25	897	Congruent melting point
34	620	Peritectic
	460	Inversion
	370	Lowered inversion and decomposition
42	410	Eutectic
44.4	445	Congruent melting point
48	390	Eutectic
50	423	Congruent melting point
	391	Inversion
54	400	Eutectic
57	447	Peritectic

The System CsF–ZrF_4. See Fig. 88. Source: G. D. Robbins, R. E. Thoma, and H. Insley, *J. Inorg. Nucl. Chem.* **27**:559 (1965). Invariant and

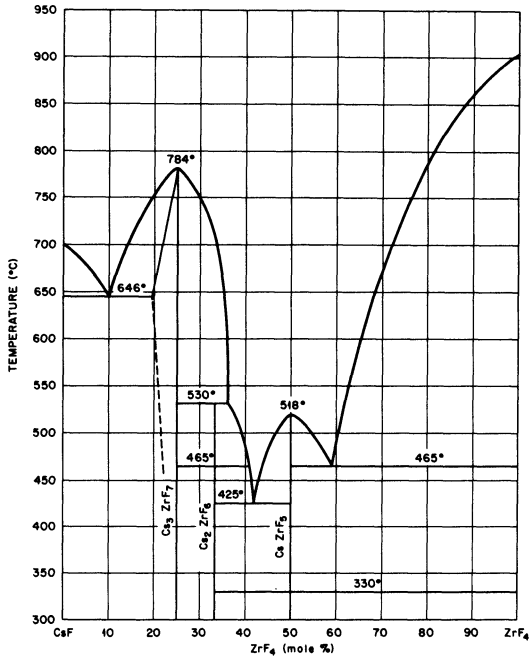


Fig. 88. The system CsF–ZrF_4.

singular equilibrium points are given in the table.

Mole % ZrF_4	Temperature	Type of equilibrium
10	646	Eutectic
25	784	Congruent melting point
37	530	Peritectic
	465	Inversion of Cs_2ZrF_6
42	425	Eutectic
50	518	Congruent melting
	330	Inversion of $CsZrF_5$
55	465	Eutectic

The compound Cs_3ZrF_7 was observed to vary in composition as shown in the phase diagram, accommodating slightly more than 20% CsF into its lattice at the solidus. Refractive indices of the solid solution were observed to vary in this composition range from 1.470 at the stoichiometric composition to 1.480 near the CsF-rich terminus of the solid solution. Cs_3ZrF_7 is face-centered cubic with $a = 9.70 \pm 0.02$; the space group is $F\bar{4}3m$, $F432$, or $Fm3m$. Although the fluoride-ion arrangement is disordered, the Zr^{4+} and Cs^+ are at or very near (0, 0, 0) + face centering and ($\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$; $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{4}$; $\frac{3}{4}$, $\frac{3}{4}$, $\frac{3}{4}$) + face centering, respectively. The x-ray density of the compound is 4.53 g/cm^{-3}. This compound is apparently isotypic with K_3ZrF_7, $(NH_4)_3ZrF_7$, and K_3UF_7 as regards the metal ions. The compound Cs_2ZrF_6 was described by V. Bode and G. Teufer, *Z. Anorg. Allgem. Chem.* **283**:18 (1956) as hexagonal, $P\text{-}3m1$, $a = 6.41$, $c = 5.01$.

The System NaF–HfF_4. See Fig. 89. Source: R. E. Thoma, C. F. Weaver, T. N. McVay, H. A. Friedman, and W. R. Grimes, ORNL, un-

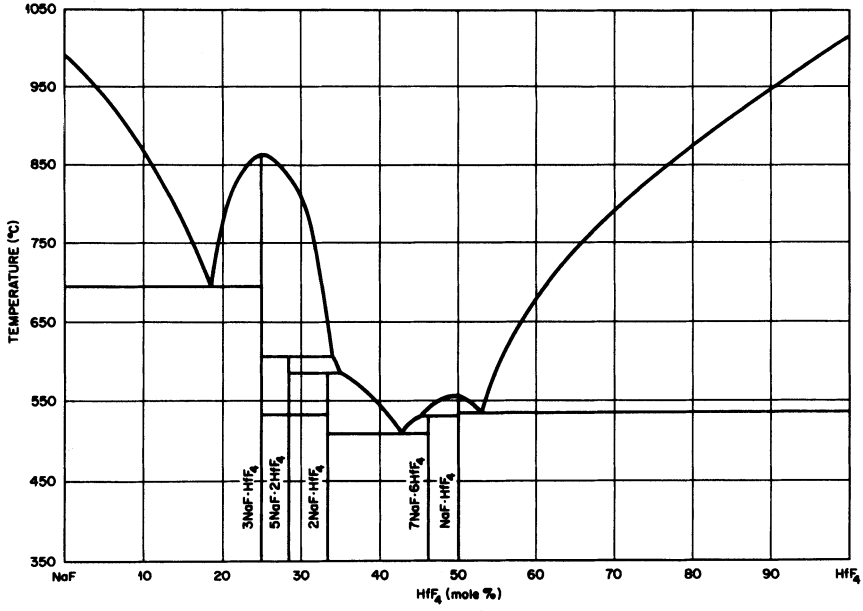


Fig. 89. The system NaF–HfF_4.

published work (1965–1968). Invariant and singular equilibrium points are given in the table.

Mole % HfF_4 in liquid	Temperature	Type of equilibrium
18.5	695	Eutectic
25	863	Congruent melting point
34	606	Peritectic
	535	Inversion
35	586	Peritectic
	Below 350	Inversion
	Below 350	Inversion
43	510	Eutectic
45	531	Peritectic
50	557	Congruent melting point
53	535	Eutectic

Structural data: $3NaF \cdot HfF_4$, hexagonal, $a = 6.1$, $c = 10.84$ [G. D. Brunton *et al.*, ORNL-3761 (1965)].

A determination of the crystal structure of $3NaF \cdot HfF_4$ has been reported by L. A. Harris *et al.*, *Acta Cryst.* **12**:172 (1959).

The System NaF–HF_4. See Fig. 90. A phase diagram of the NaF–HfF_4 system was reported by I. N. Sheiko, V. T. Mal'tsev, and G. A. Bukhalova, *Ukr. Khim. Zh.* **32**:1292 (1966), as shown in Fig. 90. The disparity between this phase diagram and that shown in Fig. 89 indicates that further investigations of the system would be useful.

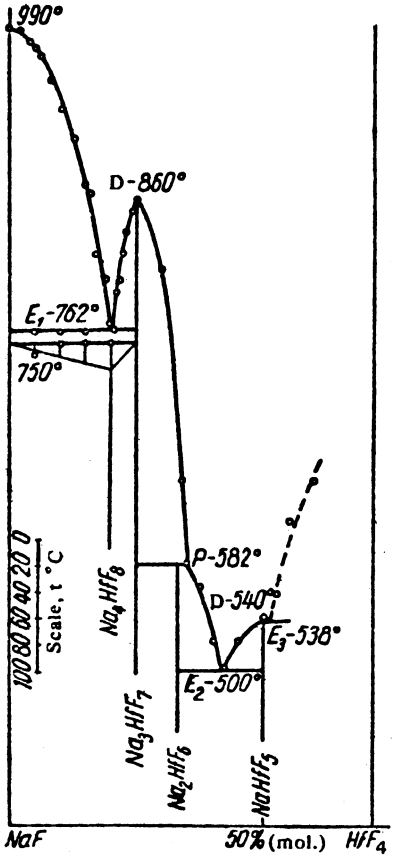


Fig. 90. The system $NaF–HfF_4$.

The System $KF–HfF_4$. See Fig. 91. Source: I. N. Sheiko, V. T. Mal'tsev, and G. A. Bukhalova, *Ukr. Khim. Zh.* **32**:1292 (1966). $E = 766°$, 13 % HfF_4; m.p. $3KF \cdot HfF_4$ 923°; $P = 608°$, 35 % HfF_4; $E = 408°$, 40 % HfF_4; m.p. $KF \cdot HfF_4$ 432°; $E = 420°$, 57 % HfF_4.

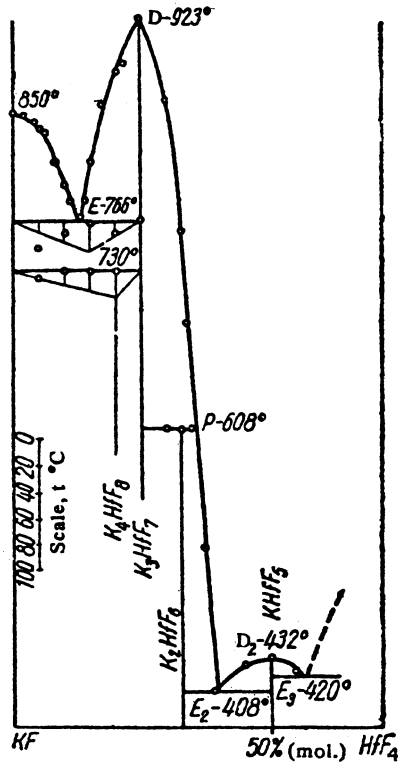


Fig. 91. The system $KF–HfF_4$.

The System $LiF–ThF_4$. See Fig. 92. Source: R. E. Thoma, H. Insley, B. S. Landau, H. A. Friedman, and W. R. Grimes, *J. Phys. Chem.*

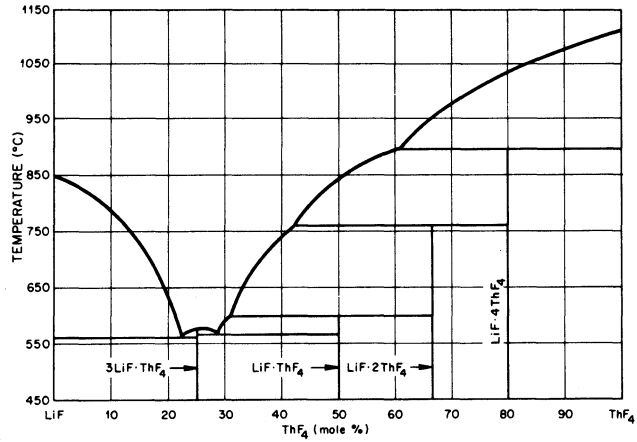


Fig. 92. The system $LiF–ThF_4$.

63, 1266 (1959). Invariant and singular equilibrium points are given in the table.

Mole % ThF_4 in liquid	Temperature	Type of equilibrium
23	568	Eutectic
25	573	Congruent melting point
29	565	Eutectic
30.5	597	Peritectic
42	762	Peritectic
62	897	Peritectic

Structural data: $3LiF \cdot ThF_4$: tetragonal, $P4/nmm$, $a = 6.206$, $c = 6.470$ [L. A. Harris, G. D. White, and R. E. Thoma, *J. Phys. Chem.* **63**:1974 (1959)]. $LiF \cdot ThF_4$: tetragonal, $141/a$, $a = 15.20$, $c = 6.60$.

The System NaF–ThF_4. See Fig. 93. Source: R. E. Thoma *et al.*, *J. Phys. Chem.* **63**:1266 (1959), and Roy E. Thoma, *J. Inorg. Nucl. Chem.*

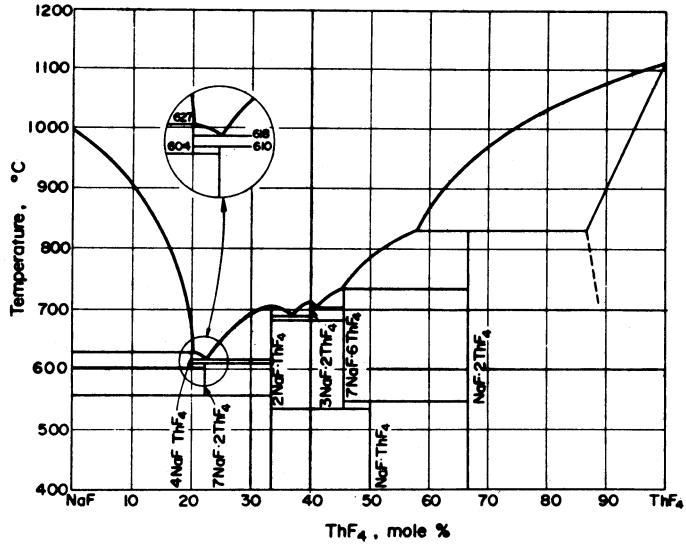


Fig. 93. The system NaF–ThF_4.

34:2747 (1972). Invariant and singular equilibrium points are given in the table.

Mole % ThF_4 in liquid	Temperature	Type of equilibrium
21.5	645	Peritectic
22.5	618	Eutectic
	604	Inversion
	558	Decomposition
33.3	705	Congruent melting point
37	690	Eutectic
40	712	Congruent melting point
41	705	Eutectic
	683	Decomposition
45.5	730	Peritectic
58	831	Peritectic

Structural data: $4NaF \cdot ThF_4$: cubic, $a = 11.04$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948). $7NaF \cdot 2ThF_4$: hexagonal, $a = 12.713$, $c = 10.377$ [G. D. Brunton *et al.*, ORNL-3761 (1965)]. $\beta_2 = 2NaF \cdot ThF_4$: hexagonal, $P32$, $a = 5.99$, $c = 3.81$ [W. H. Zachariasen, *ibid.*]. $\delta = 2NaF \cdot ThF_4$: hexagonal, $a = 6.14$, $c = 7.36$ [W. H. Zachariasen, *ibid.*]. $3NaF \cdot 2ThF_4$: cubic, $a = 5.62$ [G. D. Brunton, *ibid.*]. $7NaF \cdot 6ThF_4$: hexagonal, $R\bar{3}$, $a = 14.96$, $c = 9.912$ [G. D. Brunton, *ibid.*]. $NaF \cdot 2ThF_4$: cubic, $I\bar{4}3M$, $a = 8.722$ [W. H. Zachariasen, *Acta Cryst.* **2**:390 (1949)].

The System KF–ThF_4. See Fig. 94. Source: W. J. Asker, E. R. Segnit, and A. W. Wylie, *J. Chem. Soc.* **1952**:4470. The system has also been described by A. G. Bergman and E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **53**:753 (1941), and by V. S. Emelyanov and A. J. Evstyukhin, *J. Nucl. Energy* **5**:108 (1957). From independent examination at ORNL, the invariant equilibria listed by Asker *et al.* are considered to be the most reliable. This examination indicated that K_2ThF_6 melts congruently. Invariant and singular equilibrium points are given in the table.

Mole % ThF_4 in liquid	Temperature	Type of equilibrium
14	694	Eutectic
	635	Inversion
	712	Peritectic
25	865	Congruent melting point
	570	Decomposition
31	691	Eutectic
	747	Peritectic
	645	Inversion
50[a]	905	Congruent melting point
56	875	Eutectic
66	930	Peritectic
75	990	Congruent melting point
78	980	Eutectic

[a] It has been shown that the compound labeled by Asker *et al.* has the actual formula $7KF \cdot 6ThF_4$; cf. R. E. Thoma, Crystal Structures of Some Compounds of UF_4 and ThF_4 with Alkali Fluorides, ORNL-CF-58-12-40 (December 11, 1958).

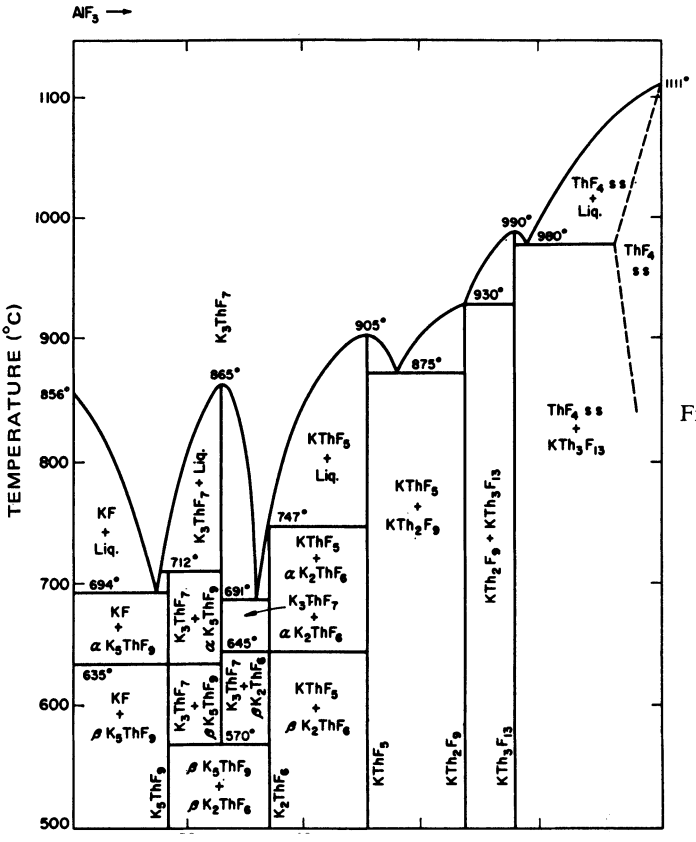


Fig. 94. The system KF–ThF_4.

Structural data: $5KF \cdot ThF_4$: orthorhombic, $Cmc2_1$, $a = 7.848$, $b = 12.840$, $c = 10.785$ [R. R. Ryan and R. A. Penneman, *Acta Cryst.* **B27**:829 (1971). $2KF \cdot ThF_4$: hexagonal, $P\bar{6}2m$, $a = 6.565$, $c = 3.815$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2157 (1948)].

The System RbF–ThF_4. See Fig. 95. Source: R. E. Thoma, H. Insley, G. M. Hebert, and G. D. Brunton, ORNL, unpublished work (1961–1965).

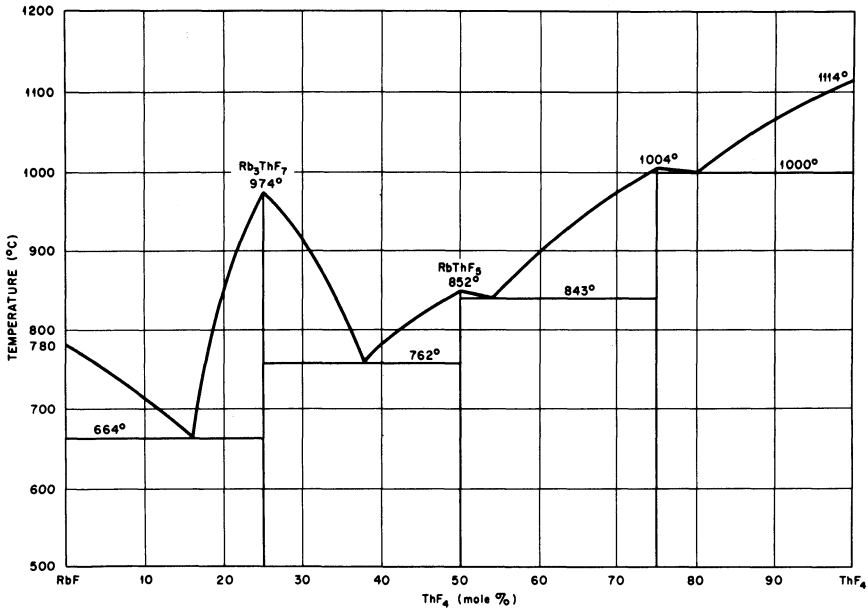


Fig. 95. The system RbF–ThF_4.

Invariant and singular equilibrium points are given in the table.

Mole % ThF_4	Temperature	Type of equilibrium
15.5	665	Eutectic
	545	Upper temp. of stability of $5RbF \cdot ThF_4$
20.5	895	Peritectic
25	973	m.p. $3RbF \cdot ThF_4$
	488	Lower temp. of stability of $3RbF \cdot ThF_4$
31	845	Eutectic
33.3	862	m.p. $2RbF \cdot ThF_4$
39.5	803	Eutectic
46.2	835	m.p. $7RbF \cdot 6ThF_4$
51.5	800	Eutectic
65.5	925	Peritectic
75	995	Peritectic

An earlier description of the system was given by E. P. Dergunov and A. G. Bergman, *Dokl. Akad. Nauk SSSR* **60**:391 (1948). See Fig. 1505, *Phase Diagrams for Ceramists* (1964). The phase diagram shown here was constructed from experimental data obtained as a means of evaluating the Dergunov and Bergman report.

Structural data: $3RbF \cdot ThF_4$: cubic, $a = 9.62$ [E. P. Dergunov and A. G. Bergman, *ibid.*]. $2RbF \cdot ThF_4$: hexagonal, $P\bar{6}2m$, $a = 6.85$, $c = 3.83$ [L. A. Harris, *Acta Cryst.* **13**:502 (1960)]. $7RbF \cdot 6ThF_4$: rhombohedral, $R\bar{3}$, $a = 9.58$; hexagonal parameters: $a = 15.39$, $c = 10.73$ [G. D. Brunton *et al.*, ORNL-3761 (1965). $RbF \cdot 3ThF_4$: $P2_1ma$; $a = 8.6490$, $b = 8.176$, $c = 7.4453$ [George Brunton, *Acta Cryst.* **B27**:1823 (1971). $RbF \cdot 6ThF_4$: hexagonal, $R3m$, $a = 8.33$, $c = 25.40$ [G. D. Brunton, *Acta Cryst.* **B28**:144 (1972)]. X-ray powder diffraction data are listed in ORNL-3761 for all of the intermediate phases shown in the phase diagram.

The System CsF–ThF_4. See Fig. 96. Source: R. E. Thoma and T. S. Carlton, *J. Inorg. Nucl. Chem.* **17**:88 (1961). Invariant and singular equilibrium points are given in the table.

Mole % ThF_4 in liquid	Invariant temperature	Type of equilibrium
9	615	Eutectic
25	980	Congruent melting point
30.5	838	Eutectic
33.3	869	Congruent melting point
40	775	Eutectic
50	839	Congruent melting point
53	797	Eutectic
55	842	Peritectic
56.5	860	Peritectic
	813	Lower limit of stability of $CsF \cdot 2ThF_4$
75	830	Upper limit of stability of $CsF \cdot 3ThF_4$
84	1010	Peritectic

Structural data: $3CsF \cdot ThF_4$: cubic, $a = 10.04$.

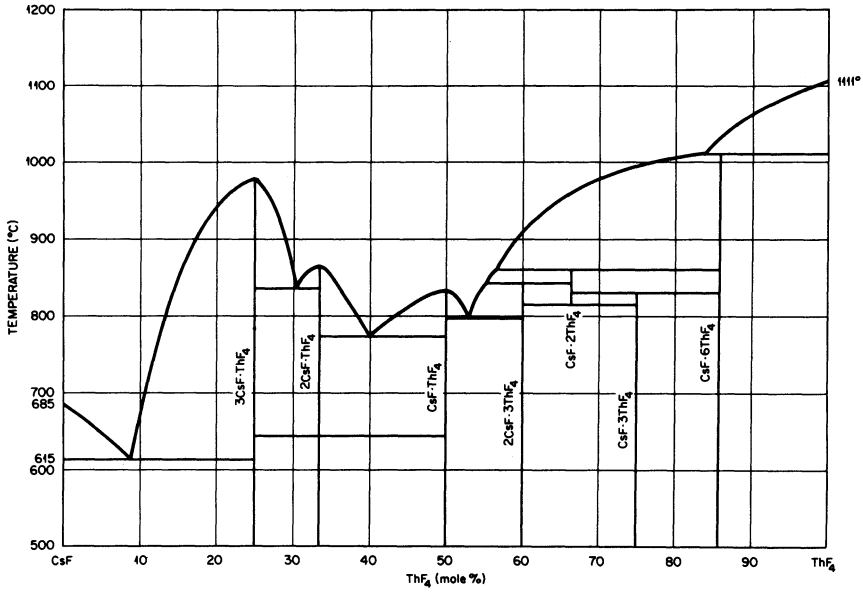


Fig. 96. The system CsF–ThF_4.

The System TlF–ThF_4. See Fig. 97. Source: D. Avignant and J. C. Cousseins, *Compt. Rend. Acad. Sci., Ser. C* **272**:2151 (1971). Invariant and singular equilibrium points are given in the table.

Mole % ThF_4	Temperature	Type of equilibrium
10	285	Eutectic
23	378	Peritectic
24	385	Peritectic
42.5	701	Peritectic
	519	Upper temp. of stability of TlF · ThF_4
60.5	902	Peritectic
65	956	Peritectic

On the basis of x-ray powder diffraction data, it was inferred that 3TlF · ThF_4 is cubic; at low temperatures TlF · UF_6 is orthorhombic;

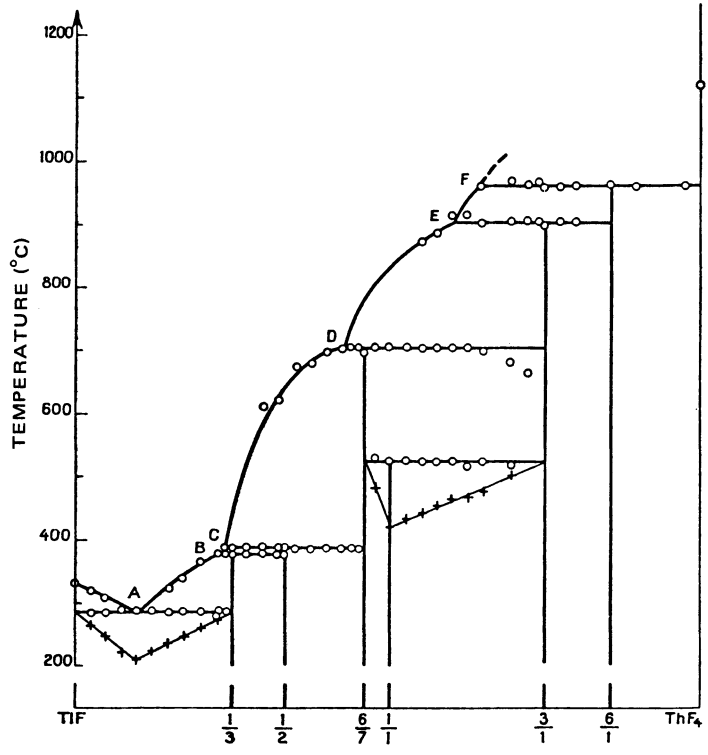


Fig. 97. The system TlF–ThF_4.

$7TlF \cdot 6ThF_4$ is rhombohedral and isostructural with $7NaF \cdot 6ZrF_4$, with hexagonal parameters $a = 15.60$, $c = 10.84$; and $TlF \cdot 6ThF_4$ is hexagonal, $P6_3/mmc$, $a = 8.31$, $c = 16.86$.

The System LiF–UF_4. See Fig. 98. Source: C. J. Barton *et al.*, *J. Am. Ceram. Soc.* **41**:63 (1958). Lower stability limit for $4LiF \cdot UF_4$ 470°; $P = 500°$, 26% UF_4; $E = 490°$, 27% UF_4; $P = 610°$, 40% UF_4; $P = 775°$, 57% UF_4.

Structural data: $4LiF \cdot UF_4$, orthorhombic, *Pnma*, $a = 9.960$, $b = 9.883$, $c = 5.986$ [G. D. Brunton, *J. Inorg. Nucl. Chem.* **29**:1631 (1967)]. $3LiF \cdot UF_4$ (metastable phase): tetragonal, $a = 6.13$, $c = 6.39$ [G. D. Brunton *et al.*, ORNL-3761]. $LiF \cdot UF_4$: tetragonal, $I4_1/c$, $a = 14.884$, $c = 6.547$ [G. D. Brunton, *Acta Cryst.* **21**:814 (1966)].

The compound $LiF \cdot UF_4$ was reported in the source reference to have the formula $7LiF \cdot 6UF_4$. The correct stoichiometry was established by determination of the crystal structure.

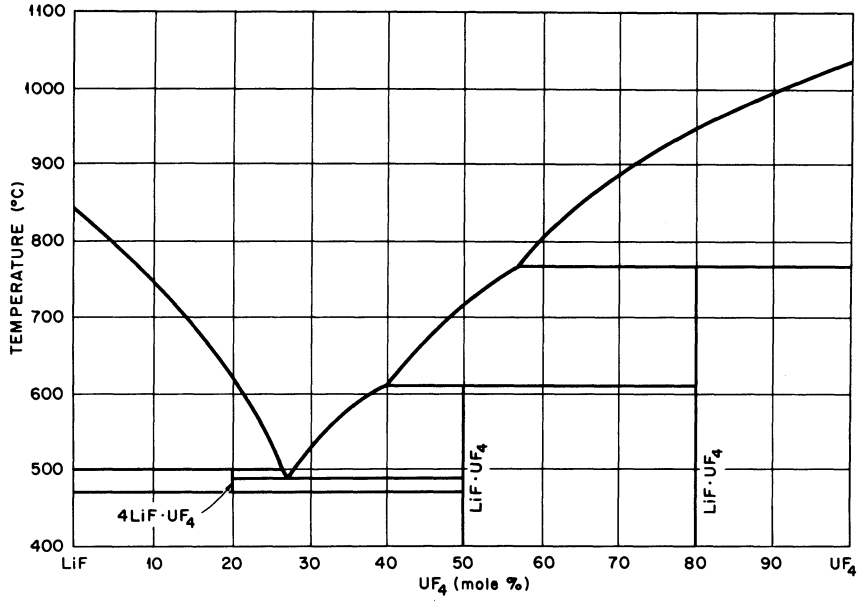


Fig. 98. The system LiF–UF_4.

The System NaF–UF_4. See Fig. 99. Source: C. J. Barton *et al.*, *J. Am. Ceram. Soc.* **41**:63 (1958). Invariant and singular equilibrium points are given in the table.

Mole % UF_4 in liquid	Temperature	Type of equilibrium	Coexisting phases
21.5	618	Eutectic	$L \rightleftarrows \alpha \cdot 3NaF \cdot UF_4 + NaF$
25	629	Congruent melting point	$L \rightleftarrows \alpha \cdot 3NaF \cdot UF_4$
	528	Inversion	$\alpha \cdot 3NaF \cdot UF_4 \rightleftarrows \beta \cdot 3NaF \cdot UF_4$
	497	Decomposition	$\beta \cdot 3NaF \cdot UF_4 \rightleftarrows NaF + 2NaF \cdot UF_4$
28	623	Eutectic	$L \rightleftarrows \alpha \cdot 3NaF \cdot UF_4 + 2NaF \cdot UF_4$
32.5	648	Peritectic	$L + 5NaF \cdot 3UF_4 \rightleftarrows 2NaF \cdot UF_4$
37	673	Peritectic	$L + 7NaF \cdot 6UF_4 \rightleftarrows 5NaF \cdot 3UF_4$
	630	Decomposition	$5NaF \cdot 3UF_4 \rightleftarrows 2NaF \cdot UF_4 + 7NaF \cdot 6UF_4$
46.2	718	Congruent melting point	$L \rightleftarrows 7NaF \cdot 6UF_4$
56	680	Eutectic	$L \rightleftarrows 7NaF \cdot 6UF_4 + UF_4$
	660	Upper stability limit of $NaF \cdot 2UF_4$	$7NaF \cdot 6UF_4 + UF_4 \rightleftarrows NaF \cdot 2UF_4$

Structural data: $7NaF \cdot 2UF_4$: orthorhombic, *Fmmm*, *Fmmm*2, or *F*222, $a = 17.7$, $b = 29.8$, $c = 12.7$ [G. D. Brunton *et al.*, ORNL-3761]. $3NaF \cdot UF_4$: tetragonal, *I*4*mmm*, $a = 5.448$, $c = 10.896$ [G. D. Brunton *et al.*, *ibid.*; see also W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948)].

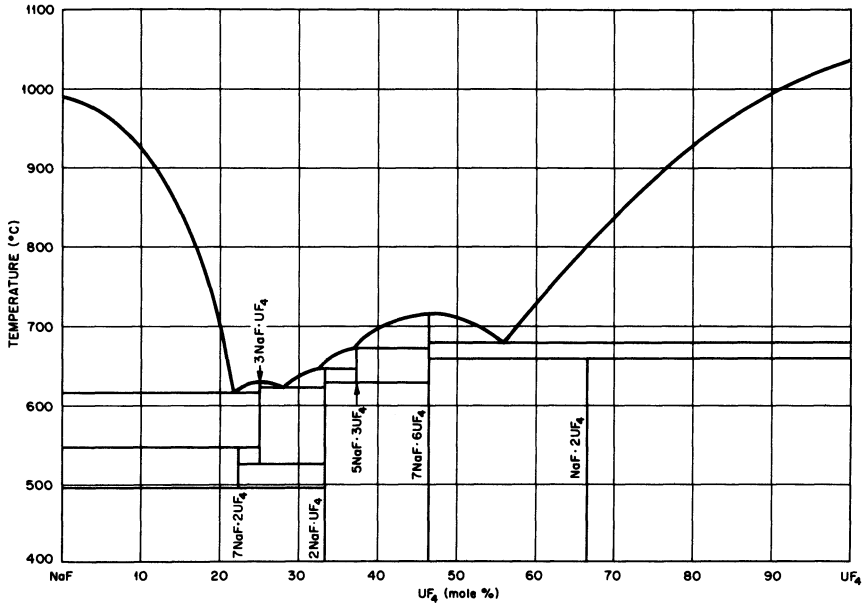


Fig. 99. The system $NaF–UF_4$.

δ-$2NaF \cdot UF_4$: hexagonal, $a = 6.11$, $c = 7.25$ [W. H. Zachariasen, *ibid.*]. γ-$2NaF \cdot UF_4$: orthorhombic, *Immm*, $a = 5.56$, $b = 4.01$, $c = 11.64$ [W. H. Zachariasen, *ibid.*]. β_2-$2NaF \cdot UF_4$: hexagonal, *P*32, $a = 5.95$, $c = 3.73$ [W. H. Zachariasen, *ibid.*]. $7NaF \cdot 6UF_4$: hexagonal, *R*3, $a = 14.72$, $c = 9.84$ [G. D. Brunton *et al.*, *ibid.*].

The System KF–UF_4. See Fig. 100. Source: R. E. Thoma, H. Insley, B. S. Landau, H. A. Friedman, and W. R. Grimes, *J. Am. Ceram. Soc.* **41**:539 (1958). Invariant and singular equilibrium points are given in the table.

Mole % UF_4 in liquid	Temperature	Type of equilibrium
15	735	Eutectic
25	957	Congruent melting point
	608	Decomposition
35	755	Peritectic
38.5	740	Eutectic
46.2	789	Congruent melting point
54.5	735	Eutectic
65	765	Peritectic

Structural data; $3KF \cdot UF_4$: orthorhombic, *Pnmm* or $Pnm2_1$, $a = 6.59$, $b = 8.30$, $c = 7.20$ [J. H. Burns, ORNL-3262 (1962)]. β_1-$2KF \cdot UF_4$: hexagonal, $P\bar{6}2m$, $a = 6.5528$, $c = 3.749$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948)]. $7KF \cdot 6UF_4$: rhombohedral, $R\bar{3}$, $a = 15.09$, $c = 10.38$ [G. D. Brunton *et al.*, ORNL-3761]. $KF \cdot 2UF_4$: orthorhombic, *Pnam*, $a = 8.7021$, $b = 11.477$, $c = 7.0350$ [G. D. Brunton, *Acta Cryst.* **B25**:1919 (1969)]. $KF \cdot 6UF_4$: hexagonal, *P6/mmc*, $a = 8.18$, $c = 16.42$ [W. H. Zachariasen, *J. Am. Chem. Soc.* **70**:2147 (1948).

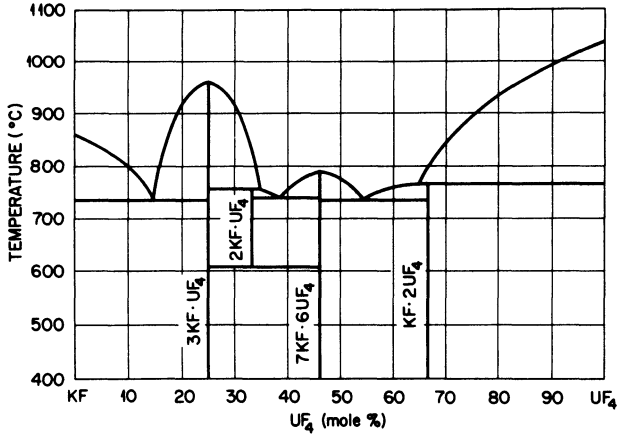


Fig. 100. The system KF–UF_4.

The System RbF–UF_4. See Fig. 101. Source: R. E. Thoma, H. Insley, B. S. Landau, H. A. Friedman, and W. R. Grimes, *J. Am. Ceram. Soc.*

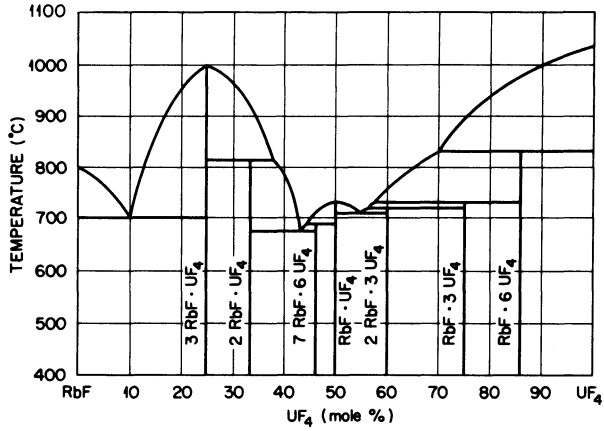


Fig. 101. The system RbF–UF_4.

41:538 (1958). Invariant and singular equilibrium points are given in the table.

Mole % UF_4 in liquid	Temperature	Type of equilibrium
10	710	Eutectic
25	995	Congruent melting point
38	818	Peritectic
43.5	675	Eutectic
44	693	Peritectic
50	735	Congruent melting point
55	714	Eutectic
56.5	722	Peritectic
57	730	Peritectic
70.5	832	Peritectic

Structural data: $3RbF \cdot UF_4$: cubic, $a = 9.5667$ [G. D. Brunton *et al.*, ORNL-3761]. $2RbF \cdot UF_4$: orthorhombic, *cmcm*, $a = 6.998$, $b = 12.098$, $c = 7.669$ [T. K. Keenan, *Inorg. Nucl. Chem. Letters* **3**:463 (1967). $7RbF \cdot 6UF_4$: rhombohedral, $a = 9.595$; hexagonal parameters: $a = 15.49$, $c = 10.42$]. $RbF \cdot 6UF_4$: hexagonal, $P6_3/mmc$, $a = 8.195$, $c = 16.37$ [G. D. Brunton *et al.*, ORNL-3761].

The System CsF–UF_4. See Fig. 102. Source: C. J. Barton, L. M. Bratcher, T. N. McVay, and R. E. Thoma, ORNL, unpublished work

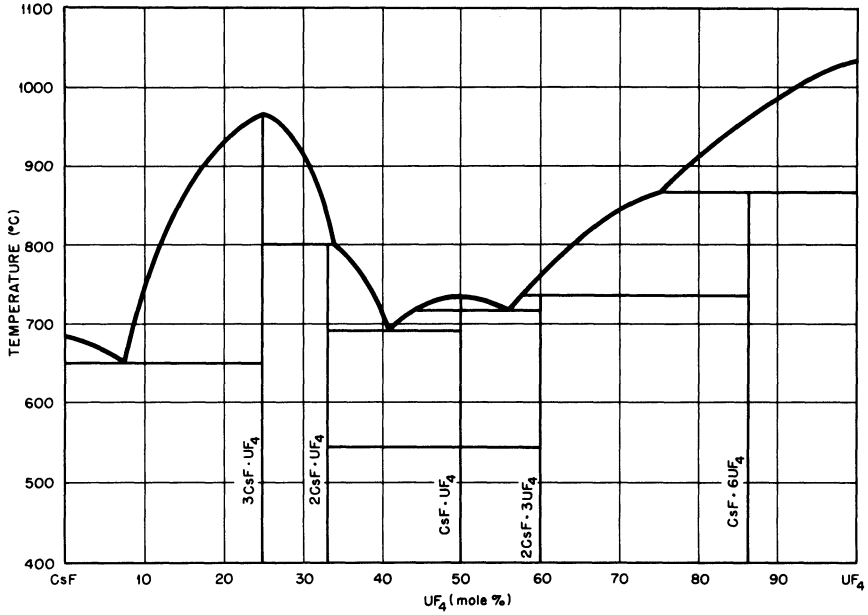


Fig. 102. The system CsF–UF_4.

(1951–1953). Invariant and singular equilibrium points are given in the table.

Mole % UF_4 in liquid	Temperature	Type of equilibrium
7.5	650	Eutectic
25	970	Congruent melting point
34	800	Peritectic
41	695	Eutectic
50	735	Congruent melting point
	550	Inversion
53	725	Eutectic

Structural data: $2CsF \cdot 3UF_4$, monoclinic, $P21/M$, $a_0 = 18.316$, $b_0 = 8.455$, $c_0 = 8.234$, $\beta = 96.12°$ [ORNL-3761]; $CsF \cdot 6UF_4$ hexagonal,

$P63/mmc$, $a_0 = 8.2424$, $c_0 = 16.412$ [G. D. Brunton, *Acta Cryst.* **B27**:225 (1971)].

The System TlF–UF_4. See Fig. 103. Source: D. Avignant and J. C. Cousseins, *Compt. Rend.* **272**:2151 (1971). $E = 274°$, 87.5 TlF; Tl_4UF_8 inversion at 227°; $P = 542°$, 77 TlF; Tl_3UF_7 lower temperature of stability, 285°; m.p. Tl_2UF_6 620°; $E = 598°$, 61 TlF; $Tl_7U_6F_3$, upper temperature of stability 548°; $P = 640$, 53.5 TlF; $P = 674°$, 40 TlF; $P = 717°$, composition undefined.

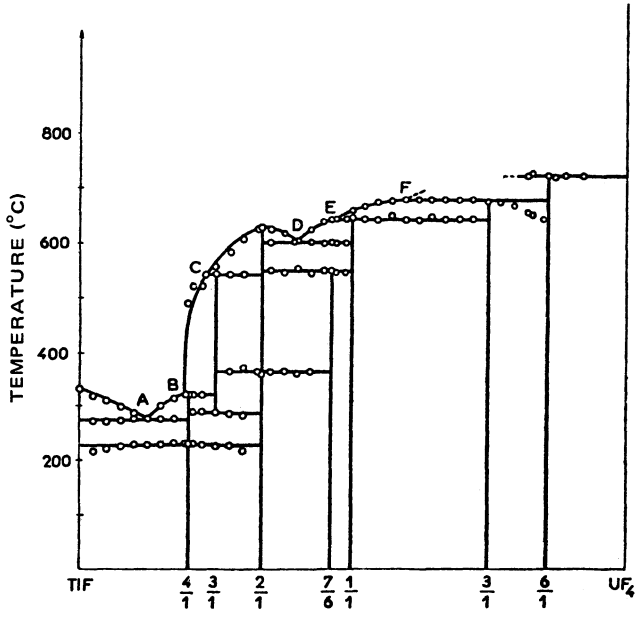


Fig. 103. The system TlF–UF_4.

Crystallographic data: Tl_3UF_7: cubic, $a = 9.434$; Tl_2UF_6: orthorhombic, $a = 4.07$, $b = 6.97$, $c = 10.80$; TlU_6F_{25}: hexagonal, $a = 8.18$, $c = 16.46$.

The System KF–NbF_5. See Fig. 104. Source: Ahmed Mukhtar and Rene Winand, *Compt. Rend.* **260**:3675 (1965). $E = 723°$, 23% K_2NbF_7; congruent m.p. of KF · K_2NbF_7 780°; $E = 706°$, 80% K_2NbF_7.

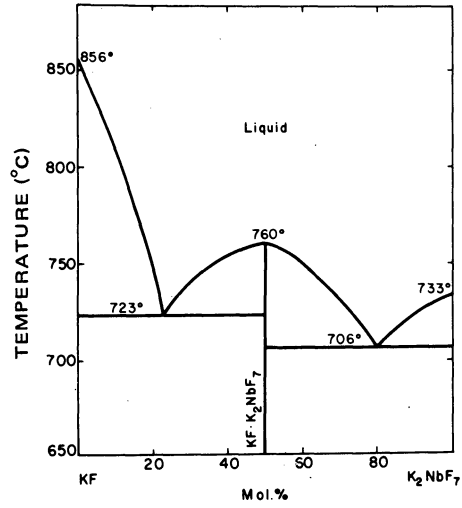


Fig. 104. The system $KF–NbF_5$.

The System $KF–TaF_5$. See Fig. 105. Source: Ts'ui Ping-hsin, N. P. Luzhnaya, and V. I. Konstantinov, *Zh. Neorg. Khim.* **8**:389 (1963). $E = 717°$, 21.5% KF; $E = 727°$, 73% KF (est.); liquidus curves for the two

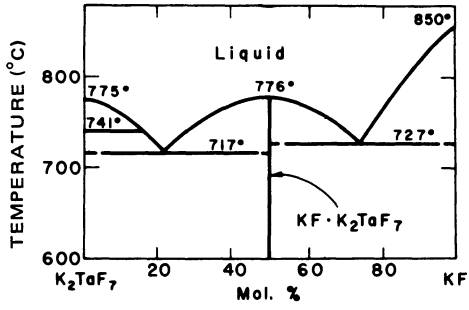


Fig. 105. The system $KF–TaF_5$.

crystalline modifications of K_2TaF_7 intersect at the transformation point located at 741° and at 18% KF.

The System $BeF_2–MgF_2$. See Fig. 106. Source: W. E. Counts, Rustum Roy, and E. F. Osborn, *J. Am. Ceram. Soc.* **36**:14 (1953). $E = 528 \pm 5°$, 5% MgF_2.

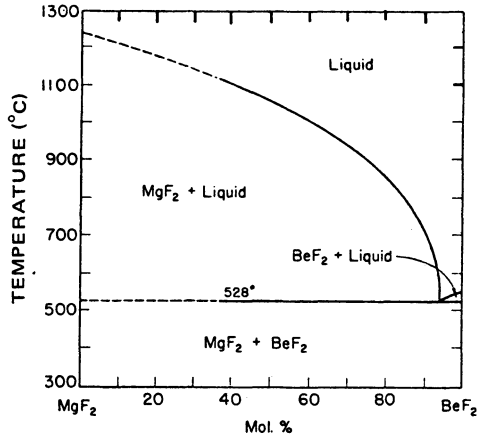


Fig. 106. The system BeF_2–MgF_2.

The System BeF_2–CaF_2. See Fig. 107. Source: W. E. Counts, Rustum Roy, and E. F. Osborn, *J. Am. Ceram. Soc.* **36**:12 (1953). $E = 495°$, 11% CaF_2; $P = 890°$, 43% CaF_2.

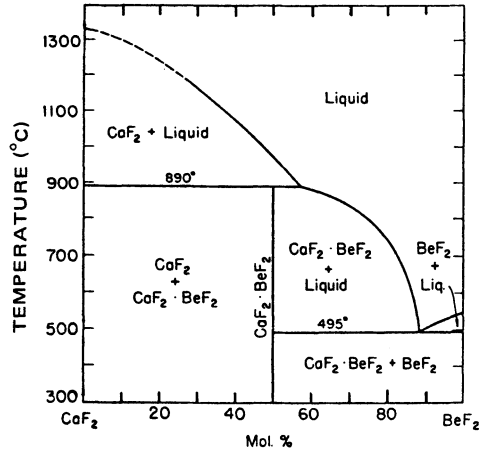


Fig. 107. The system BeF_2–CaF_2.

The System BeF_2–SrF_2. See Fig. 108. Source: O. N. Breusov, G. Trapp, A. V. Novoselova, and Yu. P. Simanov, *Zh. Neorg. Khim.* **4**:672 (1959). Invariant and singular equilibrium points are given in the table.

Mole % BeF_2	Temperature	Type of equilibrium
~100	582 ± 29	Eutectic
100	220	Polymorphic transition in crystalline BeF_2
92.5	774	Monotectic
75	774	Monotectic
50	954 ± 10	Congruent m.p.
	923	Polymorphic transition in crystalline $BeF_2 \cdot SrF_2$
60	883	Eutectic

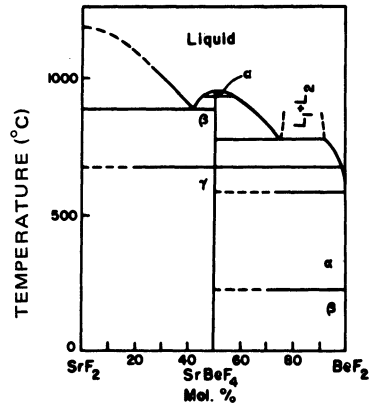


Fig. 108. The system BeF_2–SrF_2.

The System BeF_2–BaF_2. See Fig. 109. Source: D. F. Kirkina, A. V. Novoselova, and Yu. P. Simanov, *Zh. Neorg. Khim.* **1**:128 (1956). E =

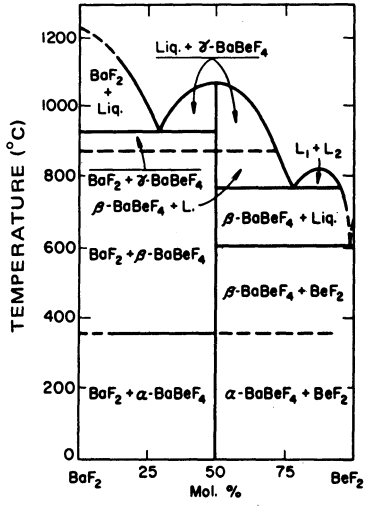


Fig. 109. The system BeF_2–BaF_2.

600°, 1–2 % BaF_2; intersections of the liquidus and binodal curves at 7 and 20 mole % BaF_2; congruent m.p. $BeF_2 \cdot BaF_2$ 1080°; $E = 920$, 72% BaF_2. $BeF_2 \cdot BaF_2$ polymorphic transformation temperatures occur at 870 and 360°.

The System BeF_2–PbF_2. See Fig. 110. Source: D. M. Roy, R. Roy, and E. F. Osborn, *J. Am. Ceram. Soc.* **37**:300 (1954). $E = 525°$, 90% BeF_2; m.p. $BeF_2 \cdot PbF_2$ 585 ± 5°; $E = 465°$, 21% BeF_2; m.p. $BeF_2 \cdot 3PbF_2$ 482 ± 5°; $E = 477°$, 79% PbF_2 (est.).

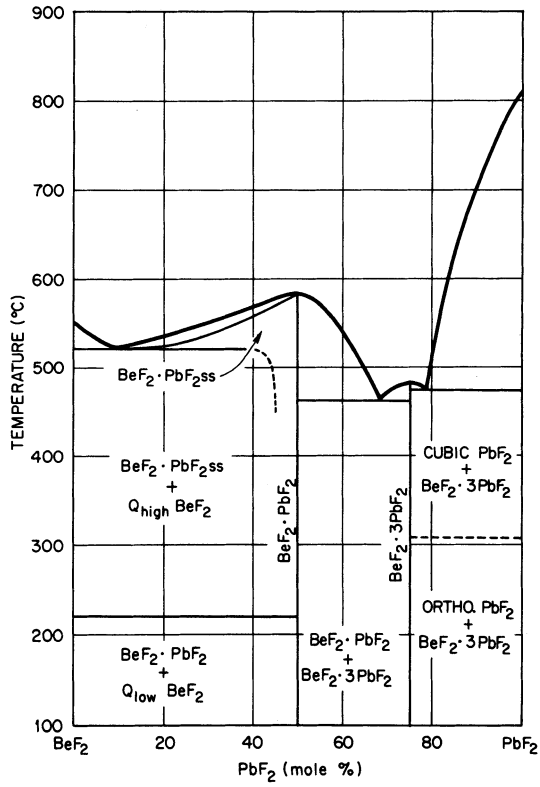


Fig. 110. The system BeF_2–PbF_2.

The System MgF_2–CaF_2. See Fig. 111. Source: C. J. Barton, L. M. Bratcher, and J. P. Blakely, ORNL, unpublished work (1955–1956). The eutectic occurs at 48 mole % CaF_2, and at 985°. The initial phase diagram of the MgF_2–CaF_2 system was reported by E. Beck, *Metallurgie* **5**:504 (1908). Phase diagrams of the ternary systems LiF–KF–MgF_2 A. G. Berg-

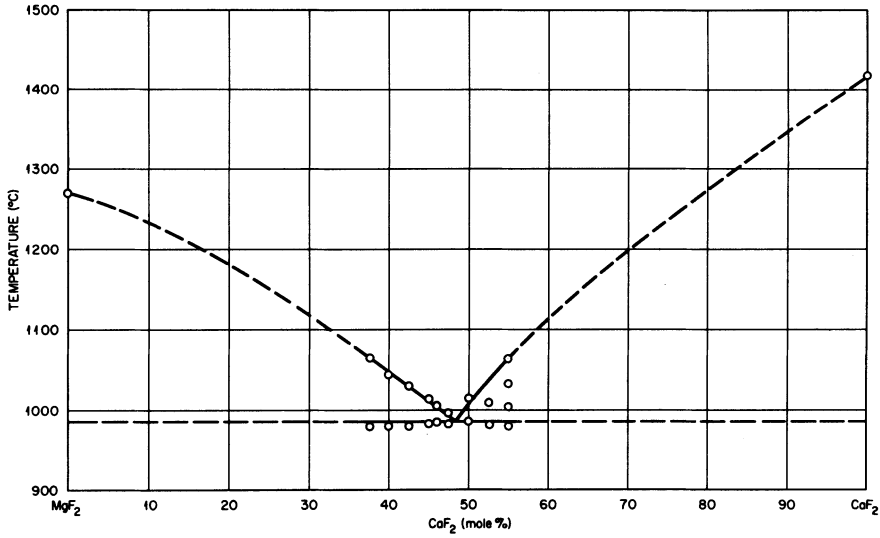


Fig. 111. The system MgF_2–CaF_2.

man and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:75 (1941)] and NaF–KF–MgF_2 [A. G. Bergman and E. P. Dergunov, *op. cit.*, **48**:330 (1945)] include information pertaining to the binary system.

The System MgF_2–BaF_2. See Fig. 112. Source: M. Okamoto and U. Nisioka, *Sci. Rept. Tohoku Univ., First Ser.* **24**:142 (1935–1936). E = 910°, 15% MgF_2; P = 930°, 19% MgF_2.

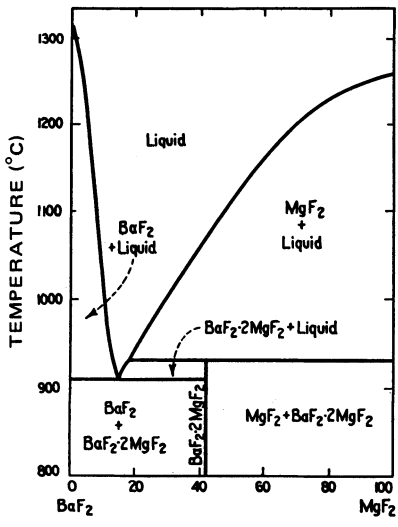


Fig. 112. The system MgF_2–BaF_2.

The System CaF_2–CdF_2. See Fig. 113. Source: M. P. O'Horo and W. B. White, *J. Am. Ceram. Soc.* **54**:588 (1971). CaF_2 and CdF_2 form a continuous series of solid solutions without a temperature minimum.

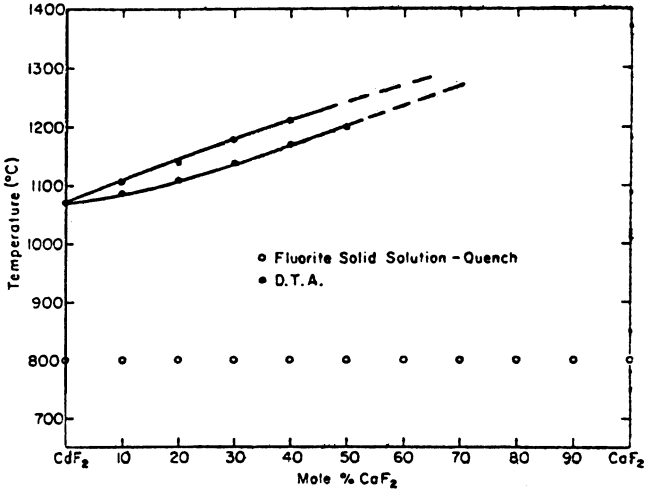


Fig. 113. The system CaF_2–CdF_2.

The System NiF_2–BaF_2. See Fig. 114. Source: J. C. Cousseins and M. Samouel, *Compt. Rend.* **265**:1121 (1967). $P = 1000°$, 39.5% BaF_2;

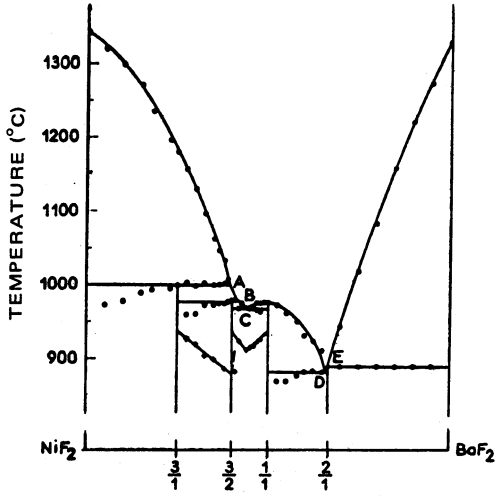


Fig. 114. The system NiF_2–BaF_2.

$P = 977°$, 42% BaF_2; m.p. $NiF_2 \cdot BaF_2$ 977°; $E = 882°$, 60% BaF_2; $P = 890°$, 65% BaF_2.

Structural data: $NiF \cdot BaF_2$: orthorhombic, $a = 14.46$, $b = 4.15$, $c = 5.80$.

The System CuF_2–BaF_2. See Fig. 115. Source: M. Samouel, *Compt. Rend.* **270**:1805 (1970). $E = 616°$, 22.5% BaF_2; $P = 624°$, 26% BaF_2; $E =$

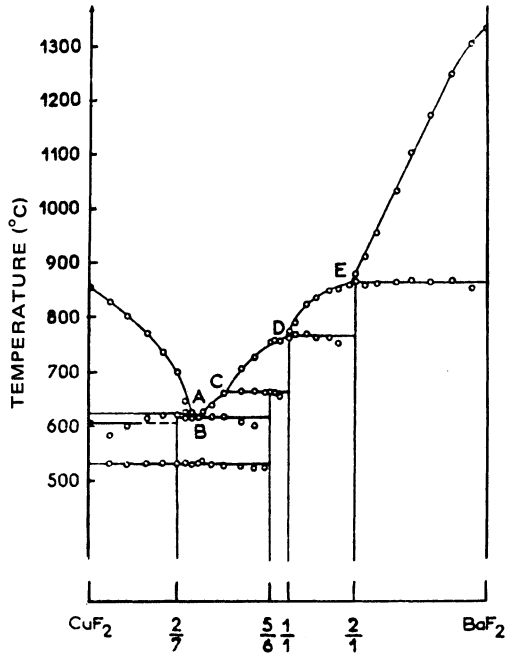


Fig. 115. The system CuF_2–BaF_2.

616°, 27.5% BaF_2; $P = 663°$, 34.5% BaF_2; $P = 764$, 49.4% BaF_2; $P = 863$, 65% BaF_2.

The System SrF_2–BaF_2. See Fig. 116. Source: Ralph H. Nafziger, *J. Am. Ceram. Soc.* **54**:467 (1971). SrF_2 and BaF_2 form a continuous series of solid solutions with a shallow minimum in the liquidus curve at 27 ± 2 wt. % SrF_2 and 1305°. Length of each vertical line indicates one standard deviation for replicate liquidus and solidus temperature measurements for each composition studied.

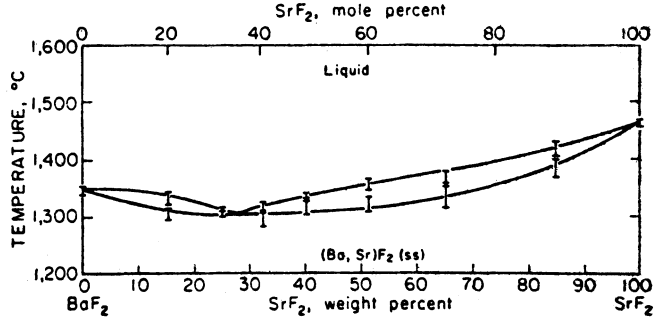


Fig. 116. The system SrF_2–BaF_2.

The System CdF_2–PbF_2. See Fig. 117. Source: M. P. O'Horo and W. B. White, *J. Am. Ceram. Soc.* **54**:588 (1971). The minimum temperature in the solid solution occurs at 60 mole % PbF_2. Its exact composition was not stated.

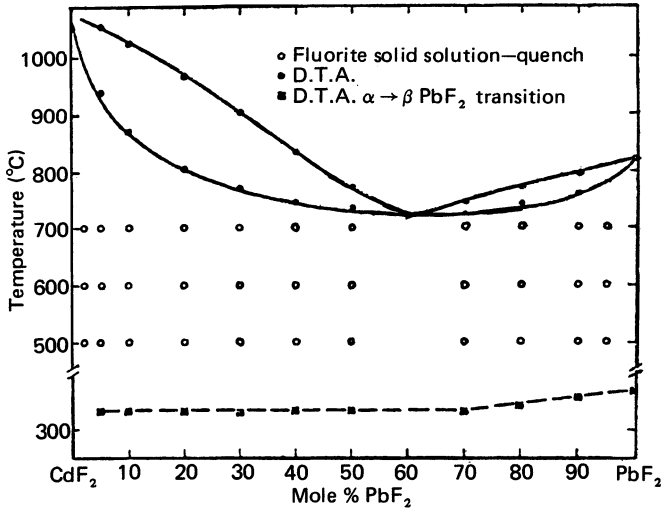


Fig. 117. The system CdF_2–PbF_2.

The System ZnF_2–CdF_2. See Fig. 118. Source: M. P. O'Horo and W. B. White, *J. Am. Ceram. Soc.* **54**:588 (1971). The invariant equilibrium points in the CdF_2–ZnF_2 system were not given in the source reference.

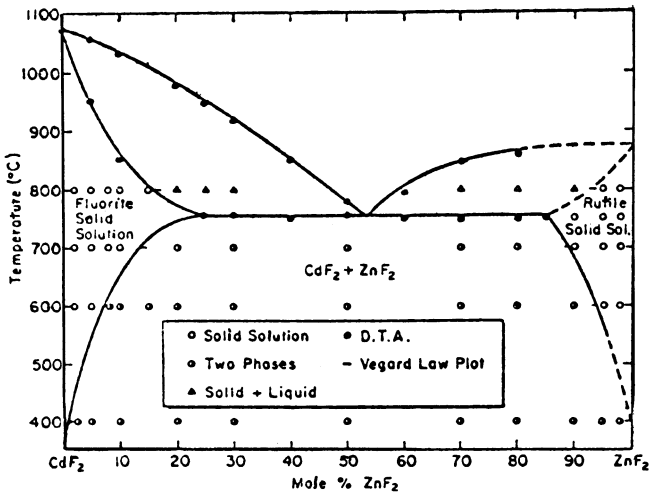


Fig. 118. The system ZnF_2–CdF_2.

The System MgF_2–ScF_3. See Fig. 119. Source: L. N. Komissarova and B. I. Pokrovskii, *Dokl. Akad. Nauk SSSR* **149**:300 (1963). $E = 1095°$, 34% ScF_3. The eutectoid invariant point at 840° occurs at 5 mole % ScF_3.

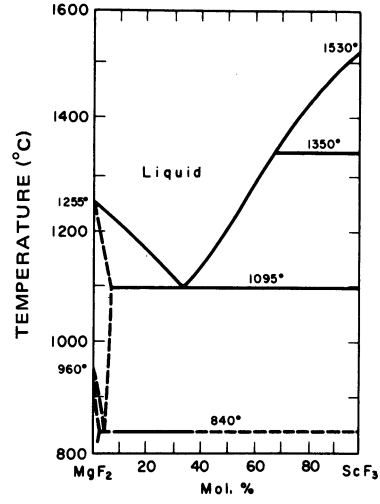


Fig. 119. The system MgF_2–ScF_3.

The System MgF_2–YF_3. See Fig. 120. Source: R. H. Nafziger, R. L. Lincoln, and R. Riazance, *J. Inorg. Nucl. Chem.* **35**:421 (1973). The authors

of the source reference describe the system as "of the simple eutectic type with a very small amount of orthorhombic (α-YF_3) solid solution and perhaps a limited amount of YF_3 solid solubility in MgF_2, but did not list the exact temperature nor composition of the eutectic.

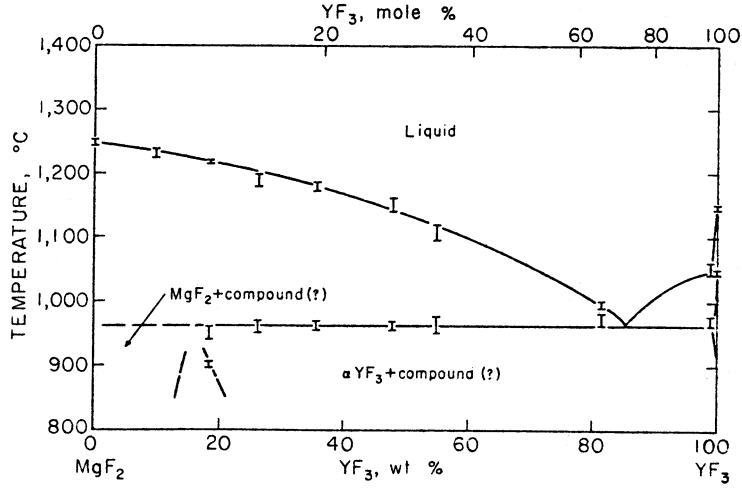


Fig. 120. The system MgF_2–YF_3.

The System CaF_2–YF_3. See Fig. 121. Source: J. Short and Rustum Roy, *J. Phys. Chem.* **67**:1861 (1963). Invariant equilibria were not fixed

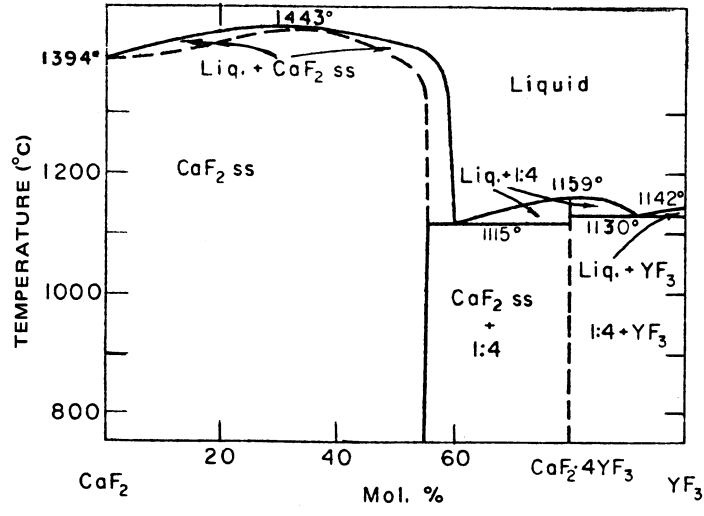


Fig. 121. The system CaF_2–YF_3.

precisely by Short and Roy. They state that the 1443° maximum in the liquidus curve occurs at about 33 mole % YF_3 and that the limit of solid solution of YF_3 in CaF_2 is about 55 mole % YF_3. The 1115° eutectic is located at approximately 58 mole % YF_3. The phase melting congruently at 1159° was tentatively assigned the formula $CaF_2 \cdot 4YF_3$, although the authors recognized that it could represent the limit of CaF_2 solubility in a "hitherto unreported hexagonal form of YF_3."

The System SrF_2–YF_3. See Fig. 122. Source: R. H. Nafziger, R. L. Lincoln, and N. Riazance, *J. Inorg. Nucl. Chem.* **35**:421 (1973). The maximum in the liquidus curve at 12 wt. % (8.05 mole %) YF_3 occurs at 1462°.

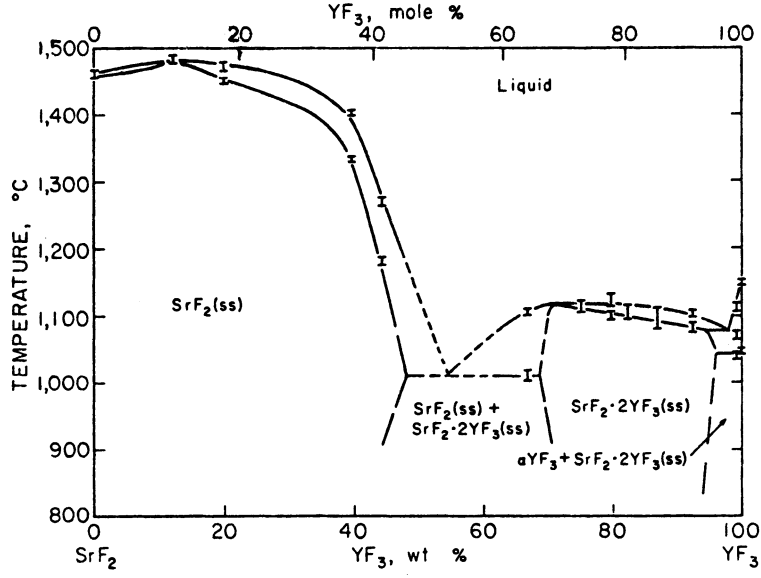
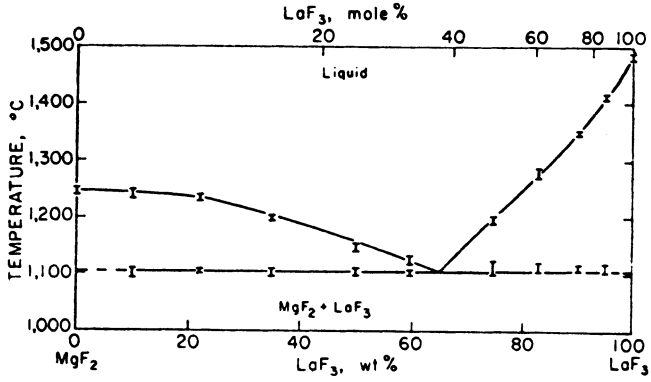


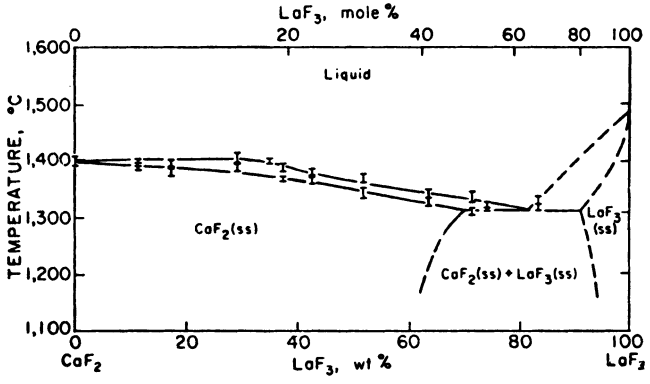
Fig. 122. The system SrF_2–YF_3.

Compositions and temperatures associated with the saturated solid solutions near the mid-composition range were not delineated in the source reference. The temperature of the eutectic at 98 wt. % (96.9 mole %) YF_3 is 1075°.

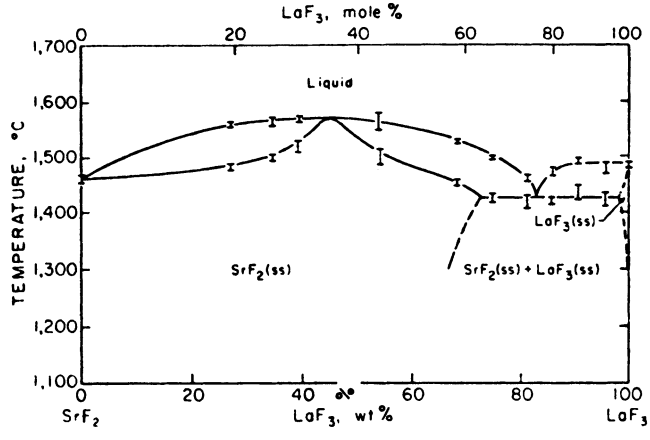
The Systems MF_2–LaF_3. See Fig. 123a–d. Source: R. H. Nafziger and Riazance, *J. Am. Ceram. Soc.* **55**:130 (1972). Composition and temperature of the invariant equilibrium points in the alkaline earth–LaF_3 systems were listed only for MgF_2–LaF_3, where the eutectic temperature



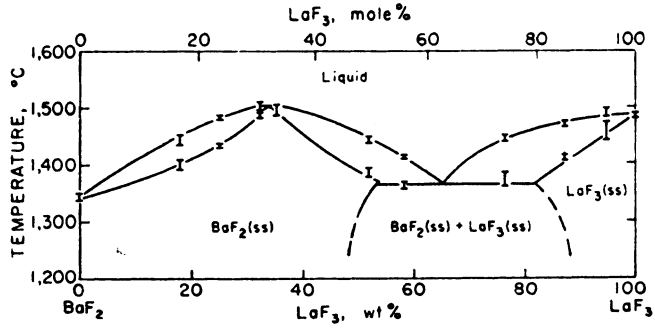
(a)



(b)



(c)



(d)

Fig. 123. The MF_2–LaF_3 systems. (a) MgF_2–LaF_3; (b) CaF_2–LaF_3; (c) SrF_2–LaF_3; (d) BaF_2–LaF_3.

was reported as 1105°, at 35 ± 1 wt. % MgF_2. Evidence of solid solution formation was observed in each of the binary systems except for MgF_2–LaF_3.

The System CaF_2–LaF_3. See Fig. 124. Source: E. G. Ippolitov, N. G. Gogadze, and B. M. Zhigarnovskii, *Russ. J. Inorg. Chem.* (*English Transl.*) **15**:1729 (1970). The eutectic in the limited solid solution series occurs at 60 mole % LaF_3, 1300°.

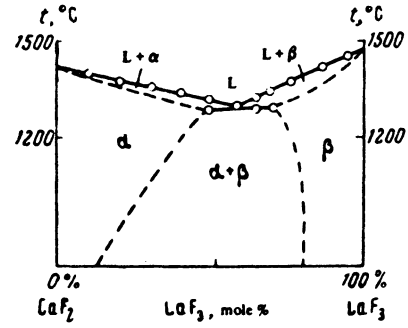


Fig. 124. The system CaF_2–LaF_3. (L) Single-phase liquid field; (α) fluorite solid solution; (β) tysonite solid solution.

The System CaF_2–NdF_3. See Fig. 125. Source: E. G. Ippolitov, N. G. Gogadze, and B. M. Zhigarnovskii, *Russ. J. Inorg. Chem.* (*English Transl.*) **15**:1729 (1970). The eutectic in the limited solid solution series occurs at 58 mole % NdF_3, 1275°.

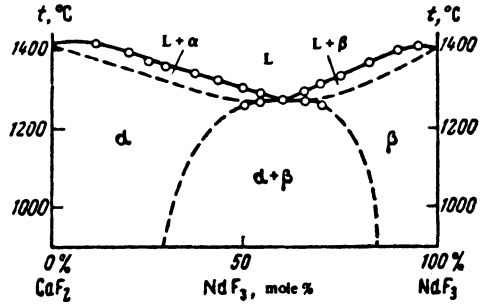


Fig. 125. The system CaF_2–NdF_3.

The System CaF_2–GdF_3. See Fig. 126. Source: N. G. Gogadze, E. G. Ippolitov, and B. M. Zhigarnovskii, *Russ. J. Inorg. Chem.* (*English Transl.*) **12**:600 (1972). E = 1210°, 58% GdF_3; maximum on liquidus, 1300°, 75–80% GdF_3.

Structural data: The cubic solid solution for the composition 75 mole % GdF_3 has a lattice constant a = 5.56: At 1020°C the cubic phase undergoes a polymorphic transformation, and below this temperature there is a hexagonal structure with lattice parameters a = 6.84, c = 7.03.

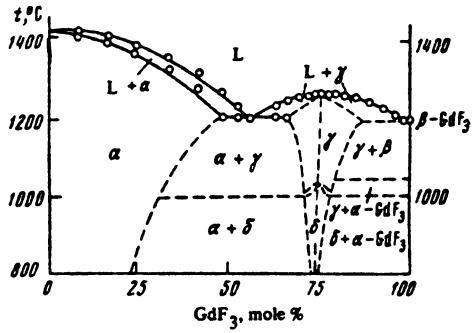


Fig. 126. The system CaF_2–GdF_3. (α) Solid solution with fluorite structure; (γ) solid solution with cubic structure; (δ) hexagonal solid solution; (β-GdF_3) high-molecular (hexagonal) modification of GdF_3; (α-GdF_3) orthorhombic modification of GdF_3.

The System CrF_2–CrF_3. See Fig. 127. Source: B. J. Sturm, *Inorg. Chem.* **1**:665 (1962). Invariant equilibrium points: $E = 831 \pm 5°$, 14% CrF_2; $P = 997 \pm 5°$, 29% CrF_2.

Structural data: $CrF_2 \cdot CrF_3$: monoclinic, $C2/c$, $a = 7.773$, $b = 7.540$, $c = 7.440$, $\beta = 124.25°$ (H. Steinfink and J. H. Burns, *Acta Cryst.* **17**:823 (1964).

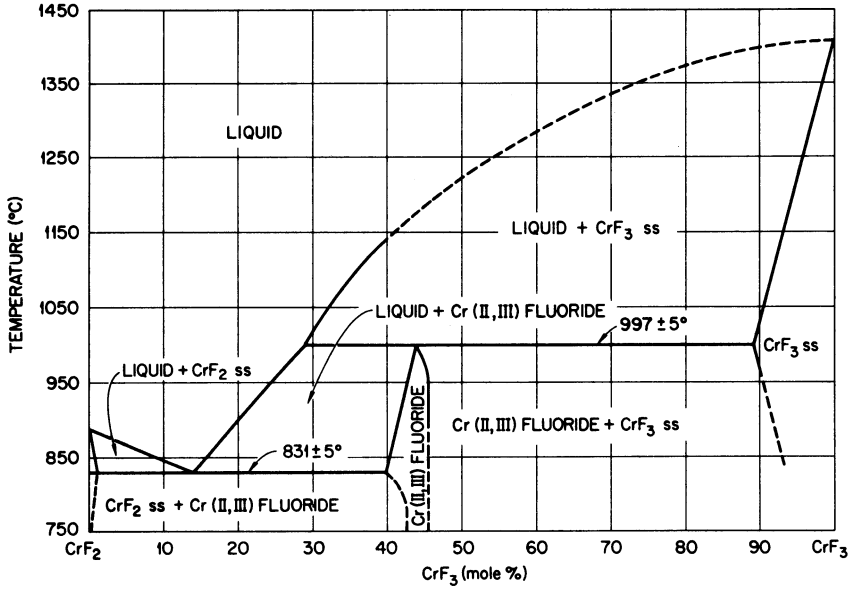


Fig. 127. The system CrF_2–CrF_3.

The System BaF_2–UF_3. See Fig. 128. Source: R. W. M. D'Eye and F. S. Martin, *J. Chem. Soc.* **1957**:1851. Authors of the source reference found

that the limit of solubility of UF_3 in BaF_2 is roughly 50 mole % and that the solubility limit of BaF_2 in UF_3 is about 20 mole %. The peritectic temperature was inferred to be about 1400°, although its composition was not deduced.

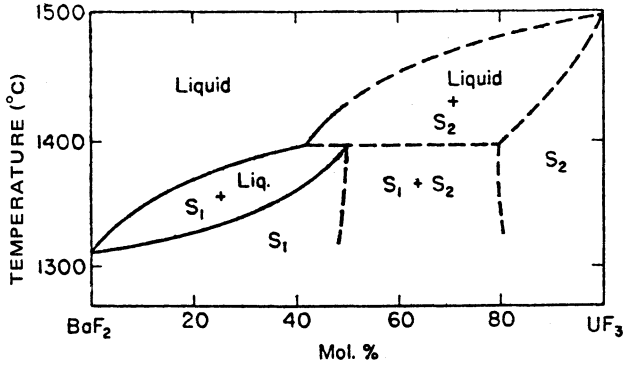


Fig. 128. The system BaF_2–UF_3.

The System PbF_2–AlF_3. See Fig. 129. Source: R. G. Shore and B. M. Wanklyn, *J. Am. Ceram. Soc.* **52**:79 (1969). $E = 570°$, 76.5% PbF_2. The composition of the 649° peritectic was not given. It occurs at approximately 69% PbF_2. The limit of solid-state solubility of AlF_3 in PbF_2 was reported to be 15%. Structure data: The compound $3PbF_2 \cdot 2AlF_3$ was reported by Shore and Wanklyn to have tetragonal structure, $I4/m$, $a = 14.23$, $c = 7.20$.

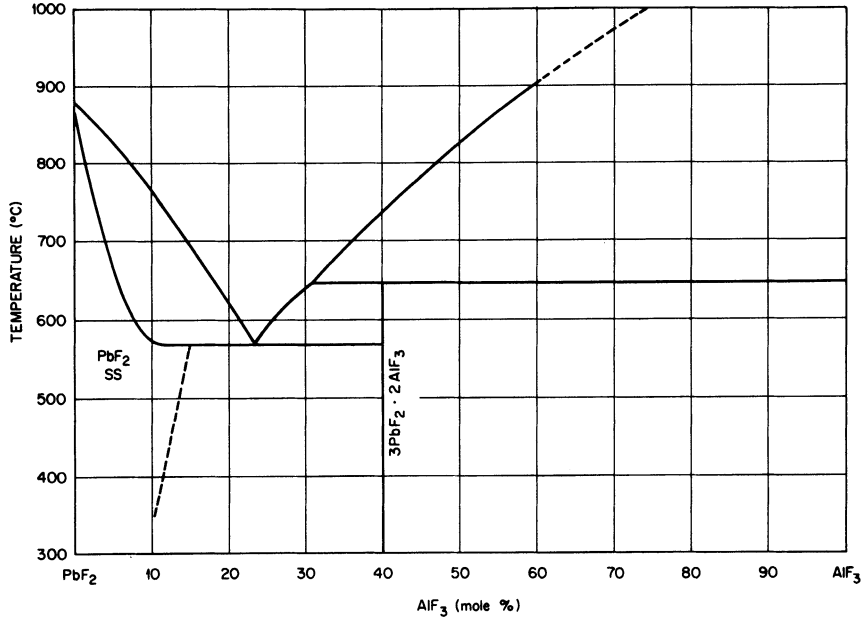


Fig. 129. The system PbF_2–AlF_3.

The System BeF_2–ZrF_4. See Fig. 130. Source: R. E. Thoma, H. Insley, H. A. Friedman, and G. M. Hebert, *J. Nucl. Mater.* **27**:166 (1968). $E = 535°$, 7.3% ZrF_4, immiscibility limits at 645°, 14, and 26% ZrF_4; critical solution point ~745°, 20% ZrF_4. A phase diagram of the system was reported earlier by U. M. Korenev and A. V. Novoselova, *Dokl. Akad. Nauk SSSR* **149**:5 (1963), which indicates the existence of an intermediate compound, $BeZrF_6$. No evidence of such a compound was discovered in the experiments reported in the source reference.

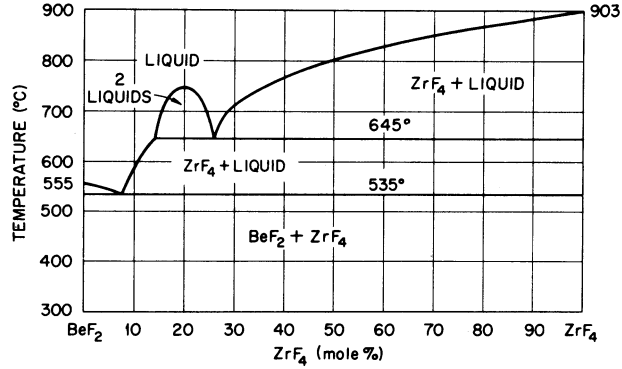


Fig. 130. The system BeF_2–ZrF_4.

The System BeF_2–ThF_4. See Fig. 131. Source: R. E. Thoma, H. Insley, H. A. Friedman, and C. F. Weaver, *J. Phys. Chem.* **64**:865 (1960). $E = 527 \pm 3°$, 2.0% ThF_4.

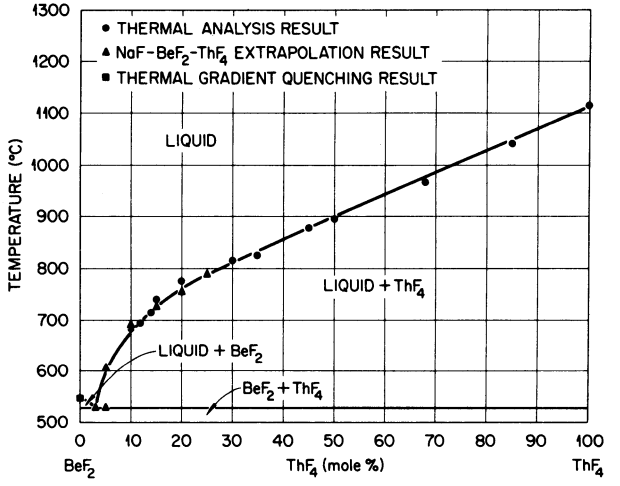


Fig. 131. The system BeF_2–ThF_4.

The System BeF_2–UF_4. See Fig. 132. Source: L. V. Jones *et al.*, *J. Am. Ceram. Soc.* **45**:79 (1962). $E = 535 \pm 2°$, 0.5% UF_4.

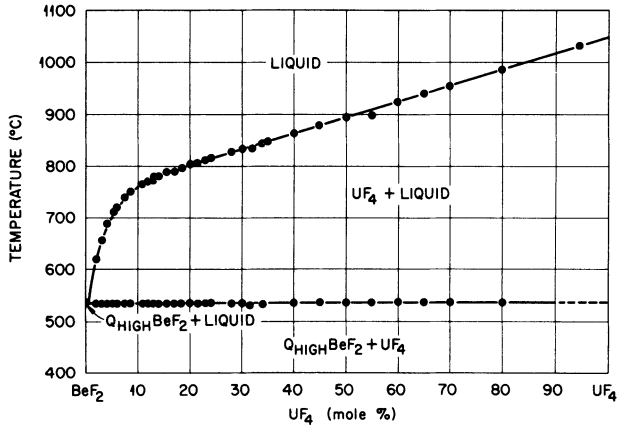


Fig. 132. The system BeF_2–UF_4.

The System MgF_2–ThF_4. See Fig. 133. Source: J. O. Blomeke, ORNL-1030 (declassified) (1951). $E = 915°$, 25% ThF_4; congruent m.p. $MgTh_2F_{10}$ 937°, $E = 925°$, 40% ThF_4.

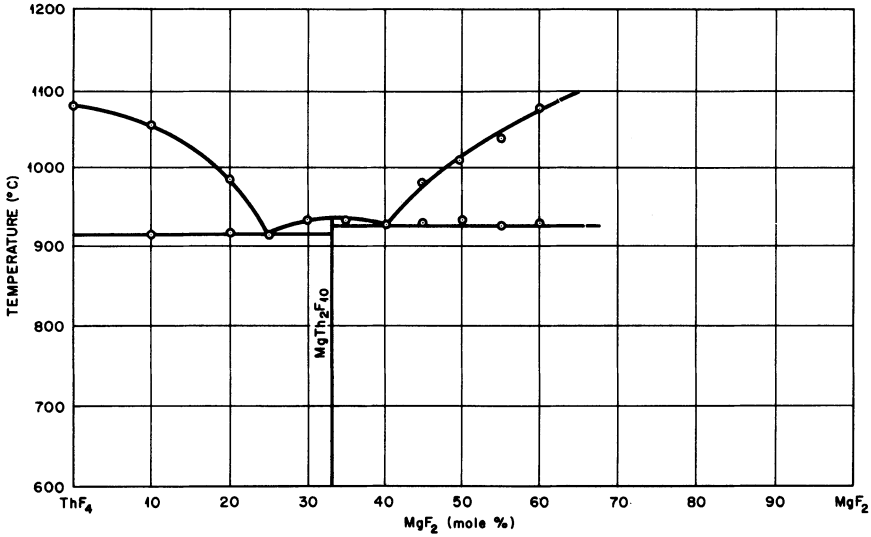


Fig. 133. The system MgF_2–ThF_4.

The System CaF_2–UF_4. See Fig. 134. Source: N. P. Nekrasova *et al.*, *At. Energ.* (*USSR*) **22**:293 (1967). Invariant and singular equilibrium

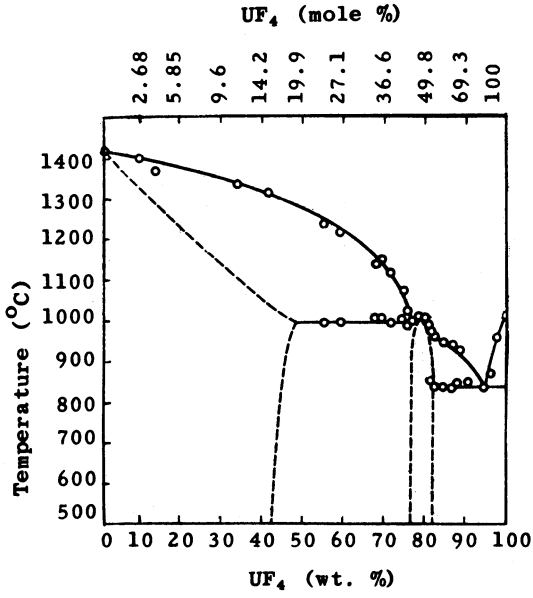


Fig. 134. The system CaF_2–UF_4.

points: $E = 997°$, 76.7% UF_4; congruent m.p. $CaUF_6$ 1005°; $E = 1008°$, 94.5% UF_4.

The System SnF_2–UF_4. See Fig. 135. Source: B. J. Thamer, USAEC Report LA-2286, July, 1959, Los Alamos Scientific Laboratory. $E = 212°$, 0.5% MF_4; $P = 340°$, 4.5% UF_4; $P = 371°$, 7.8% UF_4.

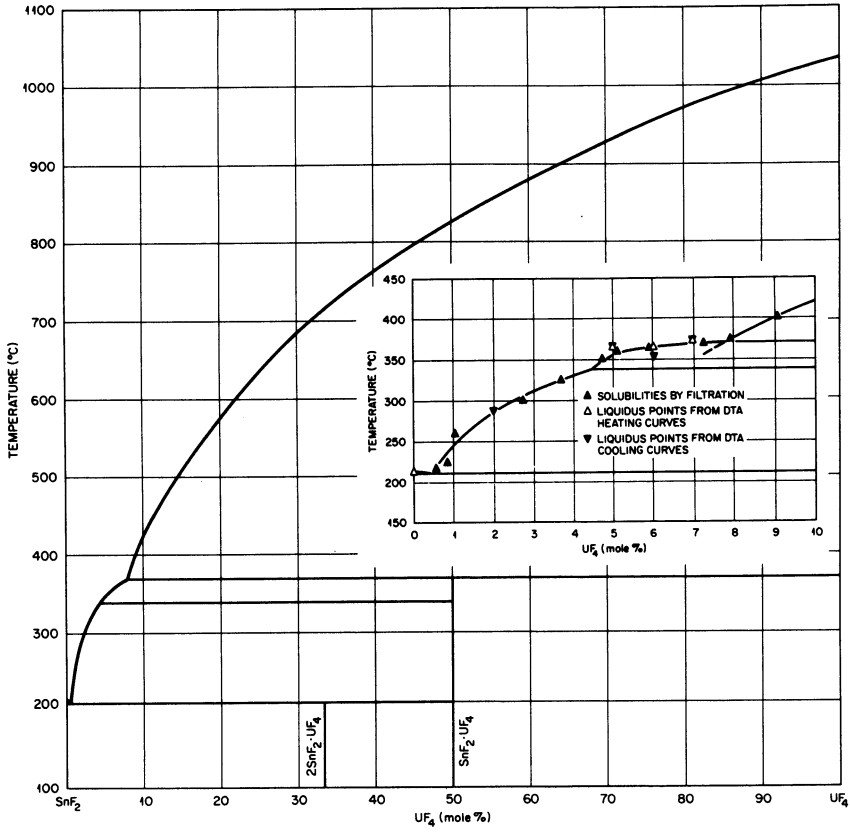


Fig. 135. The system SnF_2–UF_4.

The System PbF_2–UF_4. See Fig. 136. Source: C. J. Barton, L. M. Bratcher, J. P. Blakely, G. J. Nessle, and W. R. Grimes, ORNL, unpublished work (1950–1951). Congruent m.p. $6PbF_2 \cdot UF_4$ 920°; $E = 835°$, 35% UF_4; congruent m.p. $3PbF_2 \cdot 2UF_4$ 840°; $E = 762 \pm 10°$, 62% UF_4. The eutectic at quite high PbF_2 concentrations is not shown on this diagram. The authors examined thermal data but found no evidence of a eutectic thermal effect.

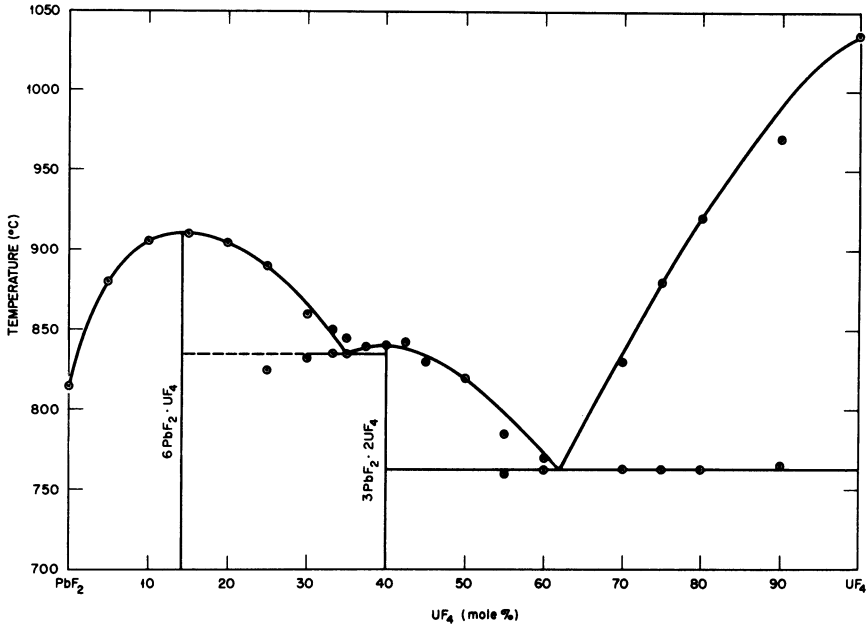
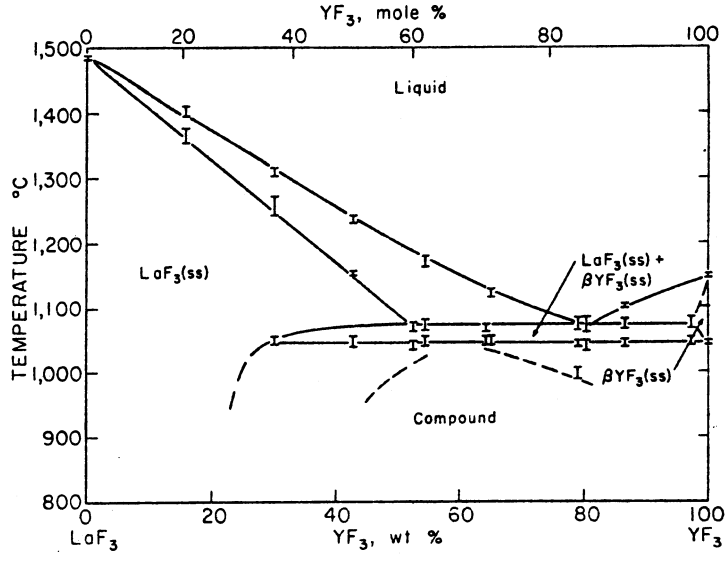


Fig. 136. The system PbF_2–UF_4.

The System YF_3–LaF_3. See Fig. 137. Source: R. H. Nafziger, R. L. Lincoln and R. Riazance, *J. Inorg. Nucl. Chem.* **35**:421 (1973). $E = 1037°$, 81 wt. % (15 mole %) YF_3.



The System CeF_3–ThF_4. See Fig. 138. Source: L. O. Gilpatrick, H. Insley, and C. J. Barton, ORNL-4622, MSR Program Semiannual Progress Report for Period Ending Aug. 31, 1970, p. 91. $E = 950°$, 75% ThF_4; $P = 960°$, 73% ThF_4; $P = 1155°$, approx. 52% ThF_4.

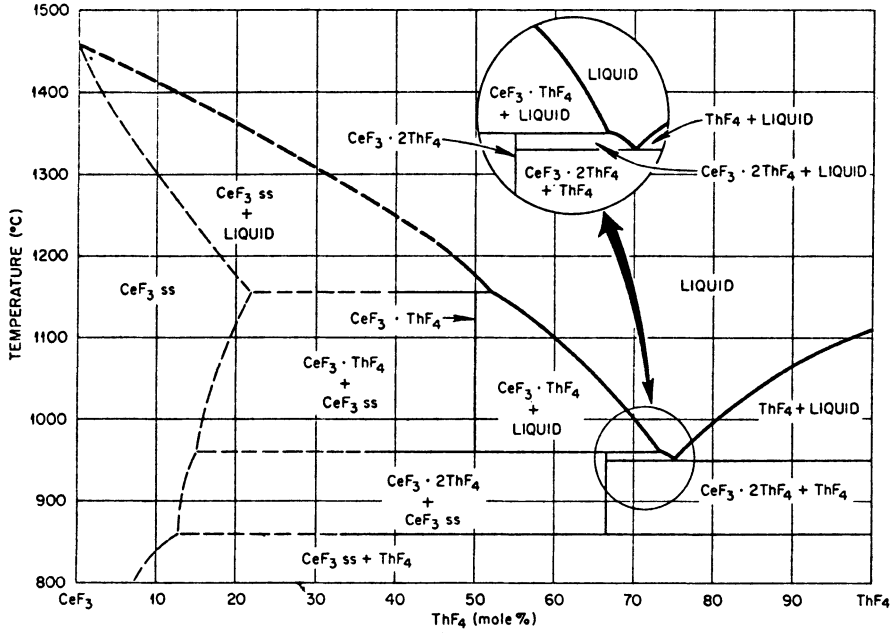


Fig. 138. The system CeF_3–ThF_4.

The System UF_3–UF_4. See Fig. 139. Source: R. E. Thoma, G. D. Brunton, and H. Insley, *J. Inorg. Nucl. Chem.* **36**:1095 (1974). $E = 865 \pm 4°$,

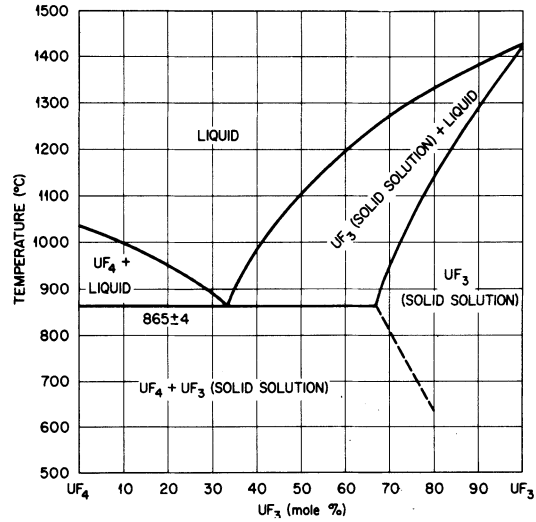


Fig. 139. The system UF_3–UF_4.

32% UF_3. The results of the investigation described in the source reference are in disagreement with those reported earlier by L. A. Khripin, S. A. Poduzova, and G. M. Zadneprovskii, *Russ. J. Inorg. Chem.* (*English Transl.*) **13**:1439 (1968).

The System ZrF_4–UF_4. See Fig. 140. Source: C. J. Barton *et al.*, *J. Phys. Chem.* **62**:665 (1958). The system ZrF_4–UF_4 forms a continuous series of solid solutions having a minimum melting temperature of 765° at 23 mole % UF_4.

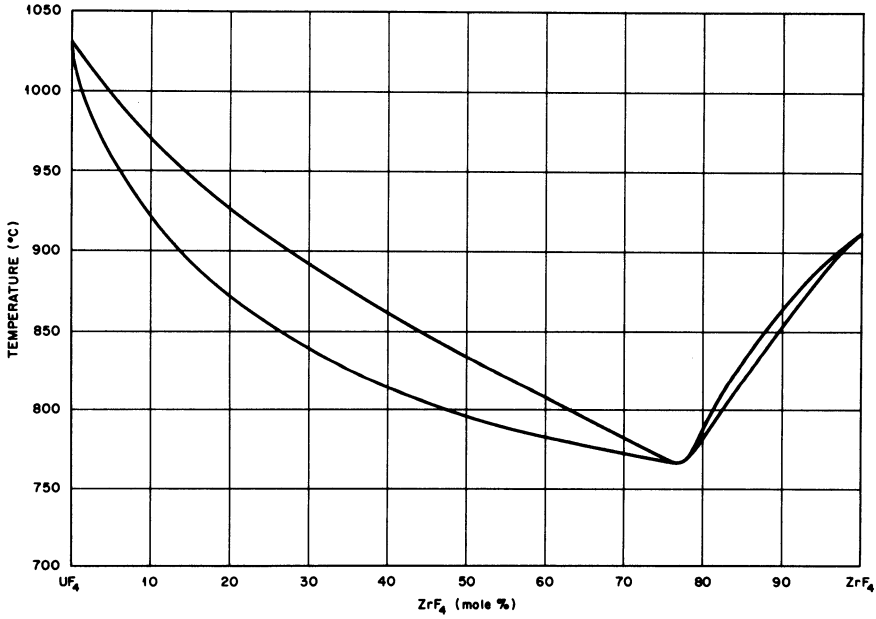


Fig. 140. The system ZrF_4–UF_4.

The System ThF_4–UF_4. See Fig. 141. Source: C. F. Weaver, R. E. Thoma, H. Insley, and H. A. Friedman, *J. Am. Ceram. Soc.* **43**:213 (1960).

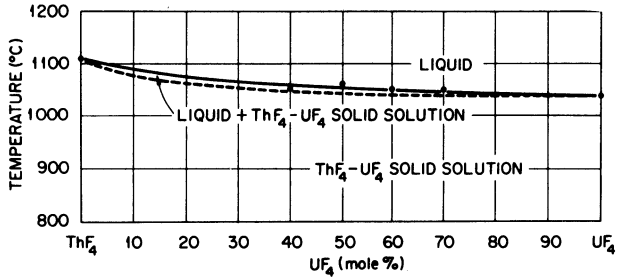


Fig. 141. The system ThF_4–UF_4.

The system ThF_4–UF_4 forms a continuous series of solid solutions without a temperature minimum.

The System LiF–SbF_5. The phase diagram for the system has not yet been described. However, the structure of an intermediate compound, $LiSbF_5$, has been determined by J. H. Burns, *Acta. Cryst.* **15**:1098 (1962), as hexagonal, $R\bar{3}$, $a = 5.18$, $c = 13.60$.

The System XeF_2–VF_5. See Fig. 142. Source: V. A. Legasov and A. S. Marinin, *Russ. J. Phys. Chem.* (*English Transl.*) **46**:1388 (1972). The intermediate compound $XeF_2 \cdot VF_5$ melts congruently at $38 \pm 1°$;

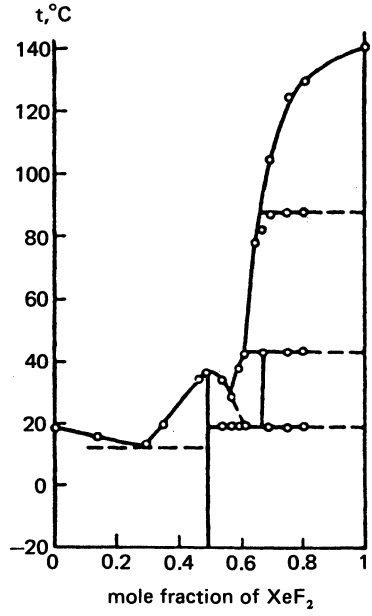


Fig. 142. The system XeF_2–VF_5.

the peritectic associated with the upper limit of stability of the intermediate compound, $2XeF_2 \cdot VF_5$, occurs at 44°. Compositions of the invariant equilibrium points were not given in the source reference.

The System XeF_2–IF_5. See Fig. 143. Source: V. A. Legasov, V. B. Sokolov, and B. B. Chaivanov, *Russ. J. Phys. Chem.* (*English Transl.*) **43**:1650 (1969). The composition of the invariant equilibrium points was not given by the authors of the source reference. The m.p. of $XeF_2 \cdot IF_5$ is $102 \pm 0.7°$.

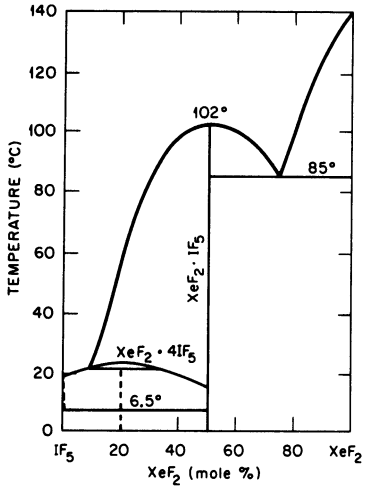


Fig. 143. The system XeF_2–IF_5.

The System BF_3–UF_6. See Fig. 144. Source: J. Fischer and R. C. Vogel, *J. Am. Chem. Soc.* **76**:4381 (1954). The eutectic occurs at −63.5° and at 3.2 ± 0.5% UF_6.

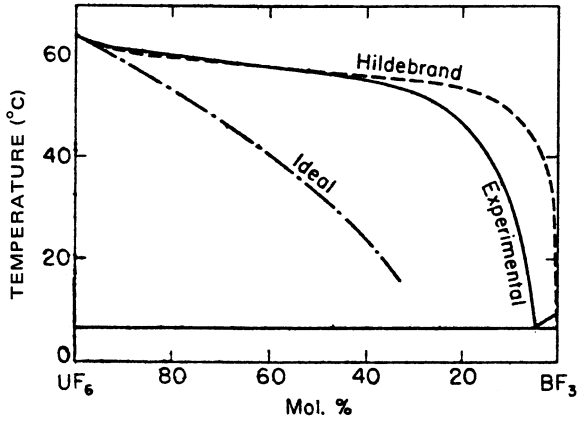


Fig. 144. The system BF_3–UF_6.

The System BF_5–UF_6. See Fig. 145. Source: J. Fischer and R. C. Vogel, *J. Am. Chem. Soc.* **76**:4829 (1954). $E = 6.4°$, 4.1% UF_6.

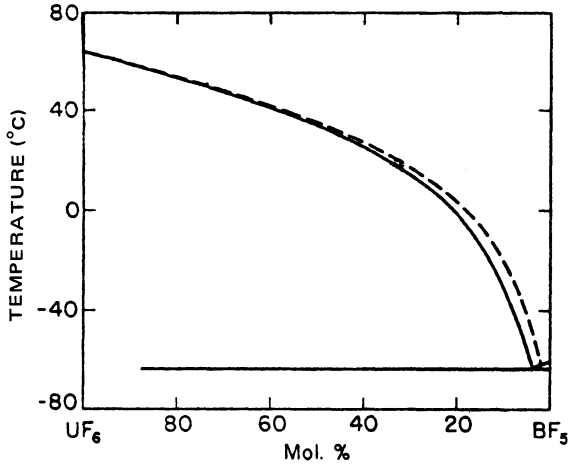


Fig. 145. The system BF_5–UF_6.

The System MoF_5–SbF_5. See Fig. 146. Source: V. K. Ezhov, *Russ. J. Inorg. Chem.* (*English Transl.*) **17** (1972). Invariant equilibrium data were not described quantitatively in the reference source, which cites the original reference as: V. N. Prusakov and V. K. Ezhov, Materialy Sim-

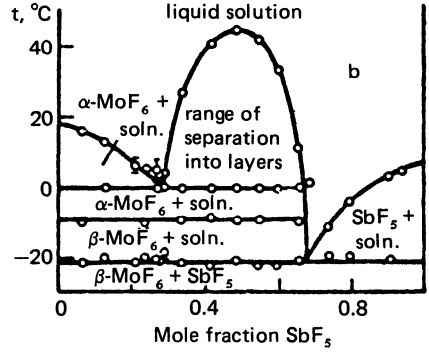


Fig. 146. The system MoF_5–SbF_5.

poziuma SEV "Issledovaniya v Oblasti Pererabotki Obluchennogo Topliva" (Proceedings of the SEV Symposium "Studies in the Processing of Irradiated Fuel"), KAE SSSR (1968), p. 331.

The System VF_5–UF_6. See Fig. 147. Source: V. K. Ezhov, *Russ. J. Inorg. Chem.* (*English Transl.*) **17**:1059 (1972). The temperature and composition of the eutectic were not given in the source reference.

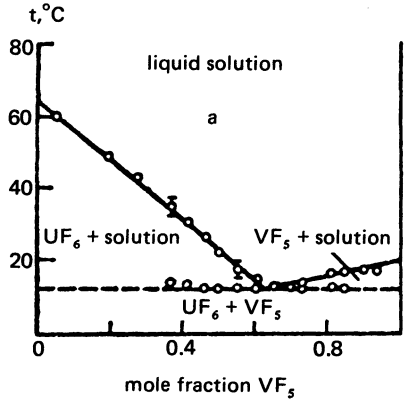
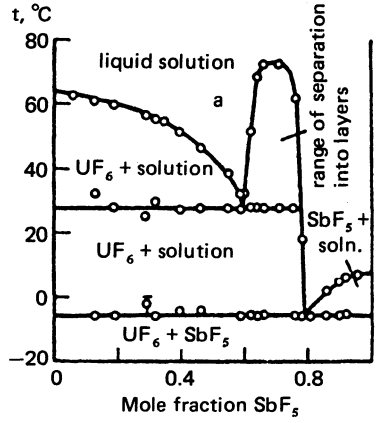


Fig. 147. The system VF_5–UF_6.

The System SbF_5–UF_6. See Fig. 148. Source: V. K. Ezhov, *Russ. J. Inorg. Chem.* **17**:1059 (1972). The region of liquid–liquid immiscibility extends from 23.3 to 39.0 mole % UF_6. The temperature of the monotectic was not specified in the source reference. The upper critical temperature of separation is 73.5°. The eutectic occurs at −5.5° and at 19.5 mole % UF_6.

Fig. 148. The system SbF_5–UF_6.



The System TaF_5–UF_6. See Fig. 149. Source: V. K. Ezhov, *Russ. J. Inorg. Chem.* (*English Transl.*) **17**:1059 (1972). The composition and temperature of the invariant equilibrium points were not given in the source reference.

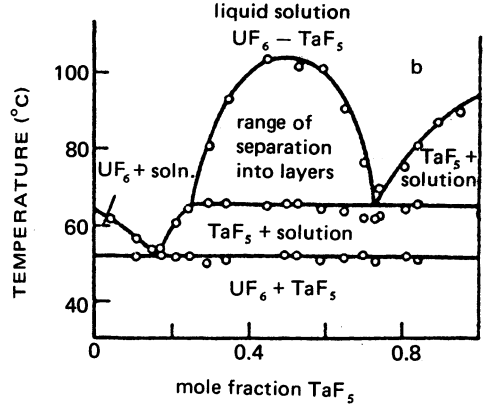


Fig. 149. The system TaF_5–UF_6.

The System WF_6–UF_6. See Fig. 150. Source: V. N. Prusakov and V. K. Eshov, *At. Energ.* (*USSR*) **25**:64 (1968). The system forms a single eutectic, at $-1.0 \pm 0.7°$ and 96 ± 2 mole %. No evidence of solid solution was observed.

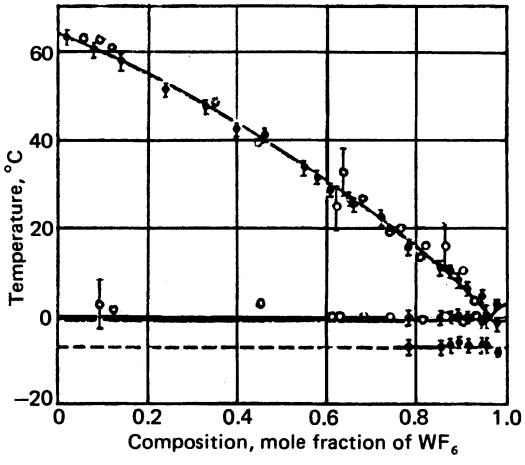


Fig. 150. The system WF_6–UF_6.

TERNARY SYSTEMS

The System LiF–NaF–KF. See Fig. 151. Source: A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:753 (1941). $E = 454°$, LiF–NaF–KF (46.5–11.5–42.0%). The isotherms shown in Fig. 151 were verified to $\pm 5°$ for 0 to 60% KF by C. J. Barton *et al.*, ORNL (1951–1954).

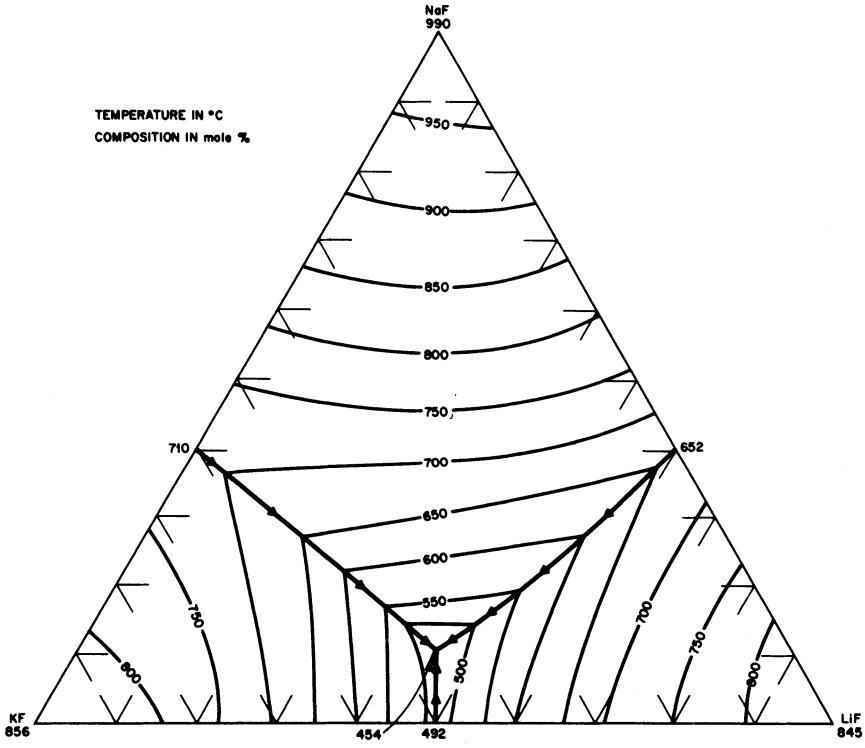


Fig. 151. The system LiF–NaF–KF.

The System LiF–NaF–RbF. See Fig. 152. Source: C. J. Barton, L. M. Bratcher, J. P. Blakely, and W. R. Grimes, ORNL, unpublished work (1951). $E = 435°$, LiF–NaF–RbF (42–6–52%). The composition and temperature of the ternary peritectic point were not determined. General

characteristics of the system were reported by E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **58**: 1369 (1947). According to this work, the composition and temperature of the eutectic is LiF–NaF–RbF (46.5–6.5–47%) and 426°.

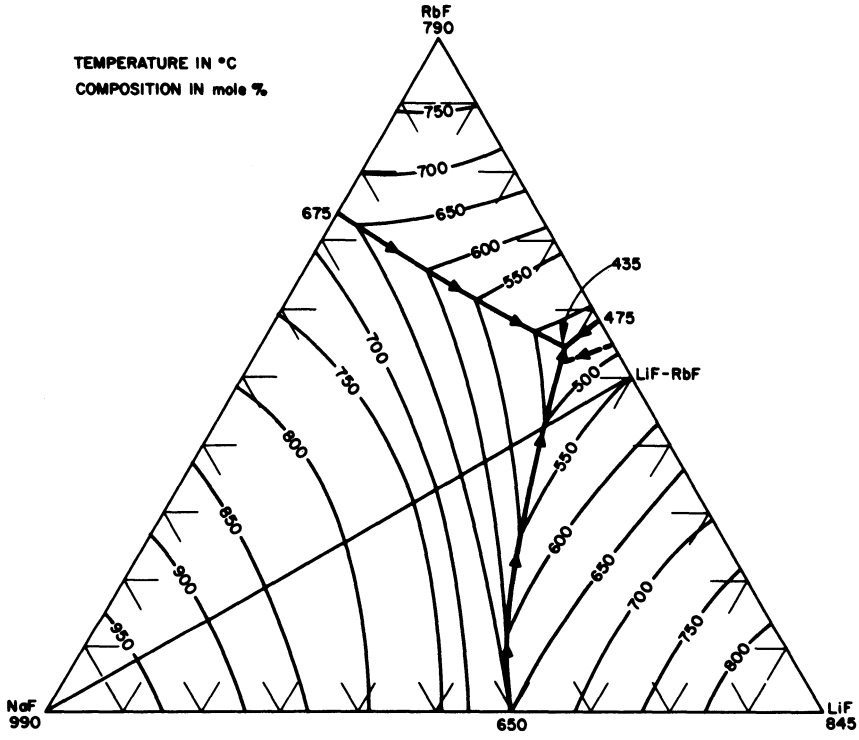


Fig. 152. The system LiF–NaF–RbF.

The System LiF–NaF–CsF. See Fig. 153. Source: G. A. Bukhalova and D. V. Sementsova, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:1880 (1965). $E = 448°$, LiF–NaF–CsF (48–11–41%) $E = 446°$, LiF–NaF–CsF (9–37–54%). The authors of the source reference emphasize that the results of their investigation of the LiF–NaF–CsF system clearly indicate that the compound LiF–CsF melts congruently.

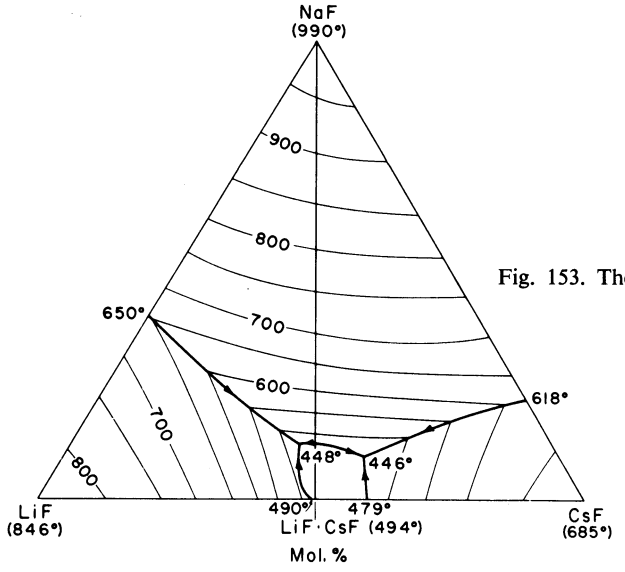
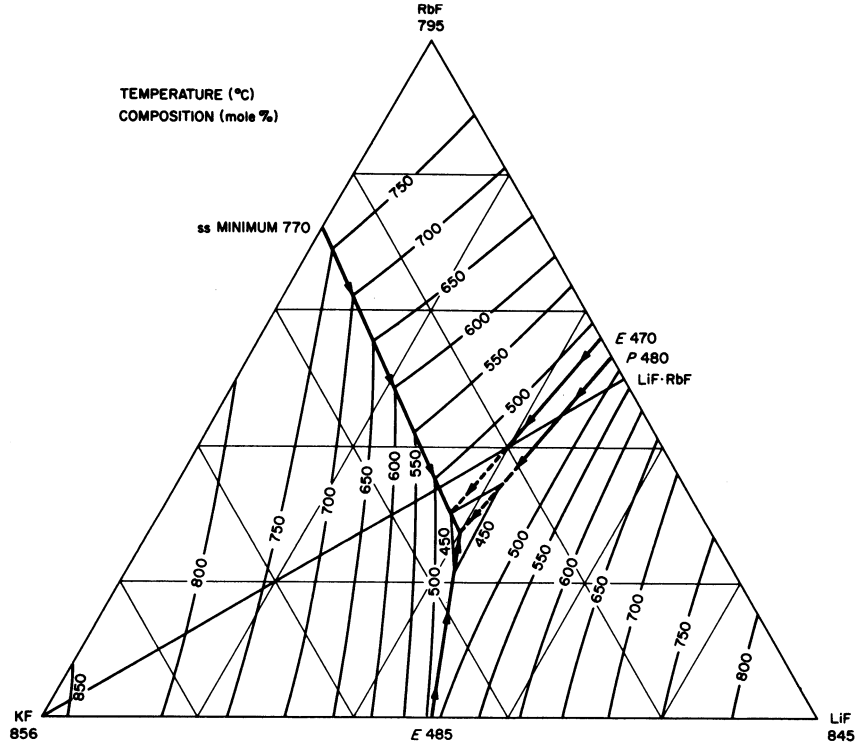


Fig. 153. The system LiF–NaF–CsF.

The System LiF–KF–RbF. See Fig. 154. Source: C. J. Barton, J. P. Blakely, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work



(1951). A single invariant point occurs within the system, $E = 440 \pm 10°$, LiF–KF–RbF (40–32.5–27.5%), m.p. 440 ± 10°. General characteristics of the system were reported in earlier work by E. P. Dergunov, *Dokl. Akad. Nauk SSSR* **58**:1369 (1947).

The System NaF–KF–RbF. See Fig. 155. Source: C. J. Barton, L. M. Bratcher, J. P. Blakely, and W. R. Grimes, ORNL, unpublished work (1951). $E = 621 \pm 10°$, NaF–KF–RbF (21–5–74%).

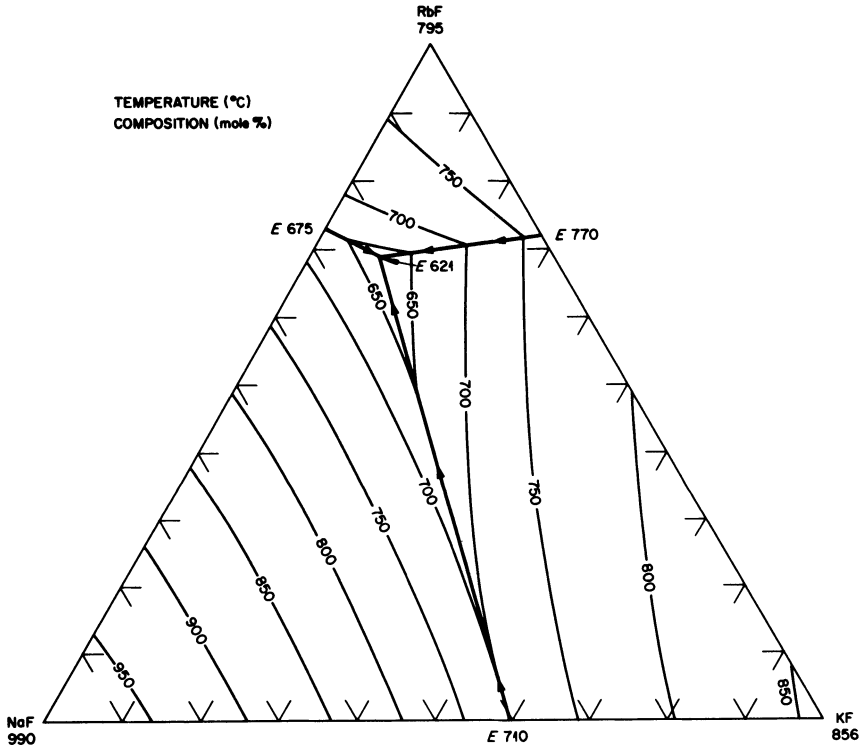


Fig. 155. The system NaF–KF–RbF.

The System LiF–NaF–BeF_2. See Fig. 156. Source: R. E. Moore, C. J. Barton, W. R. Grimes, R. E. Meadows, L. M. Bratcher, G. D. White, and T. N. McVay, ORNL, unpublished work (1951–1958). Invariant equilibria shown in the phase diagram by the intersections of dotted-line boundary curves have not been determined with sufficient precision to be listed in the table. Phase relationships of two of the ternary compounds in this system have been reported by W. Jahn [Silicate Models. V. NaLi-(BeF_4), a Model Substance for Monticellite, $CaMg(SiO_4)$, *Z. Anorg. Allgem. Chem.* **276**:113–127 (1954); Silicate Models. VI. $Na_3Li(BeF_4)_2$,

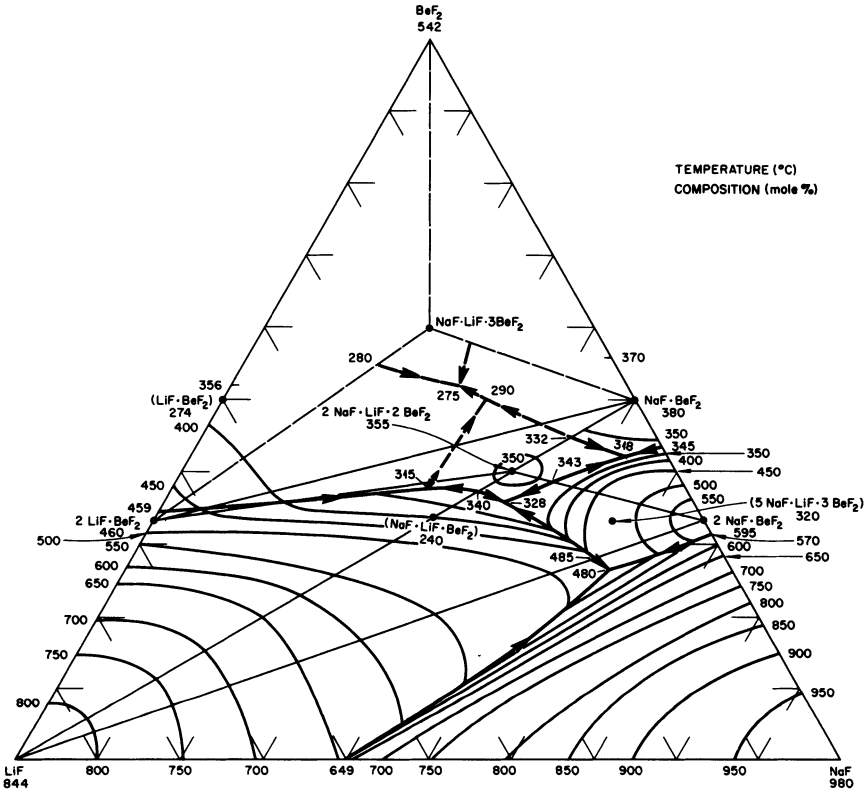


Fig. 156. The system LiF–NaF–BeF_2. The dotted lines represent incompletely defined phase boundaries and Alkemade lines.

a New Compound in the Ternary System NaF–LiF–BeF_2, and Its Relation to Merwinite, $Ca_3Mg(SiO_4)_2$, *Z. Anorg. Allgem. Chem.* **277**:274–286 (1954)].

Mole %			Temperature	Type of equilibrium
LiF	NaF	BeF_2		
15	58	27	480	Eutectic
23	41	36	328	Eutectic
20	40	40	355	Congruent melting point
5	53	42	318	Eutectic
31.5	31	37.5	315	Eutectic

(Continued)

Minimum temperature on Alkemede lines				
Mole %			Temperature	Alkemede line
LiF	NaF	BeF_2		
16	56	28	485	$2NaF \cdot BeF_2$–LiF
26	37	37	340	$2NaF \cdot LiF \cdot 2BeF_2$–LiF
11	44	45	332	$NaF \cdot BeF_2$–$2NaF \cdot LiF \cdot 2BeF$
16	45	39	343	$2NaF \cdot BeF_2$–$2NaF \cdot LiF \cdot 2BeF$
30.5	31	38.5	316	$2LiF \cdot BeF_2$–$2NaF \cdot LiF \cdot 2BeF_2$

The System LiF–RbF–BeF_2. See Fig. 157. Source: T. B. Rhinehammer, D. E. Etter, C. R. Hudgens, N. E. Rogers, and P. A. Tucker, unpub-

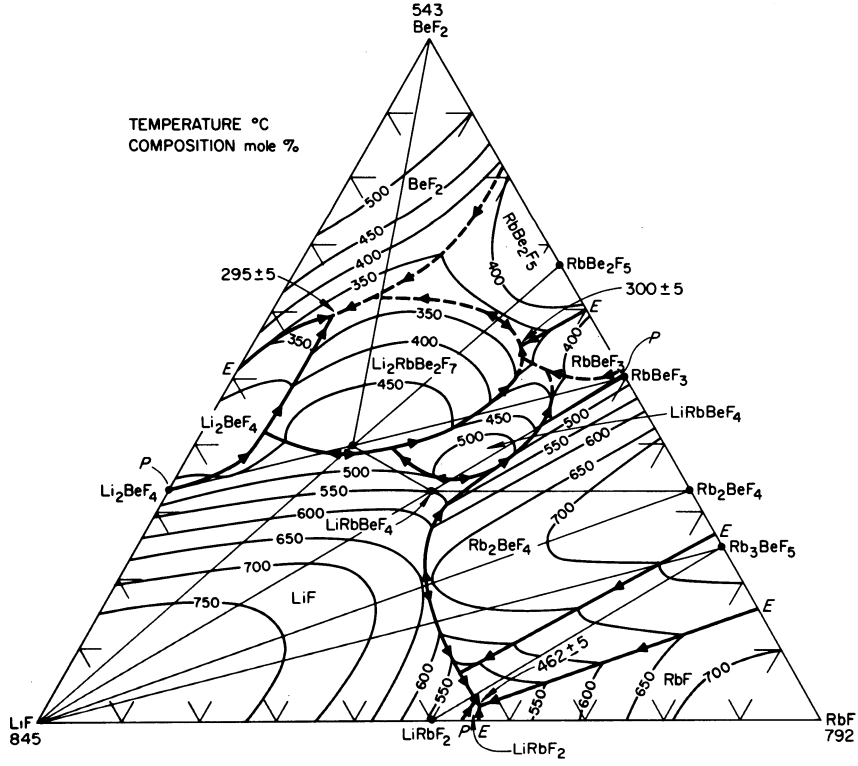


Fig. 157. The system LiF–RbF–BeF_2.

lished work, Mound Laboratory, Miamisburg, Ohio (1956–1958). Invariant and singular equilibrium points are given in the table. In a study of isotypic fluoroberyllates of the formula $Li_3Me(BeF_4)_2$, J. LeRoy, L. Pontonnier, and S. Aleonard, *Mater. Res. Bull.* **6**:267 (1971), described the section

Mole %			Temperature	Type of equilibrium
LiF	RbF	BeF_2		
40.0	20.0	40.0	485 ± 5	Congruent melting point
33.3	33.3	33.3	544 ± 5	Incongruent melting point of $LiF \cdot RbF \cdot BeF_2$
33.0	8.0	59.0	295 ± 5	Eutectic
26.0	12.0	62.0	305–320	Peritectic
12.0	29.0	59.0	334 ± 5	Quasi-binary eutectic
11.0	34.0	55.0	300 ± 5	Eutectic
11.5	35.5	53.0	335 ± 5	Peritectic
9.0	40.0	51.0	380 ± 5	Peritectic
50.0	8.0	42.0	426 ± 5	Peritectic
41.5	19.5	39.0	473 ± 5	Quasi-binary eutectic
35.0	25.5	39.5	470 ± 5	Peritectic
23.0	40.0	37.0	530 ± 5	Peritectic
30.0	35.0	35.0	544 ± 5	Maximum on the boundary
41.0	39.5	19.5	640 ± 5	Quasi-binary eutectic
43.0	50.0	7.0	527 ± 5	Peritectic
43.0	54.0	3.0	470 ± 5	Peritectic
43.0	55.0	2.0	462 ± 5	Eutectic

Li_2BeF_4–Rb_2BeF_4 to include the two ternary compounds $LiRbBeF_4$ and $Li_3Rb(BeF_4)_2$, each melting congruently, at the temperatures 560° and 500°, respectively. These results are in disagreement with Fig. 157. Experimental details are not available for either of the two proposed phase diagrams.

The System NaF–KF–BeF_2. See Fig. 158. Source: R. E. Moore, C. J. Barton, L. M. Bratcher, T. N. McVay, and W. R. Grimes, ORNL, unpublished work (1952–1955). Invariant and singular equilibrium points were not established with sufficient precision to list here.

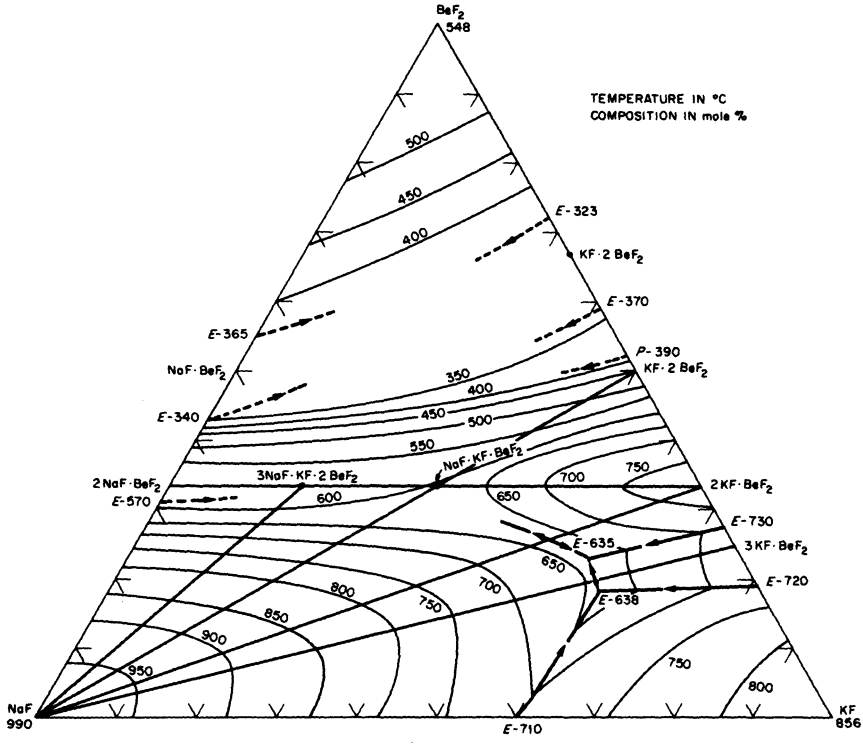


Fig. 158. The system NaF–KF–BeF_2.

The System NaF–RbF–BeF_2. See Fig. 159. Source: R. E. Moore, C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1954–1957). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	RbF	BeF_2		
20.5	68.5	11.0	610	Eutectic
50	37.5	12.5	640	Quasi-binary eutectic
46	38	16	635	Eutectic
45	37	18	650	Quasi-binary eutectic
37	33	30	570	Peritectic
52	15	33	477	Eutectic
33	33.7	33.3	580	Peritectic
51.7	15	33.3	480	Quasi-binary eutectic

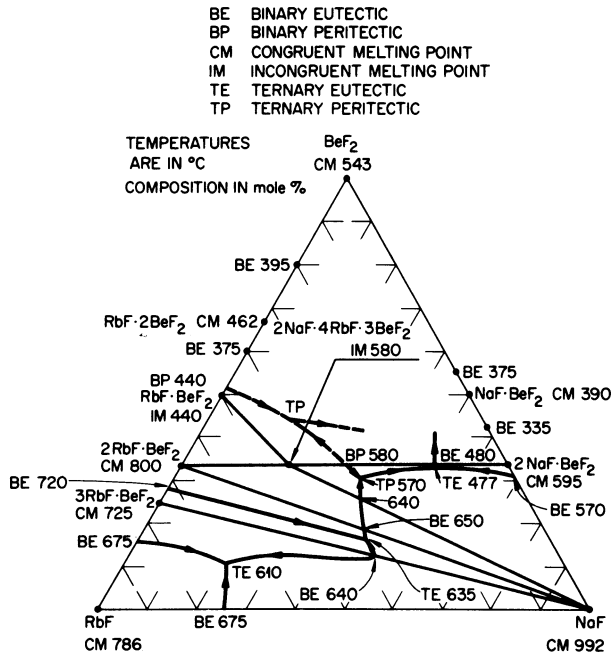


Fig. 159. The system NaF–RbF–BeF_2.

The System LiF–NaF–MgF_2. See Fig. 160. Source: A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **31**:755 (1941). $E =$ 630°, LiF–NaF–MgF_2 (47–43–10%); $E = 684°$, LiF–NaF–MgF_2 (59–12–29%).

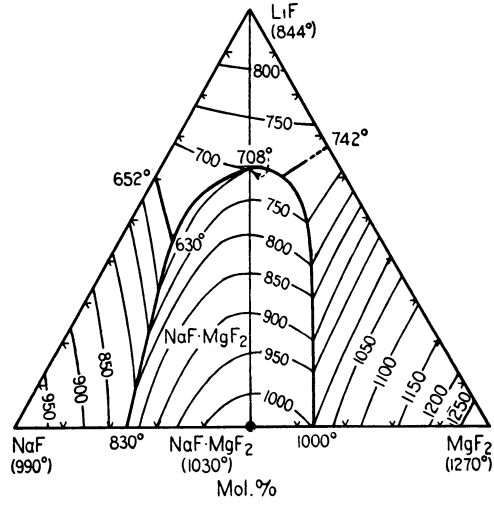


Fig. 160. The system LiF–NaF–MgF_2.

The System LiF–KF–MgF_2. See Fig. 161. Source: A. G. Bergman and S. P. Pavlenko, *Compt. Rend. Acad. Sci. URSS* **30**:818 (1941). The authors reported evidence of both a peritectic and a eutectic in the LiF–KF–2KF · MgF_2 subsystem. The eutectic, solidifying at 488° (2° lower than that for LiF–KF), occurs close to the LiF–KF binary system. Its composition and the temperature and composition of the peritectic in the subsystem were not established. The eutectic in the LiF–MgF_2–2KF · MgF_2 subsystem occurs at 713°, LiF–KF–MgF_2 (63.5–6–30.5%).

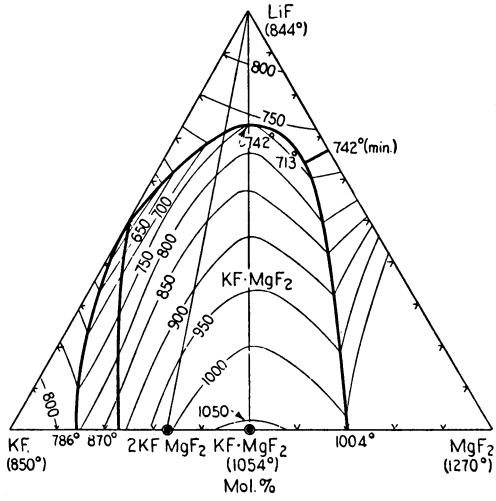


Fig. 161. The system LiF–KF–MgF_2.

The System NaF–KF–MgF_2. See Fig. 162. Source: A. G. Bergman and E. P. Dergunov, *Compt. Rend. Acad. Sci. URSS* **48**:330 (1945). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	KF	MgF_2		
33.0	13.5	53.5	975	Eutectic
62.5	15.0	22.5	798	Eutectic
34.5	59.0	6.5	685	Eutectic
39.0	50.0	11.0	710	Peritectic

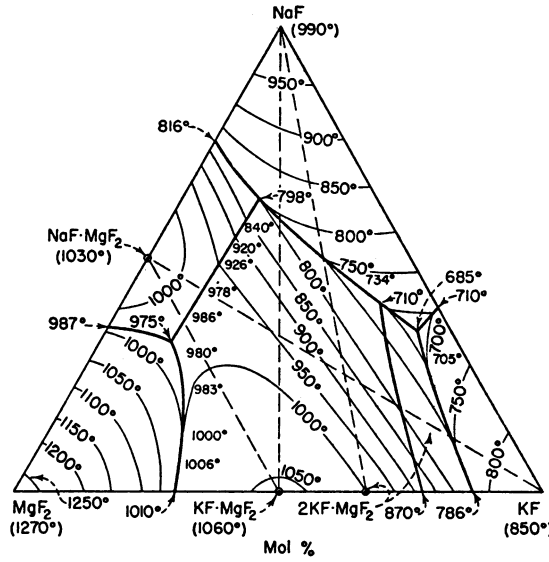
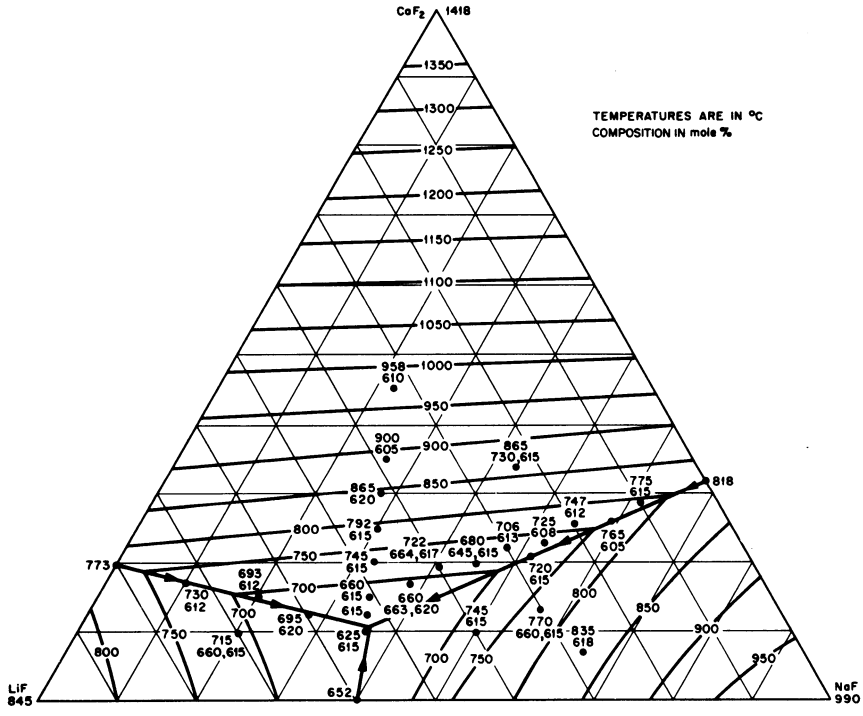


Fig. 162. The system NaF–KF–MgF_2.

The System LiF–NaF–CaF_2. See Fig. 163. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1955–



1956). $E = 616°$, LiF–NaF–CaF_2 (53–36–11%) phase diagram of the system was reported by G. A. Bukhalova, K. Sulaimankulov, and A. K. Bastandzhiyan, *Zh. Neorg. Khim.* **4**:1138 (1959).

The System LiF–CsF–CaF_2. See Fig. 164. Source: D. V. Sementsova, G. A. Bukhalova, and Z. A. Mateiko, *Russ. J. Inorg. Chem.* (*English Transl.*)

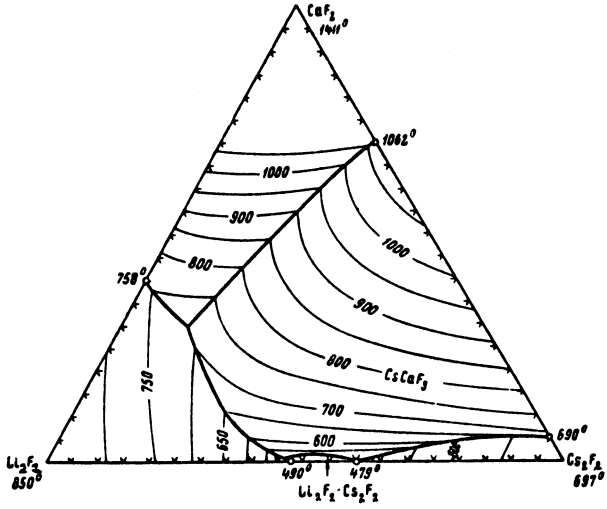


Fig. 164. The system LiF–CsF–CaF_2.

12:865 (1967). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
Li_2F_2	Cs_2F_2	CaF_2		
57	13	30	708	Eutectic
54.5	44.0	1.5	476	Eutectic
39.0	60.5	0.5	462	Eutectic

The System NaF–KF–CaF_2. See Fig. 165. Source: G. A. Bukhalova and V. T. Berezhnaya, *Russ. J. Inorg. Chem.* (*English Transl.*) **4**:1197 (1959). $E = 759°$, Na_2F_2–K_2F_2–BaF_2 (31–34–35%); $E = 621°$, Na_2F_2–K_2F_2–BaF_2 (51–36–13%).

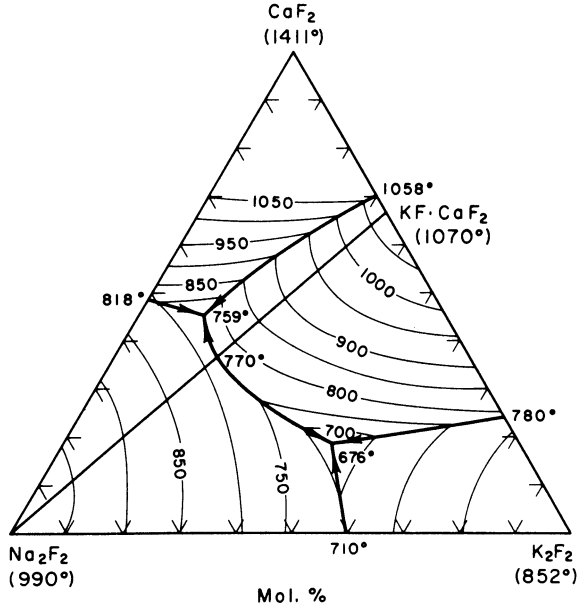


Fig. 165. The system NaF–KF–CaF_2.

The System LiF–CsF–MnF_2. See Fig. 166. Source: I. N. Belyaev and O. Ya. Revina, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:1041 (1966).

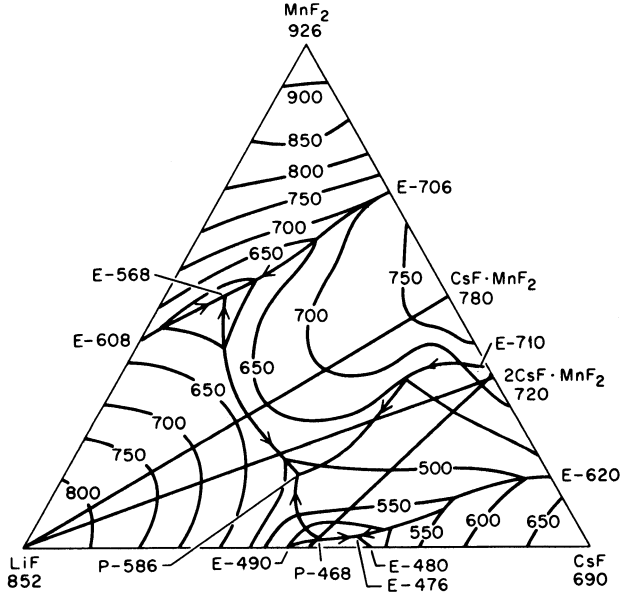


Fig. 166. The system LiF–CsF–MnF_2.

Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	CsF	MnF_2		
40	10	50	568	Eutectic
40	58	2	476	Eutectic
47	52	1	488	Peritectic
45	40	15	586	Peritectic

The System NaF–CsF–MnF_2. See Fig. 167. Source: I. N. Belyaev and O. Ya. Revina, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:1041 (1966). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	CsF	MnF_2		
18	74	8	570	Eutectic
25	44	31	634	Eutectic
41	20	39	640	Eutectic
17	16	67	650	Eutectic

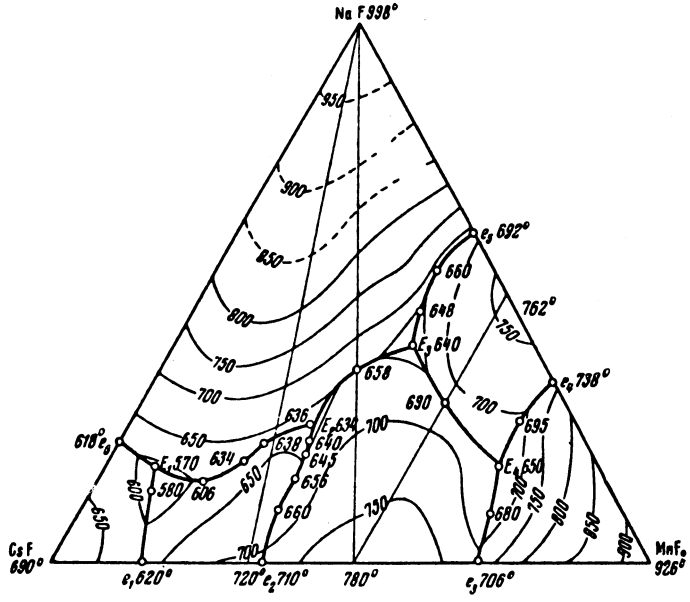


Fig. 167. The system NaF–CsF–MnF_2.

The System KF–CsF–MnF_2. See Fig. 168. Source: I. N. Belyaev and O. Ya. Revina, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:1041 (1966).

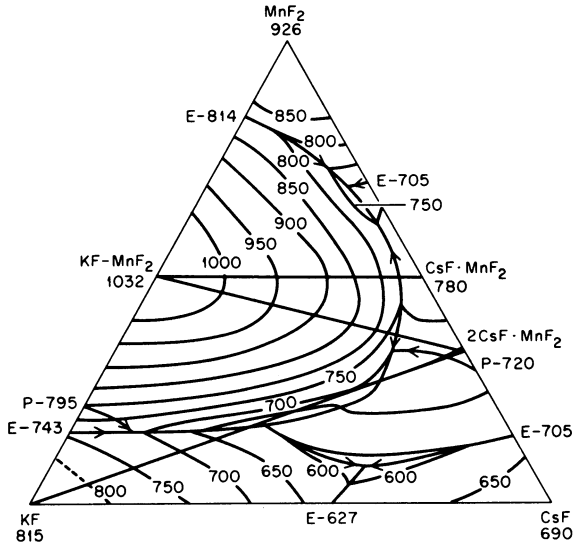


Fig. 168. The system KF–CsF–MnF_2.

Invariant and singular equilibrium points are given in the table.

Liquid mole %			Temperature	Type of equilibrium
KF	CsF	MnF_2		
3	27	70	702	Eutectic
32	60	8	560	Eutectic
13	54	38	684	Peritectic
46	37	17	610	Peritectic
73	11	16	725	Peritectic

The System LiF–NaF–SrF_2. See Fig. 169. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **5**:445 (1960). $E = 624°$, Li_2F_2–Na_2F_2–SrF_2 (51–33–16%).

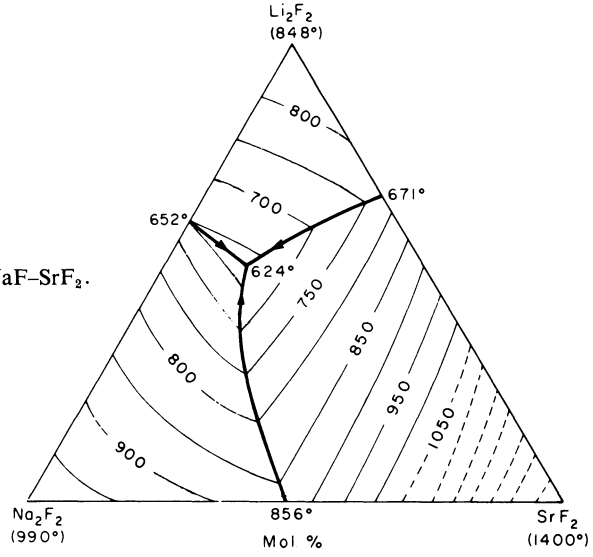


Fig. 169. The system LiF–NaF–SrF_2.

The System NaF–KF–SrF_2. See Fig. 170. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **5**:925 (1960). $E = 664°$, Na_2F_2–K_2F_2–SrF_2 (31–40–29%).

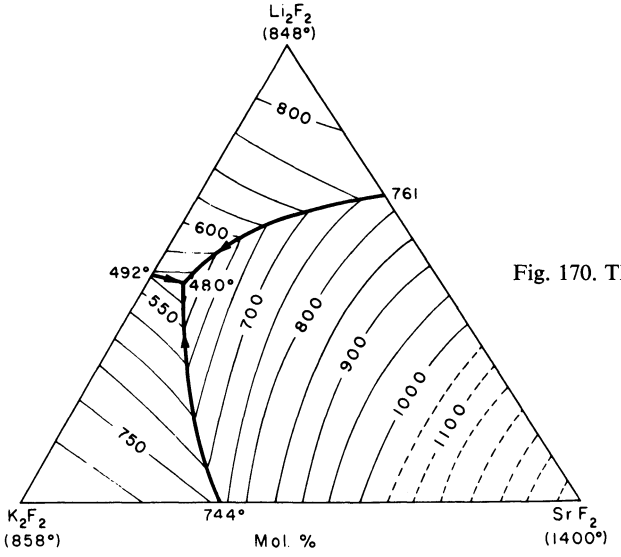


Fig. 170. The system LiF–KF–SrF_2.

The System NaF–KF–SrF_2. See Fig. 171. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **5**:925 (1960). $E = 664°$, Na_2F_2–K_2F_2–SrF_2 (31–40–29%).

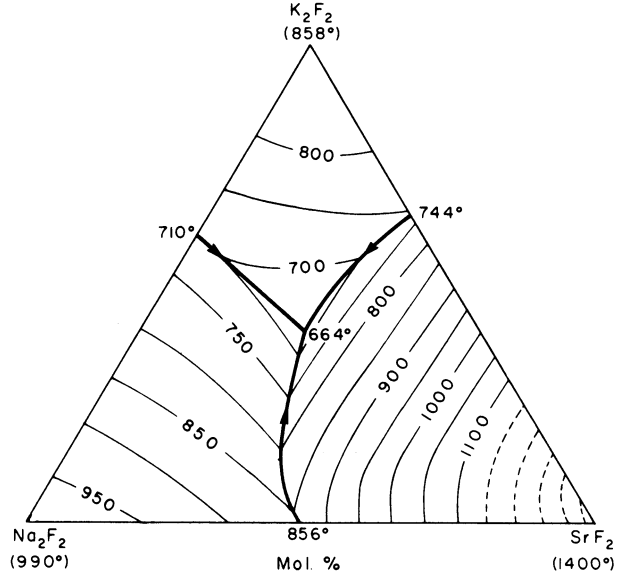


Fig. 171. The system NaF–KF–SrF_2.

The System Li_2F_2–Na_2F_2–BaF_2. See Fig. 172. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **4**:1200 (1959). $E = 621°$, Li_2F_2–Na_2F_2–BaF_2 (51–36–13%); $P = 694°$, Li_2F_2–Na_2F_2–BaF_2 (31–34–35%).

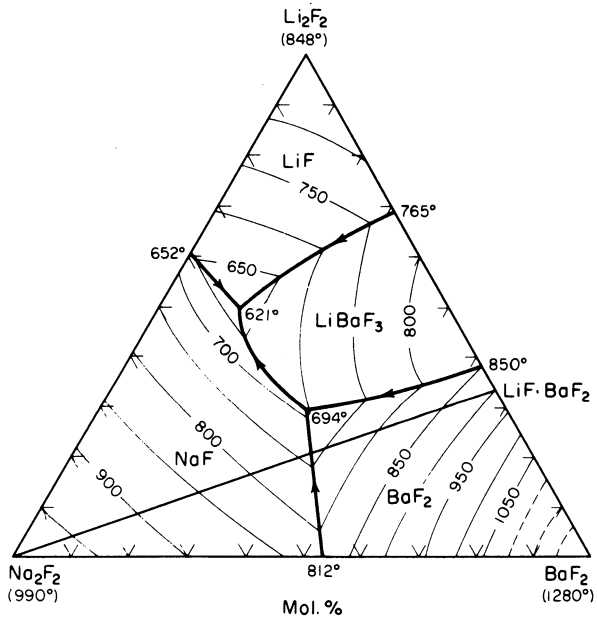


Fig. 172. The system LiF_2–Na_2F_2–BaF_2.

The System LiF–KF–BaF_2. See Fig. 173. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **4**:1199 (1959). $E = 472°$, Li_2F_2–K_2F_2–BaF_2 (49–46–5%); $P = 588°$, Li_2F_2–K_2F_2–BaF_2 (29–49–22%).

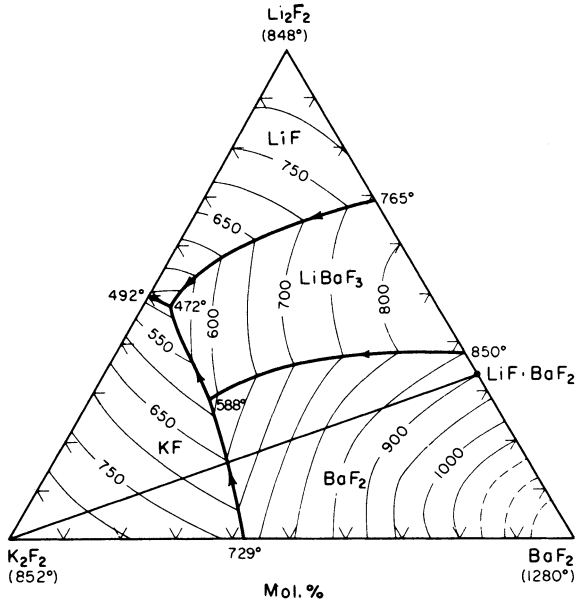


Fig. 173. The system LiF–KF–BaF_2.

The System NaF–KF–BF_3. See Fig. 174. Source: C. J. Barton, L. O. Gilpatrick, J. A. Bornmann, T. N. McVay, and H. Insley, *J. Inorg. Nucl.*

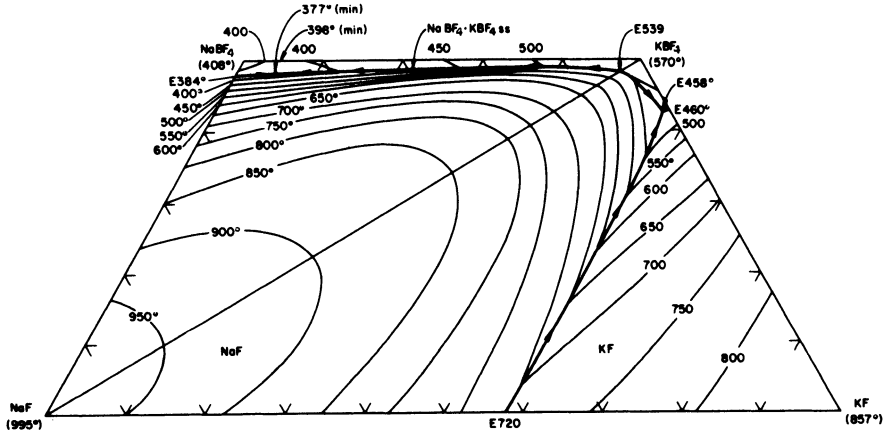


Fig. 174. The system NaF–KF–BF_3.

Chem. **33**:345 (1971). The composition of the ternary eutectic in the NaF–KBF_4–KF subsystem was not determined. It is described as being very close to the KF–KBF_4 binary eutectic in composition and melts at 458°. The shallow minimum at the liquidus, corresponding to the minimum in the $NaBF_4$–KBF_4 binary system, has the composition NaF–KF–BF_3 (48.4–3.5–48.1 %); the melting temperature is 377 ± 2°. Evidence of the existence of a subsolidus compound, $3NaBF_4 \cdot KBF_4$, stable between approximately 193 and 280°, was noted in the reference source.

The System LiF–NaF–AlF_3. See Fig. 175. Source: R. E. Thoma, B. J. Sturm, and E. H. Guinn, ORNL-3594 (1964). See also Fig. 3439 in *Phase Diagrams for Ceramists* (1969). A phase diagram of the system has also been reported by M. A. Kuvakin, *Russ. J. Inorg. Chem.* (*English Transl.*) **14**:146 (1969), that is in close agreement with the system as shown

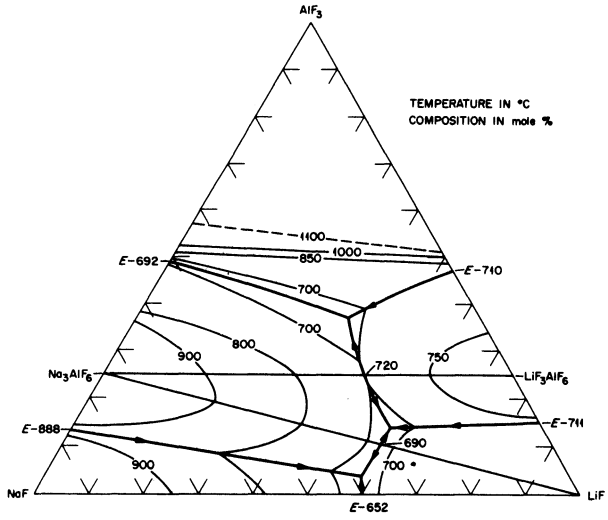


Fig. 175. The system LiF–NaF–AlF_3.

here. Thoma *et al.* did not report invariant equilibrium composition points. They were given in Kuvakin's report as follows: $E = 580°$, LiF–NaF–AlF_3 (51.4–38–10.6%); $E = 600°$, LiF–NaF–AlF_3 (40–17–43%); $P = 635°$, LiF–NaF–AlF_3 (22–19–59%); $E = 565°$, LiF–NaF–AlF_3 (18–17–65%). A phase diagram of the system was reported by M. A. Kuvakin (Fig. 176). In this source, composition–temperatures sections are shown in some detail.

The System LiF–NaF–AlF_3. See Fig. 176. Source: M. A. Kuvakin, *Russ. J. Inorg. Chem.* (*English Transl.*) **14**:146 (1969). See comments on Fig. 175.

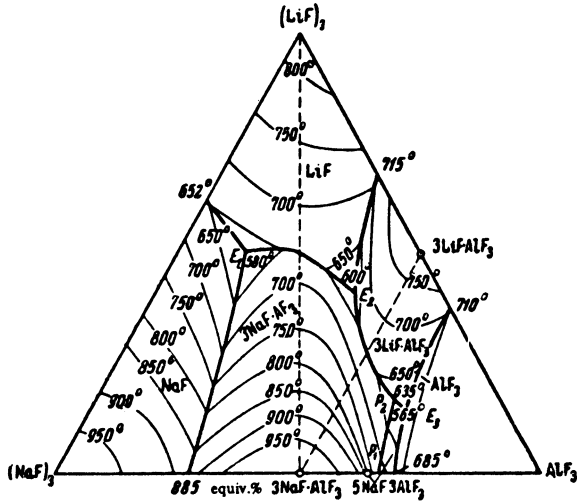


Fig. 176. The system LiF–NaF–AlF_3.

The System LiF–KF–AlF_3. See Fig. 177. Source: R. E. Thoma, B. J. Sturm, and E. H. Guinn, Molten-Salt Solvents for Fluoride Volatility

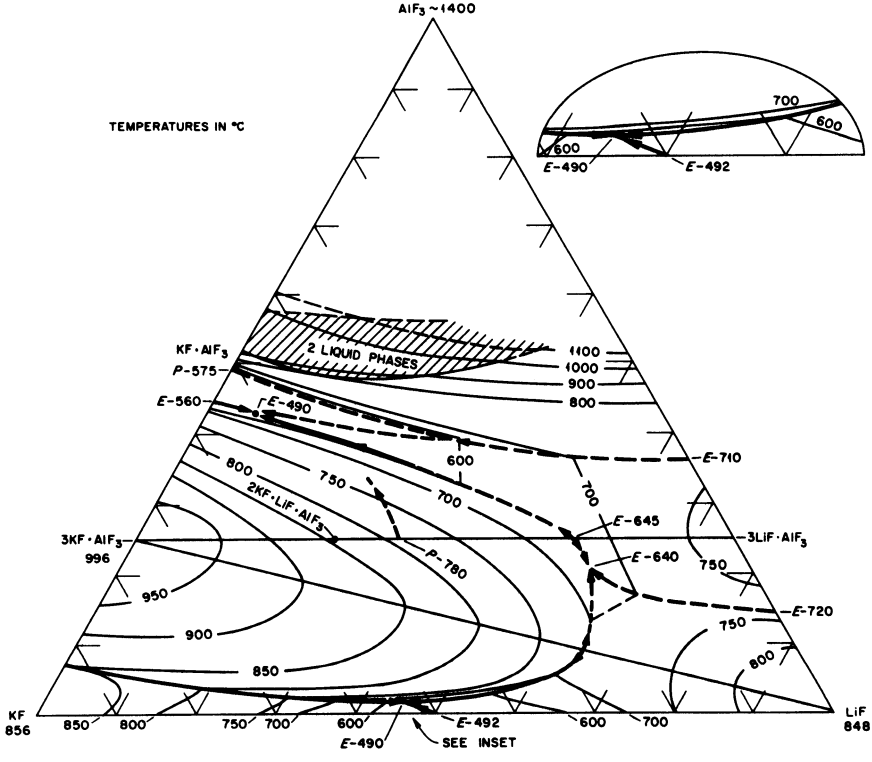


Fig. 177. The system LiF–KF–AlF_3.

Processing of Aluminum-Matrix Nuclear Fuel Elements, ORNL-3594, August, 1964. The authors did not report composition and temperature of invariant points. All primary data are listed in source report.

The System NaF–KF–AlF_3. See Fig. 178. Source: C. J. Barton, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1951–1952).

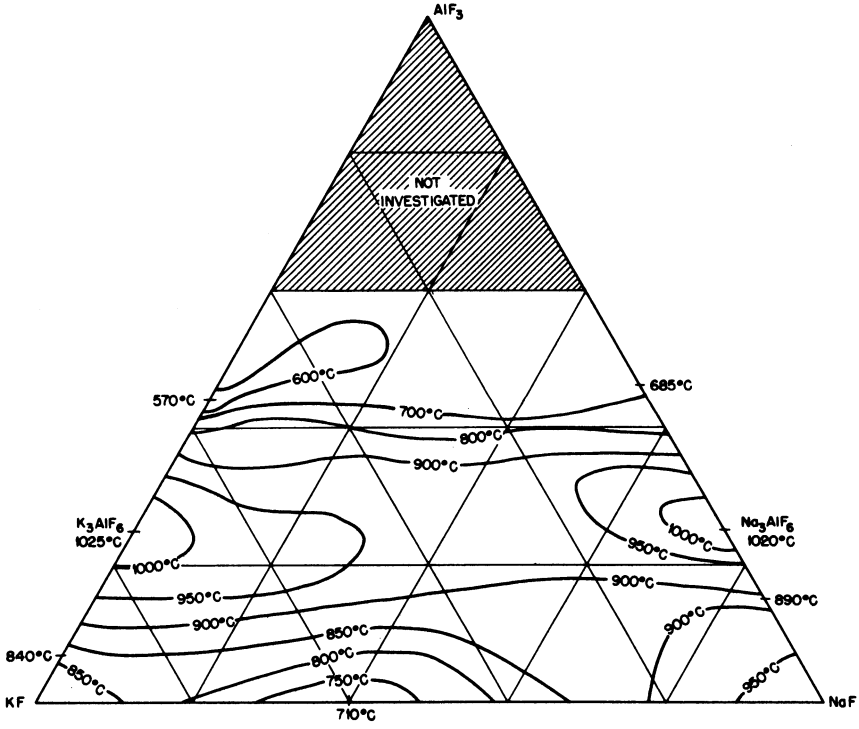


Fig. 178. The system NaF–KF–AlF_3.

No determination was made of the liquid–solid equilibria in the ternary system.

The System LiF–CsF–ScF_3. See Fig. 179. Source: E. P. Babaeva and G. A. Bukhalova, *Zh. Neorgan. Khim.* **11**:648 (1966). *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:351 (1966). Invariant equilibria: $E = 584°$, LiF–CsF–ScF_3 (51–26–23%); $E = 432°$, LiF–CsF–ScF_3 (53–46.88–0.12%); $E = 438°$, LiF–CsF–ScF_3 (39.9–60–0.1%). The diagram as constructed here supports the inference that the compound LiF · CsF melts congruently.

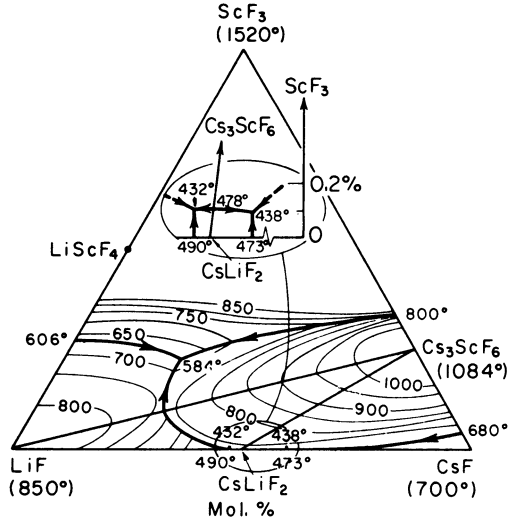


Fig. 179. The system LiF–CsF–ScF_3.

The System LiF–CsF–YF_3. See Fig. 180. Source: G. A. Bukhalova and E. P. Babaeva, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:220 (1966). Invariant and singular equilibria: $E = 650°$, LiF–CsF–YF_3 (47.5–31–21.5%); $E = 460°$, LiF–CsF–YF_3 (51.5–48–0.5%); $E = 430°$, LiF–CsF–YF_3 (39.8–60–0.2); saddle point, 710° LiF–CsF–YF_3 (57.5–32–10.5).

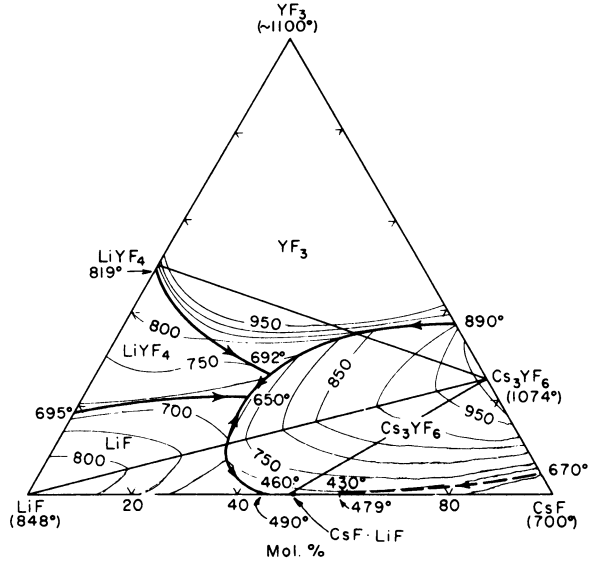


Fig. 180. The system LiF–CsF–YF_3.

The System LiF–CsF–LaF_3. See Fig. 181. Source: G. A. Bukhalova and E. P. Babaeva, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:1027 (1965).

Invariant equilibria: $E = 417°$, LiF–CsF–LaF_3 (33–61–6%); E–443°, LiF–CsF–LaF_3 (39–52–9%); $E = 480°$, LiF–CsF–LaF_3 (35–45–20%).

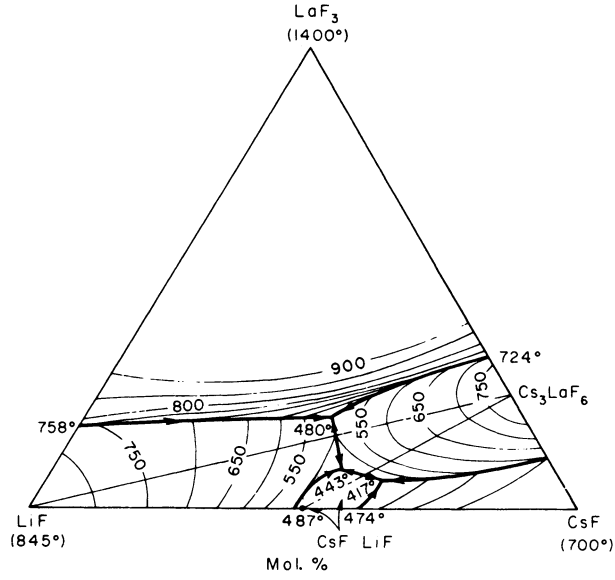


Fig. 181. The system LiF–CsF–LaF_3.

The System NaF–KF–ScF_3. See Fig. 182. Source: E. P. Babaeva and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:793 (1965). The solid-solution field of $3NaF \cdot ScF_3$–$3KF \cdot ScF_3$ meets the NaF and KF crystallization fields at $E = 674°$, NaF–KF–ScF_3 (35–60–5%) or to K_3F_3–Na_3F_3–$3NaF \cdot ScF_3$ (62.5–8.5–29%).

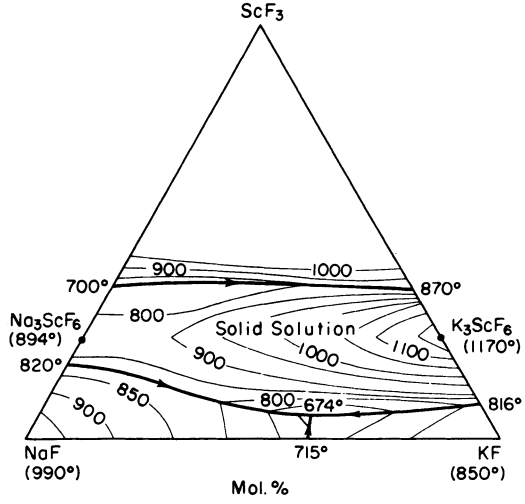


Fig. 182. The system NaF–KF–ScF_3.

The System NaF–KF–YF_3. See Fig. 183. Source: G. A. Bukhalova and E. P. Babaeva, *Russ. J. Inorg. Chem.* (*English Transl.*) **11**:350 (1966). $E = 554°$, NaF–KF–YF_3 (44–21–35%) $E = 648°$, NaF–KF–YF_3 (32–58–10%).

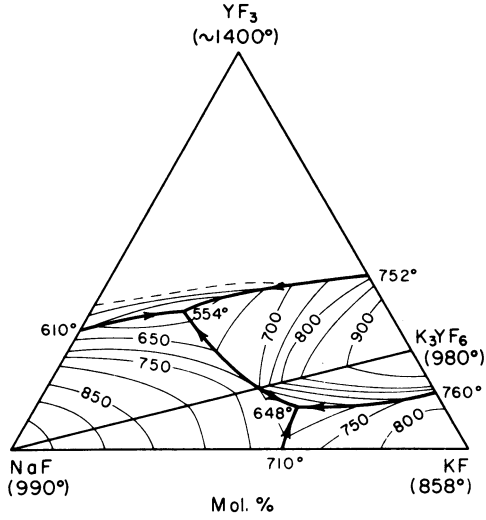


Fig. 183. The system NaF–KF–YF_3.

The System NaF–KF–LaF_3. See Fig. 184. Source: G. A. Bukhalova, E. P. Babaev, and T. M. Khliyan, *Russ. J. Inorg. Chem.* (*English Transl.*) **10**:1160 (1965). $E = 563°$, NaF–KF–LaF_3 (25–58–17%).

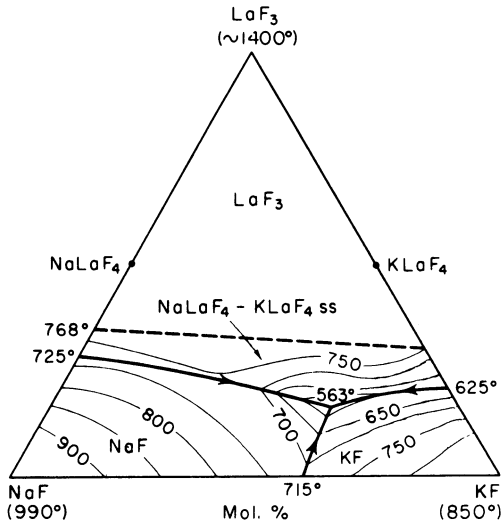


Fig. 184. The system NaF–KF–LaF_3.

The System LiF–NaF–ZrF_4. See Fig. 185. Source: R. E. Thoma, H. Insley, H. A. Friedman, and G. M. Hebert, *J. Chem. Eng. Data* **10**:219

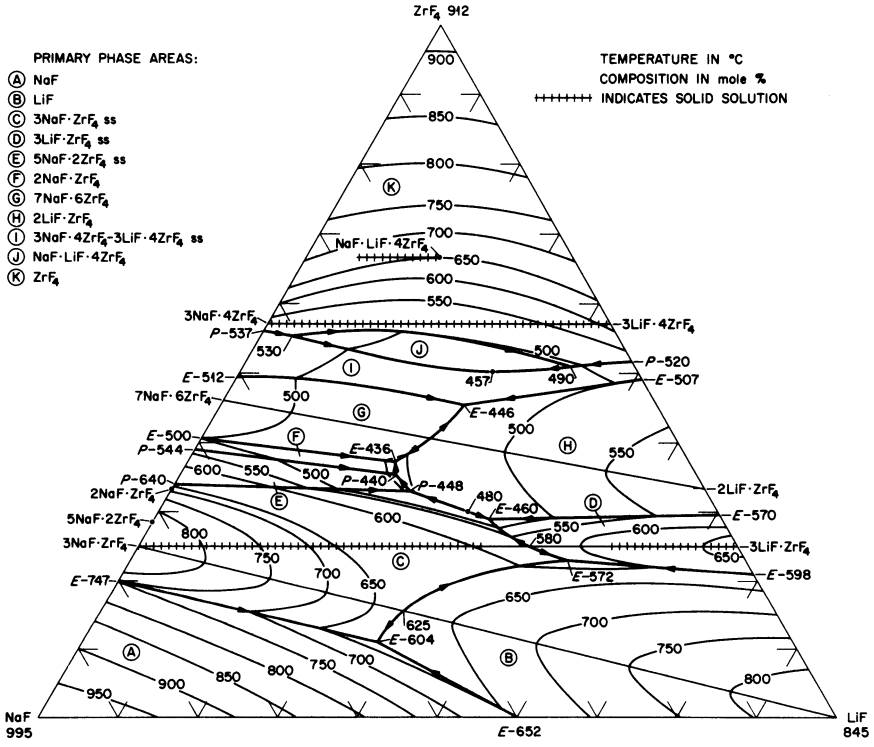


Fig. 185. The system LiF–NaF–ZrF_4.

(1965). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	NaF	ZrF_4		
79		21	598	Eutectic
75		25	662	Congruent melting point
			475	Inversion
			470	Decomposition
70.5		29.5	570	Eutectic
66.7		33.3	596	Congruent melting point
51		49	507	Eutectic
48.5		51.5	520	Peritectic
			466	Decomposition
37	52	11	694	Eutectic

(*Continued*)

Mole %			Temperature	Type of equilibrium
LiF	NaF	ZrF_4		
38	46.5	15.5	625	Boundary curve maximum
55	22	23	572	Eutectic
49	26	25	580	Boundary curve maximum
42	29	29	460	Eutectic
38.5	31.5	30	480	Boundary curve maximum
30	37	33	448	Peritectic
27	37.5	35.5	440	Peritectic
26	37	37	436	Eutectic
29	30.5	40.5	470	Boundary curve maximum
30.5	24	45.5	446	Eutectic
32	18	50	457	Boundary curve minimum
41	8	51	490	Peritectic
4	40.5	55.5	530	Peritectic

The System NaF–KF–ZrF_4. See Fig. 186. Source: R. E. Thoma, C. J. Barton, H. Insley, H. A. Friedman, and W. R. Grimes, ORNL,

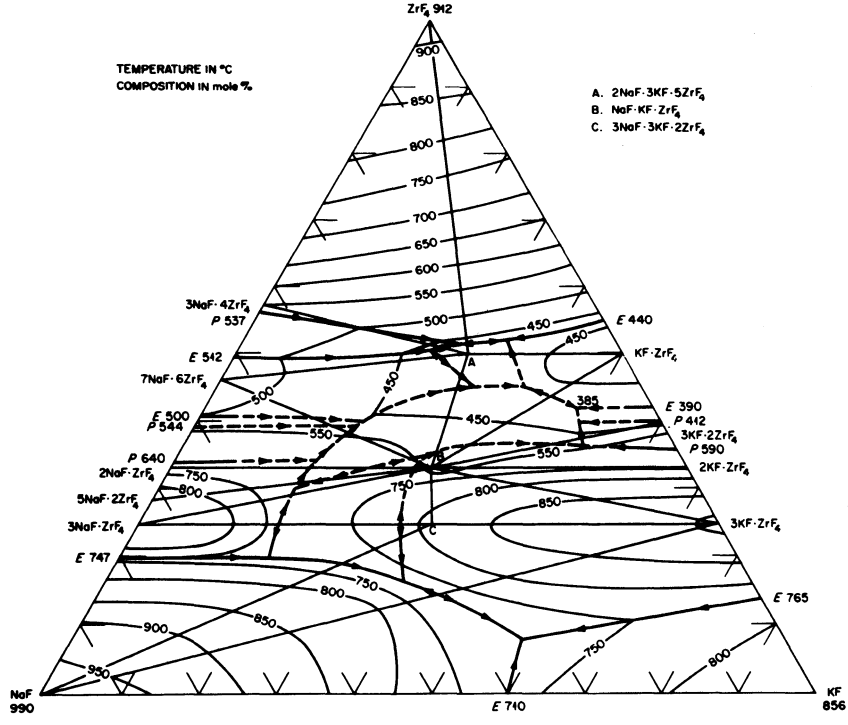


Fig. 186. The system NaF–KF–ZrF_4.

unpublished work (1951–1955). Invariant and singular equilibrium points are given in the table.

Mole %[b]			Temperature	Type of equilibrium
NaF	KF	ZrF_4		
34	58	8	695	Eutectic
45	38	17	720	Eutectic
61	19	20	710	Eutectic
52	17	31		Peritectic
48	18	34		Peritectic
33	32	35		Peritectic
12	51	37		Peritectic
40	21	39		Peritectic
39	21	40		Peritectic
11	49	40		Peritectic
10	48	42	385	Eutectic
15	40	45		Eutectic
22	33	45		Peritectic
25	25	50		Eutectic
21	29	51		Eutectic
20	30	50	432	Congruent melting point

[a] Three solid phases have been observed routinely in the composition region 30–50 mole % ZrF_4 which have not been identified as to composition. Whether these exhibit primary phases at the liquidus surface has not yet been determined.
[b] Compositions of invariant points shown by the intersection of dotted boundary curves are approximate.

The System NaF–RbF–ZrF_4. See Fig. 187. Source: R. E. Thoma, H. Insley, H. A. Friedman, and W. R. Grimes, ORNL, unpublished work (1951–1956). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	RbF	ZrF_4		
23	72	5	643	Eutectic
50	27	23	720	Eutectic
37	32	31	605	Eutectic
39	26	35	545	Peritectic
36.3	24.2	39.5	427	Peritectic
34.5	24	41.5	424	Peritectic
34	23.5	42.5	422	Peritectic
33	23.5	43.5	420	Eutectic
28.5	21.5	50	443	Eutectic
28	21.5	50.5	446	Peritectic
23.5	39.5	37	470	Peritectic
21	40	39	438	Peritectic
8	50	42	400	Eutectic
8.5	47	44.5	395	Eutectic
6.2	45.8	48	380	Eutectic
5	42	53	398	Eutectic
6.5	39	54.5	423	Peritectic
33.3	33.3	33.3	642	Congruent melting point
25	25	50	462	Congruent melting point

Minimum temperatures on Alkemede lines

Mole %			Temperature	Alkemede line
NaF	RbF	ZrF_4		
24.75	24.75	51.5	455	$NaF \cdot RbF \cdot 2ZrF_4$–$ZrF_4$
6	44	50	402	$NaF \cdot RbF \cdot 2ZrF_4$–$RbF \cdot ZrF_4$
7.5	46.5	46	405	$NaF \cdot RbF \cdot 2ZrF_4$–$5RbF \cdot 4ZrF_4$
8.5	48.5	43	405	$3NaF \cdot RbF \cdot 4ZrF_4$–$5RbF \cdot 4ZrF_4$
28	28	44	435	$3NaF \cdot RbF \cdot 4ZrF_4$–$NaF \cdot RbF \cdot 2ZrF_4$
22	40	38	442	$3NaF \cdot 3RbF \cdot 4ZrF_4$–$2RbF \cdot ZrF_4$
29	41.5	49.5	450	$NaF \cdot RbF \cdot 2ZrF_4$–$7NaF \cdot 6ZrF_4$
37	24.5	38.5	610	$NaF \cdot RbF \cdot ZrF_4$–$3NaF \cdot ZrF_4$
47	28	25	732	$3NaF \cdot ZrF_4$–$3RbF \cdot ZrF_4$
36	48	16	777	NaF–$3RbF \cdot ZrF_4$
30	36.7	33.3	598	$NaF \cdot RbF \cdot ZrF_4$–$3RbF \cdot ZrF_4$

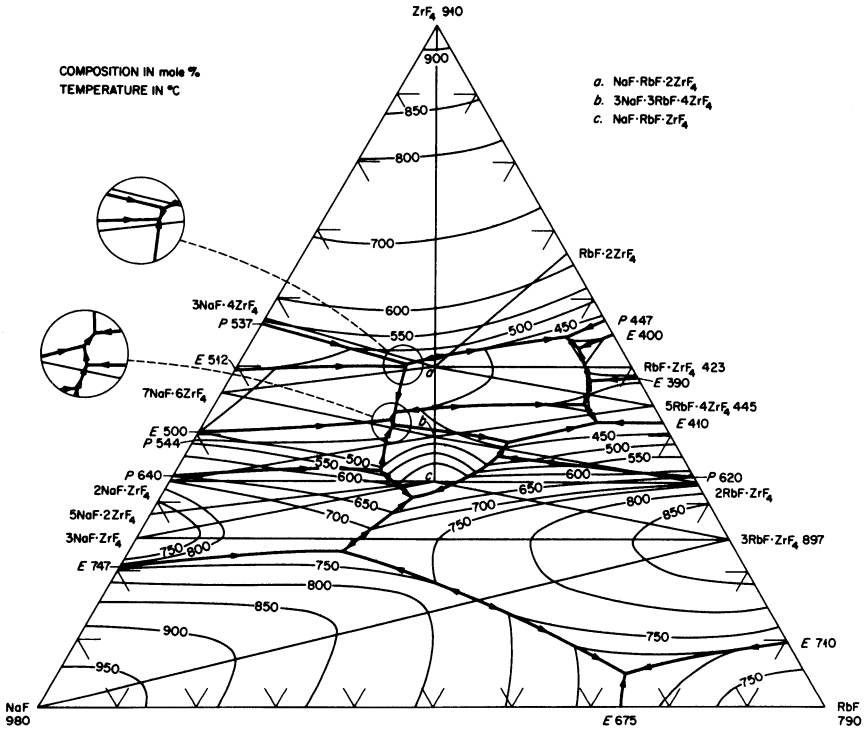


Fig. 187. The system NaF–RbF–ZrF_4.

The System LiF–NaF–ThF_4. See Fig. 188. Source: R. E. Thoma, *J. Inorg. Nucl. Chem.* **34**:2747 (1972). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	NaF	ThF_4		
30	56.5	13.5	505	Eutectic
18	65	17	558	Peritectic
46.5	35.5	18	536	Max. on boundary curve
7	72.5	20.5	604	Peritectic
2	77.5	20.5	610	Peritectic
43.5	32.5	24	509	Eutectic
53	17	30	531	Max. on boundary curve
54.5	13.5	32	525	Eutectic
54.5	12.5	33	540	Peritectic
6	56.5	37.5	683	Peritectic
50	12	38	585	Peritectic

(Continued)

Mole %			Temperature	Type of equilibrium
LiF	NaF	ThF_4		
48	12	40	643	Peritectic
44	12	44	725	Peritectic
16	27	57	825	Peritectic

Structure data: $7(Na, Li)F \cdot 6ThF_4$: hexagonal, $P\bar{3}Cl$, $a = 9.9056$, $c = 13.282$ [G. D. Brunton and D. R. Sears, *Acta Cryst.* **B25**:2519 (1969)]. This phase was regarded as a ternary compound of variable composition when the structure was determined. In the final examination of phase relations in the LiF–NaF–ThF_4 ternary system, it was concluded that the structure was a new crystal modification of $7NaF \cdot 6ThF_4$ that could accommodate extensive solid solution of Li^+.

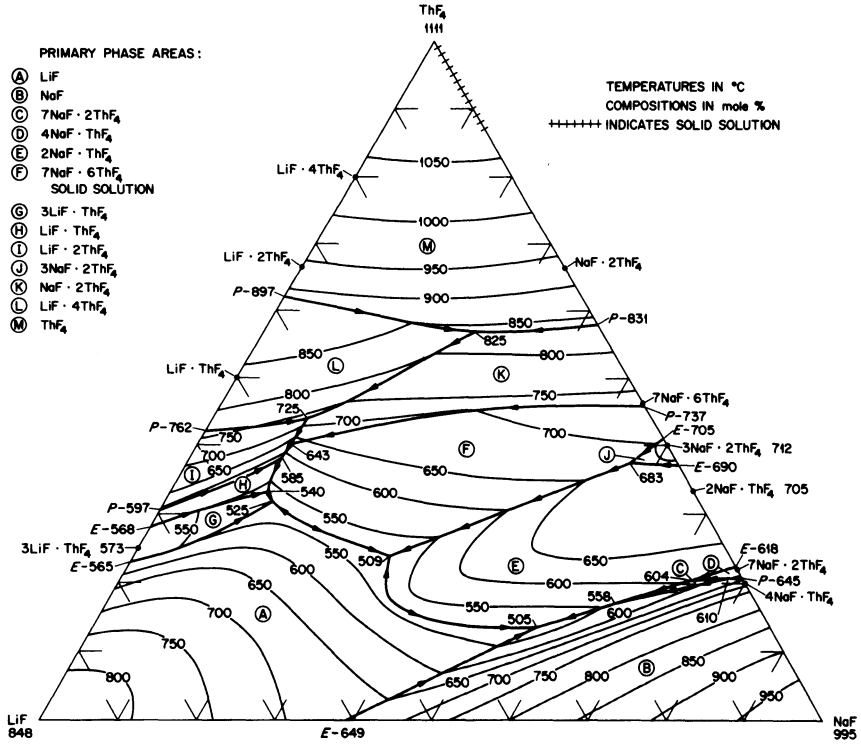


Fig. 188. The system LiF–NaF–ThF_4.

The System NaF–KF–ThF_4. Source: R. E. Thoma, H. A. Friedman, and H. Insley, ORNL, unpublished work (1955–1958). Investigations with

the system NaF–KF–ThF_4 were not carried out to the extent that a phase diagram could be constructed. Within the system, a single ternary compound NaF · KF · ThF_4 (isostructural with NaF · KF · UF_4) was found to exist. It forms extensive solid solutions with 2NaF · ThF_4. The quasibinary system, NaF–3KF · ThF_4, is a limiting system of the subsystem NaF–KF–3KF · ThF_4, in which a single eutectic occurs near the composition NaF–KF–ThF_4 (36–57–7 mole %). Within the subsystem NaF–NaF · KF · ThF_4–2NaF · ThF_4, a temperature minimum occurs near the composition NaF–KF–ThF_4 (11–67–22 mole %), and at 535°.

Structural data: NaF · KF · ThF_4: hexagonal, *P*-3, $a = 3.63073$, $c = 7.8907$ [G. D. Brunton *et al.*, ORNL-3761]. Crystals of the ternary compound NaF · KF · ThF_4 are hexagonal, *P*-3, $a = 6.32$, $c = 8.18$ [Brunton, *ibid.*].

The System KF–RbF–ThF_4. See Fig. 189. Source: E. P. Dergunov and A. G. Bergman, *Dokl. Akad. Nauk SSSR* **60**:394 (1948). No ternary invariant equilibria are found in the KF–RbF–ThF_4 system. The compounds indicated as $KThF_5$ and $RbThF_5$ are probably 7KF · 6ThF_4 and 7RbF · 6ThF_4.

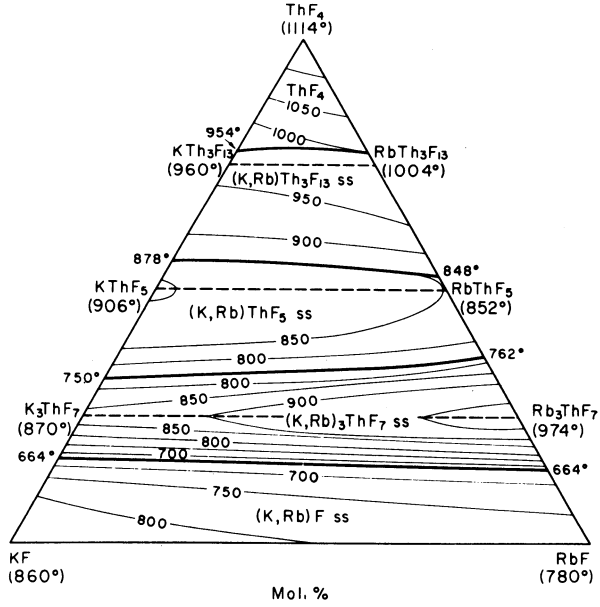


Fig. 189. The system KF–RbF–ThF_4.

The System LiF–NaF–UF_4. See Fig. 190. Source: R. E. Thoma *et al.*, *J. Am. Ceram. Soc.* **42**:21 (1959). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium	Coexisting solid phases
NaF	LiF	UF_4			
60.0	21.0	19.0	480	Eutectic	$2NaF \cdot UF_4$, NaF, LiF
65.0	13.0	22.0	497	Peritectic	β-$3NaF \cdot UF_4$, $2NaF \cdot UF_4$, NaF
7.0	65.5	27.5	470	Peritectic	$7LiF \cdot 6UF_4$ ss,[a] $4LiF \cdot UF_4$, LiF
35.0	37.0	28.0	480	Eutectic	$2NaF \cdot UF_4$, $7NaF \cdot 6UF_4$ ss, LiF
57.0	13.0	30.0	630	Peritectic	$2NaF \cdot UF_4$, $5NaF \cdot 3UF_4$, $7NaF \cdot 6UF_4$ ss
24.3	43.5	32.2	445	Eutectic	$7NaF \cdot 6UF_4$ ss, LiF, $7LiF \cdot 6UF_4$ ss
24.5	29.0	46.5	602	Eutectic	$NaF \cdot 2UF_4$, $7LiF \cdot 6UF_4$ ss, $7NaF \cdot 6UF_4$ ss
24.3	28.7	47.0	605	Peritectic	$NaF \cdot 2UF_4$, $LiF \cdot 4UF_4$, $7LiF \cdot 6UF_4$ ss
23.5	28.0	48.5	640	Peritectic	UF_4, $NaF \cdot 2UF_4$, $LiF \cdot 4UF_4$
37.5	10.5	52.0	660	Peritectic	UF_4, $NaF \cdot 2UF_4$, $7NaF \cdot 6UF_4$ ss

[a] The compounds $7LiF \cdot 6UF_4$ and $7NaF \cdot 6UF_4$ as they occur in the ternary system are members of the solid solution $7LiF \cdot 6UF_4$–$7NaF \cdot 6UF_4$.

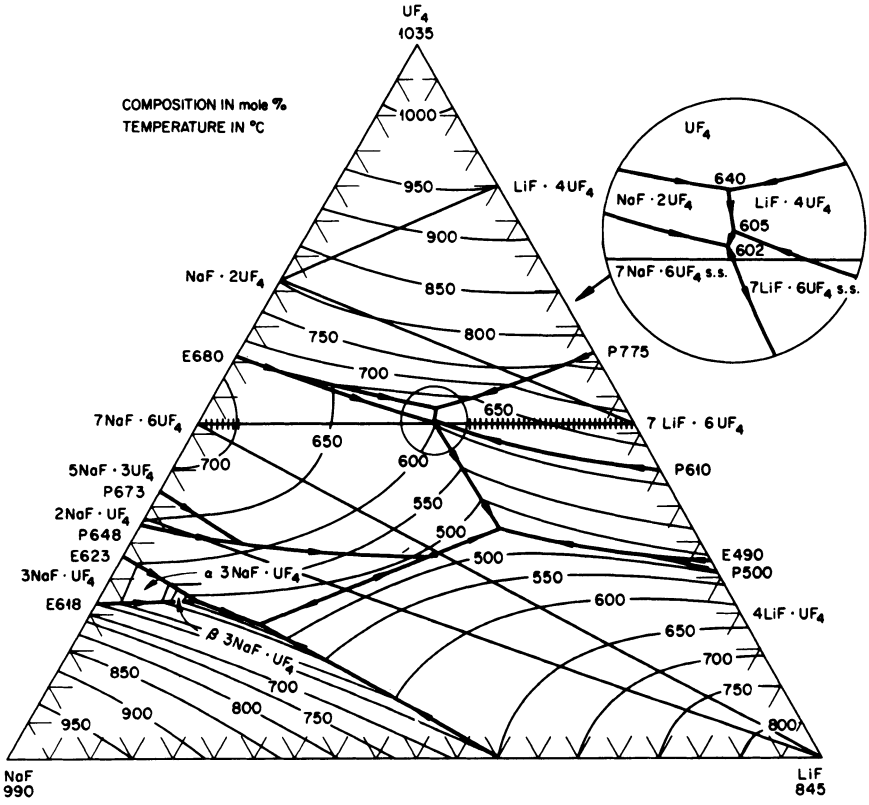


Fig. 190. The system LiF–NaF–UF_4. –|–|–|–|–|–|– indicates solid solution.

The System LiF–KF–UF_4. See Fig. 191. Source: C. J. Barton, J. P. Blakely, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work

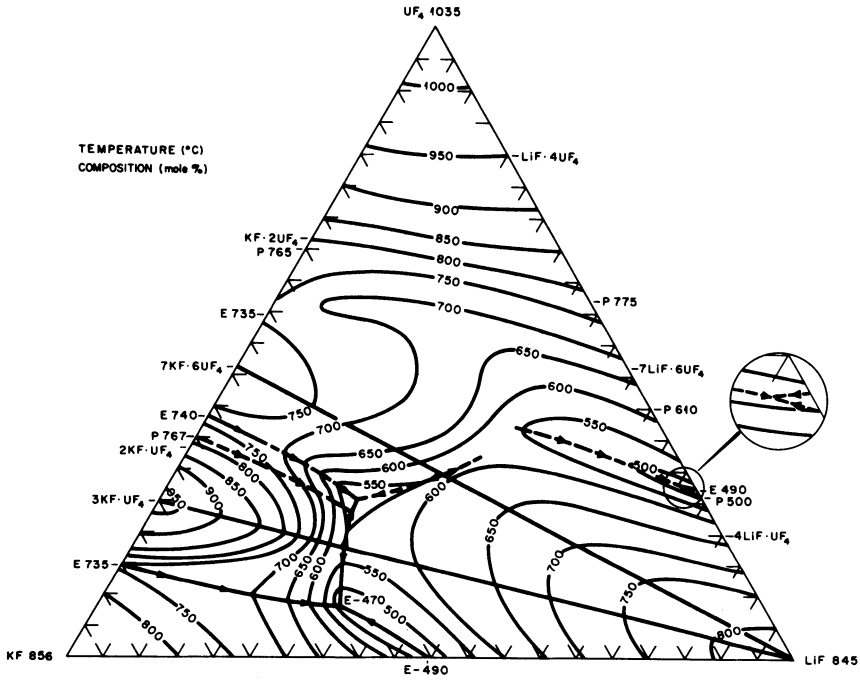


Fig. 191. The system LiF–KF–UF_4.

(1950–1951). The ternary system was not defined with sufficient precision in the work cited to list the invariant-point compositions and temperatures.

The System LiF–RbF–UF_4. See Fig. 192. Source: C. J. Barton, J. P. Blakely, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work, (1950–1951). Examination of the LiF–RbF–UF_4 system did not proceed

to the point that invariant equilibria were determined, except for the eutectic at 450°, LiF–RbF–UF_4 (39–58–3%).

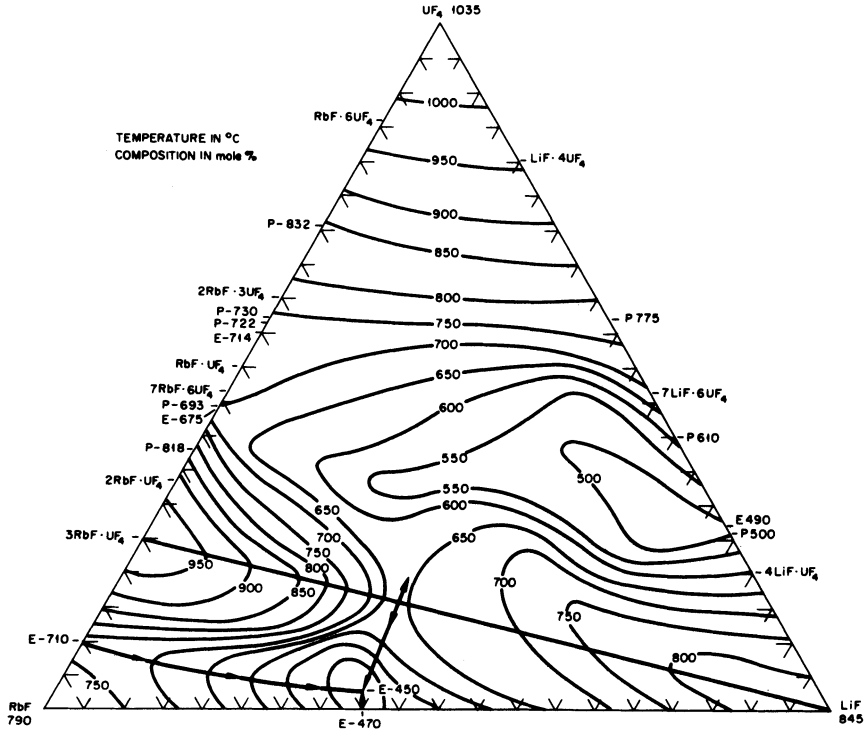


Fig. 192. The system LiF–RbF–UF_4.

The System NaF–KF–UF_4. See Fig. 193. Source: R. E. Thoma, C. J. Barton, J. P. Blakely, R. E. Moore, H. Insley, and H. A. Friedman, ORNL, unpublished work (1950–1958). Investigation of the phase diagram did not proceed to the extent that invariant equilibrium points were established accurately.

Structural data: NaF · KF · ThF_4: hexagonal, *P*-3, $a = 6.24$, $c = 7.80$ [G. D. Brunton *et al.*, ORNL-3761].

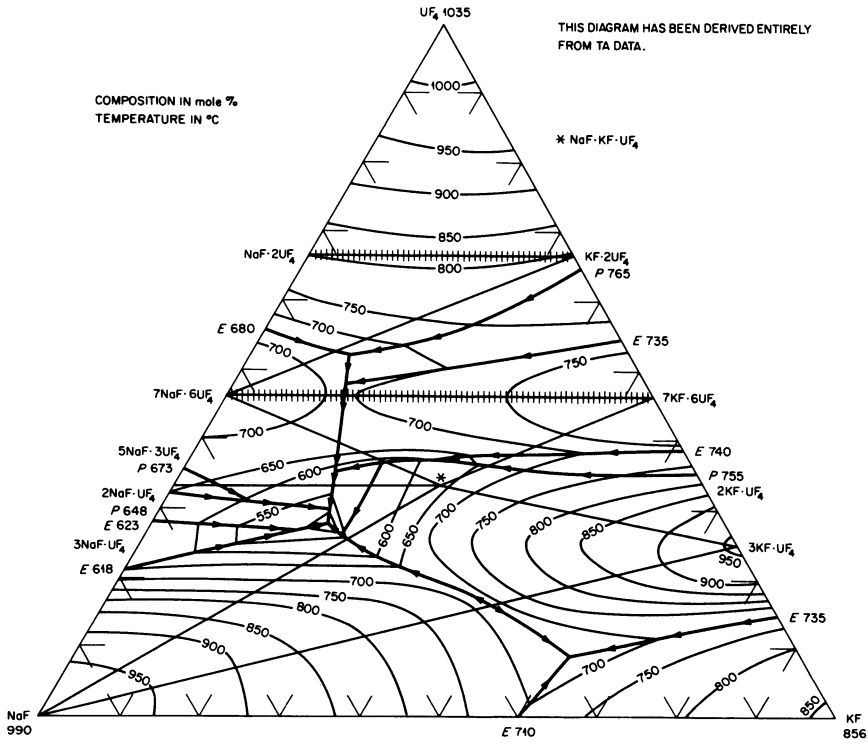


Fig. 193. The system NaF–KF–UF_4.

The System NaF–RbF–UF_4. See Fig. 194. Source: R. E. Thoma, H. Insley, H. A. Friedman, and W. R. Grimes, ORNL, unpublished work (1955–1956). Equilibrium relationships of the ternary compound along several composition–temperature sections were given in ORNL-2548, p. 105–107. Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	RbF	UF_4		
18	73	9	670	Eutectic
46	33	21	470	Eutectic
44	33	23	500	Peritectic
45	30	25	485	Peritectic
52	23	25	500	Decomposition
41	26	33	555	Peritectic
57	8	35	630	Decomposition

(Continued)

Mole %			Temperature	Type of equilibrium
NaF	RbF	UF_4		
33	30	37	535	Eutectic
32	29	39	540	Peritectic
26	27	47	620	Eutectic
25	25	50	630	Eutectic
27	22	51	633	Peritectic
33	14	53	655	Peritectic
42	3	55	678	Peritectic

Structural data: NaF · RbF · UF_4: hexagonal, P-3, $a = 6.29$, $c = 8.13$ [G. D. Brunton *et al.*, ORNL-3761 (1965)].

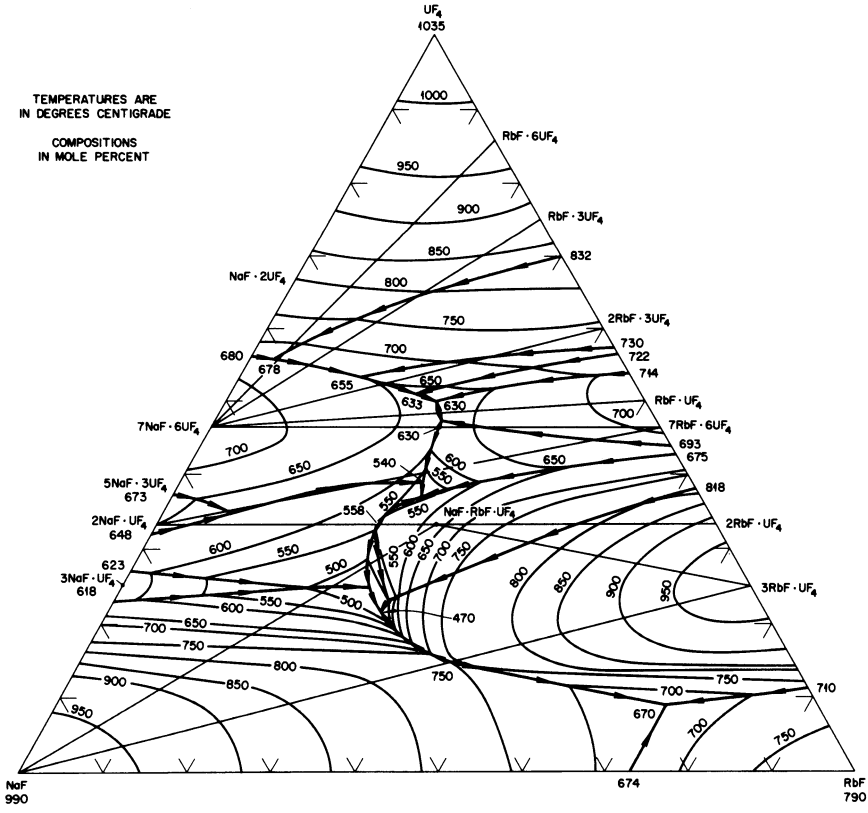


Fig. 194. The system NaF–RbF–UF_4.

The System KF–K_2NbF_7–NaF. See Fig. 195. Source: L. A. Kamenskaya, V. I. Konstantinov, and A. M. Matveev, *Russ. J. Inorg. Chem.* (*English Transl.*) **17**:1343 (1972). $E = 655°$, NaF–KF–K_2NbF_7 (25.5–54.5–

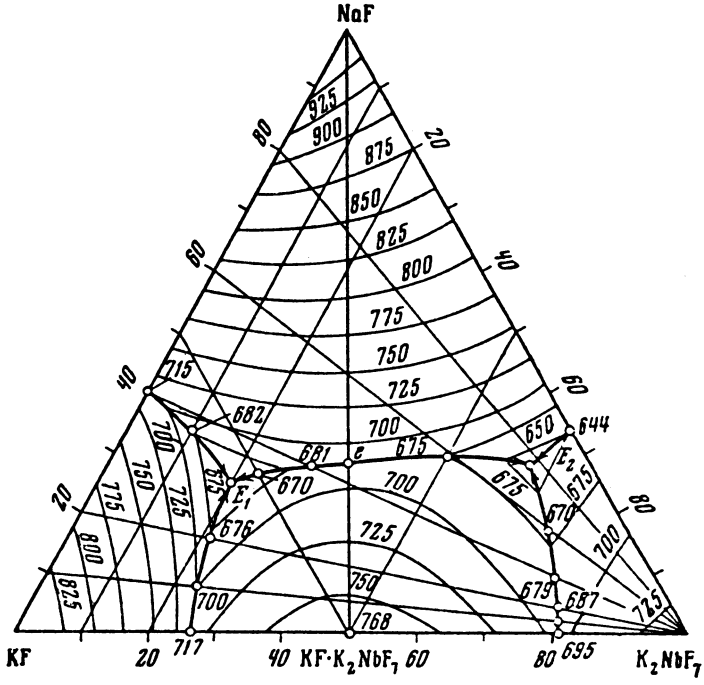


Fig. 195. The system KF–K_2NbF_7–NaF.

20%); $E = 636°$, NaF–KF–K_2NbF_7 (28–9–63%); saddle point, $E = 688°$, NaF–K_2NbF_7 (27.5–72.5%).

The System LiF–CaF_2–MgF_2. See Fig. 196. Source: W. E. Roake, *J. Electrochem. Soc.* **104**:409 (1957). A phase diagram of the system was also reported by V. T. Berezhnaya and G. A. Bukhalova, *Zh. Neorg. Khim.* **4**:903 (1959). Neither phase diagram shows evidence of the occurrence of solid solutions in the LiF–MgF_2 binary system, although the work of Counts *et al.* (Fig. 17) shows extensive solid solubility. Roake did not report the composition of the eutectic. According to Berezhnaya and Bukhalova, the eutectic in the system occurs at 676°, Li_2F_2–MgF_2–CaF_2 (47–35–18%).

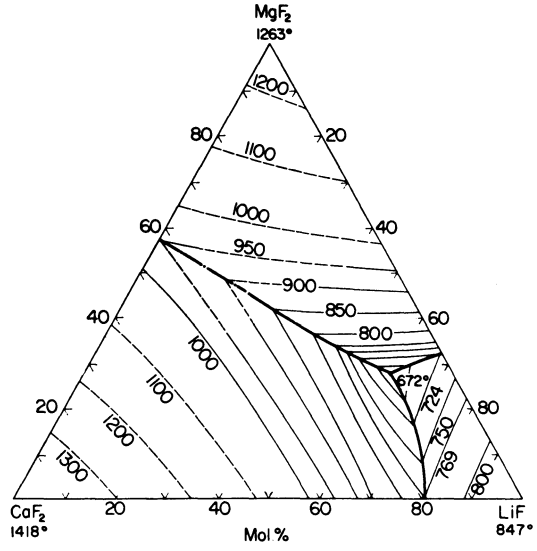


Fig. 196. The system LiF–CaF_2–MgF_2.

The System LiF–MgF_2–SrF_2. See Fig. 197. Source: Z. A. Mateiko and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **7**:84 (1962). $E = 646°$, Li_2F_2–MgF_2–SrF_2 (36–39–25%).

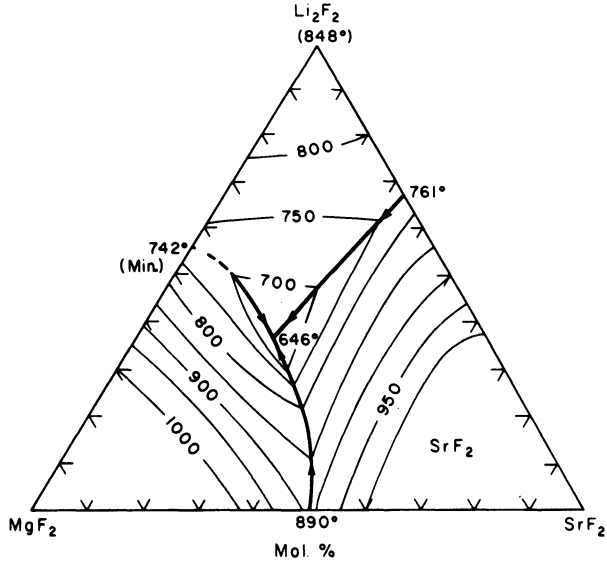


Fig. 197. The system LiF–MgF_2–SrF_2.

The System LiF–MgF_2–BaF_2. See Fig. 198. Source: G. A. Bukhalova and V. T. Berezhnaya, *Russ. J. Inorg. Chem.* (*English Transl.*) **4**:51 (1959).

$E = 748°$, 17.0 Li_2F_2–29.0 MgF_2–54.0 BaF_2; $P = 741°$, 25.0 Li_2F_2–46.5 MgF_2–28.5 BaF_2; $E = 654°$, Li_2F_2–MgF_2–BaF_2 (35–35–30%).

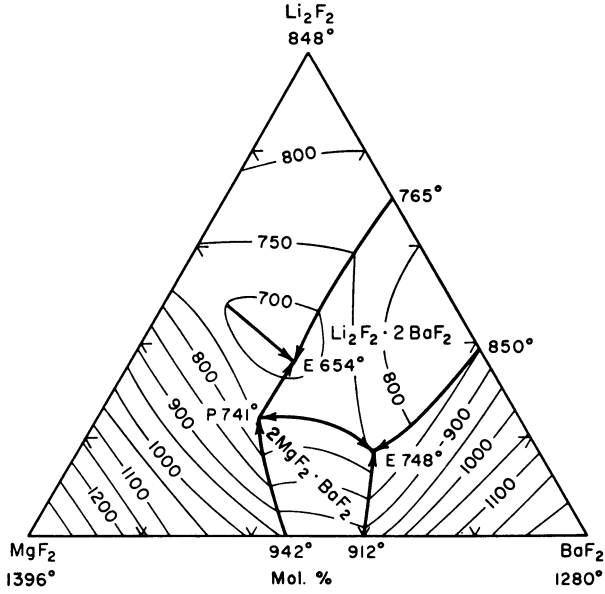


Fig. 198. The system LiF–MgF_2–BaF_2.

The System LiF–CaF_2–SrF_2. See Fig. 199. Source: Z. A. Mateiko and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **6**:882 (1961). $E = 740°$, Li_2F_2–CaF_2–SrF_2 (57–21–22%). The authors note that CaF_2 and SrF_2 form a continuous solid solution.

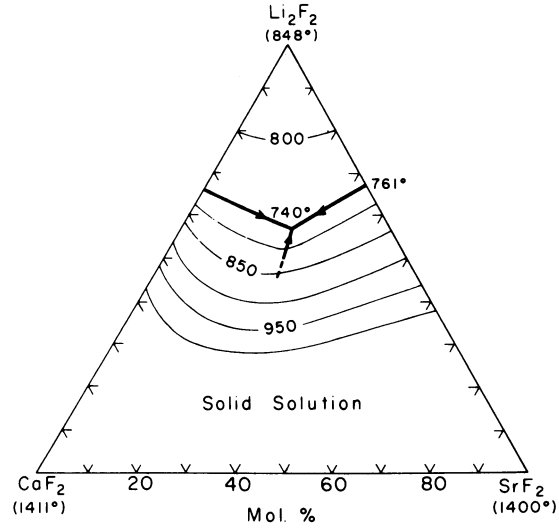


Fig. 199. The system LiF–CaF_2–SrF_2.

The System $LiF–CaF_2–BaF_2$. See Fig. 200. Source: G. A. Bukhalova and V. T. Berezhnaya, *Russ. J. Inorg. Chem.* (*English Transl.*) **2**:1409 (1957). $E = 710°$, Li_2F_2–CaF_2–BaF_2 (50.5–25.5–24%); $E = 758°$, Li_2F_2–CaF_2–BaF_2 (30.0–32.8–37.2%). The authors of the source reference note that CaF_2 and BaF_2 form continuous solid solutions with a temperature minimum at 1277° and at 50 mole % BaF_2.

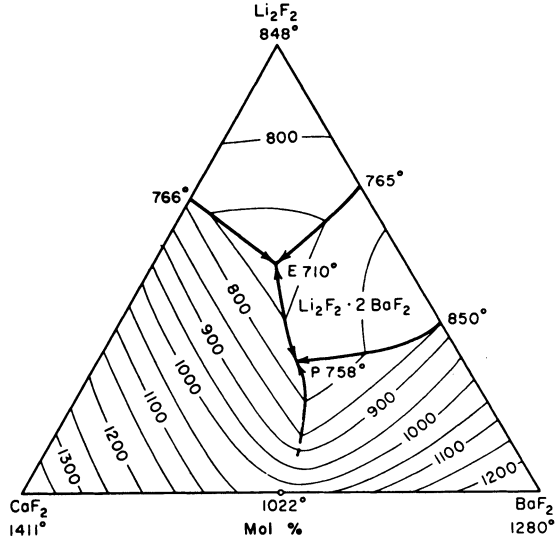


Fig. 200. The system $LiF–CaF_2–BaF_2$.

The System $LiF–SrF_2–BaF_2$. See Fig. 201. Source: Z. A. Mateiko and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **7**:85 (1962).

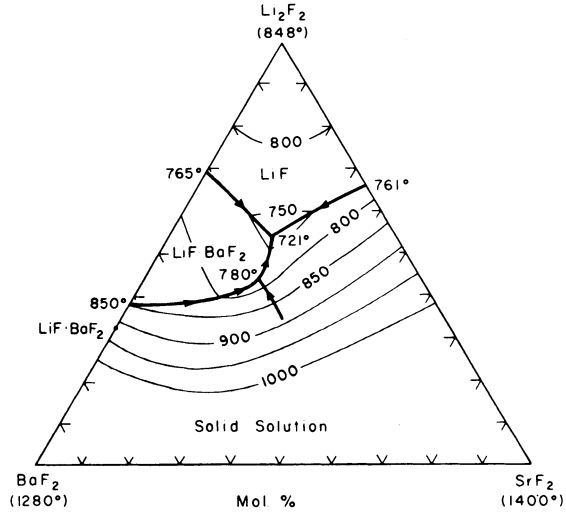


Fig. 201. The system $LiF–SrF_2–BaF_2$.

$E = 721°$, Li_2F_2–SrF_2–BaF_2 (53–21–26%); $P = 780°$, Li_2F_2–SrF_2–BaF_2 (43–24–33%).

The System NaF–MgF_2–CaF_2. See Fig. 202. Source: C. J. Barton, L. M. Bratcher, J. P. Blakely, and W. R. Grimes, ORNL, unpublished work (1956). $E = 745°$, NaF–MgF_2–CaF_2 (65–12–23%); $E = 905°$, NaF–MgF_2–CaF_2 (35–37–28%).

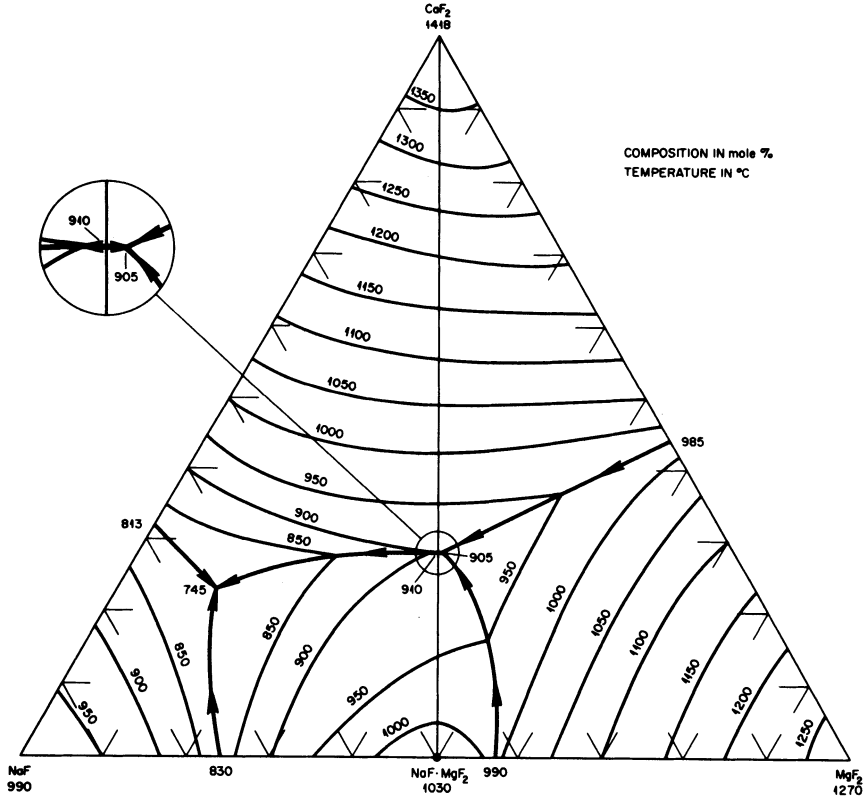


Fig. 202. The system NaF–MgF_2–CaF_2.

The System NaF–MgF_2–BaF_2. See Fig. 203. Source: G. Grube, *Z. Elektrochem.* **33**:483 (1927). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of invariant equilibrium
NaF	MgF_2	BaF_2		
32	18	50	750	Eutectic
8	21	71	800	Eutectic
8	36	56	850	Eutectic

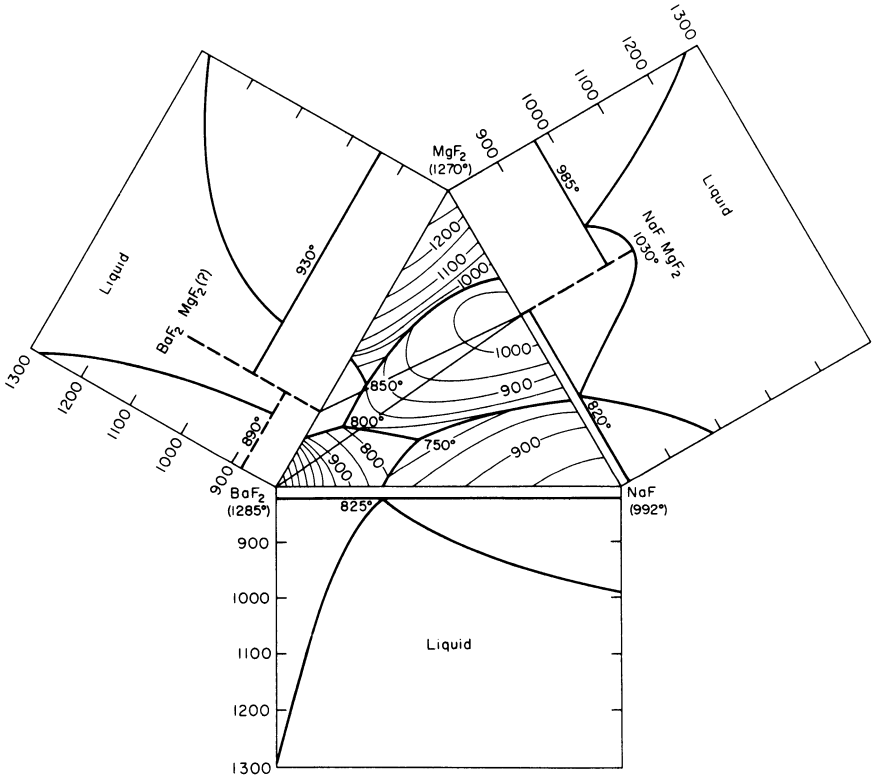


Fig. 203. The system $NaF–MgF_2–BaF_2$.

The System $NaF–CaF_2–SrF_2$. See Fig. 204. Source: G. A. Bukhalova, V. T. Berezhnaya, and Z. A. Mateiko, *Russ. J. Inorg. Chem.* (*English Transl.*) 7:1156 (1962). The eutectic invariant point separating the primary

phase field of NaF from that of the continuous solid solutions formed by CaF_2 and SrF_2 occurs at the composition Na_2F_2–CaF_2–SrF_2 (62–12.5–25.5%) and at 699°.

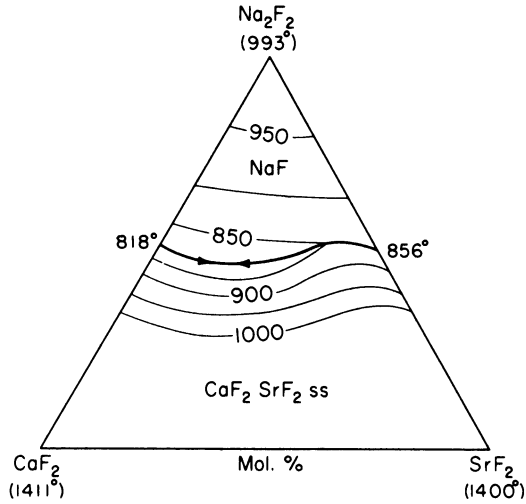


Fig. 204. The system NaF–CaF_2–SrF_2.

The System NaF–CaF_2–BaF_2. See Fig. 205. Source: G. A. Bukhalova, V. T. Berezhnaya, and A. G. Bergman, *Russ. J. Inorg. Chem.* (*English Transl.*) **6**:1197 (1961). Invariant equilibrium points: $E = 680°$, NaF–CaF_2–BaF_2 (53–14–33%); $P = 867°$, NaF–CaF_2–BaF_2 (18–35–47%).

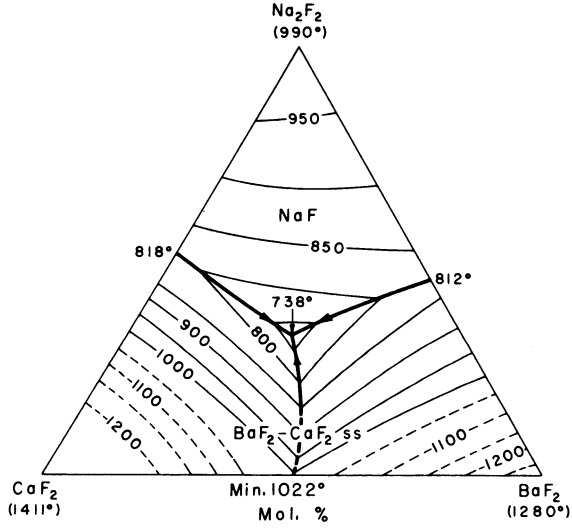


Fig. 205. The system NaF–CaF_2–BaF_2.

The System $NaF–SrF_2–BaF_2$. See Fig. 206. Source: G. A. Bukhalova, Z. A. Mateiko, and V. T. Berezhnaya, *Russ. J. Inorg. Chem.* (*English Transl.*) 7:856 (1962). Continuous solid solutions are formed by BaF_2 and SrF_2. The boundary separating the crystallization fields has a minimum at 804°, $Na_2F_2–SrF_2–BaF_2$ (44–26–30%).

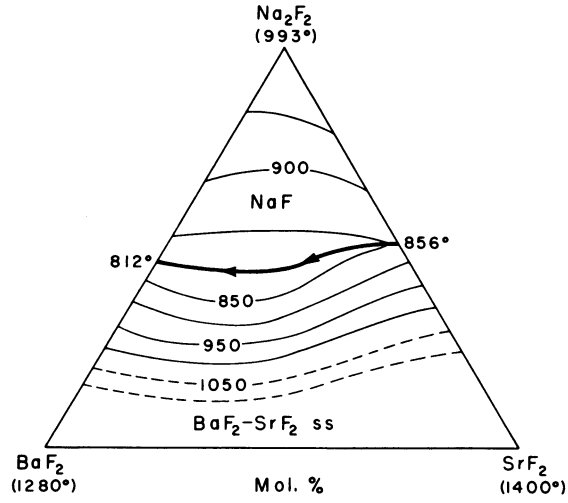


Fig. 206. The system $NaF–SrF_2–BaF_2$.

The System $KF–CaF_2–SrF_2$. See Fig. 207. Source: G. A. Bukhalova, V. T. Berezhnaya, and Z. A. Mateiko, *Russ. J. Inorg. Chem.* (*English*

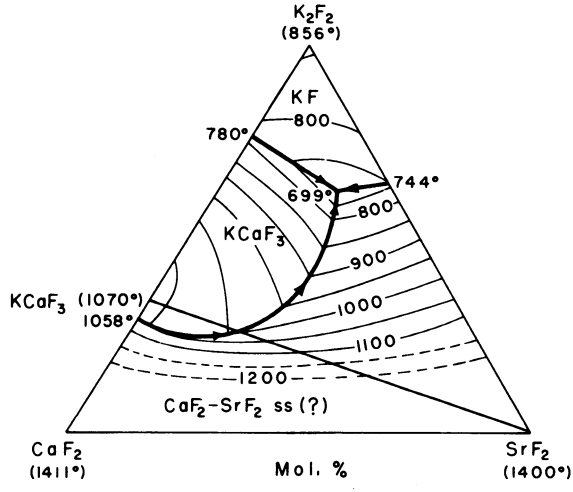


Fig. 207. The system $KF–CaF_2–SrF_2$.

Transl.) **7**:1156 (1962). The boundary separating the primary phase field of NaF from that of the continuous solid solutions formed by CaF_2 and SrF_2 has a minimum melting point of 804° at Na_2F_2–CaF_2–SrF_2 (48–36–16%). The eutectic at 699° occurs at K_2F_2–CaF_2–SrF_2 (62–12.5–25.5%).

The System KF–BaF_2–CaF_2. See Fig. 208. Source: G. A. Bukhalova, V. T. Berezhnaya, and A. G. Bergman, *Russ. J. Inorg. Chem.* (*English Transl.*) **6**:1197 (1961). $E = 680°$, K_2F_2–CaF_2–BaF_2 (53–14–33%); $P =$

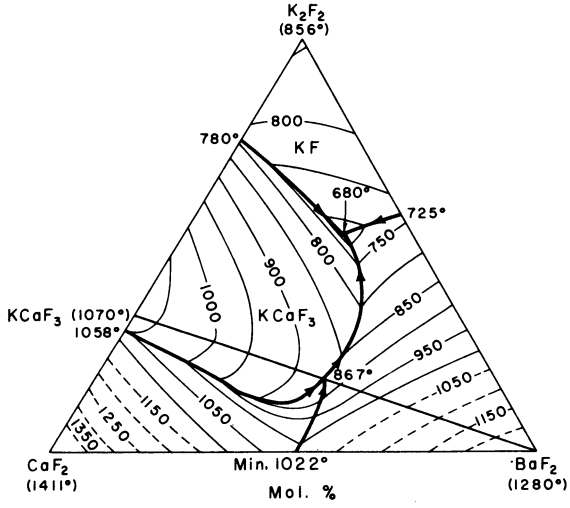


Fig. 208. The system KF–BaF_2–CaF_2.

867°, K_2F_2–CaF_2–BaF_2 (18–35–47%). It is noteworthy that the congruently melting compound $KCaF_3$ also becomes incongruently melting in the K, Ca, F, Cl system.

The System KF–SrF_2–BaF_2. See Fig. 209. Source: G. A. Bukhalova, Z. A. Mateiko, and V. T. Berezhnaya, *Russ. J. Inorg. Chem.* (*English Transl.*) **7**:1655 (1962). The boundary separating the crystallization fields of NaF and the BaF_2–SrF_2 solid solution has a minimum at 804°, Na_2F_2–SrF_2–BaF_2 (44–26–30%).

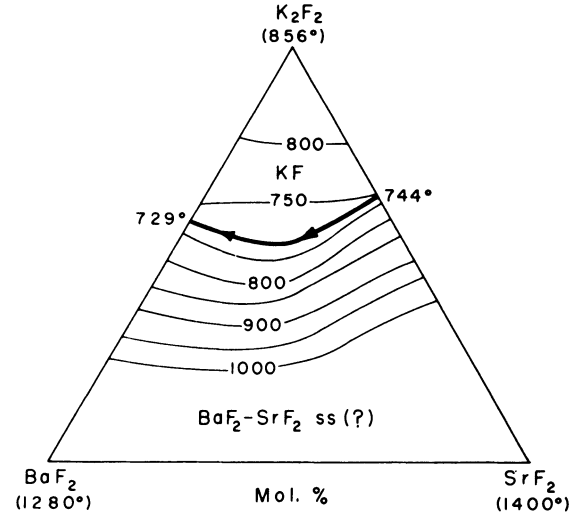


Fig. 209. The system $KF–SrF_2–BaF_2$.

The System $LiF–BeF_2–AlF_3$. See Fig. 210. Source: R. L. Boles and R. E. Thoma, Volatility Process Phase Studies—A Survey of Molten

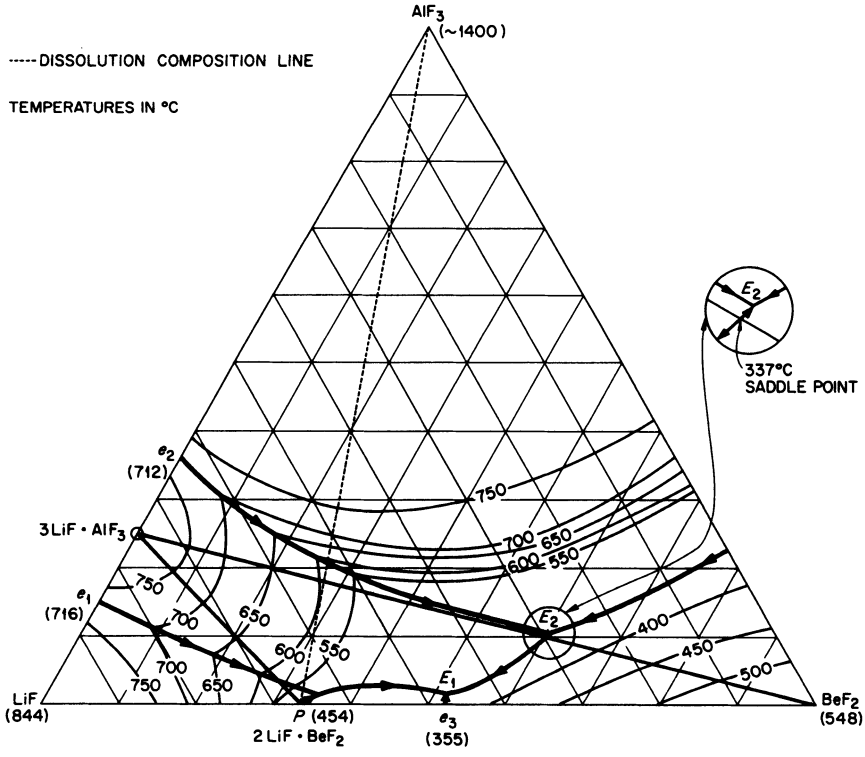


Fig. 210. The system $LiF–BeF_2–AlF_3$.

Fluoride Solvents Suitable for Dissolution of AlF_3, ORNL-TM-400, October 22, 1962. Invariant and singular equilibrium composition points were not listed by the authors. All the experimental data are included in the source reference.

The System NaF–MgF_2–AlF_3. See Fig. 211. Source: E. Vatslavik and A. I. Belyaev, *Russ. J. Inorg. Chem.* (*English Transl.*) **3**:324 (1958). No invariant equilibrium points were found to exist in the section of the phase diagram reported in the source reference. The section $3NaF \cdot AlF_3$–

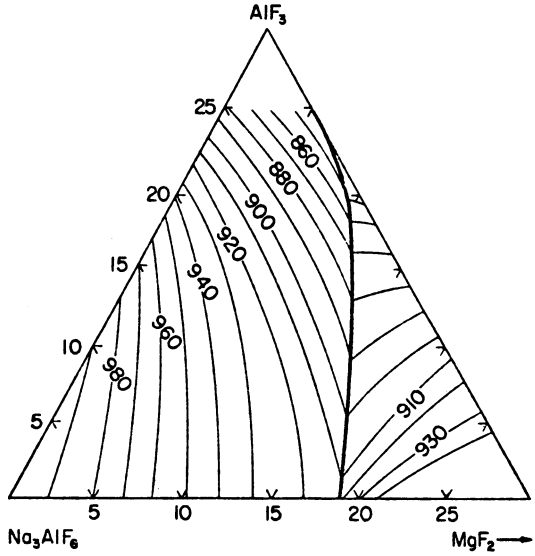


Fig. 211. The system NaF–MgF_2–AlF_3.

MgF_2, while not discussed in the source reference, is in fair agreement with previous work reported by A. M. Romanovskii and Ya. K. Berent, *Legkie Metal.* **6**:34 (1935). See Fig. 3449, *Phase Diagrams for Ceramists* (1969).

The System NaF–CaF_2–AlF_3. (See also the listing for this system immediately below.) See Fig. 212. Source: P. P. Fediotieff and W. P. Iljinskii, *Z. Anorg. Allgem. Chem.* **129**:106 (1923). The authors did not list the exact composition of the invariant points in the source reference. Later investigations by other workers established the phase relationships along the section $3NaF \cdot AlF_3$–CaF_2 (Fig. 207). The results of this work indicate that the NaF–CaF_2–AlF_3 ternary system has not yet been defined accurately.

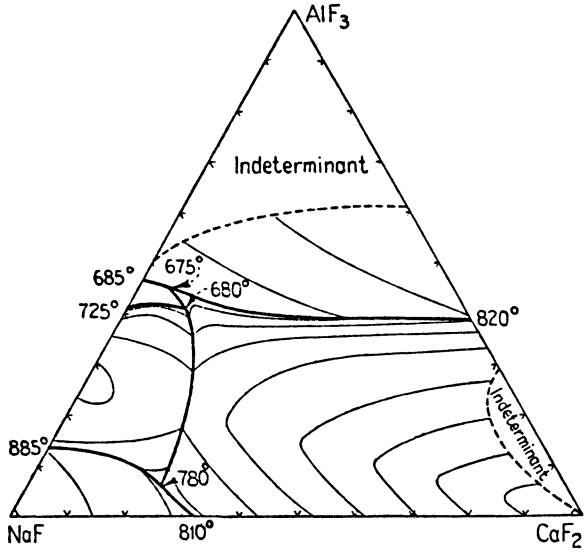


Fig. 212. The system $NaF–CaF_2–AlF_3$.

The System $NaF–CaF_2–AlF_3$. (See also the listing for this system immediately above.) See Fig. 213. Source: M. Verdan and R. Monnier,

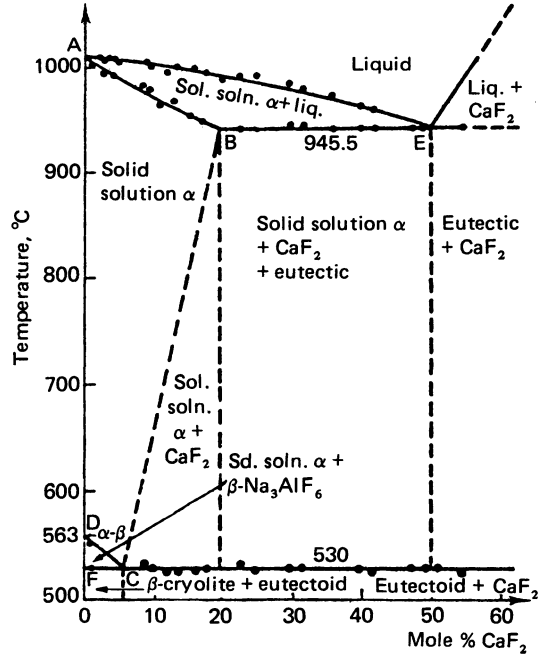


Fig. 213. The system $NaF–CaF_2–AlF_3$.

Rev. Intern. Hautes Temp. Refract. **9**:205 (1972). Invariant equilibrium compositions and temperatures are shown in the phase diagram.

The System $LiF–BeF_2–CeF_3$. See Fig. 214. Source: L. O. Gilpatrick, H. Insley, and C. J. Barton, MSR Program Semiannual Progress Report for Period Ending Aug. 30, 1970, ORNL-4622, p. 90.

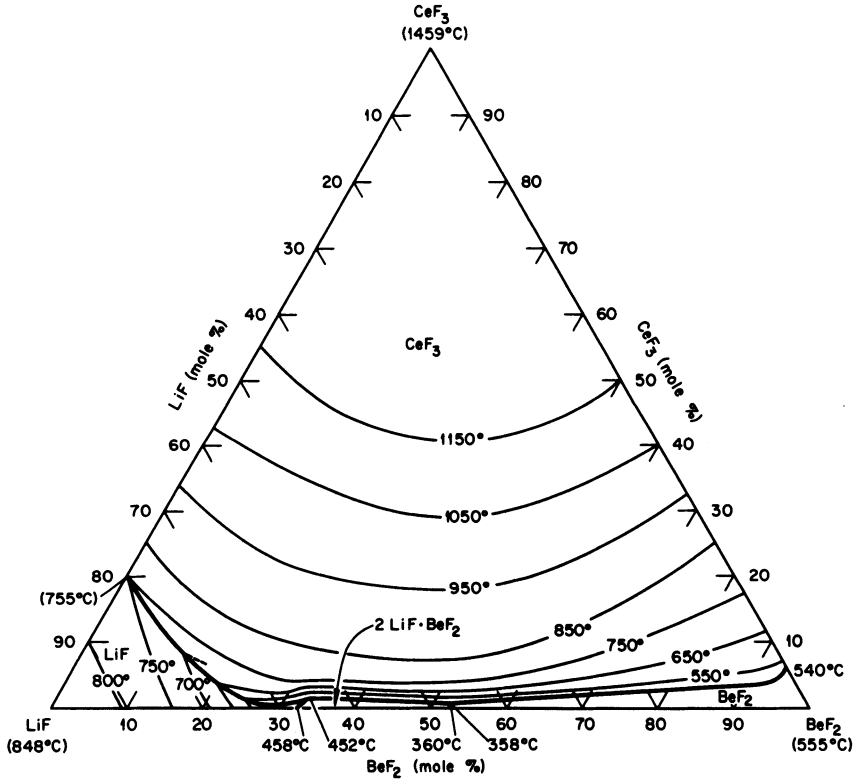


Fig. 214. The system $LiF–BeF_2–CeF_3$.

The System $KF–BeF_2–LaF_3$. See Fig. 215. Source: Hu Ch'ih-tsu and A. V. Novoselova, *Zh. Neorg. Khim.* **6**:2148 (1961). Invariant equilibrium points: $E = 610 \pm 5°$, $KF–BeF_2–LaF_3$ (78–5–17%); $P = 722 \pm 5°$, $KF–BeF_2–LaF_3$ (67.5–16.5–16%); $P = 667 \pm 5°$, $KF–BeF_2–LaF_3$ (71–12.5–16.5%).

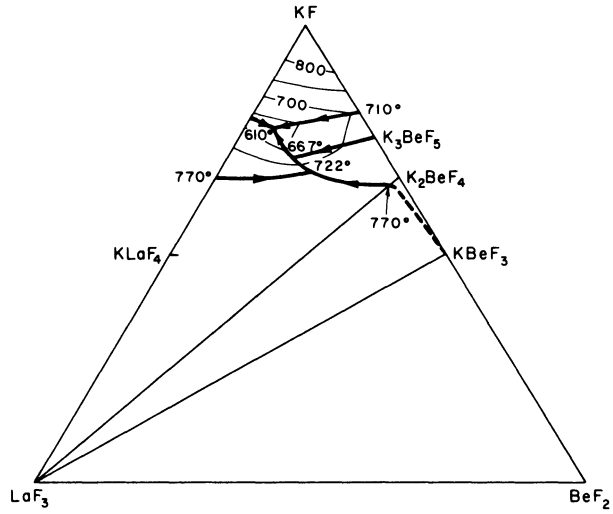


Fig. 215. The system $KF–BeF_2–LaF_3$.

The System $LiF–BeF_2–ZrF_4$. See Fig. 216. Source: R. E. Thoma, H. Insley, H. A. Friedman, and G. M. Hebert, *J. Nucl. Mater.* **27**:166 (1968). Invariant equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	BeF_2	ZrF_4		
75	5	20	480	Peritectic
73	13	14	470	Peritectic
67	29.5	3.5	445	Peritectic
64.5	30.5	5	428	Peritectic
48	50	2	355	Eutectic
47.5	10	42.5	466	Peritectic
44	18	38	460	Eutectic
27	46	27	532	Eutectic
2	88	10	532	—
27	60	13	540	—
25	54	21	955	Critical solution point
73	13.5	13.5	475	Boundary curve max.
44	34	22	470	Quasi-binary eutectic
17	79	4	~545	Boundary curve max.

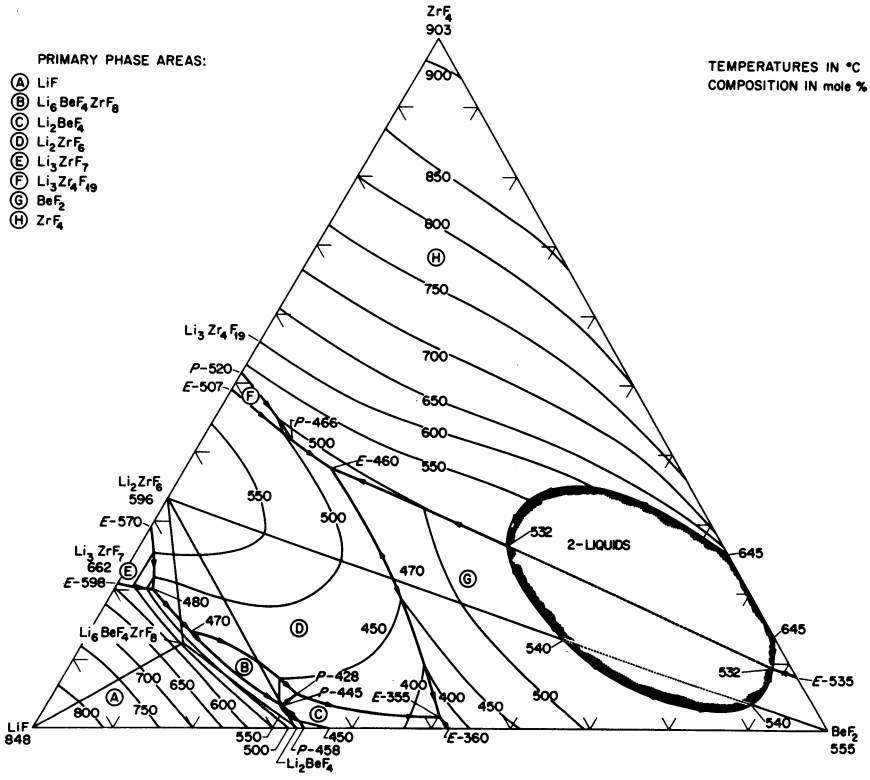


Fig. 216. The system LiF–BeF_2–ZrF_4.

Structural data: $6LiF \cdot BeF_2 \cdot ZrF_4$: tetragonal, $I4_1/amd$, $a = 6.57$, $c = 18.62$ [D. R. Sears and J. H. Burns, *J. Chem. Phys.* **41**:3478 (1964)].

The System NaF–BeF_2–ZrF_4. See Fig. 217. Source: R. E. Thoma, H. Insley, T. N. McVay, H. A. Friedman, and C. F. Weaver, ORNL, unpublished work (1960–61). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	BeF_2	ZrF_4		
70	26	4	535	Eutectic
57	40	3	335	Eutectic
55.5	39	5.5	340	Peritectic

(Continued)

Mole %			Temperature	Type of equilibrium
NaF	BeF_2	ZrF_4		
50	42.5	7.5	345	Peritectic
44	52	4	350	Peritectic
37	25	37	458	Eutectic
28	36	36	480	Peritectic

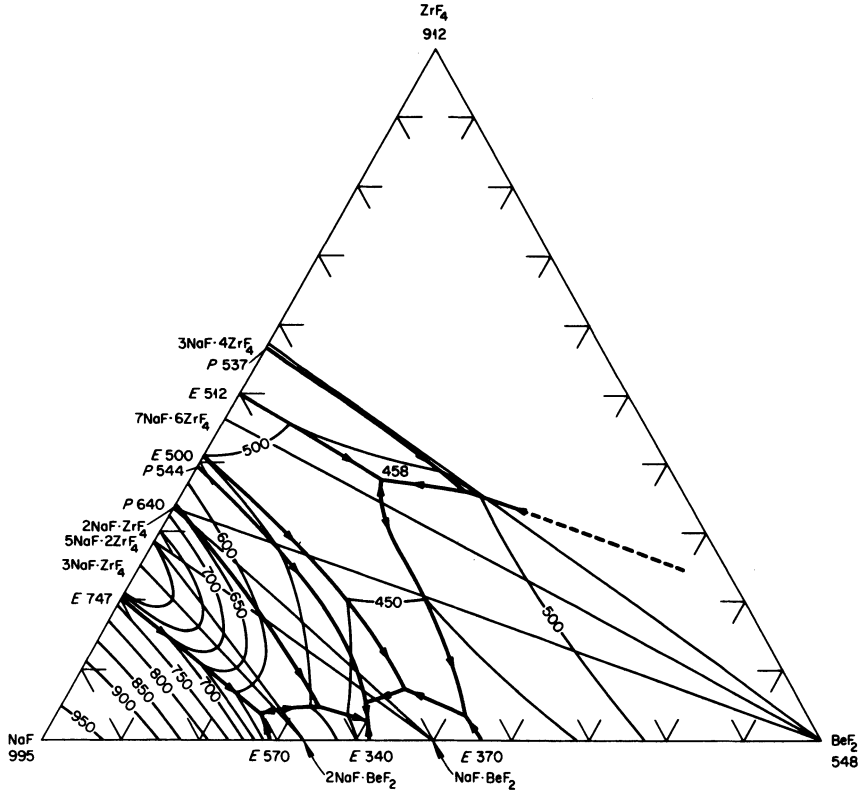


Fig. 217. The system $NaF–BeF_2–ZrF_4$.

The System $LiF–BeF_2–ThF_4$. See Fig. 218. Source: R. E. Thoma, H. Insley, H. A. Friedman, and C. F. Weaver, *J. Phys. Chem.* **64**:865 (1960). A review of the crystallization reactions occurring in the system

was reported by R. E. Thoma and J. E. Ricci, Fractional Crystallization Reactions in the System LiF–BeF_2–ThF_4, ORNL-TM-2596 (1969).

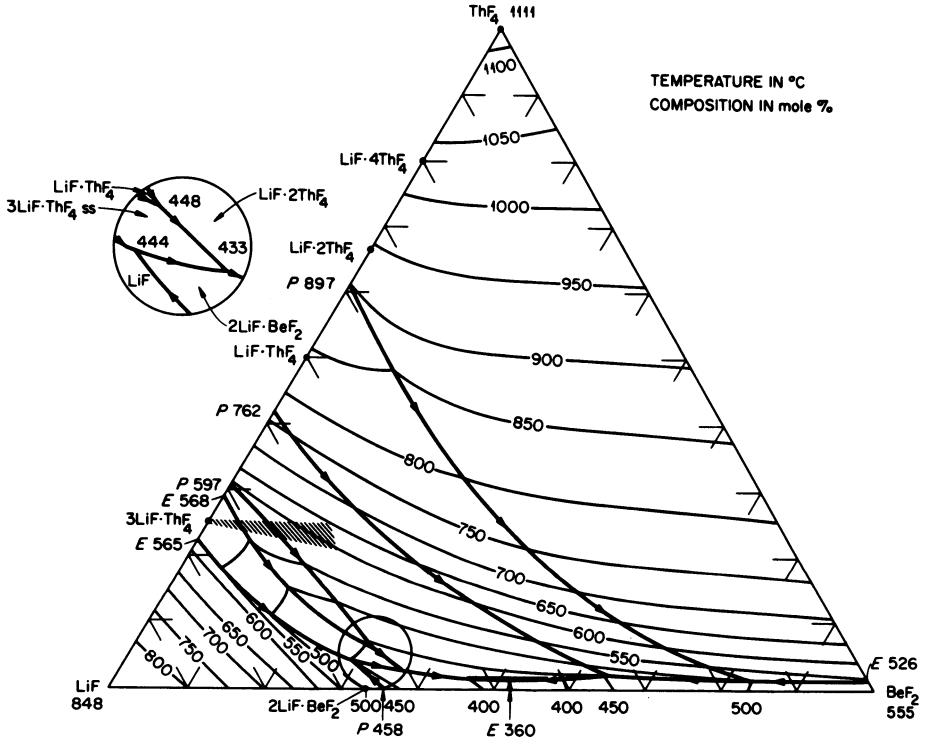


Fig. 218. The system LiF–BeF_2–ThF_4.

The System NaF–BeF_2–ThF_4. See Fig. 219. Source: R. E. Thoma, C. F. Weaver, H. Insley, and H. A. Friedman, *Nucl. Sci. Eng.* **19**:406 (1964). Invariant and singular equilibrium points are given in the table. Single crystal x-ray diffraction studies by G. D. Brunton of the phase inferred as $NaF \cdot BeF_2 \cdot 3ThF_4$ in the source reference established that its correct stoichiometry is $Na_3BeTh_{10}F_{45}$ [*Acta Cryst.* **B29**:2976 (1973)]. It is tetragonal, $P4_2/ncm$, $a = 11.803(3)$, $c = 23.420(5)$. The compound is isostructural with the phase observed in the NaF–BeF_2–UF_4 system (Fig. 73) and inferred to have the formula $NaF \cdot BeF_2 \cdot 3UF_4$. The results obtained by Brunton indicate that its correct stoichiometry is $Na_3BeU_{10}F_{45}$.

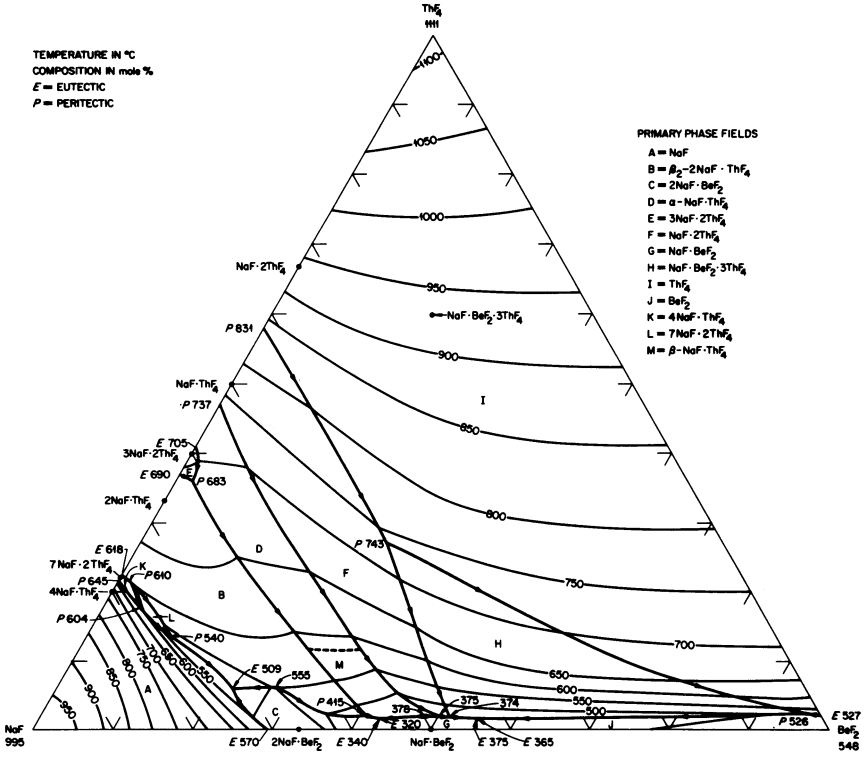


Fig. 219. The system NaF–BeF_2–ThF_4.

Mole %			Temperature	Type of equilibrium
NaF	BeF_2	ThF_4		
78	4	18	604	Peritectic
77	2.5	20.5	610	Peritectic
76	10	14	540	Peritectic
72	22	6	509	Eutectic
66.7	27	6.3	555	Eutectic
62	2	36	683	Peritectic
57	41	2	415	Peritectic
55	43	2	320	Eutectic
50	48	2	378	Maximum on boundary curve
48	50	2	375	Peritectic
47	51	2	374	Peritectic
43	55	2	365	Eutectic
42	31	27	743	Peritectic
2	96	2	526	Peritectic

The System LiF–BeF_2–UF_4. See Fig. 220. Source: L. V. Jones *et al.*, *J. Am. Ceram. Soc.* **45**:79 (1962). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	BeF_2	UF_4		
72	6	22	480	Peritectic (decomposition of $4LiF \cdot UF_4$ in the ternary system)
69	23	8	426	Eutectic
48	51.5	0.5	350	Eutectic
45.5	54	0.5	381	Peritectic
29.5	70	0.5	483	Peritectic

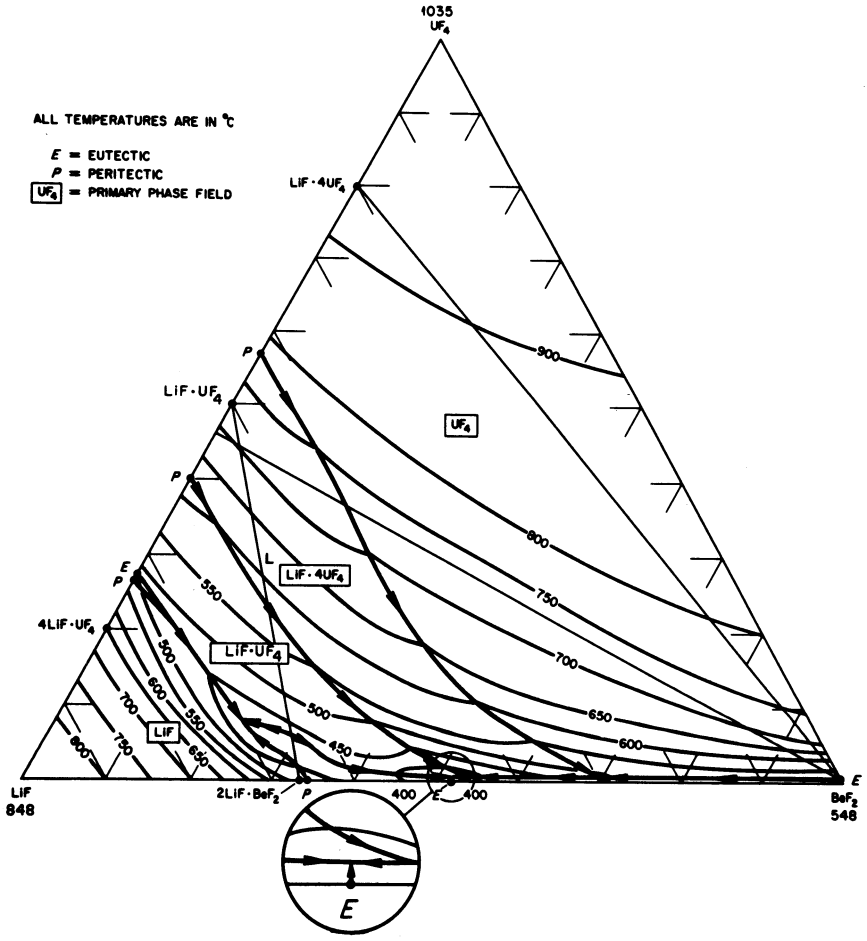


Fig. 220. The system LiF–BeF_2–UF_4.

The System $NaF–BeF_2–UF_4$. See Fig. 221. Source: J. F. Eichelberger, C. R. Hudgens, L. V. Jones, G. Pish, T. B. Rhinehammer, and L. J.

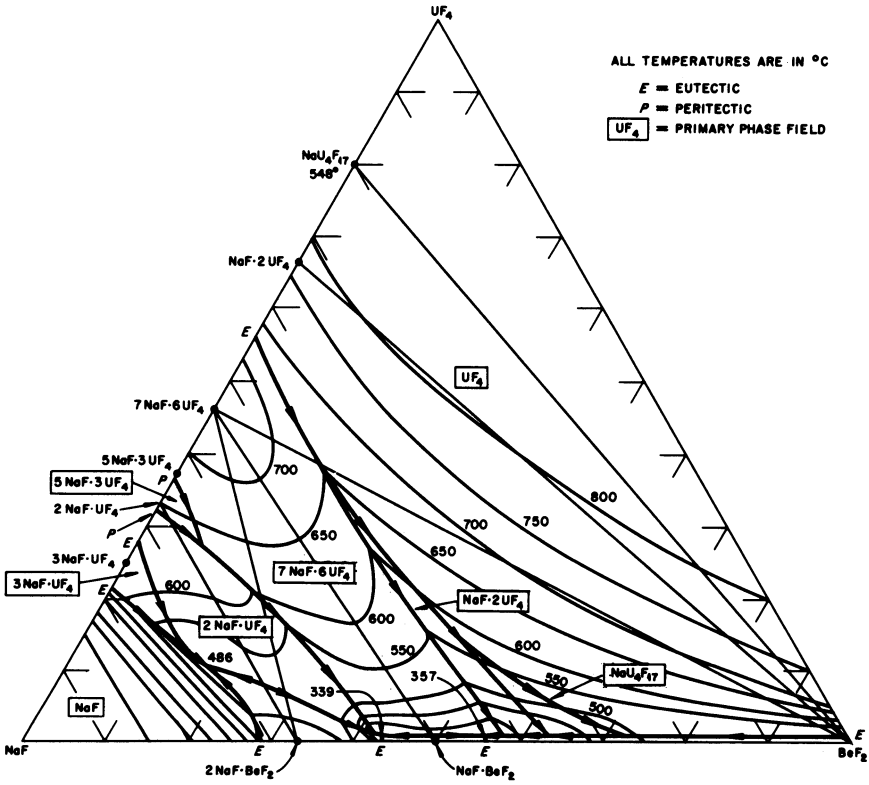


Fig. 221. The system $NaF–BeF_2–UF_4$.

Wittenberg, *J. Am. Ceram. Soc.* **46**:282 (1963). The minimum temperature in the quasibinary system $NaF \cdot BeF_2–7NaF \cdot 6UF_4$ occurs at 50.5 NaF–48.5–BeF_2–1.0 UF_4 (mole %) at 367°C. Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	BeF_2	UF_4		
74	12	14	500	Peritectic (decomposition of $3NaF \cdot UF_4$ in ternary system)
72.5	17	19.5	486	Eutectic
64.5	9	26.5	630	Peritectic (decomposition of $5NaF \cdot 3UF_4$ in ternary system)

(*Continued*)

Mole %			Temperature	Type of equilibrium
NaF	BeF_2	UF_4		
57	42	1	378	Peritectic
56	43.5	0.5	339	Eutectic
43.5	55.5	1	357	Eutectic
41	58	1	375	Peritectic
26	63	1	409	Peritectic
40	47	13	548	Peritectic
27	72	1	498	Peritectic

The System $NaF–PbF_2–UF_4$. See Fig. 222. Source: C. J. Barton, J. P. Blakely, G. J. Nessle, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1950–1951). Study of the system was discontinued before invariant equilibria were determined.

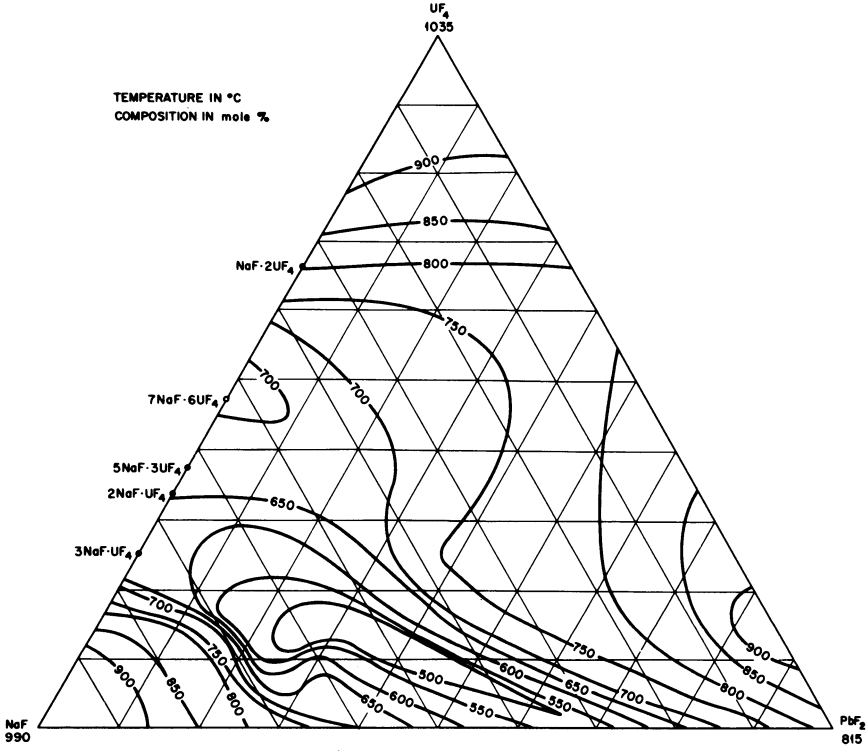


Fig. 222. The system $NaF–PbF_2–UF_4$.

The System $KF–BeF_2–ZrF_4$. See Fig. 223. Source: Chan Ngok Mai, Yu. M. Korenev, and A. V. Novoselova, *Russ. J. Inorg. Chem.* (*English*

Transl.) **10**:921 (1965). Composition–temperature sections are shown graphically in the source reference. In a subsequent report by the authors [*ibid.*, **10**:1683 (1965)], the phase relations along the limiting system $2KF \cdot BeF_2$–$3KF \cdot ZrF_4$ are shown. In that work the eutectic is listed at 720° and at 55 mole % $2KF \cdot BeF_2$.

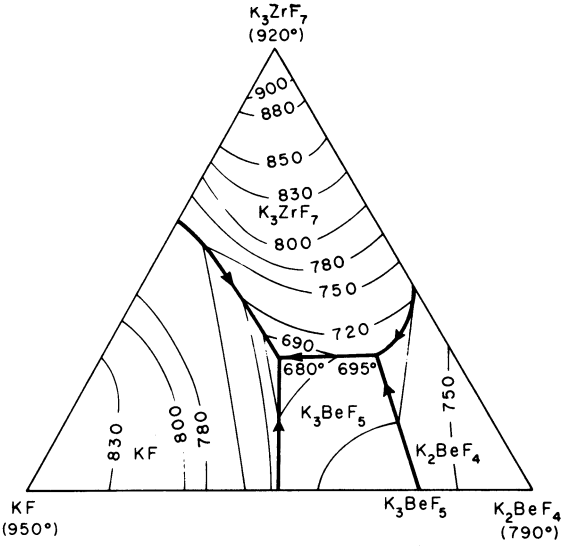


Fig. 223. The system KF–BeF_2–ZrF_4.

The System NaF–CrF_2–ZrF_4. See Fig. 224. Source: R. E. Thoma, B. S. Landau, and W. R. Grimes, ORNL, unpublished work (1957). The

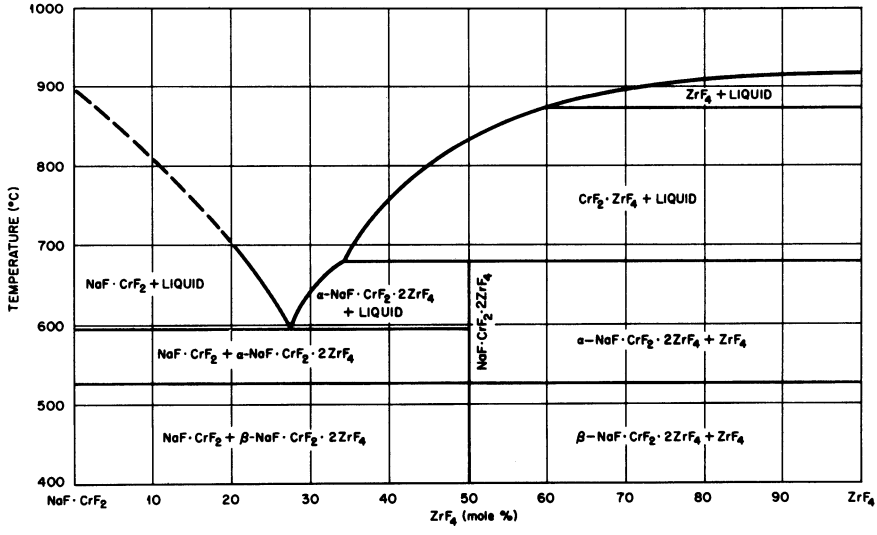


Fig. 224. The system NaF–CrF_2–ZrF_4.

ternary phase diagram of the system $NaF–CrF_2–ZrF_4$ has not been described except for the section $NaF \cdot CrF_2–ZrF_4$ shown here.

The System $KF–PbF_2–UF_4$. See Fig. 225. Source: C. J. Barton, J. P. Blakely, G. J. Nessle, L. M. Bratcher, and W. R. Grimes, ORNL, unpublished work (1950–1951). The investigation conducted by the group cited above did not progress to the point where liquid–solid phase relationships are established.

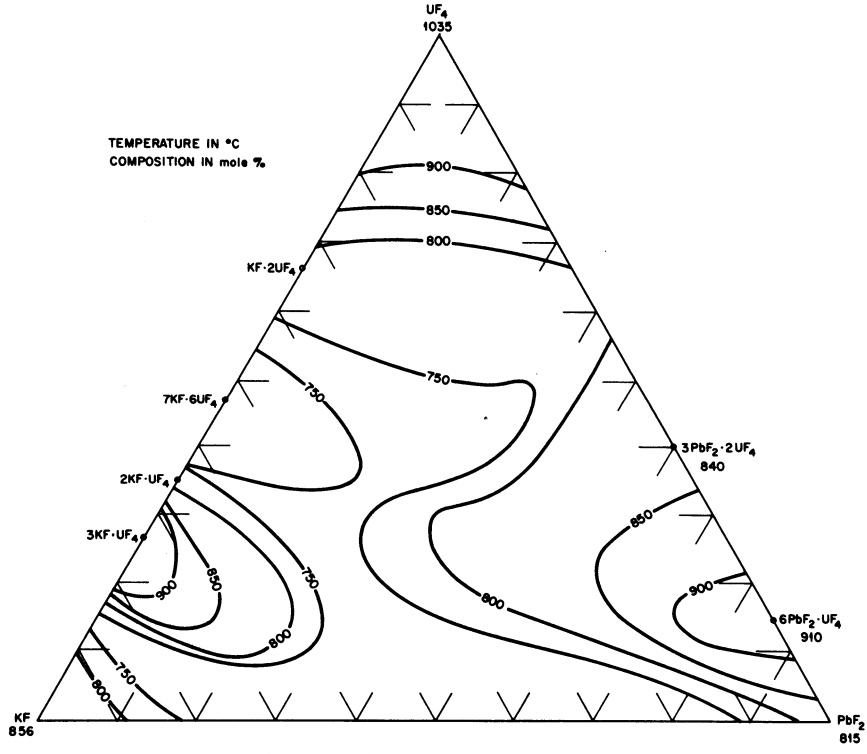


Fig. 225. The system $KF–PbF_2–UF_4$.

The System $NaF–CeF_3–ZrF_4$. See Fig. 226. Source: W. T. Ward, R. A. Strehlow, W. R. Grimes, and G. M. Watson, *J. Chem. Eng. Data* **5**:137 (1960). Enlargement indicates probable $NaF–ZrF_4$ primary phase fields.

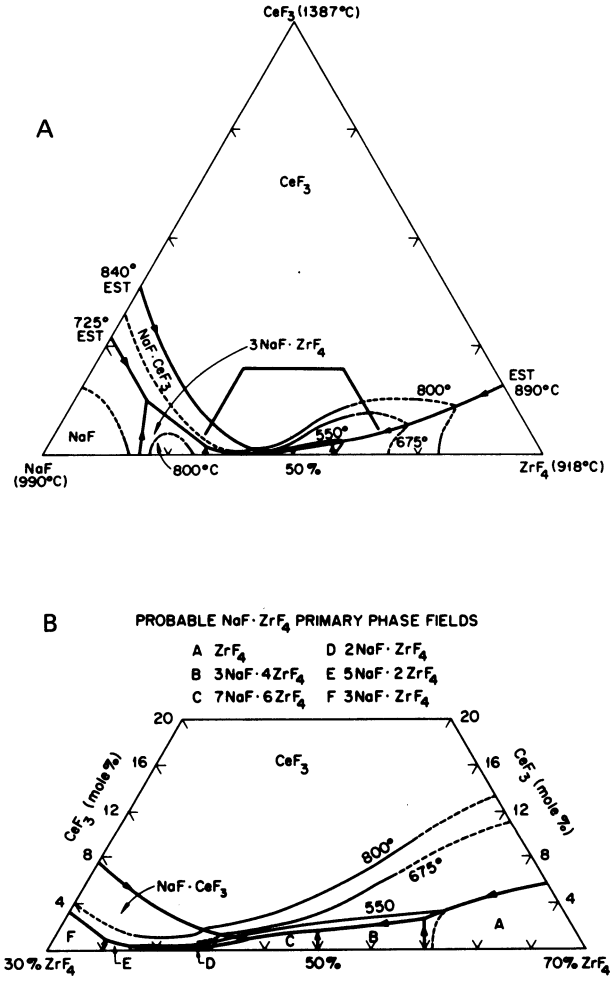


Fig. 226. The system $NaF–CeF_3–ZrF_4$. (B) gives detail of the region indicated in (A).

The System $KF–AlF_3–ZrF_4$. See Fig. 227. Source: R. E. Thoma, B. J. Sturm, and E. H. Guinn, Molten-Salt Solvents for Fluoride Volatility Processing of Aluminum-Matrix Nuclear Fuel Elements, ORNL-3594 (1964). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
KF	AlF_3	ZrF_4		
86.0		14.0	765	Eutectic
63.0		36.0	590	Peritectic
60.0		40.0	445	Peritectic
58.0		42.0	430	Eutectic
45.0		55.0	440	Eutectic
63.0	15.0	22.0	490	Eutectic
~55.0	~5.0	~40.0	400	Eutectic

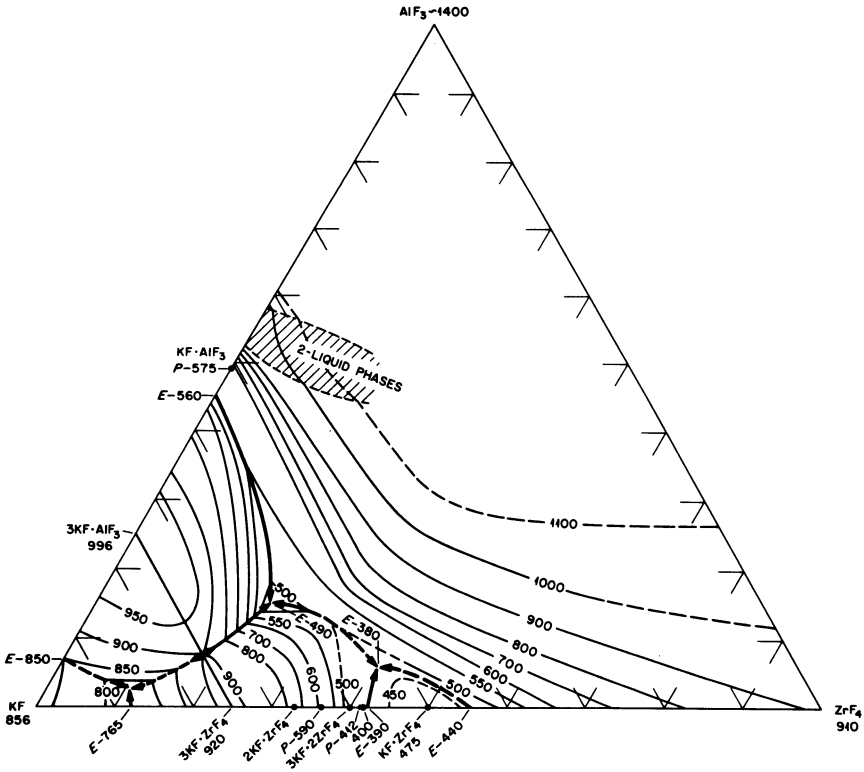


Fig. 227. The system $KF–AlF_3–ZrF_4$.

The System $NaF–ZrF_4–ThF_4$. See Fig. 228. Source: R. E. Thoma. H. A. Friedman, and H. Insley, ORNL, unpublished work (1955–1958), Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	ZrF_4	ThF_4		
77	3	20	645	Peritectic
75.5	2.5	22	618	Eutectic
63	6	31[a]	683	Decomposition point of $3NaF \cdot 2ThF_4$ in ternary system
65.5	21.5	13	622	Peritectic
65.5	24.5	10	615	Peritectic
64.5	25.5	10	593	Peritectic
58	38	4	538	Peritectic
58	40	2	495	Eutectic
53	42	4	510	Peritectic
49.2	48	2.5	505	Eutectic

[a] Approximate composition.

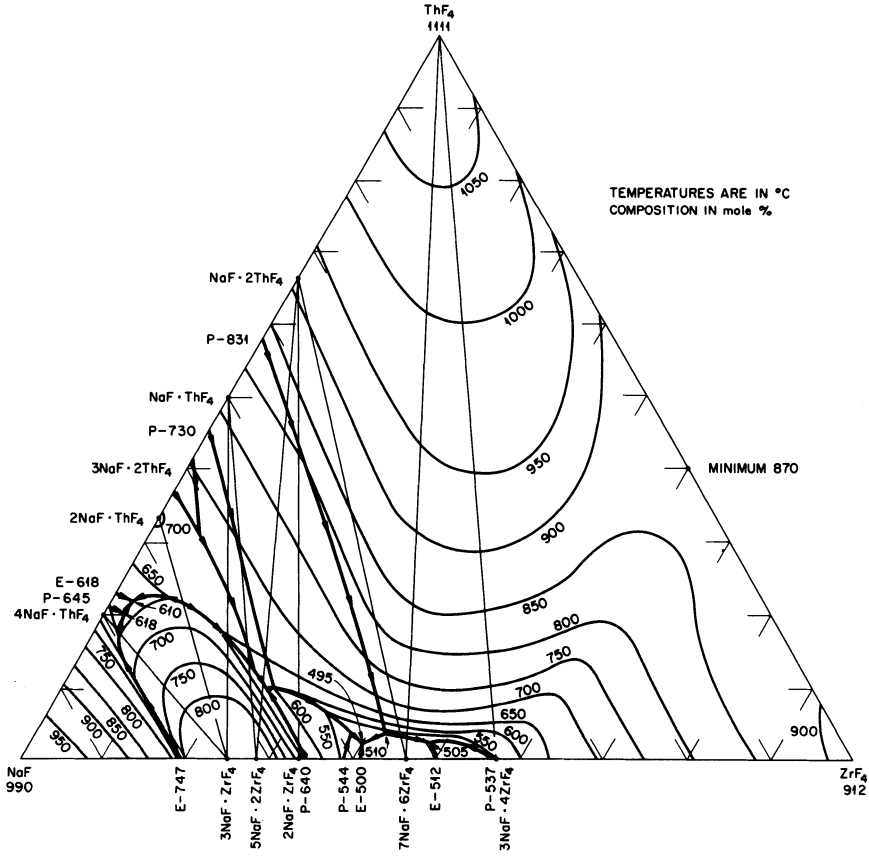


Fig. 228. The system $NaF–ZrF_4–ThF_4$.

The System $NaF–ZrF_4–UF_4$. See Fig. 229. Source: C. J. Barton, W. R. Grimes, H. Insley, R. E. Moore, and R. E. Thoma, *J. Phys. Chem.*

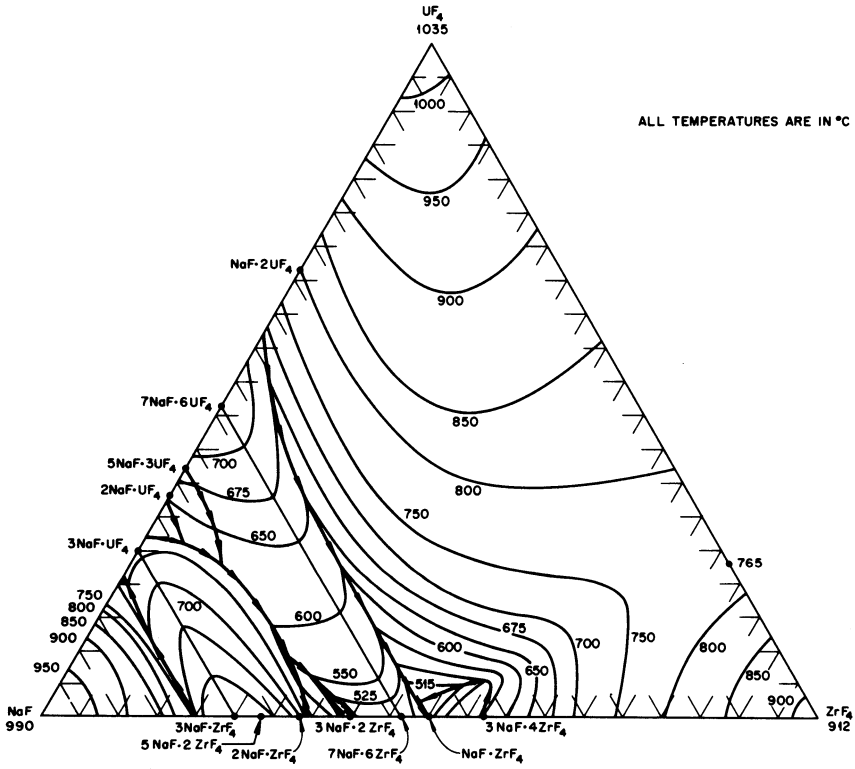


Fig. 229. The system $NaF–ZrF_4–UF_4$.

62:671 (1958). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
NaF	ZrF_4	UF_4		
69.5	4.0	26.5	646	Maximum temperature of boundary curve
68.5	5.5	26.0	640	Peritectic
65.5	12.0	22.5	613	Peritectic or decomposition
64.0	27.0	9.0	592	Peritectic
61.5	34.5	4.0	540	Peritectic
50.5	47	2.5	513	Peritectic

The System LiF–ThF_4–UF_4. See Fig. 230. Source: C. F. Weaver, R. E. Thoma, H. Insley, and H. A. Friedman, *J. Am. Ceram. Soc.* **43**:214 (1960). Complete tables of data for the system were reported in: C. F.

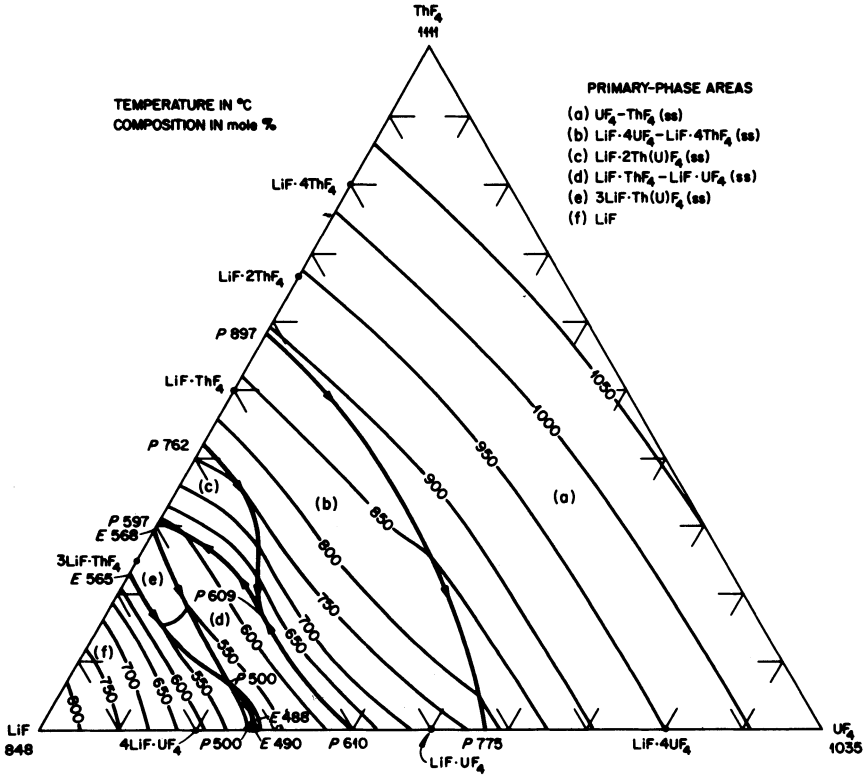


Fig. 230. The system LiF–ThF_4–UF_4.

Weaver *et al.*, Phase Equilibria in the Systems UF_4–ThF_4 and LiF–UF_4–ThF_4, ORNL-2719 (1959). Invariant and singular equilibrium points are given in the table.

Mole %			Temperature	Type of equilibrium
LiF	ThF_4	UF_4		
72.5	7.0	20.5	500	Peritectic
72.0	1.5	26.5	488	Eutectic
53	18	19	609	Peritectic

The System NaF–ThF_4–UF_4. See Fig. 231. Source: R. E. Thoma, H. Insley, G. M. Hebert, H. A. Friedman, and C. F. Weaver, *J. Am. Ceram. Soc.* **46**:37 (1963). The report by R. E. Thoma *et al.*, Phase Equilibria in the System NaF–ThF_4–UF_4, ORNL-3304 (1963), contains all of the

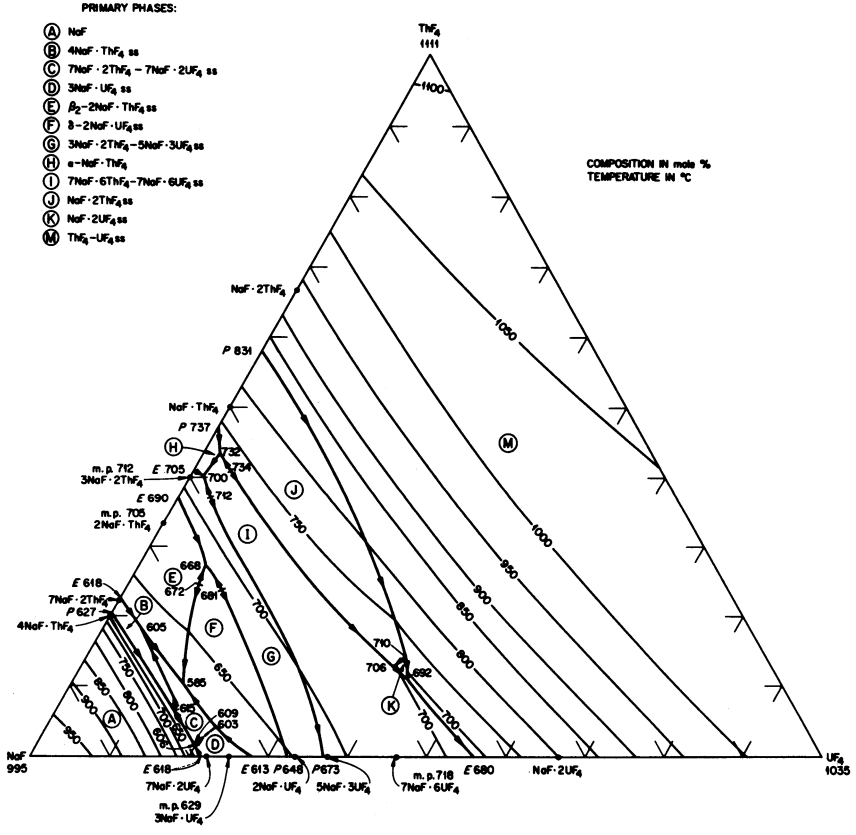


Fig. 231. The system NaF–ThF_4–UF_4.

experimental data used to construct the phase diagram. The table lists the composition and temperature of the singular and invariant equilibrium points in the system and, for clarity, the coexistent phases at each of these points.

Mole %			Temperature	Type of equilibrium	Phases present
NaF	ThF_4	UF_4			
78.5	6.5	15	615	Eutectic	NaF, $4NaF \cdot (Th, U)F_4$ ss, $7NaF \cdot 2(Th, U)F_4$ ss, liquid
78.5	5.5	16	616	Maximum on boundary curve	NaF, $7NaF \cdot 2(Th, U)F_4$ ss, liquid
78	7	15	616	Maximum on boundary curve	$4NaF \cdot (Th, U)F_4$ ss, $7NaF \cdot 2(Th, U)F_4$ ss, liquid
77	2.5	20.5	609	Maximum on boundary curve	$3NaF \cdot (U, Th)F_4$ ss, $7NaF \cdot 2(Th, U)F_4$ ss, liquid
77	18.5	4.5	605	Peritectic	$4NaF \cdot (Th, U)F_4$ ss, $7NaF \cdot 2(Th, U)F_4$ ss, β_2-$2NaF \cdot (Th, U)F_4$ ss, liquid
75.5	10.5	14	585	Eutectic	$7NaF \cdot 2(Th, U)F_4$ ss, β_2-$2NaF \cdot (Th, U)F_4$ ss, δ-$2NaF \cdot (U, Th)F_4$ ss, liquid
74.5	4	21.5	603	Peritectic	$3NaF \cdot (U, Th)F_4$ ss, $7NaF \cdot 2(Th, U)F_4$ ss, δ-$2NaF \cdot (U, Th)F_4$ ss, liquid
66.5	24.5	9	672	Maximum on boundary curve	β_2-$2NaF \cdot (Th, U)F_4$ ss, δ-$2NaF \cdot (U, Th)F_4$ ss, liquid
64.5	27	8.5	668	Eutectic	β_2-$2NaF \cdot (Th, U)F_4$ ss, δ-$2NaF \cdot (U, Th)F_4$ ss, $3NaF \cdot 2ThF_4$–$5NaF \cdot 3UF_4$ ss, liquid
64	24	12	681	Maximum on boundary curve	δ-$2NaF \cdot (U, Th)F_4$ ss, $3NaF \cdot 2ThF_4$–$5NaF \cdot 3UF_4$ ss
59	37	4	712	Maximum on boundary curve	$3NaF \cdot 2ThF_4$–$5NaF \cdot 3UF_4$ ss, $7NaF \cdot 6(Th, U)F_4$ ss, liquid
58.5	40	1.5	700	Eutectic	$3NaF \cdot 2ThF_4$–$5NaF \cdot 3UF_4$ ss, α-$NaF \cdot ThF_4$, $7NaF \cdot 6(Th, U)F_4$ ss, liquid
54.5	43.5	2	732	Peritectic	α-$NaF \cdot ThF_4$, $7NaF \cdot 6(Th, U)F_4$ ss, $NaF \cdot 2Th_{1-x}U_xF_4$ ss, liquid
54.5	41	4.5	734	Maximum on boundary curve	$7NaF \cdot 6(Th, U)F_4$ ss, $NaF \cdot 2Th_{1-x}U_xF_4$ ss, liquid

(*Continued*)

Mole %			Temperature	Type of equilibrium	Phases present
NaF	ThF_4	UF_4			
48	13	39	706	Peritectic	$7NaF \cdot 6(Th, U)F_4$ ss, $NaF \cdot 2Th_{1-x}U_xF_4$ ss, $NaF \cdot 2U_{1-x}ThF_4$ ss, liquid
47	11	42	692	Peritectic	$7NaF \cdot 6(Th, U)F_4$ ss, $NaF \cdot 2U_{1-x}Th_xF_4$ ss, $(Th, U)F_4$ ss, liquid
46	14	40	710	Peritectic	$NaF \cdot 2Th_{1-x}U_xF_4$ ss, $NaF \cdot 2U_{1-x}Th_xF_4$ ss, $(Th, U)F_4$ ss, liquid

The System BeF_2–ThF_4–UF_4. See Fig. 232. Source: C. F. Weaver, R. E. Thoma, H. A. Friedman, and G. M. Hebert, *J. Am. Ceram. Soc.*

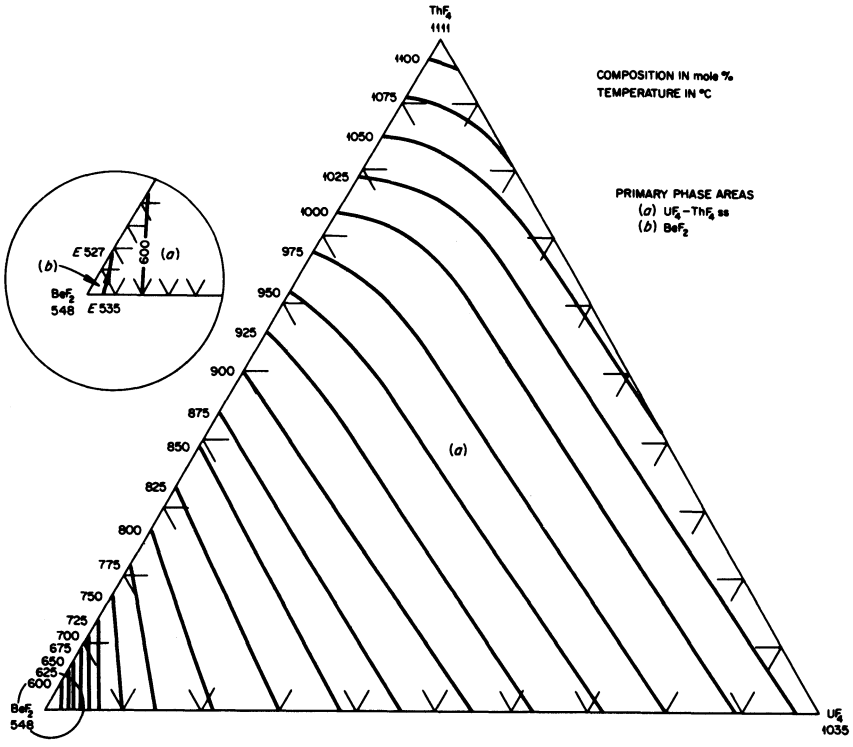


Fig. 232. The system BeF_2–ThF_4–UF_4.

44:146 (1961). No invariant equilibrium reactions occur in the system as a result of the formation of a continuous solid solution in the ThF_4–UF_4 binary system.

The System MgF_2–CaF_2–BaF_2. See Fig. 233. Source: V. T. Berezhnaya and G. A. Bukhalova, *Russ. J. Inorg. Chem.* (*English Transl.*) **6**:1091 (1961). Invariant equilibrium points: $E = 777°$, MgF_2–CaF_2–BaF_2 (27–

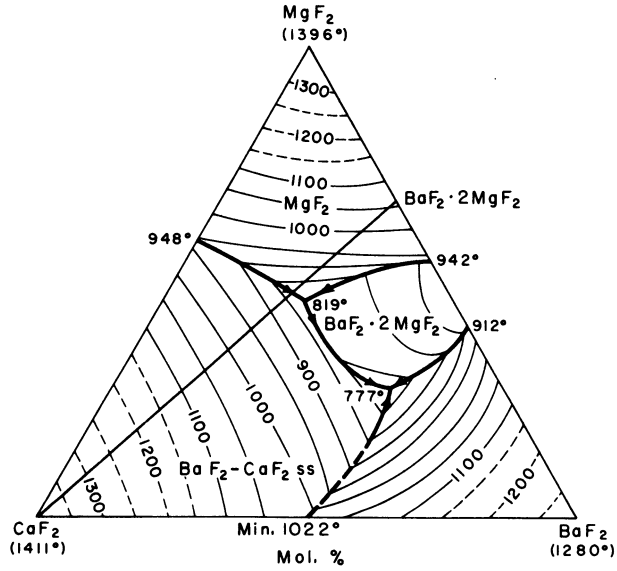


Fig. 233. The system MgF_2–CaF_2–BaF_2.

21–52%); $P = 819°$, MgF_2–CaF_2–BaF_2 (46–28–26%). A more recent version of the system was reported (in wt. %) by E. E. Lukashenko and E. A. Rentova, *Russ. J. Inorg. Chem.* (*English Transl.*) **15**:1766 (1970).

The $LiBaF_3$–KMF_3 Systems. See Figs. 234–236. Source: I. N. Belaev, S. A. Shilov, and S. Kh. Kulaeva, *Russ. J. Inorg. Chem.* (*English Transl.*) **15**:575 (1970). The systems with $KCdF_3$ and $KCaF_3$ are of the eutectic type, with the eutectics at $BaLiF_3$ 59.0 mole %, 695°, and 67.5 mole %, 703°, respectively. In the system with $KCdF_3$, there is restricted solid solubility in potassium fluorocadmate. The maximum solubility of $BaLiF_3$ at the eutectic temperature is 18 mole %. Continuous series of solid solutions are formed in systems in which the ratio of the radii of the interchanging ions does not exceed 1.34.

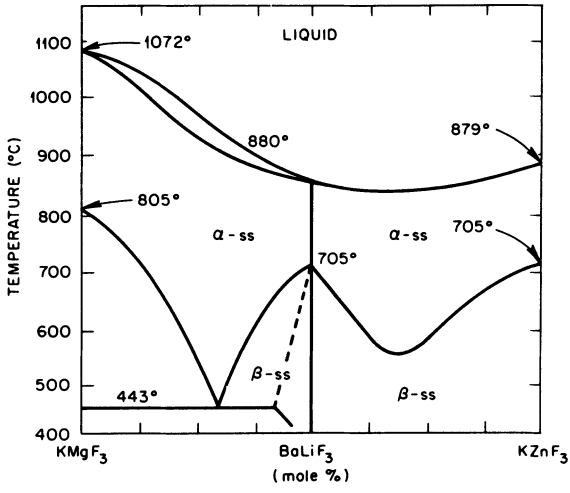


Fig. 234. The system $LiBaF_3$–$KMg(Zn)F_3$. α-s.s. = high-temperature solid solutions; β-s.s. = low-temperature solid solutions. (I) $KMgF_3$; (II) BaF_2.

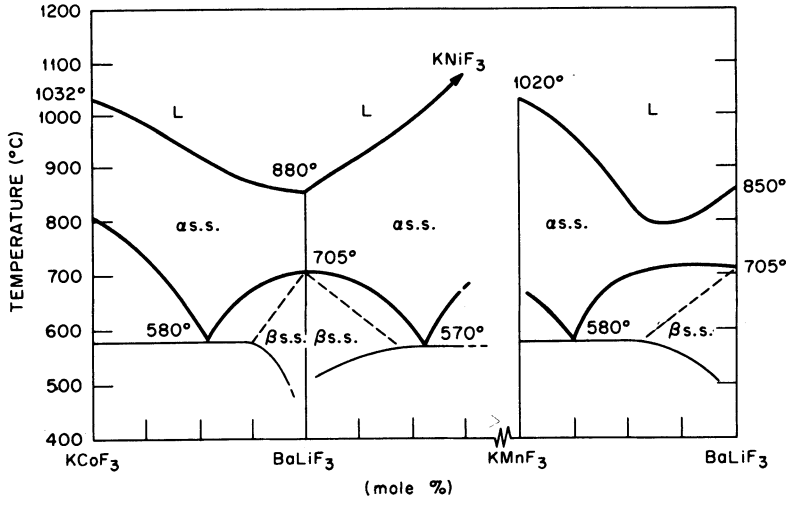


Fig. 235. The system $LiBaF_3$–$KNi(Co,Mn)F_3$. (I) $KCoF_3$; (II) $KNiF_3$; (III) $KMnF_3$; (IV) BaF_2.

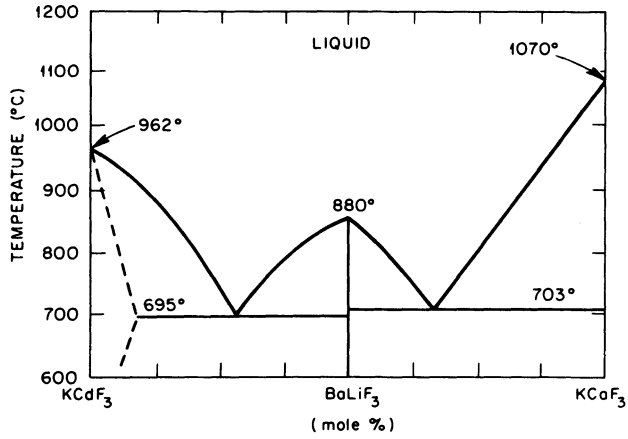


Fig. 236. The systems $LiBaF_3$–$KCdF_3$ and $LiBaF_3$–$KCaF_3$. (I) $LiBaF_3$; (II) $KCaF_3$; (III) BaF_2.

The System Li_2TiF_6–Na_2TiF_6–K_2TiF_6. See Fig. 237. Source: R. V. Chernov and N. M. Ermolenko, *Russ. J. Inorg. Chem.* (*English Transl.*) **17**:1340 (1972). The minimum crystallization temperature in this partial

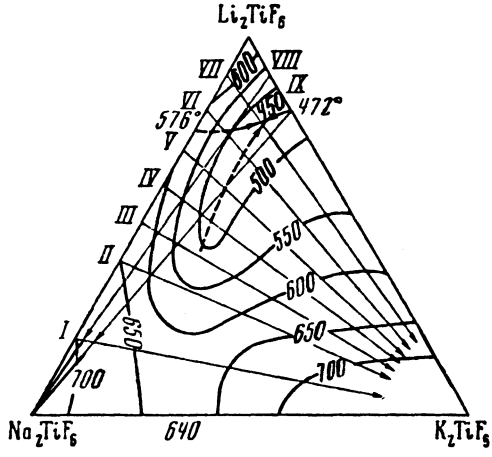


Fig. 237. The system Li_2TiF_6–Na_2TiF_6–K_2TiF_6.

section of the quaternary system was found to be 450° at the composition Li_2TiF_6–K_2TiF_6–Na_2TiF_6 (85–13–2 mole %). The authors noted that the eutectic temperature for the quaternary system is 420°.